GRADUATE LECTURE NOTES

Advanced Quantum Mechanics

Graduate Reference Text

Instructor-Directed Expanded Edition

Eight chapters with instructor supplementary material

October 16, 2026

Contents

List of Notation iii

1 Basic Concepts 1
- 1.1 Stern–Gerlach Experiment . . . 1
- 1.2 Linear Vector Spaces . . . 3
- 1.3 Eigenvalues and Eigenvectors of Operators (Matrices) . . . 6
- 1.4 Measurement . . . 8
- 1.5 Transformation of Representations . . . 15
- 1.6 Position and Momentum Representations . . . 16

2 Quantum Dynamics 23
- 2.1 Time Evolution and the Schrödinger Equation . . . 23
- 2.2 Schrödinger and Heisenberg Pictures . . . 27
- 2.3 One-Dimensional Linear Harmonic Oscillator . . . 30
- 2.4 Coherent States of the Harmonic Oscillator . . . 35
- 2.5 Squeezed States . . . 37
- 2.6 Schrödinger Equation in the Coordinate Representation and the WKB Approximation . . . 38
- 2.7 Feynman Path Integral . . . 41
- 2.8 Gauge Transformations . . . 45
- 2.9 Berry Phase . . . 49
- 2.10 Magnetic Monopoles and Charge Quantization . . . 51

3 Angular Momentum Theory 54
- 3.1 Rotations and Angular Momentum . . . 54
- 3.2 Spin Angular Momentum . . . 56
- 3.3 Ensembles and the Density Operator . . . 60
- 3.4 Composite Systems and Generalized Quantum Measurements . . . 63
- 3.5 Eigenvalues and Eigenstates of Angular Momentum . . . 67
- 3.6 Orbital Angular Momentum . . . 71
- 3.7 Coupling of Two Angular Momenta . . . 74
- 3.8 Spin Correlations and Bell's Inequality . . . 77
- 3.9 Quantum Teleportation: An Application of Entanglement and Measurement . . . 80

4 Approximation Methods 83
- 4.1 The Idea of Stationary Perturbation Theory . . . 83
- 4.2 Nondegenerate Perturbation Theory . . . 84
- 4.3 Degenerate Perturbation Theory . . . 88
- 4.4 Variational Method . . . 92
- 4.5 Strong-Coupling Schrödinger Equation . . . 93
- 4.6 Time-Dependent Perturbation Theory . . . 95

4.7 Interaction Picture and Perturbation Theory 98

5 Scattering Theory **101**
5.1 General Description 101
5.2 Green-Function Method 102
5.3 Lippmann–Schwinger Equation 105
5.4 Born Series 109
5.5 Partial-Wave Method 111

6 Identical Particles and Permutation Symmetry **115**
6.1 Identical Particles and Permutation Symmetry 115
6.2 Completely Symmetric and Antisymmetric States 117

7 Bosons, Fermions, and Second Quantization **119**
7.1 Bosons: Fock Space and Creation and Annihilation Operators 119
7.2 Fermions: Antisymmetric States and Operator Algebra 122

8 Relativistic Quantum Mechanics **125**
8.1 Relativistic Equations of Motion 125
8.2 Relativistic Invariance 128
8.3 Plane-Wave Solutions of the Free Dirac Equation 132

List of Notation

Throughout these notes, *hats* denote abstract operators, *bold symbols* denote spatial vectors, and a dagger denotes the Hermitian adjoint. Matrices in a specified basis may omit hats. Subscripts identify indices, states, or representations as indicated; local parameters are defined where introduced.

Notation	Meaning and global convention
$\|\psi\rangle\,,\ \langle\psi\|$	State ket and its dual bra.
$\langle\phi\|\psi\rangle\,,\ \left\langle\phi\middle\|\hat{A}\middle\|\psi\right\rangle$	Inner product; operator matrix element.
$\hat{A},\ A_{mn}=\left\langle m\middle\|\hat{A}\middle\|n\right\rangle$	Abstract operator; its matrix entries in a basis.
$\hat{A}^\dagger,\ \mathbf{1}$	Hermitian adjoint; identity operator.
$[A,B],\ \{A,B\}$	Commutator $AB-BA$; anticommutator $AB+BA$.
$\langle A\rangle_\psi,\ \langle A\rangle_\rho$	Expectations in a pure state and a mixed state.
$\mathrm{tr}(A)$	Trace, independent of basis.
$\delta_{mn},\ \delta^{(3)}(\boldsymbol{r})$	Kronecker and three-dimensional Dirac deltas.
$\mathrm{i},\ \hbar,\ c$	Imaginary unit; reduced Planck constant; light speed.
$\boldsymbol{r},\ \boldsymbol{p};\ \hat{\boldsymbol{r}},\ \hat{\boldsymbol{p}}$	Position and momentum vectors; corresponding operators.
$\psi(\boldsymbol{r},t)=\langle\boldsymbol{r}\|\psi\rangle$	Position-space wavefunction (time argument understood).
$\hat{H},\ \hat{U}(t,t_0)$	Hamiltonian; unitary time evolution.
$\hat{A}_{\mathrm{S}},\ \hat{A}_{\mathrm{H}},\ \hat{A}_{\mathrm{I}}$	Schrödinger, Heisenberg and interaction pictures.
$\hat{\rho},\ \rho(\boldsymbol{r},t)$	Density operator; position-space probability density.
ΔA	Standard deviation of observable A.
$\hat{\boldsymbol{J}},\ \hat{\boldsymbol{L}},\ \hat{\boldsymbol{S}}$	Total, orbital and spin angular momenta.
$\|j,m\rangle\,,\ \hat{J}_\pm$	Angular-momentum eigenstate; ladder operators.
$\sigma_i,\ \hat{\boldsymbol{s}}=\frac{\hbar}{2}\hat{\boldsymbol{\sigma}}$	Pauli matrices; spin-$\frac{1}{2}$ angular momentum.
$\hat{P}_{ij},\ \hat{\mathcal{P}}_\pm$	Particle exchange; Bose/Fermi symmetry projectors.
$\hat{a}_i,\ \hat{a}_i^\dagger$	Annihilation and creation in mode i (index optional).
$\hat{n}_i,\ \hat{N}=\sum_i\hat{n}_i$	Mode occupation and total number operators.
$\|n_1,n_2,\ldots\rangle$	Fock occupation-number basis for identical particles.
$\mathcal{T},\ \hat{T}$	Time ordering; scattering transition operator.
$f(\Omega),\ \mathrm{d}\sigma/\mathrm{d}\Omega$	Scattering amplitude; differential cross section.
$\hat{G}_0^\pm,\ \hat{G}^\pm$	Free and full resolvents; + is outgoing.
$x^\mu,\ g_{\mu\nu}$	Spacetime four-coordinate; Minkowski metric.
$\gamma^\mu,\ \overline{\psi}=\psi^\dagger\gamma^0$	Dirac gamma matrices; Dirac adjoint.

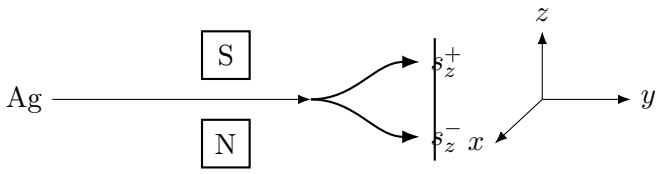
Conventions: σ_i denotes a Pauli matrix, while $\sigma(\Omega)$ may denote a scattering cross section; the index or argument distinguishes them. The relativistic discussion uses $\hbar=c=1$; factors of c elsewhere follow the units specified in the text.

Chapter 1

Basic Concepts

1.1 Stern–Gerlach Experiment

- The fundamental difference from classical mechanics is readily demonstrated.
- The central issue of quantum mechanics—the measurement problem—is readily demonstrated.
- A two-level system is the most quantum-mechanical system.



1) Results

A heated beam of silver atoms splits into two beams after passing through an inhomogeneous magnetic field.

2) Analysis

- The splitting is caused by interaction with the magnetic field and must therefore be due to the magnetic moment $\boldsymbol{M}$ of the atoms. The interaction potential is $V = -\boldsymbol{M} \cdot \boldsymbol{B}$.
- The magnetic moment is proportional to the angular momentum $\boldsymbol{J}$: $\boldsymbol{M} \propto \boldsymbol{J}$.
- The force on the atoms is

$$\boldsymbol{F} = -\boldsymbol{\nabla} V = M_z \frac{\partial B_z}{\partial z} \boldsymbol{e}_z \propto J_z \frac{\partial B_z}{\partial z} \boldsymbol{e}_z.$$

The beam separates symmetrically into an upper and a lower component. This indicates that the angular momentum along the direction of the magnetic field (the z direction) has only two components of equal magnitude and opposite directions.

If this angular momentum were produced by the rotation of the atoms themselves, the angular-momentum directions of the thermal atoms would be randomly distributed. After a large number of atoms passed through the magnetic field, their distribution on the screen would be symmetric and continuous, rather than consisting of two separated components. Thus the force does not originate from orbital angular momentum.

A silver atom has 47 electrons. Of these, 46 occupy filled shells with a spherically symmetric distribution and have no net angular momentum. The angular momentum of the silver atom is entirely due to the remaining valence electron. The separated two-component distribution shows that it is caused by the intrinsic angular momentum of the valence electron, denoted by

$\boldsymbol{s}$. The component s_z has only two values of equal magnitude and opposite direction, s_z^+ and s_z^-.

3) Quantum properties

- Spin angular momentum exists as an intrinsic physical quantity (independent of space and time).
- The values of spin angular momentum are discrete.
- The magnetic field performs a measurement. A magnetic field in the z direction measures the angular momentum in the z direction.

4) Cascaded Stern–Gerlach experiments

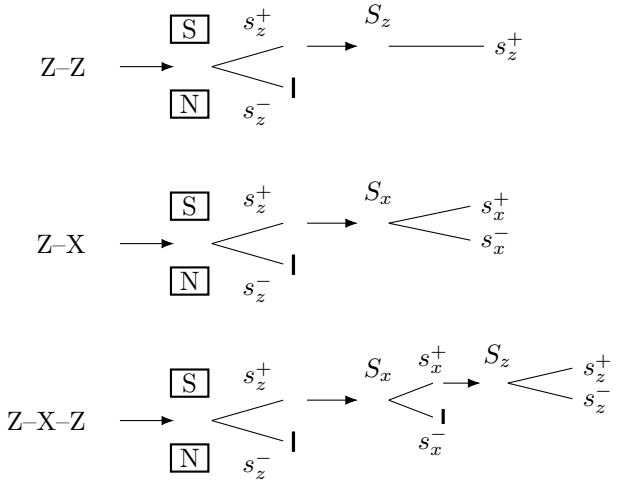


Figure 1.

The incident atomic beam passes successively through two magnetic fields in the z direction, as shown in the upper part of Fig. 1. Before the second magnetic field, s_z has the definite value s_z^+. Therefore the force on the atoms in that field is definite, and s_z still has the definite value s_z^+ after the second field.

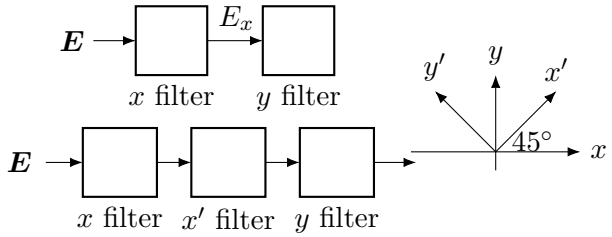
Now let the incident atomic beam pass through two magnetic fields, in the z and x directions, as shown in the middle of Fig. 1. In the second magnetic field the atoms experience a force $\boldsymbol{F} \propto J_x(\partial B_x/\partial x)\boldsymbol{e}_x$. The beam splits after the second magnetic field, indicating that s_x has two components s_x^+ and s_x^- before that field, although s_z has the definite value s_z^+.

- Quantum property: when s_z has a definite value, s_x does not. s_z and s_x cannot have definite values simultaneously.

Next, let the beam pass successively through three magnetic fields in the z, x, and z directions, as shown in the lower part of Fig. 1. Finally one observes two components s_z^+ and s_z^-. Thus s_z has two components before the third magnetic field, even though s_x has the definite value s_x^+.

- Quantum property: when s_x has a definite value, s_z likewise does not. s_x and s_z cannot have definite values simultaneously.

5) Analogy with classical electromagnetic waves (matter particles and light waves)



An electromagnetic wave propagating along the z direction disappears completely after passing successively through a filter that transmits only the x component (x filter) and a filter that transmits only the y component (y filter):

$$\begin{aligned} & \boldsymbol{E}(\boldsymbol{r},t) = E_0(\boldsymbol{e}_x + \boldsymbol{e}_y)\cos(kz-\omega t), \\ \xrightarrow{x \text{ filter}} \quad & E_0\boldsymbol{e}_x \cos(kz-\omega t), \\ \xrightarrow{y \text{ filter}} \quad & 0. \end{aligned}$$

Place an x' filter between the x and y filters, with x' at 45° to both x and y. An electromagnetic wave in the y direction can still be observed at the end:

$$\begin{aligned} & \boldsymbol{E}(\boldsymbol{r},t) = E_0(\boldsymbol{e}_x + \boldsymbol{e}_y)\cos(kz-\omega t), \\ \xrightarrow{x \text{ filter}} \quad & E_0\boldsymbol{e}_x \cos(kz-\omega t) = \frac{E_0}{\sqrt{2}}(\boldsymbol{e}_{x'} - \boldsymbol{e}_{y'})\cos(kz-\omega t), \\ \xrightarrow{x' \text{ filter}} \quad & \frac{E_0}{\sqrt{2}}\boldsymbol{e}_{x'}\cos(kz-\omega t) = \frac{E_0}{2}(\boldsymbol{e}_x + \boldsymbol{e}_y)\cos(kz-\omega t), \\ \xrightarrow{y \text{ filter}} \quad & \frac{E_0}{2}\boldsymbol{e}_y \cos(kz-\omega t). \end{aligned}$$

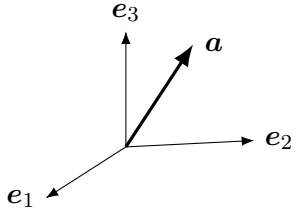
Similarity: s_x, s_y, s_z and $E_{x'}, E_{y'}$ can all be regarded as two-component vectors.

Difference: $\boldsymbol{s}$ is intrinsic angular momentum, a quantum-mechanical quantity; $\boldsymbol{E}$ is a spatially dependent mechanical quantity, a classical-mechanical quantity.

1.2 Linear Vector Spaces

From the preceding section, the projection s_n of the electron's spin angular momentum along any direction n can take only two values. We may regard s_n as a two-dimensional vector. To establish the mathematical description of quantum mechanics, we first discuss linear vector spaces.

1) Three-dimensional vector space



Arbitrary vector: $\boldsymbol{a}$.

Basis vectors: $\boldsymbol{e}_n$, $n = 1, 2, 3$.

Completeness of the basis:

$$\boldsymbol{a} = \sum_{n=1}^{3} a_n \boldsymbol{e}_n.$$

Inner product:

$$\boldsymbol{a} \cdot \boldsymbol{b} = \sum_{n,m} a_n b_m \boldsymbol{e}_n \cdot \boldsymbol{e}_m.$$

Squared norm of a vector: $\boldsymbol{a} \cdot \boldsymbol{a} \geq 0$.

If the basis vectors are orthonormal, $\boldsymbol{e}_n \cdot \boldsymbol{e}_m = \delta_{nm}$, the inner product has the matrix form

$$\boldsymbol{a} \cdot \boldsymbol{b} = \sum_{n} a_n b_n = \widetilde{a}\, b, \quad b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}, \quad a = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \quad \widetilde{a} = (a_1, a_2, a_3),$$

where $\widetilde{a}$ is the transpose of a.

Components (matrix elements) of a vector:

$$a_n = \boldsymbol{e}_n \cdot \boldsymbol{a}.$$

The vector $\boldsymbol{a}$ is its abstract or general form. The matrix a is the representation of $\boldsymbol{a}$ in a specific coordinate system (representation). Its matrix elements depend on the choice of basis vectors; for example, Cartesian and spherical coordinate systems give different representations.

Operations on a vector, such as translations and rotations (operators), give another vector in the same three-dimensional space: $\hat{T}\boldsymbol{a} = \boldsymbol{b}$.

2) Hilbert space

Extend the three-dimensional vector space to a complex vector space of arbitrary dimension:

$$\text{three dimensions} \longrightarrow \text{arbitrary finite, infinite, or continuous dimension,}$$

$$\text{constant vectors} \longrightarrow \text{complex-function vectors.}$$

Use Dirac notation (a ket) to denote an arbitrary vector: $|a\rangle$.

For a complex vector, introduce the bra $\langle a|$ to represent its complex-conjugate vector. The bra and ket are not independent; they are conjugate to each other:

$$|a\rangle \longleftrightarrow \langle a| .$$

A vector can be represented by either a ket $|a\rangle$ or a bra $\langle a|$.

In a complex-function vector space, constants are generally complex, so

$$\alpha\, |a\rangle \longleftrightarrow \langle a|\, \alpha^*.$$

For an operation (operator) $\hat{T}$ in the complex space,

$$\hat{T}\, |a\rangle \longleftrightarrow \langle a|\, \hat{T}^\dagger.$$

$\hat{T}^\dagger$ is the left operator corresponding to the right operator $\hat{T}$, called the Hermitian-conjugate operator. Note that

$$\hat{T}\hat{F}\, |a\rangle = \hat{T}(\hat{F}\, |a\rangle) \longleftrightarrow \langle a|\, \hat{F}^\dagger \hat{T}^\dagger.$$

We now choose a particular representation, taking an N-dimensional discrete space as an example.

Basis vectors: $|n\rangle$, $n = 1, 2, \ldots$.

Expansion in a complete basis (concrete representation of a vector):

$$|a\rangle = \sum_n a_n |n\rangle , \qquad \langle a| = \sum_n a_n^* \langle n| .$$

Inner product:

$$\langle a|b\rangle = \sum_{n,m} a_n^* b_m \langle n|m\rangle ,$$

which is a complex number.

Squared norm: $\langle a|a\rangle \geq 0$. (Only when $|a\rangle$ and $\langle a|$ are defined as conjugates can the squared norm be guaranteed nonnegative.)

Normalized vector: defining $|\widetilde{a}\rangle = |a\rangle / \sqrt{\langle a|a\rangle}$ gives $\langle \widetilde{a}|\widetilde{a}\rangle = 1$; the vector is said to be normalized.

Orthonormality of basis vectors: $\langle n|m\rangle = \delta_{nm}$.

Matrix form of the inner product:

$$\langle a|b\rangle = \sum_n a_n^* b_n = a^\dagger b, \qquad b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_N \end{pmatrix}, \quad a = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_N \end{pmatrix},$$

where the Hermitian conjugate (complex-conjugate transpose) of the column matrix a is the row matrix

$$a^\dagger = (a_1^*, a_2^*, \ldots, a_N^*).$$

Evidently,

$$\langle a|b\rangle = \langle b|a\rangle^* , \qquad \left\langle a\middle|\hat{T}\middle|b\right\rangle = \left\langle b\middle|\hat{T}^\dagger\middle|a\right\rangle^* .$$

Components (matrix elements) of a vector: multiplying $|a\rangle = \sum_n a_n |n\rangle$ from the left by $\langle m|$ gives

$$a_m = \langle m|a\rangle .$$

Completeness of the basis:

$$|a\rangle = \sum_n a_n |n\rangle = \sum_n |n\rangle \langle n|a\rangle = \left(\sum_n |n\rangle \langle n| \right) |a\rangle .$$

Because $|a\rangle$ is arbitrary,

$$\boxed{\sum_n |n\rangle \langle n| = \mathbf{1}},$$

which is the completeness relation for the basis.

Representation of an operator (matrix form). Let $|b\rangle = \hat{F}|a\rangle$. Then

$$\langle m|b\rangle = \left\langle m\middle|\hat{F}\middle|a\right\rangle = \sum_n \left\langle m\middle|\hat{F}\middle|n\right\rangle \langle n|a\rangle , \qquad F_{mn} = \left\langle m\middle|\hat{F}\middle|n\right\rangle ,$$

and therefore

$$b_m = \sum_n F_{mn} a_n .$$

In matrix form, $b = Fa$. The matrix F represents the operator $\hat{F}$; it is a square matrix with elements F_{mn}.

Representation of the Hermitian-conjugate operator (left operator). From $\langle b| = \langle a|\hat{F}^\dagger$,

$$\langle b|m\rangle = \left\langle a\middle|\hat{F}^\dagger\middle|m\right\rangle = \sum_n \langle a|n\rangle \left\langle n\middle|\hat{F}^\dagger\middle|m\right\rangle,$$

so

$$b_m^* = \sum_n (F^\dagger)_{nm} a_n^*, \qquad b_m = \sum_n [(F^\dagger)_{nm}]^* a_n.$$

Comparison yields

$$[(F^\dagger)_{nm}]^* = F_{mn}, \qquad (F^\dagger)_{nm} = F_{mn}^*, \qquad F^\dagger = (\widetilde{F})^*.$$

Thus $F^\dagger$ is the Hermitian-conjugate matrix of F.

Outer product. The outer product is $|a\rangle\langle b|$.

Its representation is a square matrix (a is a column matrix and $b^\dagger$ a row matrix), so the outer product is an operator. In fact,

$$(|a\rangle\langle b|)|c\rangle = |a\rangle(\langle b|c\rangle),$$

showing that $|a\rangle\langle b|$ maps $|c\rangle$ to another vector parallel to $|a\rangle$. Hence $|a\rangle\langle b|$ is indeed an operator. Its representation is a square matrix with elements

$$(|a\rangle\langle b|)_{mn} = \langle m||a\rangle\langle b||n\rangle = \langle m|a\rangle\langle b|n\rangle = a_m b_n^*.$$

1.3 Eigenvalues and Eigenvectors of Operators (Matrices)

1) Eigenvalues and eigenstates of a general operator

The eigenvalue equation of an operator is

$$\hat{T}|a\rangle = \lambda|a\rangle,$$

where λ is called an eigenvalue of $\hat{T}$ and $|a\rangle$ is called an eigenvector of $\hat{T}$.

In matrix form (prove this yourself using the completeness relation),

$$Ta = \lambda a, \qquad (T - \lambda I)a = 0.$$

The condition for a nonzero eigenvector $a \neq 0$ is

$$\det(T - \lambda I) = 0.$$

This is the secular equation in an N-dimensional space:

$$\begin{vmatrix} T_{11}-\lambda & T_{12} & \cdots & T_{1N} \\ T_{21} & T_{22}-\lambda & \cdots & T_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ T_{N1} & T_{N2} & \cdots & T_{NN}-\lambda \end{vmatrix} = 0.$$

One thereby obtains N eigenvalues λ_i $(i = 1, 2, \ldots, N)$. Substituting each into

$$(T - \lambda_i I)a = 0$$

gives the corresponding eigenvector a.

Example. Find the eigenvalues and eigenvectors of

$$T = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The secular equation is

$$\begin{vmatrix} 1-\lambda & 0 \\ 0 & -1-\lambda \end{vmatrix} = 0, \quad \text{i.e.,} \quad \lambda^2 - 1 = 0,$$

and the eigenvalues are $\lambda_\pm = \pm 1$.

Let $a = (a_1, a_2)^\mathsf{T}$. For $\lambda = 1$,

$$\begin{pmatrix} 0 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0.$$

The normalized eigenvector is $a = (1, 0)^\mathsf{T}$. Similarly, taking $\lambda = -1$ gives $a = (0, 1)^\mathsf{T}$.

2) Hermitian operators

If $\hat{T}^\dagger = \hat{T}$, or $T^\dagger = T$, the operator $\hat{T}$ is called Hermitian and T is called a Hermitian matrix.

We discuss the following properties of Hermitian operators.

a) Conjugate symmetry.

$$\left\langle a \middle| \hat{T} \middle| b \right\rangle = \left\langle b \middle| \hat{T}^\dagger \middle| a \right\rangle^* = \left\langle b \middle| \hat{T} \middle| a \right\rangle^*.$$

In particular,

$$\left\langle a \middle| \hat{T} \middle| a \right\rangle = \left\langle a \middle| \hat{T} \middle| a \right\rangle^*,$$

which shows that the expectation value $\left\langle a \middle| \hat{T} \middle| a \right\rangle$ of a Hermitian operator is real.

For an anti-Hermitian operator, $\hat{T}^\dagger = -\hat{T}$ and

$$\left\langle a \middle| \hat{T} \middle| a \right\rangle = -\left\langle a \middle| \hat{T} \middle| a \right\rangle^*,$$

so its expectation value is purely imaginary.

b) Reality of eigenvalues and orthogonality.

Suppose

$$\hat{T}\,|i\rangle = \lambda_i\,|i\rangle\,, \qquad \hat{T}\,|j\rangle = \lambda_j\,|j\rangle\,.$$

From

$$\left\langle j \middle| \hat{T} \middle| i \right\rangle = \lambda_i\,\langle j|i\rangle$$

and

$$\langle j|\,\hat{T} = \langle j|\,\lambda_j^*, \qquad \left\langle j \middle| \hat{T} \middle| i \right\rangle = \lambda_j^*\,\langle j|i\rangle\,,$$

we obtain

$$(\lambda_i - \lambda_j^*)\,\langle j|i\rangle = 0.$$

For $j = i$ and $\langle i|i\rangle \neq 0$, $\lambda_i = \lambda_i^*$: the eigenvalues of a Hermitian operator are real.

For $j \neq i$, if $\lambda_i - \lambda_j \neq 0$, then $\langle j|i\rangle = 0$. Thus eigenstates corresponding to different eigenvalues of a Hermitian operator are orthogonal. Since they can always be normalized, the orthonormality condition is

$$\langle i|j\rangle = \delta_{ij}.$$

Question: Are eigenstates corresponding to the same eigenvalue orthogonal?

c) Orthogonalization by linear superposition (Schmidt orthogonalization). Suppose several eigenstates correspond to the same eigenvalue, i.e., there is degeneracy. For example, for a g-fold degenerate eigenvalue ($g \geq 2$),

$$\hat{T}\,|i,j\rangle = \lambda_i\,|i,j\rangle\,, \qquad j = 1,\ldots,g.$$

Are these g eigenvectors orthogonal?

Redefine g new states:

$$|i,n\rangle = \sum_{j=1}^{g} C_{nj}\,|i,j\rangle\,, \qquad n = 1,2,\ldots,g.$$

Since

$$\hat{T}\,|i,n\rangle = \sum_{j=1}^{g} C_{nj}\hat{T}\,|i,j\rangle = \lambda_i \sum_{j=1}^{g} C_{nj}\,|i,j\rangle = \lambda_i\,|i,n\rangle\,,$$

each $|i,n\rangle$ is still an eigenstate of $\hat{T}$ with eigenvalue λ_i. Hence it suffices to show that the $|i,n\rangle$ can be made orthonormal.

Can one choose the coefficients C_{nj} so that these g new states satisfy

$$\langle i,m|i,n\rangle = \delta_{mn}\ ?$$

There are g normalization equations and $(g^2-g)/2$ orthogonality equations, making $g(g+1)/2$ independent equations, fewer than the g^2 coefficients C_{nj}. Thus there are various choices of coefficients satisfying the orthonormality conditions, so that the new states $|i,n\rangle$ are orthonormal:

$$\langle i,m|j,n\rangle = \delta_{ij}\delta_{mn}.$$

Here δ_{ij} follows from orthonormality of eigenstates belonging to different eigenvalues, while δ_{mn} follows from orthogonalization by linear superposition.

Conclusion: Eigenstates of a Hermitian operator can be chosen orthonormal, whether the eigenvalues are degenerate or nondegenerate.

d) Completeness. It can be shown that eigenvectors of a Hermitian operator satisfy

$$\sum_i |i\rangle\,\langle i| = \mathbf{1}.$$

Hence the eigenvectors of a Hermitian operator can form a set of orthonormal basis vectors of a Hilbert space, i.e., a linear vector space or a representation.

1.4 Measurement

We discuss the connection between quantum mechanics and Hilbert space.

1) Measurement of a single physical quantity

In the Stern–Gerlach experiment, s_z has no definite value before the magnetic field (if it did, only one beam of silver atoms would emerge). It has a definite value only after the field: one beam has s_z^+, and the other has s_z^-. Therefore the values of quantum-mechanical quantities are closely related to the state of the system. A physical quantity has no definite value in a general state; it has a definite value only in certain particular states.

Postulate of quantum mechanics: Physical quantities of a quantum-mechanical system are represented by Hermitian operators in a linear vector space, and states are represented by

vectors. Because the eigenstates of a Hermitian operator are complete, the eigenstates of any physical quantity can form a linear vector space, or a representation.

The eigenvalue equation of $\hat{F}$ is

$$\hat{F}\left|n\right\rangle = f_n \left|n\right\rangle .$$

In the F representation, the basis vectors are $|n\rangle$. An arbitrary state is

$$\left|a\right\rangle = \sum_n \left|n\right\rangle \left\langle n|a\right\rangle .$$

The coefficient $\langle n|a\rangle$ is the concrete representation of $|a\rangle$ in the F representation.

Postulate of quantum mechanics: The physical quantity $\hat{F}$ has a definite value, namely the eigenvalue f_n, only in its eigenstate $|n\rangle$. In an arbitrary state $|a\rangle$, $\hat{F}$ has no definite value, but has a definite average value

$$\langle F\rangle \equiv \left\langle a\middle|\hat{F}\middle|a\right\rangle .$$

(The eigenvalues and expectation values of Hermitian operators are real. This is why physical quantities are represented by Hermitian operators.)

$$\begin{aligned}\langle F\rangle \equiv \left\langle a\middle|\hat{F}\middle|a\right\rangle &= \sum_{n,m} \left\langle a|n\right\rangle \left\langle n\middle|\hat{F}\middle|m\right\rangle \left\langle m|a\right\rangle \\ &= \sum_{n,m} \left\langle a|n\right\rangle f_m \delta_{nm} \left\langle m|a\right\rangle = \sum_n \left\langle a|n\right\rangle \left\langle n|a\right\rangle f_n = \sum_n |\left\langle n|a\right\rangle|^2 f_n .\end{aligned}$$

Thus the probability that $\hat{F}$ takes the value f_n is $|\langle n|a\rangle|^2$. The quantity $\langle n|a\rangle$ is a probability amplitude, and $|\langle n|a\rangle|^2$ is the probability for state $|a\rangle$ to contain state $|n\rangle$.

In a general state, the value of a physical quantity is uncertain, but its probabilities and expectation value are definite.

In the eigenbasis of $\hat{F}$, the matrix elements are

$$F_{mn} = \left\langle m\middle|\hat{F}\middle|n\right\rangle = f_n \left\langle m|n\right\rangle = f_n \delta_{mn}.$$

Thus a physical quantity is represented by a diagonal matrix in its own representation, and the diagonal elements are its eigenvalues.

We now discuss measurement and collapse of the state.

Before the magnetic field in the Stern–Gerlach experiment, s_z of an Ag atom has no definite value; after passing through the field it has a definite value. Applying the magnetic field can be regarded as measuring the spin. Once measured, it has a definite value.

When $\hat{F}$ is measured in an arbitrary state $|a\rangle$, the system collapses into an eigenstate of $\hat{F}$, so $\hat{F}$ has a definite value. Which eigenstate does it collapse into? The probability for state $|a\rangle$ to contain state $|n\rangle$ is $|\langle n|a\rangle|^2$. Therefore the probability of collapse into $|n\rangle$, i.e., of obtaining the measurement value f_n, is $|\langle n|a\rangle|^2$.

Measurement allows the particle nature of the system to become manifest; one may also say that measurement produces a particle.

Question: Measure $\hat{F}$ in state $|a\rangle$ and obtain f_n. If $\hat{F}$ is measured again immediately, what value will be obtained?

First measurement: $|a\rangle \longrightarrow |n\rangle$.

At the second measurement, the system is already in the eigenstate $|n\rangle$ of $\hat{F}$; the result is still f_n. This explains why an Ag atom in the Stern–Gerlach experiment still has spin s_z^+ after passing through two magnetic fields in the z direction.

2) Spin matrices

From the Stern–Gerlach experiment, the electron has spin 1/2, and the spin operator $\hat{\boldsymbol{s}}\cdot\boldsymbol{e}_n = \hat{s}_n$ along any direction $\boldsymbol{e}_n$ takes the values $\pm\hbar/2$. Its eigenvalue equation is

$$\hat{s}_n \left|s_n^{\pm}\right\rangle = \pm\frac{\hbar}{2}\left|s_n^{\pm}\right\rangle .$$

In the representation formed by the eigenstates of $\hat{s}_z$ (the s_z representation), s_z is a diagonal matrix whose diagonal entries are $\pm\hbar/2$:

$$s_z = \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

Solving

$$s_z\begin{pmatrix}a\\ b\end{pmatrix} = \pm\frac{\hbar}{2}\begin{pmatrix}a\\ b\end{pmatrix}$$

gives the normalized eigenstates

$$\left|s_z^{+}\right\rangle = \begin{pmatrix}1\\ 0\end{pmatrix}, \qquad \left|s_z^{-}\right\rangle = \begin{pmatrix}0\\ 1\end{pmatrix}.$$

We now find the matrices for $\hat{s}_x$ and $\hat{s}_y$ in the s_z representation. From the completeness of the eigenstates of $\hat{s}_z$,

$$\left|s_x^{+}\right\rangle = \sum_{i=\pm}\left|s_z^{i}\right\rangle\left\langle s_z^{i}\middle|s_x^{+}\right\rangle = \left\langle s_z^{+}\middle|s_x^{+}\right\rangle\left|s_z^{+}\right\rangle + \left\langle s_z^{-}\middle|s_x^{+}\right\rangle\left|s_z^{-}\right\rangle .$$

The cascaded Stern–Gerlach experiment shows that a state with a definite s_x value, after passing through a z-directed magnetic field (a measurement device for s_z), separates into two beams of equal intensity, with equal probabilities for s_z^+ and s_z^-:

$$\left|\left\langle s_z^{+}\middle|s_x^{+}\right\rangle\right| = \left|\left\langle s_z^{-}\middle|s_x^{+}\right\rangle\right| = \frac{1}{\sqrt{2}}.$$

Consequently,

$$\left|s_x^{+}\right\rangle = \frac{1}{\sqrt{2}}\left|s_z^{+}\right\rangle + \frac{1}{\sqrt{2}}e^{\mathrm{i}\delta_1}\left|s_z^{-}\right\rangle .$$

Through normalization, the phase factors have been collected into δ_1.

Using the orthonormality of $|s_x^{\pm}\rangle$, $\langle s_x^i|s_x^j\rangle = \delta_{ij}$, we obtain

$$\left|s_x^{-}\right\rangle = \frac{1}{\sqrt{2}}\left|s_z^{+}\right\rangle - \frac{1}{\sqrt{2}}e^{\mathrm{i}\delta_1}\left|s_z^{-}\right\rangle .$$

Equivalently,

$$\begin{cases}\left|s_z^{+}\right\rangle = \dfrac{1}{\sqrt{2}}(\left|s_x^{+}\right\rangle + \left|s_x^{-}\right\rangle),\\ \left|s_z^{-}\right\rangle = \dfrac{1}{\sqrt{2}}e^{-\mathrm{i}\delta_1}(\left|s_x^{+}\right\rangle - \left|s_x^{-}\right\rangle).\end{cases}$$

Substituting into the matrix elements of $\hat{s}_x$ gives

$$\left\langle s_z^{+}\middle|\hat{s}_x\middle|s_z^{+}\right\rangle = 0, \quad \left\langle s_z^{+}\middle|\hat{s}_x\middle|s_z^{-}\right\rangle = \frac{\hbar}{2}e^{-\mathrm{i}\delta_1}, \quad \left\langle s_z^{-}\middle|\hat{s}_x\middle|s_z^{+}\right\rangle = \frac{\hbar}{2}e^{\mathrm{i}\delta_1}, \quad \langle s_z^{-}|\hat{s}_x|s_z^{-}\rangle = 0.$$

Thus

$$s_x = \frac{\hbar}{2}\begin{pmatrix}0 & e^{-\mathrm{i}\delta_1}\\ e^{\mathrm{i}\delta_1} & 0\end{pmatrix}.$$

Similarly,

$$\left|s_y^{\pm}\right\rangle = \frac{1}{\sqrt{2}}\left|s_z^{+}\right\rangle \pm \frac{1}{\sqrt{2}}e^{\mathrm{i}\delta_2}\left|s_z^{-}\right\rangle, \qquad s_y = \frac{\hbar}{2}\begin{pmatrix} 0 & e^{-\mathrm{i}\delta_2} \\ e^{\mathrm{i}\delta_2} & 0 \end{pmatrix}.$$

Now consider the values of δ_1, δ_2. A state with a definite s_x value, after passing through a magnetic field in the y direction (a measurement device for s_y), splits into two equally intense beams corresponding to s_y^+ and s_y^-:

$$|\left\langle s_y^{\pm}\middle|s_x^{+}\right\rangle| = \frac{1}{\sqrt{2}}.$$

That is,

$$\frac{1}{2}\left|1 \pm e^{\mathrm{i}(\delta_1-\delta_2)}\right| = \frac{1}{\sqrt{2}}.$$

Hence

$$\delta_1 - \delta_2 = \pm\frac{\pi}{2}.$$

Choosing $\delta_1 = 0$ and $\delta_2 = \pi/2$ yields

$$s_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad s_y = \frac{\hbar}{2}\begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}.$$

For s_x, s_y, s_z, one can show that

$$[s_i, s_j] = \mathrm{i}\epsilon_{ijk}\hbar s_k, \qquad \{s_i, s_j\} = \frac{\hbar^2}{2}\delta_{ij},$$

where

$$[A, B] = AB - BA, \qquad \{A, B\} = AB + BA.$$

Define

$$s^2 = \sum_i s_i^2 = \frac{3}{4}\hbar^2\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Because s^2 is proportional to the identity matrix, every state is its eigenstate, and it has the definite value $3\hbar^2/4$ in every state. Clearly, $\hat{s^2}$ commutes with every physical quantity $\hat{F}$ and may be regarded as a classical quantity.

3) Conditions for simultaneous measurement of several physical quantities

Can two physical quantities $\hat{A}$ and $\hat{B}$ be measured simultaneously?

Theorem. If $\hat{A}$ and $\hat{B}$ have a complete set of common eigenstates, then $\hat{A}$ and $\hat{B}$ commute.

Proof. Let the complete common eigenstates be $|n\rangle$, with

$$\hat{A}\left|n\right\rangle = a_n\left|n\right\rangle, \qquad \hat{B}\left|n\right\rangle = b_n\left|n\right\rangle.$$

In the representation formed by $|n\rangle$, an arbitrary state is

$$\left|\psi\right\rangle = \sum_n \left|n\right\rangle\left\langle n|\psi\right\rangle.$$

Then

$$\begin{aligned}(\hat{A}\hat{B} - \hat{B}\hat{A})\left|\psi\right\rangle &= \sum_n (\hat{A}\hat{B} - \hat{B}\hat{A})\left|n\right\rangle\left\langle n|\psi\right\rangle \\ &= \sum_n (\hat{A}b_n - \hat{B}a_n)\left|n\right\rangle\left\langle n|\psi\right\rangle \\ &= \sum_n (a_n b_n - b_n a_n)\left|n\right\rangle\left\langle n|\psi\right\rangle = 0.\end{aligned}$$

Since $|\psi\rangle$ is arbitrary, $[\hat{A}, \hat{B}] = 0$.

Converse theorem. If $\hat{A}$ and $\hat{B}$ commute, then they have a complete set of common eigenstates.

Proof. Let $|n\rangle$ be an eigenstate of $\hat{A}$:

$$\hat{A}\,|n\rangle = a_n\,|n\rangle\,.$$

From $[\hat{A}, \hat{B}] = 0$,

$$\hat{A}\hat{B}\,|n\rangle = \hat{B}\hat{A}\,|n\rangle = a_n\hat{B}\,|n\rangle\,.$$

Thus $\hat{B}\,|n\rangle$ is also an eigenstate of $\hat{A}$ with eigenvalue a_n. If $\hat{A}$ is nondegenerate, $\hat{B}\,|n\rangle$ and $|n\rangle$ are the same state up to a constant:

$$\hat{B}\,|n\rangle = b_n\,|n\rangle\,.$$

Therefore $|n\rangle$ is also an eigenstate of $\hat{B}$, and $\hat{A}$ and $\hat{B}$ have a complete common eigenbasis.

If $\hat{A}$ is degenerate, one can use the Schmidt method to prove the same result (exercise).

Conclusion. When several physical quantities commute, they can have common eigenstates. In those common eigenstates, all these quantities have definite values, and they can be measured simultaneously.

Note that $[\hat{A}, \hat{B}] = 0$ and $[\hat{B}, \hat{C}] = 0$ mean that A, B can have definite values simultaneously and that B, C can have definite values simultaneously. However, A, C need not have definite values simultaneously.

When $\hat{A}$ and $\hat{B}$ do not commute, they cannot have a complete set of common eigenstates and cannot always be measured simultaneously. They may, however, be simultaneously measurable in a subspace. For example, for orbital angular momentum,

$$[\hat{\boldsymbol{L}}^2, \hat{L}_x] = 0, \qquad [\hat{\boldsymbol{L}}^2, \hat{L}_y] = 0, \qquad [\hat{L}_x, \hat{L}_y] \neq 0.$$

Thus L^2, L_x can have definite values simultaneously, and L^2, L_y can have definite values simultaneously, while L_x, L_y cannot generally have definite values simultaneously. In the state with $L = 0$, however, L_x, L_y are simultaneously definite, with $L_x = L_y = 0$.

4) Complete set of physical quantities

When eigenvalues and eigenstates of a physical quantity are degenerate,

$$\hat{F}\,|i, j\rangle = f_i\,|i, j\rangle\,, \qquad j = 1, \ldots, g,$$

Schmidt orthogonalization can be used mathematically to obtain orthonormal eigenstates. In physics, however, one generally introduces a complete set of physical quantities to achieve orthonormality.

That several eigenstates share one eigenvalue indicates that this physical quantity is insufficient to characterize the states completely; another physical quantity is needed to distinguish them. Introduce $\hat{T}$ and require that the eigenstates of $\hat{F}$ also be eigenstates of $\hat{T}$:

$$\hat{F}\,|i, j\rangle = f_i\,|i, j\rangle\,, \qquad \hat{T}\,|i, j\rangle = t_j\,|i, j\rangle\,.$$

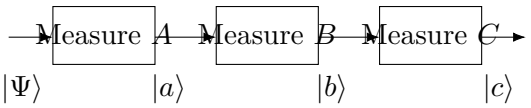
For the set $\{\hat{F}, \hat{T}\}$, if the pair of eigenvalues $\{f_i, t_j\}$ corresponds one-to-one to the eigenstate $|i, j\rangle$, there is no degeneracy, and the orthonormality condition is

$$\langle i, j|i', j'\rangle = \delta_{ii'}\delta_{jj'}.$$

5) Cascaded measurements

For three physical quantities $\hat{A}, \hat{B}, \hat{C}$, consider the following two measurement procedures.

First procedure:



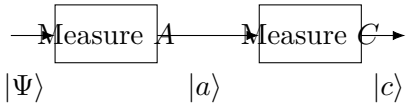
The state sequence is $|\Psi\rangle \to |a\rangle \to |b\rangle \to |c\rangle$; the values of the physical quantities are initially indefinite and then are a, b, c after their respective measurements.

After three measurements, the probability for $\hat{C}$ to take the value c is

$$P(c) = \sum_{a,b} |\langle a|\Psi\rangle|^2 |\langle b|a\rangle|^2 |\langle c|b\rangle|^2.$$

Here $|\langle a|\Psi\rangle|^2$ is the probability of collapse from $|\Psi\rangle$ to $|a\rangle$; $|\langle b|a\rangle|^2$ is the probability from $|a\rangle$ to $|b\rangle$; and $|\langle c|b\rangle|^2$ is the probability from $|b\rangle$ to $|c\rangle$.

Second procedure:



After two measurements, the probability for $\hat{C}$ to take the value c is

$$P'(c) = \sum_{a} |\langle a|\Psi\rangle|^2 |\langle c|a\rangle|^2.$$

Clearly, in general,

$$P(c) \neq P'(c).$$

However, if $[\hat{A}, \hat{B}] = 0$, $\hat{A}$ and $\hat{B}$ have common eigenstates $|n\rangle$, which replace $|a\rangle$ and $|b\rangle$:

$$P(c) = \sum_{n} |\langle n|\Psi\rangle|^2 |\langle c|n\rangle|^2 = P'(c).$$

Or if $[\hat{B}, \hat{C}] = 0$, $\hat{B}$ and $\hat{C}$ have common eigenstates $|m\rangle$, which replace $|b\rangle$ and $|c\rangle$:

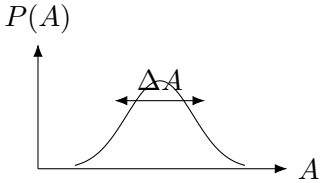
$$P(c) = \sum_{a} |\langle a|\Psi\rangle|^2 |\langle m|a\rangle|^2 = P'(c).$$

Conclusion. Since physical quantities have definite values only in their eigenstates, the order of measurements changes the final measurement results.

6) Uncertainty relation

If $[\hat{A}, \hat{B}] \neq 0$, the physical quantities $\hat{A}, \hat{B}$ cannot have definite values simultaneously. How large are their uncertainties?

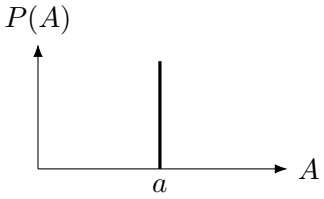
In an arbitrary state $|\psi\rangle$, $\hat{A}$ has only a definite expectation value $\langle \hat{A} \rangle \equiv \left\langle \psi \middle| \hat{A} \middle| \psi \right\rangle$ and a distribution $P(A)$.



The uncertainty is represented by the variance:

$$\langle(\Delta\hat{A})^2\rangle \equiv \left\langle\psi\middle|(\hat{A}-\langle\hat{A}\rangle)^2\middle|\psi\right\rangle \geq 0.$$

If $|\psi\rangle$ is an eigenstate of $\hat{A}$, the expectation value equals its eigenvalue, the variance is zero, and the distribution is a delta function.



Define the uncertainty between $\hat{A}$ and $\hat{B}$ using the product of variances $\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle$. How can we estimate it?

For arbitrary states $|\alpha\rangle, |\beta\rangle$ and arbitrary complex number λ, the norm of the linear-combination state is nonnegative:

$$(\langle\alpha| + \lambda^* \langle\beta|)(|\alpha\rangle + \lambda|\beta\rangle) \geq 0.$$

Choosing

$$\lambda = -\frac{\langle\beta|\alpha\rangle}{\langle\beta|\beta\rangle}$$

gives the Schwarz inequality,

$$\langle\alpha|\alpha\rangle\ \langle\beta|\beta\rangle \geq |\langle\alpha|\beta\rangle|^2.$$

Set

$$|\alpha\rangle = \Delta\hat{A}|\Psi\rangle, \qquad |\beta\rangle = \Delta\hat{B}|\Psi\rangle, \qquad \Delta\hat{A} = \hat{A} - \langle\hat{A}\rangle, \quad \Delta\hat{B} = \hat{B} - \langle\hat{B}\rangle.$$

Then

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \geq |\langle\Delta\hat{A}\Delta\hat{B}\rangle|^2.$$

Since

$$\begin{aligned}|\langle\Delta\hat{A}\Delta\hat{B}\rangle|^2 &= (\mathrm{Re}\langle\Delta\hat{A}\Delta\hat{B}\rangle)^2 + (\mathrm{Im}\langle\Delta\hat{A}\Delta\hat{B}\rangle)^2 \\ &\geq (\mathrm{Im}\langle\Delta\hat{A}\Delta\hat{B}\rangle)^2 \\ &= \left[\frac{1}{2\mathrm{i}}\left(\langle\Delta\hat{A}\Delta\hat{B}\rangle - \langle\Delta\hat{B}\Delta\hat{A}\rangle\right)\right]^2,\end{aligned}$$

we obtain

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \geq \left[\frac{1}{2\mathrm{i}}\left(\langle\Delta\hat{A}\Delta\hat{B}\rangle - \langle\Delta\hat{B}\Delta\hat{A}\rangle\right)\right]^2.$$

Equality requires equality in the Schwarz inequality and a vanishing real part of $\langle\Delta\hat{A}\Delta\hat{B}\rangle$.

From

$$\begin{aligned}\langle\Delta\hat{A}\Delta\hat{B}\rangle &= \left\langle\psi\middle|(\hat{A}-\langle\hat{A}\rangle)(\hat{B}-\langle\hat{B}\rangle)\middle|\psi\right\rangle = \langle\hat{A}\hat{B}\rangle - \langle\hat{A}\rangle\langle\hat{B}\rangle, \\ \langle\Delta\hat{B}\Delta\hat{A}\rangle &= \left\langle\psi\middle|(\hat{B}-\langle\hat{B}\rangle)(\hat{A}-\langle\hat{A}\rangle)\middle|\psi\right\rangle = \langle\hat{B}\hat{A}\rangle - \langle\hat{B}\rangle\langle\hat{A}\rangle,\end{aligned}$$

it follows that

$$\boxed{\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \geq \left(\frac{1}{2\mathrm{i}}\langle[\hat{A},\hat{B}]\rangle\right)^2.}$$

This shows that the uncertainties of two physical quantities are controlled by their commutation relation and the state $|\psi\rangle$.

1.5 Transformation of Representations

Selecting a suitable representation can simplify a calculation. The transformation between basis vectors of different representations is called a transformation of representation.

Consider two representations:

$$\begin{array}{ll} I \text{ representation:} & \text{basis } |i\rangle\,, \quad \hat{I}\,|i\rangle = i\,|i\rangle\,, \\ M \text{ representation:} & \text{basis } |m\rangle\,, \quad \hat{M}\,|m\rangle = m\,|m\rangle\,. \end{array}$$

We have

$$|m\rangle = \sum_i |i\rangle\,\langle i|m\rangle = \sum_i S_{im}\,|i\rangle\,.$$

The transformation matrix S is defined by

$$S_{im} = \langle i|m\rangle\,.$$

It represents the basis vectors of the M representation in the I representation.

1) Transformation of an arbitrary vector

In the I representation,

$$|a\rangle = \sum_i |i\rangle\,\langle i|a\rangle\,.$$

The coefficient $\langle i|a\rangle$ is the component of $|a\rangle$ along $|i\rangle$, or its representation in the I basis.

In the M representation,

$$|a\rangle = \sum_m |m\rangle\,\langle m|a\rangle\,.$$

The coefficient $\langle m|a\rangle$ is the component along $|m\rangle$, or the representation of $|a\rangle$ in the M basis.

Using completeness,

$$\langle m|a\rangle = \sum_i \langle m|i\rangle\,\langle i|a\rangle = \sum_i S^*_{im}\,\langle i|a\rangle = \sum_i S^\dagger_{mi}\,\langle i|a\rangle\,.$$

In matrix form,

$$\boxed{a_M = S^\dagger a_I.}$$

2) Transformation of an arbitrary operator

In the I representation,

$$(T_I)_{ij} = \left\langle i\middle|\hat{T}\middle|j\right\rangle.$$

In the M representation,

$$\begin{aligned}(T_M)_{mn} = \left\langle m\middle|\hat{T}\middle|n\right\rangle &= \sum_{ij} \langle m|i\rangle\left\langle i\middle|\hat{T}\middle|j\right\rangle\langle j|n\rangle \\ &= \sum_{ij} S^*_{im}(T_I)_{ij}S_{jn} = \sum_{ij}(S^\dagger)_{mi}(T_I)_{ij}S_{jn}.\end{aligned}$$

In matrix form,

$$\boxed{T_M = S^\dagger T_I S.}$$

3) The representation-transformation matrix is unitary

$$(SS^\dagger)_{ij} = \sum_m S_{im}(S^\dagger)_{mj} = \sum_m S_{im}S^*_{jm} = \sum_m \langle i|m\rangle \langle m|j\rangle = \langle i|j\rangle = \delta_{ij}.$$

Therefore $SS^\dagger = \mathbf{1}$. Similarly, $S^\dagger S = \mathbf{1}$, and hence

$$\boxed{S^\dagger = S^{-1}.}$$

A representation-transformation matrix is unitary.

Note that S is not necessarily Hermitian: $S^\dagger \neq S$.

4) Transformation of representation does not change eigenvalues

Suppose

$$T_I a_I = \lambda_I a_I.$$

Then

$$\begin{aligned} T_M a_M &= S^\dagger T_I S S^\dagger a_I = S^\dagger T_I a_I = S^\dagger \lambda_I a_I \\ &= \lambda_I S^\dagger a_I = \lambda_I a_M. \end{aligned}$$

Thus the eigenvalue in the M representation is still λ_I. A transformation of representation does not change the values of physical quantities.

Example 1: A transformation of representation does not change commutation relations.

If $[A_I, B_I] = C_I$ in the I representation, then in the M representation,

$$\begin{aligned} [A_M, B_M] &= A_M B_M - B_M A_M \\ &= S^\dagger A_I S S^\dagger B_I S - S^\dagger B_I S S^\dagger A_I S \\ &= S^\dagger (A_I B_I - B_I A_I) S = S^\dagger C_I S = C_M. \end{aligned}$$

Commutation relations are fundamental relations of quantum mechanics and are unchanged by the transformation of representation.

Example 2: A transformation of representation does not change the trace of a matrix:

$$\mathrm{tr}(T_M) = \mathrm{tr}(S^\dagger T_I S) = \mathrm{tr}(SS^\dagger T_I) = \mathrm{tr}(T_I).$$

1.6 Position and Momentum Representations

1) Continuous spectrum

Unlike the spin angular momentum $\hat{s}_z$, which takes discrete values, position $\hat{x}$ and momentum $\hat{p}$ take continuous values.

The eigenstates of position or momentum constitute a continuous position or momentum representation.

Eigenvalue equations:

$$\hat{x}\,|x\rangle = x\,|x\rangle\,, \qquad \hat{p}\,|p\rangle = p\,|p\rangle\,.$$

Basis vectors: $|x\rangle$, $|p\rangle$.

Orthonormality:

$$\langle x|x'\rangle = \delta(x - x'), \qquad \langle p|p'\rangle = \delta(p - p').$$

Completeness:

$$\int \mathrm{d}x\ |x\rangle \langle x| = \mathbf{1}, \qquad \int \mathrm{d}p\ |p\rangle \langle p| = \mathbf{1}.$$

For an arbitrary state $|\psi\rangle$,

$$|\psi\rangle = \int \mathrm{d}x\ |x\rangle\,\langle x|\psi\rangle = \int \mathrm{d}p\ |p\rangle\,\langle p|\psi\rangle\,.$$

The functions $\langle x|\psi\rangle$ and $\langle p|\psi\rangle$ are the concrete forms of $|\psi\rangle$ in the position and momentum representations; they are continuous column matrices.

In three-dimensional space,

$$\hat{\boldsymbol{x}}\,|\boldsymbol{x}\rangle = \boldsymbol{x}\,|\boldsymbol{x}\rangle\,, \qquad \hat{\boldsymbol{p}}\,|\boldsymbol{p}\rangle = \boldsymbol{p}\,|\boldsymbol{p}\rangle\,.$$

Postulate of quantum mechanics: The fundamental commutation relations are

$$[\hat{x}_i, \hat{x}_j] = 0, \qquad [\hat{p}_i, \hat{p}_j] = 0, \qquad [\hat{x}_i, \hat{p}_j] = \mathrm{i}\hbar\delta_{ij}, \quad i,j = 1,2,3.$$

The position–momentum uncertainty relation is

$$\langle(\Delta\hat{x})^2\rangle\langle(\Delta\hat{p}_x)^2\rangle \geq \left(\frac{1}{2\mathrm{i}}\langle[\hat{x},\hat{p}_x]\rangle\right)^2 = \frac{\hbar^2}{4}.$$

2) Spatial translations

Define the spatial-translation operator by

$$\hat{T}(\mathrm{d}\boldsymbol{x})\,|\boldsymbol{x}\rangle = |\boldsymbol{x} + \mathrm{d}\boldsymbol{x}\rangle\,.$$

For an arbitrary state $|\psi\rangle$,

$$\begin{aligned}
\hat{T}(\mathrm{d}\boldsymbol{x})\,|\psi\rangle &= \hat{T}(\mathrm{d}\boldsymbol{x}) \int \mathrm{d}^3x\ |\boldsymbol{x}\rangle\,\langle\boldsymbol{x}|\psi\rangle \\
&= \int \mathrm{d}^3x\ |\boldsymbol{x} + \mathrm{d}\boldsymbol{x}\rangle\,\langle\boldsymbol{x}|\psi\rangle \qquad \text{(the operator acts only on vectors)} \\
&= \int \mathrm{d}^3x\ |\boldsymbol{x}\rangle\,\langle\boldsymbol{x} - \mathrm{d}\boldsymbol{x}|\psi\rangle \qquad \text{(change of integration variable).}
\end{aligned}$$

For an infinitesimal translation, take

$$\hat{T}(\mathrm{d}\boldsymbol{x}) = \mathbf{1} - \mathrm{i}\hat{\boldsymbol{K}}\cdot\mathrm{d}\boldsymbol{x},$$

where the generator $\hat{\boldsymbol{K}}$ is Hermitian.

A translation is a unitary operator:

$$\hat{T}(\mathrm{d}\boldsymbol{x})\hat{T}^\dagger(\mathrm{d}\boldsymbol{x}) = (\mathbf{1} - \mathrm{i}\hat{\boldsymbol{K}}\cdot\mathrm{d}\boldsymbol{x})(\mathbf{1} + \mathrm{i}\hat{\boldsymbol{K}}\cdot\mathrm{d}\boldsymbol{x}) = \mathbf{1},$$

neglecting second-order infinitesimals.

Composition law for translations:

$$\begin{aligned}
\hat{T}(\mathrm{d}\boldsymbol{x}_1)\hat{T}(\mathrm{d}\boldsymbol{x}_2) &= (\mathbf{1} - \mathrm{i}\hat{\boldsymbol{K}}\cdot\mathrm{d}\boldsymbol{x}_1)(\mathbf{1} - \mathrm{i}\hat{\boldsymbol{K}}\cdot\mathrm{d}\boldsymbol{x}_2) \\
&= \mathbf{1} - \mathrm{i}\hat{\boldsymbol{K}}\cdot(\mathrm{d}\boldsymbol{x}_1 + \mathrm{d}\boldsymbol{x}_2) = \hat{T}(\mathrm{d}\boldsymbol{x}_1 + \mathrm{d}\boldsymbol{x}_2).
\end{aligned}$$

A finite translation is obtained by repeated infinitesimal translations.

For an arbitrary state $|\psi\rangle$,

$$\begin{aligned}
\hat{\boldsymbol{x}}\hat{T}(\mathrm{d}\boldsymbol{x})\,|\psi\rangle &= \int \mathrm{d}^3x\,\hat{\boldsymbol{x}}\,|\boldsymbol{x} + \mathrm{d}\boldsymbol{x}\rangle\,\langle\boldsymbol{x}|\psi\rangle \\
&= \int \mathrm{d}^3x\,(\boldsymbol{x} + \mathrm{d}\boldsymbol{x})\,|\boldsymbol{x} + \mathrm{d}\boldsymbol{x}\rangle\,\langle\boldsymbol{x}|\psi\rangle\,, \\
\hat{T}(\mathrm{d}\boldsymbol{x})\hat{\boldsymbol{x}}\,|\psi\rangle &= \int \mathrm{d}^3x\,\hat{T}(\mathrm{d}\boldsymbol{x})\boldsymbol{x}\,|\boldsymbol{x}\rangle\,\langle\boldsymbol{x}|\psi\rangle \\
&= \int \mathrm{d}^3x\,\boldsymbol{x}\,|\boldsymbol{x} + \mathrm{d}\boldsymbol{x}\rangle\,\langle\boldsymbol{x}|\psi\rangle\,.
\end{aligned}$$

Hence

$$[\hat{\boldsymbol{x}}, \hat{T}(\mathrm{d}\boldsymbol{x})]\,|\psi\rangle = \int \mathrm{d}^3x\,\mathrm{d}\boldsymbol{x}\;|\boldsymbol{x}+\mathrm{d}\boldsymbol{x}\rangle\,\langle\boldsymbol{x}|\psi\rangle$$
$$= \int \mathrm{d}^3x\,\mathrm{d}\boldsymbol{x}\;|\boldsymbol{x}\rangle\,\langle\boldsymbol{x}|\psi\rangle = \mathrm{d}\boldsymbol{x}\,|\psi\rangle\,,$$

where second-order infinitesimals have been neglected. Thus

$$[\hat{\boldsymbol{x}}, \hat{T}(\mathrm{d}\boldsymbol{x})] = \mathrm{d}\boldsymbol{x}, \qquad [\hat{\boldsymbol{x}}, \hat{\boldsymbol{K}}\cdot\mathrm{d}\boldsymbol{x}] = \mathrm{i}\mathrm{d}\boldsymbol{x}.$$

If the translation is taken in the x_j direction, $\mathrm{d}\boldsymbol{x} = \epsilon\boldsymbol{e}_j$, then

$$[\hat{x}_i, \hat{K}_j] = \mathrm{i}\delta_{ij}.$$

Comparing with the fundamental quantum-mechanical commutation relation,

$$[\hat{x}_i, \hat{p}_j] = \mathrm{i}\hbar\delta_{ij},$$

we have

$$\hat{\boldsymbol{p}} = \hbar\hat{\boldsymbol{K}}.$$

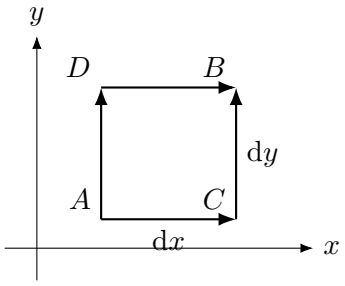
Momentum is therefore the generator of translations, and

$$\boxed{\hat{T}(\mathrm{d}\boldsymbol{x}) = \mathbf{1} - \frac{\mathrm{i}}{\hbar}\hat{\boldsymbol{p}}\cdot\mathrm{d}\boldsymbol{x}.}$$

Since $[\hat{p}_x, \hat{p}_y] = 0$,

$$[\hat{T}(\mathrm{d}x), \hat{T}(\mathrm{d}y)] = \left[\mathbf{1} - \frac{\mathrm{i}}{\hbar}\hat{p}_x\mathrm{d}x, \mathbf{1} - \frac{\mathrm{i}}{\hbar}\hat{p}_y\mathrm{d}y\right] = 0.$$

This shows that the order of translations is immaterial. Translating $A \to C$ by $\mathrm{d}x$ and then $C \to B$ by $\mathrm{d}y$ gives the same result as translating $A \to D$ by $\mathrm{d}y$ and then $D \to B$ by $\mathrm{d}x$.



3) Canonical quantization

Dirac, 1925: quantization of classical mechanical systems.

The Poisson bracket of analytical mechanics is replaced by a commutation relation in quantum mechanics:

$$[A, B]_{\mathrm{P}} = \sum_s \left(\frac{\partial A}{\partial q_s}\frac{\partial B}{\partial p_s} - \frac{\partial A}{\partial p_s}\frac{\partial B}{\partial q_s}\right) \longrightarrow \frac{1}{\mathrm{i}\hbar}[\hat{A}, \hat{B}] = \frac{1}{\mathrm{i}\hbar}(\hat{A}\hat{B} - \hat{B}\hat{A}).$$

For example,

$$[x_i, p_j]_{\mathrm{P}} = \delta_{ij} \longrightarrow [\hat{x}_i, \hat{p}_j] = \mathrm{i}\hbar\delta_{ij}.$$

4) Position representation

The matrix elements of $\hat{\boldsymbol{x}}$ are

$$\langle \boldsymbol{x}_1|\hat{\boldsymbol{x}}|\boldsymbol{x}_2\rangle = \boldsymbol{x}_2\delta^{(3)}(\boldsymbol{x}_1-\boldsymbol{x}_2) = \boldsymbol{x}_1\delta^{(3)}(\boldsymbol{x}_1-\boldsymbol{x}_2),$$

so the position operator is diagonal.

From

$$\hat{T}(\mathrm{d}\boldsymbol{x})\,|\beta\rangle = \left(\mathbf{1} - \frac{\mathrm{i}}{\hbar}\hat{\boldsymbol{p}}\cdot\mathrm{d}\boldsymbol{x}\right)|\beta\rangle$$

and

$$\begin{aligned}\hat{T}(\mathrm{d}\boldsymbol{x})\,|\beta\rangle &= \int \mathrm{d}^3x\ |\boldsymbol{x}\rangle\,\langle\boldsymbol{x}-\mathrm{d}\boldsymbol{x}|\beta\rangle \\ &= \int \mathrm{d}^3x\ |\boldsymbol{x}\rangle\,(\langle\boldsymbol{x}|\beta\rangle - \mathrm{d}\boldsymbol{x}\cdot\boldsymbol{\nabla}_{\boldsymbol{x}}\langle\boldsymbol{x}|\beta\rangle) \\ &= |\beta\rangle - \mathrm{d}\boldsymbol{x}\cdot\int \mathrm{d}^3x\ |\boldsymbol{x}\rangle\,\boldsymbol{\nabla}_{\boldsymbol{x}}\langle\boldsymbol{x}|\beta\rangle\,,\end{aligned}$$

we obtain

$$\hat{\boldsymbol{p}}\,|\beta\rangle = \int \mathrm{d}^3x\ |\boldsymbol{x}\rangle\,(-\mathrm{i}\hbar\boldsymbol{\nabla}_{\boldsymbol{x}})\,\langle\boldsymbol{x}|\beta\rangle\,,$$

and

$$\langle\alpha|\hat{\boldsymbol{p}}|\beta\rangle = \int \mathrm{d}^3x\ \langle\alpha|\boldsymbol{x}\rangle\,(-\mathrm{i}\hbar\boldsymbol{\nabla}_{\boldsymbol{x}})\,\langle\boldsymbol{x}|\beta\rangle\,.$$

Taking $|\alpha\rangle = |\boldsymbol{x}_1\rangle$ and $|\beta\rangle = |\boldsymbol{x}_2\rangle$ gives the matrix elements of momentum in the position representation:

$$\langle\boldsymbol{x}_1|\hat{\boldsymbol{p}}|\boldsymbol{x}_2\rangle = \int \mathrm{d}^3x\ \langle\boldsymbol{x}_1|\boldsymbol{x}\rangle\,(-\mathrm{i}\hbar\boldsymbol{\nabla}_{\boldsymbol{x}})\,\langle\boldsymbol{x}|\boldsymbol{x}_2\rangle = -\mathrm{i}\hbar\boldsymbol{\nabla}_{\boldsymbol{x}_1}\delta^{(3)}(\boldsymbol{x}_1-\boldsymbol{x}_2).$$

Now take $|\alpha\rangle = |\boldsymbol{x}_1\rangle$ and $|\beta\rangle = |\boldsymbol{p}\rangle$:

$$\langle\boldsymbol{x}_1|\hat{\boldsymbol{p}}|\boldsymbol{p}\rangle = \int \mathrm{d}^3x\ \langle\boldsymbol{x}_1|\boldsymbol{x}\rangle\,(-\mathrm{i}\hbar\boldsymbol{\nabla}_{\boldsymbol{x}})\,\langle\boldsymbol{x}|\boldsymbol{p}\rangle = -\mathrm{i}\hbar\boldsymbol{\nabla}_{\boldsymbol{x}_1}\langle\boldsymbol{x}_1|\boldsymbol{p}\rangle\,.$$

But the momentum eigenvalue equation gives

$$\boldsymbol{p}\,\langle\boldsymbol{x}_1|\boldsymbol{p}\rangle = -\mathrm{i}\hbar\boldsymbol{\nabla}_{\boldsymbol{x}_1}\langle\boldsymbol{x}_1|\boldsymbol{p}\rangle\,.$$

Solving this first-order differential equation, we find that the momentum eigenstate in the position representation is a plane wave:

$$\langle\boldsymbol{x}|\boldsymbol{p}\rangle = N\exp\left(\frac{\mathrm{i}}{\hbar}\boldsymbol{p}\cdot\boldsymbol{x}\right).$$

Using

$$\begin{aligned}\langle\boldsymbol{x}_1|\boldsymbol{x}_2\rangle &= \int \mathrm{d}^3p\ \langle\boldsymbol{x}_1|\boldsymbol{p}\rangle\,\langle\boldsymbol{p}|\boldsymbol{x}_2\rangle \\ &= |N|^2\int \mathrm{d}^3p\,\exp\left[\frac{\mathrm{i}}{\hbar}\boldsymbol{p}\cdot(\boldsymbol{x}_1-\boldsymbol{x}_2)\right],\end{aligned}$$

and

$$\delta^{(3)}(\boldsymbol{x}_1-\boldsymbol{x}_2) = |N|^2(2\pi\hbar)^3\delta^{(3)}(\boldsymbol{x}_1-\boldsymbol{x}_2),$$

we obtain

$$N = \frac{1}{(2\pi\hbar)^{3/2}}, \qquad \boxed{\langle\boldsymbol{x}|\boldsymbol{p}\rangle = \frac{1}{(2\pi\hbar)^{3/2}}\exp\left(\frac{\mathrm{i}}{\hbar}\boldsymbol{p}\cdot\boldsymbol{x}\right).}$$

5) Momentum representation

The matrix elements of the momentum operator are

$$\langle \boldsymbol{p}_1|\hat{\boldsymbol{p}}|\boldsymbol{p}_2\rangle = \boldsymbol{p}_1\delta^{(3)}(\boldsymbol{p}_1-\boldsymbol{p}_2),$$

so the momentum operator is diagonal.

For the position operator,

$$\begin{aligned}
\langle \boldsymbol{p}_1|\hat{\boldsymbol{x}}|\boldsymbol{p}_2\rangle &= \int \mathrm{d}^3x_1\mathrm{d}^3x_2\ \langle \boldsymbol{p}_1|\boldsymbol{x}_1\rangle\langle \boldsymbol{x}_1|\hat{\boldsymbol{x}}|\boldsymbol{x}_2\rangle\langle \boldsymbol{x}_2|\boldsymbol{p}_2\rangle \\
&= \frac{1}{(2\pi\hbar)^3}\int \mathrm{d}^3x_1\mathrm{d}^3x_2\ \exp\left(-\frac{\mathrm{i}}{\hbar}\boldsymbol{p}_1\cdot\boldsymbol{x}_1\right)\boldsymbol{x}_2\delta^{(3)}(\boldsymbol{x}_1-\boldsymbol{x}_2)\exp\left(\frac{\mathrm{i}}{\hbar}\boldsymbol{p}_2\cdot\boldsymbol{x}_2\right) \\
&= \frac{1}{(2\pi\hbar)^3}\int \mathrm{d}^3x_1\,\boldsymbol{x}_1\exp\left[\frac{\mathrm{i}}{\hbar}(\boldsymbol{p}_2-\boldsymbol{p}_1)\cdot\boldsymbol{x}_1\right] \\
&= \mathrm{i}\hbar\boldsymbol{\nabla}_{\boldsymbol{p}_1}\frac{1}{(2\pi\hbar)^3}\int \mathrm{d}^3x_1\ \exp\left[\frac{\mathrm{i}}{\hbar}(\boldsymbol{p}_2-\boldsymbol{p}_1)\cdot\boldsymbol{x}_1\right] \\
&= \mathrm{i}\hbar\boldsymbol{\nabla}_{\boldsymbol{p}_1}\delta^{(3)}(\boldsymbol{p}_1-\boldsymbol{p}_2).
\end{aligned}$$

Note that differentiation here is with respect to momentum.

The form of the position eigenstate in the momentum representation is

$$\boxed{\langle \boldsymbol{p}|\boldsymbol{x}\rangle = \langle \boldsymbol{x}|\boldsymbol{p}\rangle^* = \frac{1}{(2\pi\hbar)^{3/2}}\exp\left(-\frac{\mathrm{i}}{\hbar}\boldsymbol{p}\cdot\boldsymbol{x}\right).}$$

Example 1: Prove that $\hat{x}$ and $\hat{p}$ are Hermitian operators in the position representation. For position,

$$\langle x_1|\hat{x}|x_2\rangle = \delta(x_1-x_2)x_1,$$

and

$$\begin{aligned}
\left\langle x_1\middle|\hat{x}^\dagger\middle|x_2\right\rangle &= \langle x_2|\hat{x}|x_1\rangle^* = [\delta(x_2-x_1)x_2]^* = \delta(x_2-x_1)x_2 \\
&= \delta(x_1-x_2)x_1 = \langle x_1|\hat{x}|x_2\rangle\,.
\end{aligned}$$

For momentum,

$$\langle x_1|\hat{p}|x_2\rangle = -\mathrm{i}\hbar\frac{\partial}{\partial x_1}\delta(x_1-x_2) = -\mathrm{i}\hbar\frac{\partial}{\partial(x_1-x_2)}\delta(x_1-x_2).$$

Also,

$$\begin{aligned}
\left\langle x_1\middle|\hat{p}^\dagger\middle|x_2\right\rangle &= \langle x_2|\hat{p}|x_1\rangle^* = \left[-\mathrm{i}\hbar\frac{\partial}{\partial(x_2-x_1)}\delta(x_2-x_1)\right]^* \\
&= \mathrm{i}\hbar\frac{\partial}{\partial(x_2-x_1)}\delta(x_2-x_1) = -\mathrm{i}\hbar\frac{\partial}{\partial(x_1-x_2)}\delta(x_1-x_2) \\
&= \langle x_1|\hat{p}|x_2\rangle\,.
\end{aligned}$$

Thus both $\hat{x}$ and $\hat{p}$ are Hermitian.

Example 2: Compute $[\hat{x},\hat{p}]$ in the position representation.

$$\begin{aligned}
\hat{x}\hat{p} &= \int \mathrm{d}x_1\mathrm{d}x_2\, \hat{x}\, |x_1\rangle\, \langle x_1|\hat{p}|x_2\rangle\, \langle x_2| \\
&= \int \mathrm{d}x_1\mathrm{d}x_2\ |x_1\rangle\ x_1 \left(-\mathrm{i}\hbar\frac{\partial}{\partial x_1}\right) \delta(x_1-x_2)\, \langle x_2| \\
&= -\int \mathrm{d}x_1\mathrm{d}x_2\ |x_1\rangle \left(-\mathrm{i}\hbar\frac{\partial}{\partial x_1}\right) [x_1\, \langle x_1|]\, \delta(x_1-x_2) \quad \text{(integration by parts)} \\
&= -\int \mathrm{d}x_1\ |x_1\rangle \left(-\mathrm{i}\hbar\frac{\partial}{\partial x_1}\right) [x_1\, \langle x_1|] \\
&= \int \mathrm{d}x_1\, x_1\, |x_1\rangle \left(-\mathrm{i}\hbar\frac{\partial}{\partial x_1}\right) \langle x_1| \quad \text{(integration by parts).}
\end{aligned}$$

Similarly,

$$\hat{p}\hat{x} = -\mathrm{i}\hbar + \int \mathrm{d}x_1\, x_1\, |x_1\rangle \left(-\mathrm{i}\hbar\frac{\partial}{\partial x_1}\right) \langle x_1|\,.$$

Therefore

$$[\hat{x},\hat{p}] = \mathrm{i}\hbar.$$

This shows that commutation relations do not depend on the representation.

According to the uncertainty relation, position and momentum cannot simultaneously have definite values. For example, in a momentum eigenstate the momentum has a definite value, $\langle p|(\Delta\hat{p})^2|p\rangle = 0$, but the probability for position to take value x is

$$|\,\langle x|p\rangle\,|^2 = \left|\frac{1}{\sqrt{2\pi\hbar}} e^{\mathrm{i}px/\hbar}\right|^2 = \text{const.}$$

The particle therefore has equal probability of appearing anywhere in $-\infty < x < \infty$; position is completely indefinite, with variance $\langle p|(\Delta\hat{x})^2|p\rangle = \infty$.

Example 3: Minimum-uncertainty wave packet. For equality to hold in the uncertainty relation

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \geq \left(\frac{1}{2\mathrm{i}}\langle[\hat{A},\hat{B}]\rangle\right)^2,$$

so that the uncertainty is minimal, the state must satisfy:

1) Equality in the Schwarz inequality:

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle = |\langle\Delta\hat{A}\Delta\hat{B}\rangle|^2.$$

2) $\mathrm{Re}\langle\Delta\hat{A}\Delta\hat{B}\rangle = 0$.

The solution of condition 1 is

$$\Delta\hat{B}\,|\psi\rangle = c\Delta\hat{A}\,|\psi\rangle\,,$$

where c is a constant. Condition 2 implies

$$\mathrm{Re}\left(c\langle(\Delta\hat{A})^2\rangle\right) = 0.$$

Since $\langle(\Delta\hat{A})^2\rangle$ is real, $c = \mathrm{i}a$, with a real. Hence

$$(\hat{B} - \langle\hat{B}\rangle)\,|\psi\rangle = \mathrm{i}a(\hat{A} - \langle\hat{A}\rangle)\,|\psi\rangle\,.$$

This is the constraint on $|\psi\rangle$ for minimum uncertainty. Choose

$$\hat{A} = \hat{x}, \qquad \hat{B} = \hat{p},$$

so

$$(\hat{p} - \langle\hat{p}\rangle)\,|\psi\rangle = \mathrm{i}a(\hat{x} - \langle\hat{x}\rangle)\,|\psi\rangle\,.$$

In the position representation,

$$\left(-\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}x} - \langle\hat{p}\rangle\right)\psi(x) = \mathrm{i}a(x - \langle\hat{x}\rangle)\psi(x), \qquad \psi(x) = \langle x|\psi\rangle\,.$$

The solution is

$$\psi(x) = A\exp\left[-\frac{a(x-\langle\hat{x}\rangle)^2}{2\hbar}\right]\exp\left(\frac{\mathrm{i}}{\hbar}\langle\hat{p}\rangle x\right),$$

a Gaussian wave packet in position space.

Example 4: The ground state of the hydrogen atom. Given the state in the position representation

$$\langle\boldsymbol{r}|\psi\rangle = \frac{1}{\sqrt{\pi a_0^3}}e^{-r/a_0} \qquad \text{(hydrogen-atom ground state)},$$

find the expectation value of the momentum.

Since $\langle\boldsymbol{r}|\psi\rangle$ is not a plane wave, momentum has no definite value. The probability for momentum $\boldsymbol{p}$ is $|\,\langle\boldsymbol{p}|\psi\rangle\,|^2$, where

$$\begin{aligned}\langle\boldsymbol{p}|\psi\rangle &= \int \mathrm{d}^3r\ \langle\boldsymbol{p}|\boldsymbol{r}\rangle\,\langle\boldsymbol{r}|\psi\rangle = \int \mathrm{d}^3r\ \langle\boldsymbol{r}|\boldsymbol{p}\rangle^*\,\langle\boldsymbol{r}|\psi\rangle \\ &= \int \mathrm{d}^3r\,\frac{1}{(2\pi\hbar)^{3/2}}\exp\left(-\frac{\mathrm{i}}{\hbar}\boldsymbol{p}\cdot\boldsymbol{r}\right)\frac{1}{\sqrt{\pi a_0^3}}e^{-r/a_0} \\ &= \frac{(2a_0\hbar)^{3/2}\hbar}{\pi(a_0^2p^2+\hbar^2)^2}.\end{aligned}$$

Since $\langle\boldsymbol{p}|\psi\rangle$ depends only on the magnitude of momentum, the expectation value of the momentum magnitude is

$$\langle p\rangle = \frac{\displaystyle\int \mathrm{d}p\,p^3|\,\langle\boldsymbol{p}|\psi\rangle\,|^2}{\displaystyle\int \mathrm{d}p\,p^2|\,\langle\boldsymbol{p}|\psi\rangle\,|^2}.$$

Chapter 2

Quantum Dynamics

2.1 Time Evolution and the Schrödinger Equation

This chapter discusses the time evolution of states and physical quantities in quantum mechanics. Unlike other physical quantities, which are represented by operators, time in nonrelativistic quantum mechanics is only a parameter; it remains a classical quantity and is not an operator.

1) Time-evolution (translation) operator

Initial state: $|\alpha, t_0\rangle$.

Evolution from t_0 to t:

$$|\alpha, t\rangle = \hat{U}(t, t_0)\,|\alpha, t_0\rangle\,.$$

Expand $|\alpha, t_0\rangle$ and $|\alpha, t\rangle$ in eigenstates $|a\rangle$ of an operator $\hat{A}$:

$$|\alpha, t_0\rangle = \sum_a |a\rangle\,\langle a|\alpha, t_0\rangle\,, \qquad |\alpha, t\rangle = \sum_a |a\rangle\,\langle a|\alpha, t\rangle\,.$$

Clearly, the probability of occupying a particular eigenstate is not necessarily conserved:

$$|\,\langle a|\alpha, t_0\rangle\,|^2 \neq |\,\langle a|\alpha, t\rangle\,|^2.$$

However, the normalization condition

$$\langle \alpha, t|\alpha, t\rangle = \langle \alpha, t_0|\alpha, t_0\rangle = 1$$

implies conservation of the total probability,

$$\sum_a |\,\langle a|\alpha, t\rangle\,|^2 = \sum_a |\,\langle a|\alpha, t_0\rangle\,|^2 = 1,$$

and the time-evolution operator is unitary:

$$\hat{U}^\dagger(t, t_0)\hat{U}(t, t_0) = \mathbf{1}.$$

Require the composition law for time evolution:

$$\hat{U}(t_2, t_0) = \hat{U}(t_2, t_1)\hat{U}(t_1, t_0), \qquad t_2 > t_1 > t_0.$$

An infinitesimal time-evolution operator can be written

$$\hat{U}(t_0 + \mathrm{d}t, t_0) = \mathbf{1} - \mathrm{i}\hat{\Omega}\,\mathrm{d}t.$$

Unitarity, $\hat{U}^\dagger\hat{U} = \mathbf{1}$, requires $\hat{\Omega}$ to be Hermitian:

$$\hat{\Omega}^\dagger = \hat{\Omega}.$$

Since $\hat{U}$ is dimensionless, $\hat{\Omega}$ has dimensions of frequency. By the Planck–Einstein energy–frequency relation $E = \hbar\omega$, we may set

$$\hat{\Omega} = \frac{\hat{H}}{\hbar}, \qquad \boxed{\hat{U}(t_0 + \mathrm{d}t, t_0) = \mathbf{1} - \frac{\mathrm{i}}{\hbar}\hat{H}\,\mathrm{d}t.}$$

2) Equation of time evolution

Using the composition law,

$$\hat{U}(t+\mathrm{d}t,t_0)=\hat{U}(t+\mathrm{d}t,t)\hat{U}(t,t_0)=\left(\mathbf{1}-\frac{\mathrm{i}}{\hbar}\hat{H}\mathrm{d}t\right)\hat{U}(t,t_0).$$

Hence

$$\frac{\hat{U}(t+\mathrm{d}t,t_0)-\hat{U}(t,t_0)}{\mathrm{d}t}=-\frac{\mathrm{i}}{\hbar}\hat{H}\hat{U}(t,t_0),$$

so that

$$\boxed{\mathrm{i}\hbar\frac{\partial}{\partial t}\hat{U}(t,t_0)=\hat{H}\hat{U}(t,t_0).}$$

This is the fundamental equation of time evolution in quantum mechanics.

Multiplying the equation on the right by the initial state $|\alpha,t_0\rangle$ gives

$$\mathrm{i}\hbar\frac{\partial}{\partial t}\hat{U}(t,t_0)\,|\alpha,t_0\rangle=\hat{H}\hat{U}(t,t_0)\,|\alpha,t_0\rangle\,,$$

$$\boxed{\mathrm{i}\hbar\frac{\partial}{\partial t}\,|\alpha,t\rangle=\hat{H}\,|\alpha,t\rangle\,.}$$

This is the equation governing the time evolution of a state, the Schrödinger equation.

We now discuss solutions of this fundamental equation. If the Hamiltonian has no explicit time dependence, for example the magnetic interaction $\hat{H}\sim\hat{J}_zB$ in a constant magnetic field directed along z,

$$\boxed{\hat{U}(t,t_0)=\exp\left[-\frac{\mathrm{i}}{\hbar}\hat{H}(t-t_0)\right].}$$

If the Hamiltonian depends on time but the Hamiltonians at different times commute, for example $\hat{H}(t)\sim\hat{J}_zB(t)$ when the magnitude but not the direction of the magnetic field changes,

$$\hat{U}(t,t_0)=\exp\left[-\frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\hat{H}(t')\mathrm{d}t'\right].$$

If the Hamiltonian depends on time and the Hamiltonians at different times do not commute, for example $\hat{H}(t)\sim\boldsymbol{J}\cdot\boldsymbol{B}(t)$ when both the magnitude and direction of the field vary,

$$\hat{U}(t,t_0)=\mathbf{1}+\sum_{n=1}^{\infty}\left(-\frac{\mathrm{i}}{\hbar}\right)^n\int_{t_0}^{t}\mathrm{d}t_1\int_{t_0}^{t_1}\mathrm{d}t_2\cdots\int_{t_0}^{t_{n-1}}\mathrm{d}t_n \times\hat{H}(t_1)\hat{H}(t_2)\cdots\hat{H}(t_n).$$

This is called the Dyson series.

Below we mainly consider Hamiltonians that do not depend explicitly on time.

3) Stationary states

Suppose initially the system is in an eigenstate of $\hat{H}$:

$$|\alpha,t_0\rangle=|E\rangle\,,\qquad\hat{H}\,|E\rangle=E\,|E\rangle\,.$$

Then

$$|\alpha,t\rangle=e^{-\mathrm{i}\hat{H}(t-t_0)/\hbar}\,|\alpha,t_0\rangle=e^{-\mathrm{i}E(t-t_0)/\hbar}\,|\alpha,t_0\rangle\,.$$

The state changes only by a phase. The expectation value of any physical quantity $\hat{A}$ is

$$\left\langle\alpha,t\middle|\hat{A}\middle|\alpha,t\right\rangle=\left\langle\alpha,t_0\middle|e^{\mathrm{i}E(t-t_0)/\hbar}\hat{A}e^{-\mathrm{i}E(t-t_0)/\hbar}\middle|\alpha,t_0\right\rangle=\left\langle\alpha,t_0\middle|\hat{A}\middle|\alpha,t_0\right\rangle$$

and does not change with time. Therefore, energy eigenstates are called stationary states.

If the system is initially not in a stationary state, its amplitude for occupying the stationary state $|E\rangle$ at time t is

$$\langle E|\alpha,t\rangle = \langle E|\, e^{-\mathrm{i}\hat{H}(t-t_0)/\hbar}\,|\alpha,t_0\rangle = e^{-\mathrm{i}E(t-t_0)/\hbar}\,\langle E|\alpha,t_0\rangle\,,$$

and its probability satisfies

$$|\,\langle E|\alpha,t\rangle\,|^2 = |\,\langle E|\alpha,t_0\rangle\,|^2.$$

The probability of occupying any stationary state is the same at all times as it was initially. Thus, if the system initially occupies a stationary state, it remains in that state; if its initial state is not stationary, it remains nonstationary.

4) Electron-spin precession

Consider the interaction of an electron's spin magnetic moment with an external magnetic field directed along z:

$$\hat{H} = -\left(\frac{e}{mc}\right)\hat{\boldsymbol{s}}\cdot\boldsymbol{B} = -\frac{eB}{mc}\hat{s}_z = \omega\hat{s}_z, \qquad \omega = -\frac{eB}{mc} > 0.$$

The time-evolution operator is

$$\hat{U}(t,0) = e^{-\mathrm{i}\hat{H}t/\hbar} = e^{-\mathrm{i}\omega\hat{s}_z t/\hbar}.$$

The operators $\hat{s}_z$ and $\hat{H}$ have common eigenstates $|s_z^{\pm}\rangle$:

$$\hat{s}_z\,|s_z^{\pm}\rangle = \pm\frac{\hbar}{2}\,|s_z^{\pm}\rangle\,, \qquad \hat{H}\,|s_z^{\pm}\rangle = \pm\frac{\hbar\omega}{2}\,|s_z^{\pm}\rangle\,.$$

Suppose the system initially occupies the positive eigenstate of $\hat{s}_x$:

$$|\alpha,0\rangle = \left|s_x^{+}\right\rangle = \frac{1}{\sqrt{2}}\left|s_z^{+}\right\rangle + \frac{1}{\sqrt{2}}\,|s_z^{-}\rangle\,.$$

At an arbitrary time,

$$|\alpha,t\rangle = \hat{U}(t,0)\,|\alpha,0\rangle\,.$$

The probability of finding the system in $|s_x^{+}\rangle$ is

$$\begin{aligned}|\left\langle s_x^{+}\right|\alpha,t\rangle\,|^2 &= \left|\left(\frac{\langle s_z^{+}| + \langle s_z^{-}|}{\sqrt{2}}\right)e^{-\mathrm{i}\omega\hat{s}_z t/\hbar}\left(\frac{|s_z^{+}\rangle + |s_z^{-}\rangle}{\sqrt{2}}\right)\right|^2 \\ &= \left|\frac{e^{-\mathrm{i}\omega t/2} + e^{\mathrm{i}\omega t/2}}{2}\right|^2 = \cos^2\frac{\omega t}{2}.\end{aligned}$$

By conservation of probability,

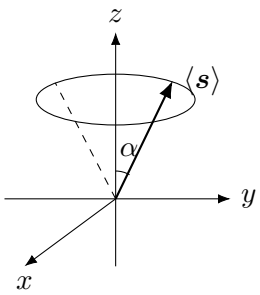
$$|\,\langle s_x^{-}|\alpha,t\rangle\,|^2 = 1 - |\left\langle s_x^{+}\right|\alpha,t\rangle\,|^2 = \sin^2\frac{\omega t}{2}.$$

The expectation value at an arbitrary time is

$$\begin{aligned}\langle s_x\rangle &= \langle\alpha,t|\hat{s}_x|\alpha,t\rangle = \sum_{s_x} s_x|\,\langle s_x|\alpha,t\rangle\,|^2 \\ &= \frac{\hbar}{2}\left(\cos^2\frac{\omega t}{2} - \sin^2\frac{\omega t}{2}\right) = \frac{\hbar}{2}\cos\omega t.\end{aligned}$$

Similarly,

$$\langle s_y\rangle = \frac{\hbar}{2}\sin\omega t, \qquad \langle s_z\rangle = 0.$$



The expectation value $\langle \boldsymbol{s} \rangle$ precesses about the z axis with time; the trajectory is a cone. This is the spin Larmor precession, with ω the Larmor frequency.

5) Energy–time uncertainty relation

From

$$\langle (\Delta \hat{x})^2 \rangle \langle (\Delta \hat{p})^2 \rangle \geq \frac{\hbar^2}{4},$$

we obtain $\Delta x \, \Delta p \geq \hbar/2$, where

$$\Delta A = \sqrt{\langle (\Delta \hat{A})^2 \rangle}.$$

By classical analogy, $x = x(t)$ and $p = p(E, p)$, one may suppose

$$\Delta t \, \Delta E \geq \frac{\hbar}{2}.$$

This is the energy–time uncertainty relation. However, unlike x, p, E, $\hat{x}$, $\hat{p}$, and $\hat{H}$ in quantum mechanics, time t remains a classical quantity. What does Δt mean?

What are $\langle (\Delta \hat{H})^2 \rangle$ and $\langle (\Delta t)^2 \rangle$?

For an arbitrary observable $\hat{O}$,

$$\langle O \rangle (t) = \left\langle \alpha, t \middle| \hat{O} \middle| \alpha, t \right\rangle .$$

Its time derivative is

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle O \rangle = \left(\frac{\partial}{\partial t} \langle \alpha, t | \right) \hat{O} \, | \alpha, t \rangle + \langle \alpha, t | \, \frac{\partial \hat{O}}{\partial t} \, | \alpha, t \rangle + \langle \alpha, t | \, \hat{O} \left(\frac{\partial}{\partial t} \, | \alpha, t \rangle \right).$$

Since

$$\mathrm{i}\hbar \frac{\partial}{\partial t} \, | \alpha, t \rangle = \hat{H} \, | \alpha, t \rangle , \qquad -\mathrm{i}\hbar \frac{\partial}{\partial t} \, \langle \alpha, t | = \langle \alpha, t | \, \hat{H},$$

we have

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle O \rangle = \frac{1}{\mathrm{i}\hbar} \left\langle \alpha, t \middle| [\hat{O}, \hat{H}] \middle| \alpha, t \right\rangle + \left\langle \frac{\partial \hat{O}}{\partial t} \right\rangle .$$

If $\hat{O}$ has no explicit time dependence,

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle O \rangle = \frac{1}{\mathrm{i}\hbar} \langle [\hat{O}, \hat{H}] \rangle .$$

Consequently, the uncertainty relation between $\hat{O}$ and $\hat{H}$ gives

$$\langle (\Delta \hat{O})^2 \rangle \langle (\Delta \hat{H})^2 \rangle \geq \left| \frac{1}{2\mathrm{i}} \langle [\hat{O}, \hat{H}] \rangle \right|^2 = \frac{\hbar^2}{4} \left(\frac{\mathrm{d} \langle O \rangle}{\mathrm{d}t} \right)^2 .$$

Therefore

$$\Delta O \, \Delta E \geq \frac{\hbar}{2} \left| \frac{\mathrm{d} \langle O \rangle}{\mathrm{d}t} \right| .$$

If we define

$$\Delta t = \Delta O \left| \frac{\mathrm{d}\langle O \rangle}{\mathrm{d}t} \right|^{-1},$$

then

$$\boxed{\Delta t \, \Delta E \geq \frac{\hbar}{2}.}$$

Thus Δt is the time needed for the expectation value of a physical quantity $\hat{O}$ to change by an amount equal to one standard deviation ΔO. Clearly, both Δt and ΔE are averages and uncertainties of physical quantities.

Example. In a stationary state $\Delta E = 0$, and $\langle \hat{O} \rangle$ is time independent:

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle O \rangle = 0, \qquad \Delta t = \Delta O \left| \frac{\mathrm{d}\langle O \rangle}{\mathrm{d}t} \right|^{-1} \longrightarrow \infty.$$

2.2 Schrödinger and Heisenberg Pictures

1) Two kinds of unitary transformations

The spatial-translation operator $\hat{U}(\mathrm{d}\boldsymbol{x})$ and time-translation operator $\hat{U}(\mathrm{d}t)$ discussed earlier both act on the state:

$$|\alpha\rangle \longrightarrow \hat{U}(\mathrm{d}\boldsymbol{x})\,|\alpha\rangle = \left(\mathbf{1} - \frac{\mathrm{i}}{\hbar}\hat{\boldsymbol{p}} \cdot \mathrm{d}\boldsymbol{x}\right)|\alpha\rangle\,,$$

$$|\alpha\rangle \longrightarrow \hat{U}(\mathrm{d}t)\,|\alpha\rangle = \left(\mathbf{1} - \frac{\mathrm{i}}{\hbar}\hat{H}\,\mathrm{d}t\right)|\alpha\rangle\,.$$

The physical quantities remain unchanged; observable differences arise from changes in the state. Under a unitary change of the state,

$$\left\langle \beta \middle| \hat{A} \middle| \alpha \right\rangle \longrightarrow \langle \beta |\, \hat{U}^\dagger \hat{A} \hat{U}\, |\alpha\rangle\,.$$

Equivalently, one may leave the state $|\alpha\rangle$ unchanged and transform the physical quantity:

$$\boxed{\hat{A} \longrightarrow \hat{U}^\dagger \hat{A} \hat{U}, \qquad |\alpha\rangle \text{ unchanged.}}$$

The two methods are completely equivalent: the time dependence may be assigned either to the state or to the observable. For example, under a spatial displacement,

$$\hat{x} \longrightarrow \hat{U}^\dagger(\mathrm{d}\boldsymbol{x})\hat{x}\hat{U}(\mathrm{d}\boldsymbol{x}) = \left(\mathbf{1} + \frac{\mathrm{i}}{\hbar}\hat{\boldsymbol{p}} \cdot \mathrm{d}\boldsymbol{x}\right)\hat{x}\left(\mathbf{1} - \frac{\mathrm{i}}{\hbar}\hat{\boldsymbol{p}} \cdot \mathrm{d}\boldsymbol{x}\right) = \hat{x} + \frac{\mathrm{i}}{\hbar}[\hat{\boldsymbol{p}} \cdot \mathrm{d}\boldsymbol{x}, \hat{x}] = \hat{x} + \mathrm{d}\boldsymbol{x}.$$

2) Schrödinger and Heisenberg pictures of time evolution

In the *Schrödinger picture*, the state evolves as

$$|\alpha, t\rangle_{\mathrm{S}} = \hat{U}(t)\,|\alpha, 0\rangle_{\mathrm{S}}\,, \qquad \hat{U}(t) = e^{-\mathrm{i}\hat{H}t/\hbar},$$

while the observable $\hat{A}_{\mathrm{S}}$ remains fixed in time.

In the *Heisenberg picture*, the observable evolves as

$$\hat{A}_{\mathrm{H}}(t) = \hat{U}^\dagger(t)\hat{A}_{\mathrm{H}}(0)\hat{U}(t),$$

while the state $|\alpha\rangle_{\mathrm{H}}$ remains fixed.

For the same initial conditions,

$$\hat{A}_{\mathrm{H}}(0) = \hat{A}_{\mathrm{S}}, \qquad |\alpha, 0\rangle_{\mathrm{S}} = |\alpha\rangle_{\mathrm{H}}\,.$$

These are the relations between the two pictures.

Notes.

1. Inner products (including matrix elements, probability amplitudes and expectation values) are independent of picture:

$$_{\rm S}\langle\alpha,t|\beta,t\rangle_{\rm S} = {}_{\rm S}\langle\alpha,0|\,\hat{U}^\dagger\hat{U}\,|\beta,0\rangle_{\rm S} = {}_{\rm S}\langle\alpha,0|\beta,0\rangle_{\rm S} = {}_{\rm H}\langle\alpha|\beta\rangle_{\rm H}\,.$$

2. Since a unitary transformation preserves commutators, if the Schrödinger-picture commutation relation is

$$[\hat{A}_{\rm S},\hat{B}_{\rm S}] = \hat{C}_{\rm S},$$

or equivalently initially $[\hat{A}_{\rm H}(0),\hat{B}_{\rm H}(0)] = \hat{C}_{\rm H}(0)$, then the equal-time relation is

$$[\hat{A}_{\rm H}(t),\hat{B}_{\rm H}(t)] = \hat{C}_{\rm H}(t).$$

3. The Hamiltonian in the previously introduced time-evolution operator $\hat{U}(t) = e^{-{\rm i}\hat{H}t/\hbar}$ is the Schrödinger-picture Hamiltonian $\hat{H} = \hat{H}_{\rm S}$. For a time-independent Hamiltonian,

$$\hat{H}_{\rm H}(t) = \hat{U}^\dagger(t)\hat{H}_{\rm S}\hat{U}(t) = e^{{\rm i}\hat{H}_{\rm S}t/\hbar}\hat{H}_{\rm S}e^{-{\rm i}\hat{H}_{\rm S}t/\hbar} = \hat{H}_{\rm S} = \hat{H}_{\rm H}.$$

So far we have considered time evolution of states and observables. Do basis vectors also change with time, as general vectors do? Basis vectors are eigenstates of observables; their time dependence is determined by that of the observables.

In the Schrödinger picture,

$$\hat{A}_{\rm S}\,|a\rangle_{\rm S} = a\,|a\rangle_{\rm S}\,.$$

Since $\hat{A}_{\rm S}$ is time independent, its eigenbasis $|a\rangle_{\rm S}$ is also time independent.

In the Heisenberg picture,

$$\hat{A}_{\rm H}(t)\,|a,t\rangle_{\rm H} = a\,|a,t\rangle_{\rm H}\,.$$

Since $\hat{A}_{\rm H}$ depends on time, its basis $|a,t\rangle_{\rm H}$ also depends on time. Using

$$\hat{U}^\dagger(t)\hat{A}_{\rm S}\hat{U}(t)\,|a,t\rangle_{\rm H} = a\,|a,t\rangle_{\rm H}\,,$$

we obtain

$$\hat{A}_{\rm S}\hat{U}(t)\,|a,t\rangle_{\rm H} = a\hat{U}(t)\,|a,t\rangle_{\rm H}\,.$$

Comparison with the Schrödinger-picture eigenvalue equation gives

$$\hat{U}(t)\,|a,t\rangle_{\rm H} = |a\rangle_{\rm S}\,,\quad |a,t\rangle_{\rm H} = \hat{U}^\dagger(t)\,|a\rangle_{\rm S} = \hat{U}^\dagger(t)\,|a,0\rangle_{\rm H}\,.$$

In summary:

	Schrödinger picture	Heisenberg picture
State	$\|\alpha,t\rangle_{\rm S} = \hat{U}(t)\,\|\alpha,0\rangle_{\rm S}$	$\|\alpha\rangle_{\rm H}$ fixed
Observable	$\hat{A}_{\rm S}$ fixed	$\hat{A}_{\rm H}(t) = \hat{U}^\dagger\hat{A}_{\rm H}(0)\hat{U}$
Basis	$\|a\rangle_{\rm S}$ fixed	$\|a,t\rangle_{\rm H} = \hat{U}^\dagger(t)\,\|a,0\rangle_{\rm H}$

3) Equation of motion in the Heisenberg picture

In classical mechanics, only physical quantities evolve with time and the notion of a state vector is not introduced. The Heisenberg picture of quantum mechanics is therefore easier to compare with classical mechanics.

Differentiate

$$\hat{A}_{\rm H}(t) = e^{{\rm i}\hat{H}t/\hbar}\hat{A}_{\rm H}(0)e^{-{\rm i}\hat{H}t/\hbar}$$

with respect to time. We obtain the Heisenberg equation of motion:

$$\boxed{\frac{\mathrm{d}\hat{A}_{\mathrm{H}}(t)}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}[\hat{A}_{\mathrm{H}}(t), \hat{H}].}$$

It plays a role similar to the Schrödinger-picture equation for the state,

$$\mathrm{i}\hbar\frac{\partial}{\partial t}\left|\alpha, t\right\rangle_{\mathrm{S}} = \hat{H}\left|\alpha, t\right\rangle_{\mathrm{S}}.$$

Compare the equation of motion of analytical classical mechanics,

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \{A, H\}_{\mathrm{Poisson}}.$$

We obtain the route from classical mechanics to quantum mechanics, canonical quantization:

$$\boxed{\{\ ,\ \}_{\mathrm{Poisson}} \quad \xrightarrow{\text{quantum mechanics}} \quad \frac{1}{\mathrm{i}\hbar}[\ ,\ ].}$$

4) Ehrenfest theorem

How is $\hat{H}$ constructed? For a quantum system with a classical counterpart, the Hamiltonian $H(\boldsymbol{x}, \boldsymbol{p})$ of analytical mechanics is taken to the quantum Hamiltonian $\hat{H}(\hat{\boldsymbol{x}}, \hat{\boldsymbol{p}})$.

For a free particle,

$$\hat{H} = \frac{\hat{\boldsymbol{p}}^2}{2m}.$$

Heisenberg equations of motion (we henceforth omit the picture index H on observables) give

$$\frac{\mathrm{d}\hat{\boldsymbol{p}}}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}[\hat{\boldsymbol{p}}, \hat{H}] = 0, \qquad \hat{\boldsymbol{p}}(t) = \hat{\boldsymbol{p}}(0),$$

$$\frac{\mathrm{d}\hat{\boldsymbol{x}}}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}[\hat{\boldsymbol{x}}, \hat{H}] = \frac{1}{2\mathrm{i}\hbar m}[\hat{\boldsymbol{x}}, \hat{\boldsymbol{p}}^2] = \frac{\hat{\boldsymbol{p}}(t)}{m} = \frac{\hat{\boldsymbol{p}}(0)}{m}, \quad \hat{\boldsymbol{x}}(t) = \hat{\boldsymbol{x}}(0) + \frac{\hat{\boldsymbol{p}}(0)}{m}t.$$

For a general system,

$$\hat{H} = \frac{\hat{\boldsymbol{p}}^2}{2m} + V(\hat{\boldsymbol{x}}).$$

The Heisenberg equations are

$$\frac{\mathrm{d}\hat{\boldsymbol{p}}}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}[\hat{\boldsymbol{p}}, \hat{H}] = \frac{1}{\mathrm{i}\hbar}[\hat{\boldsymbol{p}}, V(\hat{\boldsymbol{x}})].$$

Expand $V(\hat{\boldsymbol{x}})$ as a power series in $\hat{\boldsymbol{x}}$ and use $[\hat{x}_i, \hat{p}_j] = \mathrm{i}\hbar\delta_{ij}$. Then

$$\frac{\mathrm{d}\hat{\boldsymbol{p}}}{\mathrm{d}t} = -\boldsymbol{\nabla}V(\hat{\boldsymbol{x}}),$$

$$\frac{\mathrm{d}\hat{\boldsymbol{x}}}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}[\hat{\boldsymbol{x}}, \hat{H}] = \frac{1}{2\mathrm{i}\hbar m}[\hat{\boldsymbol{x}}, \hat{\boldsymbol{p}}^2] = \frac{\hat{\boldsymbol{p}}}{m},$$

$$m\frac{\mathrm{d}^2\hat{\boldsymbol{x}}}{\mathrm{d}t^2} = \frac{\mathrm{d}\hat{\boldsymbol{p}}}{\mathrm{d}t} = -\boldsymbol{\nabla}V(\hat{\boldsymbol{x}}).$$

Taking expectation values on both sides,

$$\boxed{m\frac{\mathrm{d}^2\langle\hat{\boldsymbol{x}}\rangle}{\mathrm{d}t^2} = \frac{\mathrm{d}\langle\hat{\boldsymbol{p}}\rangle}{\mathrm{d}t} = -\langle\boldsymbol{\nabla}V(\hat{\boldsymbol{x}})\rangle.}$$

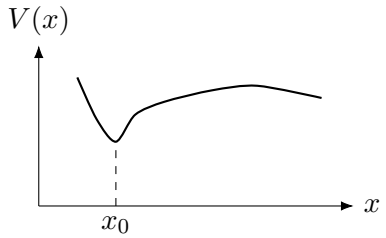
This is the Ehrenfest theorem, corresponding to Newton's law in classical mechanics. Since inner products are picture-independent, the Ehrenfest theorem for expectation values is also picture-independent.

Unless otherwise stated, problems below are generally discussed in the Schrödinger picture.

2.3 One-Dimensional Linear Harmonic Oscillator

For an arbitrary potential, expand it in a Taylor series near a minimum x_0:

$$V(x) = V(x_0) + V'(x_0)(x - x_0) + \frac{1}{2}V''(x_0)(x - x_0)^2 + \cdots .$$



The constant term $V(x_0)$ can be incorporated into the energy. At the potential minimum $V'(x_0) = 0$. Omitting higher-order terms,

$$V(x) \simeq \frac{1}{2}V''(x_0)(x - x_0)^2.$$

This approximates a harmonic-oscillator potential; therefore the harmonic oscillator has general significance.

Classically,

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2.$$

Quantum mechanically,

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2, \qquad [\hat{x}, \hat{p}] = \mathrm{i}\hbar.$$

Because

$$\hat{H} = \frac{m\omega^2}{2}\left(\hat{x} - \frac{\mathrm{i}\hat{p}}{m\omega}\right)\left(\hat{x} + \frac{\mathrm{i}\hat{p}}{m\omega}\right) + \frac{1}{2}\hbar\omega,$$

define new operators

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{\mathrm{i}\hat{p}}{m\omega}\right), \qquad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{\mathrm{i}\hat{p}}{m\omega}\right).$$

Then

$$\boxed{\hat{H} = \left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)\hbar\omega, \qquad [\hat{a}, \hat{a}^\dagger] = \mathbf{1}.}$$

Clearly, $\hat{a}$ is not Hermitian, $\hat{a}^\dagger \neq \hat{a}$. However, $\hat{a}^\dagger\hat{a}$ is Hermitian:

$$(\hat{a}^\dagger\hat{a})^\dagger = \hat{a}^\dagger\hat{a}.$$

Since $\hat{H}$ differs from $\hat{a}^\dagger\hat{a}$ only by a constant, they have common eigenstates. Let

$$\hat{a}^\dagger\hat{a}\,|n\rangle = n\,|n\rangle\,.$$

Then

$$\hat{H}\,|n\rangle = \left(n + \frac{1}{2}\right)\hbar\omega\,|n\rangle\,, \qquad E_n = \left(n + \frac{1}{2}\right)\hbar\omega.$$

Questions: What are the eigenvalues n? What are the coordinate-representation eigenstates $\langle x|n\rangle$?

1) Nonnegativity of the eigenvalues

Let

$$\hat{a}\,|n\rangle = |b\rangle\,.$$

Then

$$\langle n|\,\hat{a}^\dagger = \langle b|\,, \qquad \left\langle n\middle|\hat{a}^\dagger\hat{a}\middle|n\right\rangle = \langle b|b\rangle\,, \quad n\,\langle n|n\rangle = \langle b|b\rangle\,.$$

Because $\langle b|b\rangle \geq 0$ and $\langle n|n\rangle \geq 0$, we have $n \geq 0$.

2) Raising and lowering operators

From $[\hat{a}, \hat{a}^\dagger] = 1$,

$$\begin{aligned}(\hat{a}^\dagger\hat{a})\hat{a}\,|n\rangle &= \hat{a}(\hat{a}\hat{a}^\dagger - 1)\,|n\rangle = \hat{a}(\hat{a}^\dagger\hat{a} - 1)\,|n\rangle \\ &= (n-1)\hat{a}\,|n\rangle\,,\end{aligned}$$

$$\begin{aligned}(\hat{a}^\dagger\hat{a})\hat{a}^\dagger\,|n\rangle &= \hat{a}^\dagger(\hat{a}\hat{a}^\dagger)\,|n\rangle = \hat{a}^\dagger(\hat{a}^\dagger\hat{a} + 1)\,|n\rangle \\ &= (n+1)\hat{a}^\dagger\,|n\rangle\,.\end{aligned}$$

If $|n\rangle$ is an eigenstate of $\hat{a}^\dagger\hat{a}$, so are $\hat{a}\,|n\rangle$ and $\hat{a}^\dagger\,|n\rangle$, with the following relationships:

Eigenstate	Eigenvalue of $\hat{a}^\dagger\hat{a}$
$(\hat{a}^\dagger)^2\,\lvert n\rangle$	$n+2$
$\hat{a}^\dagger\,\lvert n\rangle$	$n+1$
$\lvert n\rangle$	n
$\hat{a}\,\lvert n\rangle$	$n-1$
$\hat{a}^2\,\lvert n\rangle$	$n-2$

Hence $\hat{a}$ is called the lowering (annihilation) operator and $\hat{a}^\dagger$ the raising (creation) operator. Together with $n \geq 0$, the possible eigenvalues have the form

$$n = n_0, n_0 + 1, n_0 + 2, \ldots, \qquad n_0 \geq 0.$$

3) Lowest eigenvalue

For the lowest eigenvalue n_0, necessarily

$$\hat{a}\,|n_0\rangle = 0.$$

Otherwise $\hat{a}\,|n_0\rangle$ would have eigenvalue $n_0 - 1$, contradicting the assumption that n_0 is the smallest eigenvalue. From

$$\hat{a}^\dagger\hat{a}\,|n_0\rangle = n_0\,|n_0\rangle = 0$$

we obtain $n_0 = 0$. Therefore

$$\boxed{E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \quad n = 0, 1, 2, \ldots.}$$

Notice that so far we have used only the commutation relation, without introducing any specific representation.

4) Matrix elements of creation and annihilation operators

Since $\hat{a}\,|n\rangle$ has eigenvalue $n-1$, and in the absence of degeneracy (one-dimensional bound states are nondegenerate),

$$\hat{a}\,|n\rangle = a_n\,|n-1\rangle\,.$$

Similarly,

$$\hat{a}^\dagger\,|n\rangle = b_n\,|n+1\rangle\,.$$

Because

$$\langle n|\,\hat{a}^\dagger = a_n^*\,\langle n-1|\,, \qquad \langle n|\,\hat{a} = b_n^*\,\langle n+1|\,,$$

we obtain

$$\left\langle n\middle|\hat{a}^\dagger\hat{a}\middle|n\right\rangle = |a_n|^2\,\langle n-1|n-1\rangle\,, \quad n = |a_n|^2, \quad |a_n| = \sqrt{n},$$

$$\left\langle n\middle|\hat{a}\hat{a}^\dagger\middle|n\right\rangle = |b_n|^2\,\langle n+1|n+1\rangle\,, \quad n+1 = |b_n|^2, \quad |b_n| = \sqrt{n+1}.$$

Choose the phases so that $a_n = \sqrt{n}$ and $b_n = \sqrt{n+1}$. Thus

$$\boxed{\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle\,, \qquad \hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle\,.}$$

5) Representation common to $\hat{H}$ and $\hat{a}^\dagger\hat{a}$

$$(a^\dagger a)_{mn} = \left\langle m\middle|\hat{a}^\dagger\hat{a}\middle|n\right\rangle = n\,\langle m|n\rangle = n\delta_{mn},$$

$$H_{mn} = \left\langle m\middle|\hat{H}\middle|n\right\rangle = \left(n+\frac{1}{2}\right)\hbar\omega\delta_{mn}.$$

Both are diagonal matrices, as expected in their own representation. Meanwhile,

$$a_{mn} = \langle m|\hat{a}|n\rangle = \sqrt{n}\,\langle m|n-1\rangle = \sqrt{n}\delta_{m,n-1},$$

$$(a^\dagger)_{mn} = \left\langle m\middle|\hat{a}^\dagger\middle|n\right\rangle = \sqrt{n+1}\,\langle m|n+1\rangle = \sqrt{n+1}\delta_{m,n+1}.$$

Since

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}^\dagger + \hat{a}), \qquad \hat{p} = \mathrm{i}\sqrt{\frac{m\hbar\omega}{2}}(\hat{a}^\dagger - \hat{a}),$$

it follows that

$$x_{mn} = \sqrt{\frac{\hbar}{2m\omega}}(\sqrt{n}\delta_{m,n-1} + \sqrt{n+1}\delta_{m,n+1}),$$

$$p_{mn} = \mathrm{i}\sqrt{\frac{m\hbar\omega}{2}}(-\sqrt{n}\delta_{m,n-1} + \sqrt{n+1}\delta_{m,n+1}).$$

They are nondiagonal matrices. Note that $\langle n|\hat{x}|n\rangle = \langle n|\hat{p}|n\rangle = 0$.

6) Coordinate representation

For the ground state $|0\rangle$,

$$\hat{a}\,|0\rangle = 0, \qquad \left(\hat{x} + \frac{\mathrm{i}}{m\omega}\hat{p}\right)|0\rangle = 0.$$

Multiplying from the left by $\langle x|$ and inserting the completeness relation,

$$\int \mathrm{d}x'\ \langle x|\left(\hat{x} + \frac{\mathrm{i}}{m\omega}\hat{p}\right)|x'\rangle\,\langle x'|0\rangle = 0.$$

Using

$$\langle x|\hat{x}|x'\rangle = x\delta(x-x'), \qquad \langle x|\hat{p}|x'\rangle = -\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}x}\delta(x-x'), \qquad \langle x'|0\rangle = \psi_0(x'),$$

we obtain

$$\left(x + \frac{\hbar}{m\omega}\frac{\mathrm{d}}{\mathrm{d}x}\right)\psi_0(x) = 0.$$

The solution of this first-order differential equation is

$$\psi_0(x) = C\exp\left(-\frac{m\omega x^2}{2\hbar}\right).$$

With the normalization condition

$$\int_{-\infty}^{\infty}|\psi_0(x)|^2\mathrm{d}x = 1,$$

we have

$$\boxed{\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\exp\left(-\frac{m\omega x^2}{2\hbar}\right).}$$

For an excited state,

$$\begin{aligned}\psi_n(x) = \langle x|n\rangle &= \frac{1}{\sqrt{n}}\langle x|\,\hat{a}^\dagger\,|n-1\rangle = \frac{1}{\sqrt{n(n-1)}}\langle x|\,(\hat{a}^\dagger)^2\,|n-2\rangle \\ &= \frac{1}{\sqrt{n!}}\langle x|\,(\hat{a}^\dagger)^n\,|0\rangle = \frac{1}{\sqrt{n!}}\left(\frac{m\omega}{2\hbar}\right)^{n/2}\langle x|\left(\hat{x} - \frac{\mathrm{i}}{m\omega}\hat{p}\right)^n|0\rangle.\end{aligned}$$

Insert n position completeness relations:

$$\begin{aligned}\psi_n(x) = \frac{1}{\sqrt{n!}}\left(\frac{m\omega}{2\hbar}\right)^{n/2}\int \mathrm{d}x_1\cdots\mathrm{d}x_n\ &\left\langle x\middle|\hat{x} - \frac{\mathrm{i}}{m\omega}\hat{p}\middle|x_1\right\rangle\left\langle x_1\middle|\hat{x} - \frac{\mathrm{i}}{m\omega}\hat{p}\middle|x_2\right\rangle \\ &\cdots\left\langle x_{n-1}\middle|\hat{x} - \frac{\mathrm{i}}{m\omega}\hat{p}\middle|x_n\right\rangle\langle x_n|0\rangle.\end{aligned}$$

Using the position and momentum matrix elements gives

$$\boxed{\psi_n(x) = \frac{1}{\sqrt{n!}}\left(\frac{m\omega}{2\hbar}\right)^{n/2}\left(x - \frac{\hbar}{m\omega}\frac{\mathrm{d}}{\mathrm{d}x}\right)^n\psi_0(x).}$$

Thus the one-dimensional harmonic-oscillator problem has been solved completely.

7) Physical meaning of $\hat{a}^\dagger\hat{a}$

The harmonic-oscillator energies are

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \qquad n = 0, 1, 2, 3, \ldots.$$

The ground-state energy $E_0 = \hbar\omega/2 > 0$ is the zero-point energy. An excited state involves quanta of energy $\varepsilon = \hbar\omega$. When the oscillator is in the n-th excited state, n quanta have been excited.

n: quantum number, particle number.

$\hat{N} = \hat{a}^\dagger\hat{a}$ is the particle-number operator; $\hat{a}$ is a particle-annihilation operator and $\hat{a}^\dagger$ a particle-creation operator.

The common representation of $\hat{H}$ and $\hat{N}$ is the particle-number representation.

To summarize the algebraic solution: bound states → discrete spectrum → identify lowering and raising operators for the discrete quantum numbers. The particle-number representation is the foundation of the second-quantization scheme.

8) Casimir effect

Does the zero-point energy have observable effects? Consider the vacuum energy of the quantized electromagnetic field.

According to Planck's studies of blackbody radiation, the electromagnetic field consists of harmonic oscillations of different frequencies ω_k (energy $\hbar\omega_k$). Here k is the wave number, $\omega_k = ck$, and n is the number of photons of energy $\hbar\omega_k$. The total electromagnetic-field energy is written

$$E = 2\int \frac{\mathrm{d}^3k}{(2\pi)^3} \sum_{n=0}^{\infty} \left(n + \frac{1}{2}\right) \hbar\omega_k.$$

Boundary conditions change the distribution of frequencies, hence the distribution of wave numbers k.

The vacuum is the unexcited state, $n = 0$, with vacuum energy

$$E = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \hbar\omega_k,$$

which is infinite. In ordinary calculations it is subtracted as the background to excitations. The question is whether the vacuum energy itself has observable effects.

Place two parallel plates in the electromagnetic radiation field, each of area L^2 with L sufficiently large and separation $a \ll L$. Boundary conditions restrict the component of wave number perpendicular to the plates to $k_z = n\pi/a$; in the transverse direction there is no significant restriction because L is large. The vacuum energy between the plates, in volume L^2a, is

$$E(a) = \hbar c \int \frac{\mathrm{d}^2k_\perp}{(2\pi/L)^2} \sum_{n=0}^{\infty} \sqrt{k_\perp^2 + \left(\frac{n\pi}{a}\right)^2}.$$

Here $\mathrm{d}^2k_\perp/(2\pi/L)^2$ is the number of transverse modes in the element $\mathrm{d}^2k_\perp$ near $\boldsymbol{k}_\perp$.

The change in vacuum energy associated with inserting the plates into this volume is

$$\Delta E(a) = E(a) - E_0(a) = \hbar c \int \frac{\mathrm{d}^2k_\perp}{(2\pi/L)^2} \Bigg[\sum_{n=0}^{\infty} \sqrt{k_\perp^2 + \left(\frac{n\pi}{a}\right)^2} - \int_0^{\infty} \mathrm{d}n \sqrt{k_\perp^2 + \left(\frac{n\pi}{a}\right)^2} \Bigg].$$

This change produces a force per unit area between the plates,

$$F(a) = -\frac{1}{L^2} \frac{\partial \Delta E(a)}{\partial a}.$$

Both $E(a)$ and $E_0(a)$ diverge (vacuum divergence), but $\Delta E(a)$ is finite. After calculation (see standard quantum-mechanics textbooks, for example those by Su Ru-Keng and Zhang Yong-De),

$$\boxed{F(a) = -\frac{\hbar c \pi^2}{240 a^4}.}$$

The attractive force caused by the change (fluctuation) in vacuum energy when the plates are inserted, and its dependence on a, were experimentally observed in 1958. This is an observable effect of the vacuum energy.

9) Time evolution of observables in the Heisenberg picture

Omitting the H subscript, the Heisenberg equations of motion give

$$\frac{\mathrm{d}\hat{a}}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}[\hat{a}, \hat{H}] = -\mathrm{i}\omega\hat{a}, \qquad \frac{\mathrm{d}\hat{a}^\dagger}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}[\hat{a}^\dagger, \hat{H}] = \mathrm{i}\omega\hat{a}^\dagger.$$

The solutions are

$$\hat{a}(t) = \hat{a}(0)e^{-\mathrm{i}\omega t}, \qquad \hat{a}^\dagger(t) = \hat{a}^\dagger(0)e^{\mathrm{i}\omega t}.$$

Alternatively,

$$\hat{x}(t) = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}(t) + \hat{a}^\dagger(t)\right) = \hat{x}(0)\cos\omega t + \frac{\hat{p}(0)}{m\omega}\sin\omega t,$$
$$\hat{p}(t) = \mathrm{i}\sqrt{\frac{m\hbar\omega}{2}}\left(\hat{a}^\dagger(t) - \hat{a}(t)\right) = \hat{p}(0)\cos\omega t - m\omega\hat{x}(0)\sin\omega t.$$

Although $\hat{x}(t)$ and $\hat{p}(t)$ oscillate, their expectation values in an energy eigenstate are

$$\langle n|\hat{x}(t)|n\rangle = \langle n|\hat{x}(0)|n\rangle\cos\omega t + \frac{1}{m\omega}\langle n|\hat{p}(0)|n\rangle\sin\omega t = 0,$$
$$\langle n|\hat{p}(t)|n\rangle = \langle n|\hat{p}(0)|n\rangle\cos\omega t - m\omega\langle n|\hat{x}(0)|n\rangle\sin\omega t = 0$$

and thus show no oscillation.

2.4 Coherent States of the Harmonic Oscillator

In energy eigenstates of the harmonic oscillator, the expectation values of $\hat{x}$ and $\hat{p}$ are zero, and no classical oscillation is observed. However, consider a linear combination of energy eigenstates, for example

$$|a\rangle = c_0|0\rangle + c_1|1\rangle.$$

In the Heisenberg picture, the expectation value is

$$\begin{aligned}\langle a|\hat{x}(t)|a\rangle &= (c_0^*\langle 0| + c_1^*\langle 1|)\hat{x}(t)(c_0|0\rangle + c_1|1\rangle) \\ &= c_0^*c_1\langle 0|\hat{x}(t)|1\rangle + c_1^*c_0\langle 1|\hat{x}(t)|0\rangle \\ &= \sqrt{\frac{\hbar}{2m\omega}}\left(c_0^*c_1e^{-\mathrm{i}\omega t} + c_1^*c_0e^{\mathrm{i}\omega t}\right).\end{aligned}$$

Oscillations appear. Can one find a state whose oscillation is closest to classical oscillation, that is, the oscillation of $\hat{x}(t)$? This leads to the coherent state $|\alpha\rangle$, defined as an eigenstate of the lowering operator:

$$\boxed{\hat{a}|\alpha\rangle = \alpha|\alpha\rangle.}$$

Note that $\hat{a}$ is not Hermitian; its eigenvalue α is generally complex.

1) Coherent states satisfy the minimum-uncertainty relation

The uncertainty relation is

$$(\Delta x)^2(\Delta p)^2 \geq \frac{\hbar^2}{4}.$$

In the coherent state $|\alpha\rangle$,

$$\langle\alpha|\hat{x}|\alpha\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left\langle\alpha\middle|\hat{a}^\dagger + \hat{a}\middle|\alpha\right\rangle = \sqrt{\frac{\hbar}{2m\omega}}(\alpha^* + \alpha),$$
$$\left\langle\alpha\middle|\hat{x}^2\middle|\alpha\right\rangle = \frac{\hbar}{2m\omega}[1 + (\alpha + \alpha^*)^2],$$
$$\langle\alpha|\hat{p}|\alpha\rangle = -\mathrm{i}\sqrt{\frac{\hbar m\omega}{2}}(\alpha - \alpha^*),$$
$$\left\langle\alpha\middle|\hat{p}^2\middle|\alpha\right\rangle = \frac{\hbar m\omega}{2}[1 - (\alpha - \alpha^*)^2].$$

It follows that

$$(\Delta x)^2(\Delta p)^2 = \left(\langle\hat{x}^2\rangle - \langle\hat{x}\rangle^2\right)\left(\langle\hat{p}^2\rangle - \langle\hat{p}\rangle^2\right) = \frac{\hbar^2}{4}.$$

Thus coherent states satisfy the minimum-uncertainty relation and are the quantum states closest to classical states.

2) Expansion of a coherent state

Expand $|\alpha\rangle$ in energy eigenstates:

$$|\alpha\rangle = \sum_n c_n |n\rangle .$$

Since

$$c_n = \langle n|\alpha\rangle = \frac{1}{\sqrt{n!}}\langle 0|\hat{a}^n|\alpha\rangle = \frac{\alpha^n}{\sqrt{n!}}\langle 0|\alpha\rangle = \frac{\alpha^n}{\sqrt{n!}}c_0,$$

we have

$$|\alpha\rangle = c_0 \sum_n \frac{\alpha^n}{\sqrt{n!}}|n\rangle .$$

Normalization gives

$$1 = \langle\alpha|\alpha\rangle = \sum_n |c_n|^2 = |c_0|^2 \sum_n \frac{|\alpha|^{2n}}{n!} = |c_0|^2 e^{|\alpha|^2}.$$

Choose $c_0 = e^{-|\alpha|^2/2}$; then

$$\boxed{|\alpha\rangle = \sum_{n=0}^{\infty} e^{-|\alpha|^2/2}\frac{\alpha^n}{\sqrt{n!}}|n\rangle .}$$

3) The ground state is a coherent state

For the ground state $|0\rangle$, $\hat{a}|0\rangle = 0$. Thus $|0\rangle$ is an eigenstate of the annihilation operator, with eigenvalue zero.

4) Time evolution of the eigenvalue

In the Heisenberg picture,

$$\hat{a}(t) = \hat{a}(0)e^{-\mathrm{i}\omega t},$$
$$\hat{a}(t)|\alpha\rangle = \hat{a}(0)e^{-\mathrm{i}\omega t}|\alpha\rangle = \alpha e^{-\mathrm{i}\omega t}|\alpha\rangle .$$

Hence the eigenvalue evolves as

$$\boxed{\alpha(t) = \alpha e^{-\mathrm{i}\omega t}.}$$

2.5 Squeezed States

A coherent state has the minimum uncertainty product and exhibits equal fluctuations of its two *dimensionless* oscillator quadratures. Minimum uncertainty does not require these two fluctuations to be equal. A squeezed state redistributes the fluctuations between conjugate quadratures while maintaining the uncertainty bound.

Define the Hermitian quadratures

$$\hat{X} = \frac{\hat{a} + \hat{a}^\dagger}{\sqrt{2}}, \qquad \hat{P} = \frac{\hat{a} - \hat{a}^\dagger}{\mathrm{i}\sqrt{2}}, \qquad [\hat{X}, \hat{P}] = \mathrm{i}\hat{I}.$$

For real r introduce the *single-mode squeezing operator*

$$\boxed{\hat{S}(r) = \exp\left[\frac{r}{2}(\hat{a}^2 - \hat{a}^{\dagger 2})\right], \qquad |0, r\rangle = \hat{S}(r)\,|0\rangle\,.}$$

Its exponent is anti-Hermitian; $\hat{S}(r)$ is unitary. The transformation of the lowering operator is the Bogoliubov relation

$$\boxed{\hat{S}^\dagger(r)\hat{a}\hat{S}(r) = \hat{a}\cosh r - \hat{a}^\dagger \sinh r.}$$

To derive it, set $\hat{a}(r) = \hat{S}^\dagger(r)\hat{a}\hat{S}(r)$. Differentiate with respect to r and use the commutators $[\hat{a}, \hat{a}^{\dagger 2}] = 2\hat{a}^\dagger$ and $[\hat{a}, \hat{a}^2] = 0$. Together with the adjoint equation this gives

$$\frac{\mathrm{d}\hat{a}(r)}{\mathrm{d}r} = -\hat{a}^\dagger(r), \qquad \frac{\mathrm{d}\hat{a}^\dagger(r)}{\mathrm{d}r} = -\hat{a}(r),$$

whose solution with $\hat{a}(0) = \hat{a}$ is the displayed transformation.

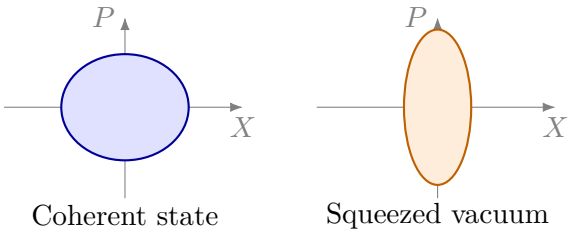
Consequently,

$$\hat{S}^\dagger(r)\hat{X}\hat{S}(r) = e^{-r}\hat{X}, \qquad \hat{S}^\dagger(r)\hat{P}\hat{S}(r) = e^{r}\hat{P}.$$

The vacuum has $\langle\hat{X}\rangle = \langle\hat{P}\rangle = 0$ and $\langle\hat{X}^2\rangle = \langle\hat{P}^2\rangle = 1/2$. Evaluating the transformed variances in $|0, r\rangle$ gives

$$\boxed{(\Delta X)^2 = \frac{1}{2}e^{-2r}, \qquad (\Delta P)^2 = \frac{1}{2}e^{2r}, \qquad \Delta X\,\Delta P = \frac{1}{2}.}$$

For $r > 0$, the X quadrature is squeezed and the P quadrature is anti-squeezed. Both coherent and squeezed vacuum states saturate the Heisenberg uncertainty relation; they have different distributions of the fluctuations.



Coherent state Squeezed vacuum

The ellipses schematically indicate quantum uncertainties in phase space. The squeezed state retains the minimum uncertainty *area*, but its shape changes. A displaced squeezed state is $\hat{D}(\alpha)\hat{S}(r)\,|0\rangle$, where $\hat{D}(\alpha) = \exp(\alpha\hat{a}^\dagger - \alpha^*\hat{a})$; the displacement moves the center without altering these variances.

Ten-minute lecture route. Recall equal quadrature fluctuations of a coherent state; write the squeezing operator and its transformation of $\hat{a}$; derive the two variances and draw the circular and elliptical uncertainty regions. The Bogoliubov differentiation and displaced case provide optional follow-up reading.

2.6 Schrödinger Equation in the Coordinate Representation and the WKB Approximation

Consider the abstract state $|\psi, t\rangle$, whose Schrödinger equation is

$$\mathrm{i}\hbar\frac{\partial}{\partial t}|\psi,t\rangle = \hat{H}|\psi,t\rangle\,.$$

Multiply from the left by the eigenbra $\langle\boldsymbol{x}|$ of the position operator $\hat{\boldsymbol{x}}$. Since these eigenstates are time independent in the Schrödinger picture,

$$\mathrm{i}\hbar\frac{\partial}{\partial t}\langle\boldsymbol{x}|\psi,t\rangle = \left\langle\boldsymbol{x}\middle|\hat{H}\middle|\psi,t\right\rangle = \int \mathrm{d}^3x'\,\left\langle\boldsymbol{x}\middle|\hat{H}\middle|\boldsymbol{x}'\right\rangle\langle\boldsymbol{x}'|\psi,t\rangle\,.$$

Let

$$\hat{H} = \frac{\hat{\boldsymbol{p}}^2}{2m} + V(\hat{\boldsymbol{x}}).$$

Since

$$\left\langle\boldsymbol{x}\middle|\hat{H}\middle|\boldsymbol{x}'\right\rangle = \delta^{(3)}(\boldsymbol{x}-\boldsymbol{x}')\left[-\frac{\hbar^2}{2m}\nabla^2 + V(\boldsymbol{x})\right],$$

we obtain

$$\mathrm{i}\hbar\frac{\partial}{\partial t}\langle\boldsymbol{x}|\psi,t\rangle = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\boldsymbol{x})\right]\langle\boldsymbol{x}|\psi,t\rangle\,.$$

Define the wavefunction $\psi(\boldsymbol{x},t) = \langle\boldsymbol{x}|\psi,t\rangle$. This is the Schrödinger equation in the coordinate representation.

1) Time-independent Schrödinger equation

Put

$$\psi(\boldsymbol{x},t) = \varphi(\boldsymbol{x})T(t).$$

If $\hat{H}$ has no time dependence, inserting this into the Schrödinger equation separates it into two equations. The time-dependent factor is

$$T(t) = e^{-\mathrm{i}Et/\hbar}.$$

The energy E and $\varphi(\boldsymbol{x})$ are determined by the other equation, the energy eigenvalue equation:

$$\boxed{\left[-\frac{\hbar^2}{2m}\nabla^2 + V(\boldsymbol{x})\right]\varphi_E(\boldsymbol{x}) = E\varphi_E(\boldsymbol{x}).}$$

A stationary state is

$$\psi_E(\boldsymbol{x},t) = \varphi_E(\boldsymbol{x})e^{-\mathrm{i}Et/\hbar}.$$

For bound states the bound-state condition

$$\lim_{|\boldsymbol{x}|\to\infty}\varphi_E(\boldsymbol{x}) = 0$$

leads to discrete values of E.

Stationary states are particular solutions when $\hat{H}$ is time independent. The general solution is a linear superposition of stationary states:

$$\psi(\boldsymbol{x},t) = \sum_E c_E\psi_E(\boldsymbol{x},t).$$

2) Probability interpretation of the wavefunction

The probability density of finding the particle at a given point is

$$\rho(\boldsymbol{x},t) = |\,\langle \boldsymbol{x}|\psi(t)\rangle\,|^2 = |\psi(\boldsymbol{x},t)|^2 = \psi^*(\boldsymbol{x},t)\psi(\boldsymbol{x},t).$$

When $V(\boldsymbol{x})$ is real, the Schrödinger equation can be written as the continuity equation,

$$\boxed{\frac{\partial \rho(\boldsymbol{x},t)}{\partial t} + \boldsymbol{\nabla}\cdot\boldsymbol{j}(\boldsymbol{x},t) = 0,}$$

$$\boldsymbol{j}(\boldsymbol{x},t) = -\frac{\mathrm{i}\hbar}{2m}\left[\psi^*(\boldsymbol{x},t)\boldsymbol{\nabla}\psi(\boldsymbol{x},t) - \psi(\boldsymbol{x},t)\boldsymbol{\nabla}\psi^*(\boldsymbol{x},t)\right].$$

What is the physical meaning of $\boldsymbol{j}$? Integrate the continuity equation over a finite volume V:

$$\int_V \mathrm{d}^3r\,\frac{\partial\rho(\boldsymbol{r},t)}{\partial t} + \int_V \mathrm{d}^3r\,\boldsymbol{\nabla}\cdot\boldsymbol{j}(\boldsymbol{r},t) = 0,$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \mathrm{d}^3r\,\rho(\boldsymbol{r},t) = -\oint_{\partial V}\mathrm{d}\boldsymbol{S}\cdot\boldsymbol{j}(\boldsymbol{r},t).$$

Local probability conservation says that the change in the probability inside V equals the negative of the outward probability flux across its boundary S. Thus $\boldsymbol{j}$ is called the probability-current density.

If we integrate over all space,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^3} \mathrm{d}^3r\,\rho(\boldsymbol{r},t) = -\oint_{\infty}\mathrm{d}\boldsymbol{S}\cdot\boldsymbol{j}(\boldsymbol{r},t) = 0.$$

Therefore $\int \mathrm{d}^3r\,\rho(\boldsymbol{r},t)$ is independent of time and is a constant. This implies:

a) Total probability is conserved; particles are neither created nor annihilated. This describes nonrelativistic quantum mechanics.

b) The normalization of a wavefunction is independent of time. Schrödinger evolution preserves normalization.

c) If V is complex, probability is not conserved, and this can effectively describe creation and annihilation of particles.

3) How does a quantum system approach a classical system?

An operator in a general state has no definite value, whereas its expectation value does. Thus expectation values of quantum observables correspond to physical quantities in classical mechanics.

The classical equation of motion is

$$\frac{\mathrm{d}O}{\mathrm{d}t} = \{O,H\}_{\text{Poisson bracket}},$$

while the quantum equation of motion for the expectation value is

$$\frac{\mathrm{d}\langle O\rangle}{\mathrm{d}t} = \frac{\langle[\hat{O},\hat{H}]\rangle}{\mathrm{i}\hbar}.$$

In the classical limit, observables commute, and $[\hat{O},\hat{H}]\to 0$. To recover the classical equation of motion, one requires

$$\hbar \longrightarrow 0, \qquad \frac{\langle[\hat{O},\hat{H}]\rangle}{\mathrm{i}\hbar} \longrightarrow \{O,H\}_{\text{Poisson bracket}}.$$

4) WKB approximation to the time-independent equation

Only a few Schrödinger equations can be solved exactly. Expanding the Schrödinger equation in powers of $\hbar$ and solving it by successive approximations gives the WKB approximation (Wentzel, Kramers and Brillouin).

Write a stationary-state wavefunction as

$$\Psi(\boldsymbol{r},t) = \psi(\boldsymbol{r})e^{-\mathrm{i}Et/\hbar} = e^{\mathrm{i}s(\boldsymbol{r})/\hbar}e^{-\mathrm{i}Et/\hbar} = e^{\mathrm{i}[s(\boldsymbol{r})-Et]/\hbar}.$$

Insert this into the stationary Schrödinger equation,

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V(\boldsymbol{r})\right]\psi(\boldsymbol{r}) = E\psi(\boldsymbol{r}).$$

We find

$$\frac{(\boldsymbol{\nabla}s)^2}{2m} + V(\boldsymbol{r}) - E - \frac{\mathrm{i}\hbar}{2m}\nabla^2 s = 0.$$

Expand $s(\boldsymbol{r})$ in $\hbar/\mathrm{i}$:

$$s(\boldsymbol{r}) = s_0(\boldsymbol{r}) + \frac{\hbar}{\mathrm{i}}s_1(\boldsymbol{r}) + \left(\frac{\hbar}{\mathrm{i}}\right)^2 s_2(\boldsymbol{r}) + \cdots.$$

Equate like powers of $\hbar$:

$$\begin{aligned}\hbar^0: &\quad (\boldsymbol{\nabla}s_0)^2 + 2m[V(\boldsymbol{r}) - E] = 0,\\ \hbar^1: &\quad \boldsymbol{\nabla}s_1\cdot\boldsymbol{\nabla}s_0 + \tfrac{1}{2}\nabla^2 s_0 = 0,\\ &\quad\ \vdots\end{aligned}$$

For one dimension,

$$s_0(x) = \pm\int^x p(x')\mathrm{d}x' + C, \qquad s_1(x) = -\ln\sqrt{p(x)} + \widetilde{C},$$

$$\boxed{p(x) = \sqrt{2m[E - V(x)]}.}$$

Keeping terms through first order in $\hbar$,

$$\begin{aligned}\psi_{\mathrm{II}}(x) &= \exp\left[\frac{\mathrm{i}}{\hbar}\left(s_0(x) + \frac{\hbar}{\mathrm{i}}s_1(x)\right)\right]\\ &= \frac{C_1}{\sqrt{p(x)}}\exp\left(\frac{\mathrm{i}}{\hbar}\int^x p(x')\mathrm{d}x'\right) + \frac{C_2}{\sqrt{p(x)}}\exp\left(-\frac{\mathrm{i}}{\hbar}\int^x p(x')\mathrm{d}x'\right).\end{aligned}$$

This solution is valid where $p(x) = \sqrt{2m[E - V(x)]} > 0$, that is, $E > V$. In classical mechanics, $E = T + V$; the condition $E > V$ therefore identifies a classically allowed region.

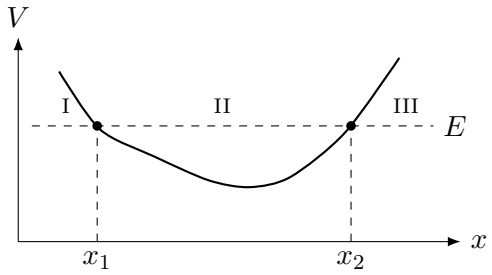
In the classically forbidden region, $E < V$, put

$$p(x) = \mathrm{i}\widetilde{p}(x) = \mathrm{i}\sqrt{2m[V(x) - E]}.$$

Then

$$\psi_{\mathrm{I,III}}(x) = \frac{C_3}{\sqrt{|p(x)|}}\exp\left(\frac{1}{\hbar}\int^x \widetilde{p}(x')\mathrm{d}x'\right) + \frac{C_4}{\sqrt{|p(x)|}}\exp\left(-\frac{1}{\hbar}\int^x \widetilde{p}(x')\mathrm{d}x'\right).$$

At $E = V$, the momentum $p = 0$ and the solution diverges: WKB is no longer valid. A point x_0 satisfying $E - V(x_0) = 0$ is a classical turning point. How can one solve the equation near x_0?



Expand the potential to linear order near x_0:

$$V(x) \simeq V(x_0) + \left.\frac{\mathrm{d}V}{\mathrm{d}x}\right|_{x_0} (x - x_0).$$

Substitution into the stationary Schrödinger equation gives

$$\frac{\mathrm{d}^2\varphi}{\mathrm{d}x^2} - \frac{2m}{\hbar^2} \left.\frac{\mathrm{d}V}{\mathrm{d}x}\right|_{x_0} (x - x_0)\varphi = 0.$$

The exact solutions are Bessel functions of order $\pm\frac{1}{3}$. Consider their asymptotic solutions when $|x - x_0|$ is large and require continuity with the classically allowed and forbidden solutions on either side of x_0. In this way the constants C_1, C_2, C_3, C_4 can be fixed.

2.7 Feynman Path Integral

1) Propagator in the Schrödinger picture

Start with the time evolution of the state:

$$|\psi, t\rangle = e^{-\mathrm{i}\hat{H}(t-t_0)/\hbar} |\psi, t_0\rangle .$$

In the coordinate representation,

$$\begin{aligned} \langle \boldsymbol{x}'' | \psi, t\rangle &= \langle \boldsymbol{x}'' | \, e^{-\mathrm{i}\hat{H}(t-t_0)/\hbar} \, |\psi, t_0\rangle \\ &= \int \mathrm{d}^3x' \, \langle \boldsymbol{x}'' | \, e^{-\mathrm{i}\hat{H}(t-t_0)/\hbar} \, |\boldsymbol{x}'\rangle \langle \boldsymbol{x}' | \psi, t_0\rangle . \end{aligned}$$

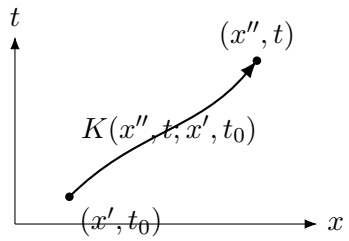
Define the matrix element of the evolution operator,

$$\boxed{K(\boldsymbol{x}'', t; \boldsymbol{x}', t_0) = \langle \boldsymbol{x}'' | \, e^{-\mathrm{i}\hat{H}(t-t_0)/\hbar} \, |\boldsymbol{x}'\rangle .}$$

Then

$$\psi(\boldsymbol{x}'', t) = \int \mathrm{d}^3x' \, K(\boldsymbol{x}'', t; \boldsymbol{x}', t_0)\psi(\boldsymbol{x}', t_0).$$

The function $K(\boldsymbol{x}'', t; \boldsymbol{x}', t_0)$ is the propagator: it propagates the initial wavefunction $\psi(\boldsymbol{x}', t_0)$ to the final wavefunction $\psi(\boldsymbol{x}'', t)$. Integration means that every possible position of the particle in the initial wavefunction contributes to its final state.



We now derive the propagator's relationship to the equation of motion. Take $t \geq t_0$ and define a related propagator incorporating the initial condition:

$$K(\boldsymbol{x}'', t; \boldsymbol{x}', t_0) = \langle \boldsymbol{x}'' | \, e^{-\mathrm{i}\hat{H}(t-t_0)/\hbar} \, | \boldsymbol{x}' \rangle \; \theta(t - t_0).$$

Differentiate with respect to t:

$$\begin{aligned}
\mathrm{i}\hbar \frac{\partial K}{\partial t} &= \langle \boldsymbol{x}'' | \, \hat{H} e^{-\mathrm{i}\hat{H}(t-t_0)/\hbar} \, | \boldsymbol{x}' \rangle \, \theta(t - t_0) \\
&\quad + \mathrm{i}\hbar \, \langle \boldsymbol{x}'' | \, e^{-\mathrm{i}\hat{H}(t-t_0)/\hbar} \, | \boldsymbol{x}' \rangle \, \delta(t - t_0) \\
&= \int \mathrm{d}^3 x''' \, \left\langle \boldsymbol{x}'' \middle| \hat{H} \middle| \boldsymbol{x}''' \right\rangle K(\boldsymbol{x}''', t; \boldsymbol{x}', t_0) \\
&\quad + \mathrm{i}\hbar \delta^{(3)}(\boldsymbol{x}'' - \boldsymbol{x}') \delta(t - t_0) \\
&= \hat{H}(\boldsymbol{x}'') K(\boldsymbol{x}'', t; \boldsymbol{x}', t_0) + \mathrm{i}\hbar \delta^{(3)}(\boldsymbol{x}'' - \boldsymbol{x}') \delta(t - t_0).
\end{aligned}$$

Thus

$$\boxed{\left(\mathrm{i}\hbar \frac{\partial}{\partial t} - \hat{H}(\boldsymbol{x}'') \right) K(\boldsymbol{x}'', t; \boldsymbol{x}', t_0) = \mathrm{i}\hbar \delta^{(3)}(\boldsymbol{x}'' - \boldsymbol{x}') \delta(t - t_0).}$$

Notice that taking the spatial trace of the propagator yields

$$\begin{aligned}
\int \mathrm{d}^3 x \, K(\boldsymbol{x}, t; \boldsymbol{x}, 0) &= \int \mathrm{d}^3 x \; \langle \boldsymbol{x} | \, e^{-\mathrm{i}\hat{H}t/\hbar} \, | \boldsymbol{x} \rangle \\
&= \sum_E \int \mathrm{d}^3 x \; \langle \boldsymbol{x} | E \rangle \, \langle E | \boldsymbol{x} \rangle \; e^{-\mathrm{i}Et/\hbar} \\
&= \sum_E e^{-\mathrm{i}Et/\hbar}.
\end{aligned}$$

With the imaginary-time substitution $t = -\mathrm{i}\hbar\beta$, this becomes the statistical-mechanical partition function $Z = \sum_E e^{-\beta E}$, showing a relation between imaginary-time propagators and statistical mechanics.

Example 1: propagator of a one-dimensional free particle.

$$\begin{aligned}
K(x'', t; x', t_0) &= \langle x'' | \, e^{-\mathrm{i}\hat{H}(t-t_0)/\hbar} \, | x' \rangle \\
&= \int \mathrm{d}p \; \langle x'' | p \rangle \, e^{-\mathrm{i}p^2(t-t_0)/(2m\hbar)} \, \langle p | x' \rangle \\
&= \int \frac{\mathrm{d}p}{2\pi\hbar} \exp\left[\frac{\mathrm{i}}{\hbar} \left(p(x'' - x') - \frac{p^2(t - t_0)}{2m} \right) \right] \\
&= \sqrt{\frac{m}{2\pi\mathrm{i}\hbar(t - t_0)}} \exp\left[\frac{\mathrm{i}m(x'' - x')^2}{2\hbar(t - t_0)} \right].
\end{aligned}$$

Once this propagator is known, the time evolution of an initial plane wave (stationary state), $\psi(x', t_0) = (2\pi)^{-1} \exp\left[\mathrm{i}(px' - p^2 t_0/(2m))/\hbar\right]$, is

$$\psi(x'', t) = \int \mathrm{d}x' \, K(x'', t; x', t_0) \psi(x', t_0),$$

which is again a plane wave.

Example 2: propagator of a one-dimensional harmonic oscillator.

$$\begin{aligned}
K(x'', t; x', t_0) &= \langle x'' | \, e^{-\mathrm{i}\hat{H}(t-t_0)/\hbar} \, | x' \rangle \\
&= \sum_n \langle x'' | n \rangle \, e^{-\mathrm{i}E_n(t-t_0)/\hbar} \, \langle n | x' \rangle \\
&= \sum_n \psi_n(x'') \psi_n^*(x') e^{-\mathrm{i}E_n(t-t_0)/\hbar}.
\end{aligned}$$

Here E_n and $\psi_n(x)$ are the oscillator's energies and its coordinate-representation wavefunctions.

2) Transition amplitude in the Heisenberg picture

In the Heisenberg picture,

$$\begin{aligned} K(\boldsymbol{x}'', t; \boldsymbol{x}', t_0) &= \langle \boldsymbol{x}'' | \, e^{-\mathrm{i}\hat{H}(t-t_0)/\hbar} \, | \boldsymbol{x}' \rangle \\ &= \left(\langle \boldsymbol{x}'' | \, e^{-\mathrm{i}\hat{H}t/\hbar} \right) \left(e^{\mathrm{i}\hat{H}t_0/\hbar} \, | \boldsymbol{x}' \rangle \right) \\ &= \left(\langle \boldsymbol{x}'' | \, \hat{U}(t,0) \right) \left(\hat{U}^\dagger(t_0,0) \, | \boldsymbol{x}' \rangle \right) = \langle \boldsymbol{x}'', t | \boldsymbol{x}', t_0 \rangle \, . \end{aligned}$$

This is the amplitude for the position eigenstate $|\boldsymbol{x}', t_0\rangle$ at time t_0 to make a transition to the position eigenstate $|\boldsymbol{x}'', t\rangle$ at time t. Thus the propagator is also called a transition amplitude.

The completeness relation in the Heisenberg picture must involve basis states at the same time:

$$\int \mathrm{d}^3 x \; |\boldsymbol{x}, t\rangle \, \langle \boldsymbol{x}, t| = \mathbf{1}.$$

It follows that

$$\begin{aligned} K(\boldsymbol{x}_3, t_3; \boldsymbol{x}_1, t_1) &= \langle \boldsymbol{x}_3, t_3 | \boldsymbol{x}_1, t_1 \rangle \\ &= \int \mathrm{d}^3 x_2 \; \langle \boldsymbol{x}_3, t_3 | \boldsymbol{x}_2, t_2 \rangle \, \langle \boldsymbol{x}_2, t_2 | \boldsymbol{x}_1, t_1 \rangle \\ &= \int \mathrm{d}^3 x_2 \, K(\boldsymbol{x}_3, t_3; \boldsymbol{x}_2, t_2) K(\boldsymbol{x}_2, t_2; \boldsymbol{x}_1, t_1), \quad t_3 > t_2 > t_1. \end{aligned}$$

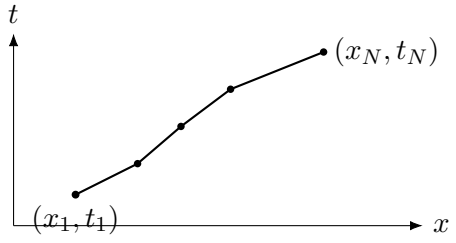
This is the composition law of propagators: propagation from $(\boldsymbol{x}_1, t_1)$ to $(\boldsymbol{x}_3, t_3)$ is the sum over propagation first to $(\boldsymbol{x}_2, t_2)$ and then to $(\boldsymbol{x}_3, t_3)$.

For simplicity, consider one spatial dimension. Divide the interval from the initial to final time into $N-1$ equal parts:

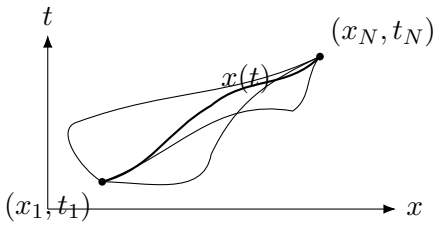
$$\mathrm{d}t = \frac{t_N - t_1}{N-1}.$$

The full propagator is

$$\begin{aligned} K(x_N, t_N; x_1, t_1) = \int & \mathrm{d}x_{N-1} \mathrm{d}x_{N-2} \cdots \mathrm{d}x_2 \; K(x_N, t_N; x_{N-1}, t_{N-1}) \\ & \times K(x_{N-1}, t_{N-1}; x_{N-2}, t_{N-2}) \cdots K(x_2, t_2; x_1, t_1). \end{aligned}$$



As $N \to \infty$, or $\mathrm{d}t \to 0$, this multiple integral becomes a sum over paths.



Unlike quantum mechanics, where every path contributes to evolution in space-time, in classical mechanics a particular initial and final state is connected by a unique path determined by the principle of least action. For example, the unique path of free fall is

$$x = h - \frac{1}{2}gt^2,$$

where h is the height and g the acceleration of gravity.

3) Path integral

The basic postulate of Feynman's path-integral theory is that the probability amplitude for propagation from (x_1, t_1) to (x_N, t_N) is the sum of contributions from all possible paths $x(t)$:

$$\boxed{K(x_N, t_N; x_1, t_1) = C \int_{x_1}^{x_N} [\mathrm{d}x(t)]\, e^{\mathrm{i}S[x(t)]/\hbar} = C \int_{x_1}^{x_N} [\mathrm{d}x(t)] \exp\left[\frac{\mathrm{i}}{\hbar}\int_{t_1}^{t_N} L_c(x, \dot{x})\mathrm{d}t\right].}$$

The contribution of each path is $Ce^{\mathrm{i}S[x(t)]/\hbar}$. The action is

$$S[x(t)] = \int_{t_1}^{t_N} L_c(x, \dot{x})\mathrm{d}t, \qquad L_c = \frac{1}{2}m\dot{x}^2 - V(x).$$

To return the path integral to a multiple integral, take $\mathrm{d}t = (t_N - t_1)/N$ and obtain

$$K(x_N, t_N; x_1, t_1) = C \lim_{N\to\infty} \int \mathrm{d}x_{N-1} \cdots \mathrm{d}x_2 \, \exp\left[\frac{\mathrm{i}}{\hbar}\sum_{n=2}^{N} L_c(x_n, \dot{x}_n)\mathrm{d}t\right].$$

If L_c is quadratic in x, the path integral is Gaussian and exactly integrable. For example, for a free particle,

$$L_c = \frac{1}{2}m\dot{x}^2, \qquad L_c(x_n, \dot{x}_n)\mathrm{d}t = \frac{1}{2}m\left(\frac{x_n - x_{n-1}}{\mathrm{d}t}\right)^2 \mathrm{d}t = \frac{m(x_n - x_{n-1})^2}{2\mathrm{d}t}.$$

Then

$$\begin{aligned} K(x_N, t_N; x_1, t_1) &= C \lim_{N\to\infty} \int \mathrm{d}x_{N-1} \cdots \mathrm{d}x_2 \, \exp\left[\frac{\mathrm{i}m}{2\hbar\mathrm{d}t}\sum_{n=2}^{N}(x_n - x_{n-1})^2\right] \\ &= C \exp\left[\frac{\mathrm{i}m(x_N - x_1)^2}{2\hbar(t_N - t_1)}\right]. \end{aligned}$$

The position eigenstates' orthogonality and normalization require

$$\lim_{t_N \to t_1} K(x_N, t_N; x_1, t_1) = \langle x_N, t_1 | x_1, t_1 \rangle = \delta(x_N - x_1).$$

Thus

$$\boxed{K(x_N, t_N; x_1, t_1) = \sqrt{\frac{m}{2\pi\mathrm{i}\hbar(t_N - t_1)}} \exp\left[\frac{\mathrm{i}m(x_N - x_1)^2}{2\hbar(t_N - t_1)}\right].}$$

This agrees with the calculation in the Schrödinger picture.

As another example, for the harmonic oscillator the classical Lagrangian is

$$L_c = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}m\omega^2 x^2.$$

After evaluating the Gaussian integral,

$$K(x_N,t_N;x_1,t_1)=\sqrt{\frac{m\omega}{2\pi\mathrm{i}\hbar\sin[\omega(t_N-t_1)]}}\times\exp\left\{\frac{\mathrm{i}m\omega}{2\hbar\sin[\omega(t_N-t_1)]}\left[(x_N^2+x_1^2)\cos[\omega(t_N-t_1)]-2x_1x_N\right]\right\}.$$

Discussion.

(1) The phase factor contributed by each path is different. In a classical system, when the action S_{cl} along the classical path is much greater than $\hbar$, the actions on nearby paths differ, and their contributions oscillate rapidly and cancel; the contribution from the stationary-action path remains.

(2) Only Gaussian path integrals can be evaluated exactly.

(3) In the Schrödinger picture, the probability amplitude refers to a transition of the state. In Feynman's path integral, the probability amplitude is linked to the particle's motion, namely to paths. Feynman's path-integral theory can be proved equivalent to the Schrödinger equation (see p. 122 of the original source).

(4) A merit of the path integral is that the difference from classical mechanics is apparent and easily understood. Operators and other q-numbers need not be introduced; all calculations take place using c-numbers. Path integrals are relatively cumbersome in nonrelativistic quantum mechanics, but important in gauge field theory.

2.8 Gauge Transformations

1) Quantum coherence caused by a constant potential difference

In classical mechanics, $\boldsymbol{F}=-\boldsymbol{\nabla}V(\boldsymbol{x})$, so adding a spatially independent constant to the potential produces no effect.

In quantum mechanics, suppose a particle moves in a potential $V_1(\boldsymbol{x})$. The evolution from the initial state $|\alpha,t_0\rangle$ is

$$|\alpha,t\rangle_1=e^{-\mathrm{i}\hat{H}(t-t_0)/\hbar}\,|\alpha,t_0\rangle\,.$$

If instead the particle moves in $V_2(\boldsymbol{x},t)=V_1(\boldsymbol{x})+\Delta V(t)$, the same initial state evolves as

$$\begin{aligned}|\alpha,t\rangle_2&=e^{-\mathrm{i}\hat{H}(t-t_0)/\hbar}\exp\left[-\frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\Delta V(t')\mathrm{d}t'\right]|\alpha,t_0\rangle\\&=\exp\left[-\frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\Delta V(t')\mathrm{d}t'\right]|\alpha,t\rangle_1\,.\end{aligned}$$

The two states differ by a phase factor $e^{\mathrm{i}(\varphi_2-\varphi_1)}$:

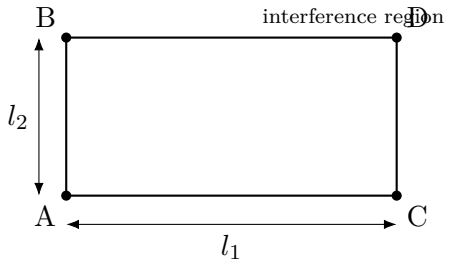
$$\varphi_2-\varphi_1=-\frac{1}{\hbar}\int_{t_0}^{t}\Delta V(t')\mathrm{d}t'\quad\xrightarrow{\Delta V\text{ constant}}\quad-\frac{\Delta V}{\hbar}(t-t_0).$$

This may seem not to lead to any measurable result. However, interference between the two wavefunctions has a term $\cos(\varphi_2-\varphi_1)$ that depends on the phase difference and is observable.

2) Quantum coherence caused by a gravitational potential difference

Consider the motion of thermal neutrons (uncharged) in a gravitational field.

At point A, the neutron beam is split into two: one travels through A–C–D to D, the other through A–B–D to D. If the frame ABCD is parallel to the ground, the two beams experience the same gravitational potential, and there is no relative gravitational phase at D.



If the frame is rotated upward by an angle δ around AC, the particle in the BD segment has a constant potential difference relative to AC:

$$\Delta V = mgl_2 \sin\delta.$$

Although the AB and CD segments also have a gravitational potential difference from AC that depends on spatial position, their contributions are the same along both paths and do not affect the relative phase. Let the neutron's travel time from B to D be T. The resulting phase difference at D is

$$\varphi_{ABD} - \varphi_{ACD} = -\frac{m}{\hbar} g l_2 T \sin\delta.$$

Since $T = l_1/v$, where v is the neutron (wavepacket) velocity, and $mv = p = \hbar/\lambda$ with λ the de Broglie wavelength in the source's convention, we have $T = l_1 m\lambda/\hbar$ and

$$\boxed{\varphi_{ABD} - \varphi_{ACD} = -\left(\frac{m}{\hbar}\right)^2 g l_1 l_2 \lambda \sin\delta.}$$

As the angle δ changes, the interference term $\cos(\varphi_{ABD} - \varphi_{ACD})$ oscillates. Such oscillations have indeed been observed experimentally (see Fig. 2.6 in the source).

This discussion shows that a spatially uniform potential difference may leave the force unchanged while producing a quantum-interference effect.

3) Gauge transformations in electromagnetism

Scalar and vector potentials:

$$\varphi(\boldsymbol{x}, t), \qquad \boldsymbol{A}(\boldsymbol{x}, t).$$

Fields:

$$\boldsymbol{E} = -\boldsymbol{\nabla}\varphi - \frac{1}{c}\frac{\partial \boldsymbol{A}}{\partial t}, \qquad \boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}.$$

From the Hamiltonian of classical electrodynamics, the quantum Hamiltonian of a charged particle in an electromagnetic field is

$$\boxed{\hat{H} = \frac{1}{2m}\left(\hat{\boldsymbol{p}} - \frac{e}{c}\boldsymbol{A}\right)^2 + e\varphi, \quad \varphi = \varphi(\hat{\boldsymbol{x}}, t),\ \boldsymbol{A} = \boldsymbol{A}(\hat{\boldsymbol{x}}, t).}$$

Using the fundamental commutation relations

$$[\hat{x}_i, \hat{x}_j] = [\hat{p}_i, \hat{p}_j] = 0, \qquad [\hat{x}_i, \hat{p}_j] = \mathrm{i}\hbar\delta_{ij},$$

we have

$$\left[\hat{p}_i - \frac{e}{c}A_i, \hat{p}_j - \frac{e}{c}A_j\right] = \frac{\mathrm{i}\hbar e}{c}\epsilon_{ijk}B_k.$$

The Heisenberg equation of motion is

$$\begin{aligned}\frac{\mathrm{d}\hat{x}_i}{\mathrm{d}t} &= \frac{1}{\mathrm{i}\hbar}[\hat{x}_i, \hat{H}] \\ &= \frac{1}{2mi\hbar}\left[\hat{x}_i, \left(\hat{p}_j - \frac{e}{c}A_j\right)\left(\hat{p}_j - \frac{e}{c}A_j\right)\right] \\ &= \frac{1}{m}\left(\hat{p}_i - \frac{e}{c}A_i\right).\end{aligned}$$

Consequently,

$$\begin{aligned}m\frac{\mathrm{d}^2\hat{x}_i}{\mathrm{d}t^2} &= \frac{\mathrm{d}}{\mathrm{d}t}\left(\hat{p}_i - \frac{e}{c}A_i\right) \\ &= \frac{1}{\mathrm{i}\hbar}\left[\hat{p}_i - \frac{e}{c}A_i, \hat{H}\right] - \frac{e}{c}\frac{\partial A_i}{\partial t} \\ &= \frac{e}{2mc}\epsilon_{ijk}\left(\hat{p}_j - \frac{e}{c}A_j\right)B_k - \frac{e}{2mc}\epsilon_{ijk}B_j\left(\hat{p}_k - \frac{e}{c}A_k\right) + eE_i.\end{aligned}$$

In vector notation,

$$\boxed{m\frac{\mathrm{d}^2\hat{\boldsymbol{x}}}{\mathrm{d}t^2} = e\left[\boldsymbol{E} + \frac{1}{2c}\left(\frac{\mathrm{d}\hat{\boldsymbol{x}}}{\mathrm{d}t}\times\boldsymbol{B} - \boldsymbol{B}\times\frac{\mathrm{d}\hat{\boldsymbol{x}}}{\mathrm{d}t}\right)\right].}$$

This is the Ehrenfest theorem in the Heisenberg picture. Note that the velocity operator $\mathrm{d}\hat{\boldsymbol{x}}/\mathrm{d}t$ and $\boldsymbol{B} = \boldsymbol{\nabla}\times\boldsymbol{A}$ generally do not commute; operator ordering matters in this expression.

In the coordinate representation of the Schrödinger picture,

$$\mathrm{i}\hbar\frac{\partial\psi(\boldsymbol{x},t)}{\partial t} = \left[\frac{1}{2m}\left(-\mathrm{i}\hbar\boldsymbol{\nabla} - \frac{e}{c}\boldsymbol{A}(\boldsymbol{x},t)\right)^2 + e\varphi(\boldsymbol{x},t)\right]\psi(\boldsymbol{x},t).$$

The continuity equation is

$$\frac{\partial\rho(\boldsymbol{x},t)}{\partial t} + \boldsymbol{\nabla}\cdot\boldsymbol{j}(\boldsymbol{x},t) = 0,$$

where

$$\rho(\boldsymbol{x},t) = |\psi(\boldsymbol{x},t)|^2,$$

$$\boldsymbol{j}(\boldsymbol{x},t) = \frac{\hbar}{m}\,\mathrm{Im}\left[\psi^*(\boldsymbol{x},t)\left(\boldsymbol{\nabla} - \frac{\mathrm{i}e}{\hbar c}\boldsymbol{A}(\boldsymbol{x},t)\right)\psi(\boldsymbol{x},t)\right].$$

Now transform the potentials as

$$\varphi(\boldsymbol{x},t) \longrightarrow \varphi(\boldsymbol{x},t), \qquad \boldsymbol{A}(\boldsymbol{x},t) \longrightarrow \boldsymbol{A}(\boldsymbol{x},t) + \boldsymbol{\nabla}\Lambda(\boldsymbol{x}),$$

where $\Lambda(\boldsymbol{x})$ is an arbitrary scalar function. It is clear that the fields $\boldsymbol{E}$ and $\boldsymbol{B}$ remain unchanged. A transformation of potentials leaving the field strengths unchanged is called a *gauge transformation*; distinct functions $\Lambda(\boldsymbol{x})$ specify different gauges.

Let the state corresponding to the potentials $\varphi, \boldsymbol{A}$ be $|\psi,t\rangle_1$, and that corresponding to $\varphi, \boldsymbol{A}+\boldsymbol{\nabla}\Lambda$ be $|\psi,t\rangle_2$. They satisfy respectively

$$\mathrm{i}\hbar\frac{\partial}{\partial t}|\psi,t\rangle_1 = \left[\frac{(\hat{\boldsymbol{p}} - e\boldsymbol{A}/c)^2}{2m} + e\varphi\right]|\psi,t\rangle_1,$$

$$\mathrm{i}\hbar\frac{\partial}{\partial t}|\psi,t\rangle_2 = \left[\frac{[\hat{\boldsymbol{p}} - e(\boldsymbol{A}+\boldsymbol{\nabla}\Lambda)/c]^2}{2m} + e\varphi\right]|\psi,t\rangle_2.$$

Take

$$|\psi,t\rangle_2 = \exp\left(\frac{\mathrm{i}e}{\hbar c}\Lambda(\hat{\boldsymbol{x}})\right)|\psi,t\rangle_1.$$

Because

$$e^{-\mathrm{i}e\Lambda/(\hbar c)}\left[\hat{\boldsymbol{p}}-\frac{e}{c}(\boldsymbol{A}+\boldsymbol{\nabla}\Lambda)\right]e^{\mathrm{i}e\Lambda/(\hbar c)}=\hat{\boldsymbol{p}}-\frac{e}{c}\boldsymbol{A},$$

it is easy to show that the two Schrödinger equations are consistent. Thus, under a gauge transformation the states differ by a phase factor:

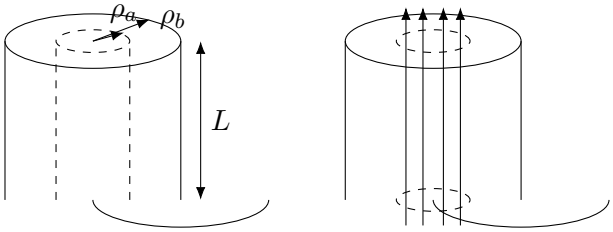
$$\boxed{|\psi,t\rangle\longrightarrow\exp\left(\frac{\mathrm{i}e}{\hbar c}\Lambda(\hat{\boldsymbol{x}})\right)|\psi,t\rangle\,.}$$

Does a gauge transformation have observable quantum effects?

4) Aharonov–Bohm (AB) effect

Consider a long hollow cylindrical shell with inner radius ρ_a and outer radius ρ_b. The particle moves in the shell $\rho_a<\rho<\rho_b$; it cannot enter $\rho<\rho_a$ or $\rho>\rho_b$. The wavefunction vanishes at $\rho=\rho_a$ and $\rho=\rho_b$, so this is a bound-state problem.

Now place a long solenoid in the central hollow region, with its magnetic field confined inside the solenoid, ensuring that the field outside vanishes.



Assume

$$\boldsymbol{B}=\begin{cases}0, & \rho>\rho_a,\\ B\boldsymbol{e}_z, & \rho<\rho_a.\end{cases}$$

In cylindrical coordinates,

$$\boldsymbol{B}=\boldsymbol{\nabla}\times\boldsymbol{A}=\frac{1}{\rho}\begin{vmatrix}\boldsymbol{e}_\rho & \rho\boldsymbol{e}_\phi & \boldsymbol{e}_z\\ \partial/\partial\rho & \partial/\partial\phi & \partial/\partial z\\ A_\rho & \rho A_\phi & A_z\end{vmatrix}.$$

One finds

$$\boxed{\boldsymbol{A}=\begin{cases}\dfrac{B\rho_a^2}{2\rho}\boldsymbol{e}_\phi, & \rho>\rho_a,\\ \dfrac{B\rho}{2}\boldsymbol{e}_\phi, & \rho<\rho_a.\end{cases}}$$

Introducing the field into the region $\rho<\rho_a$ changes the vector potential in the particle's accessible region $\rho_a<\rho<\rho_b$ from $\boldsymbol{A}=0$ to $\boldsymbol{A}\neq 0$, while the magnetic field there remains $\boldsymbol{B}=0$.

The stationary Schrödinger equation is

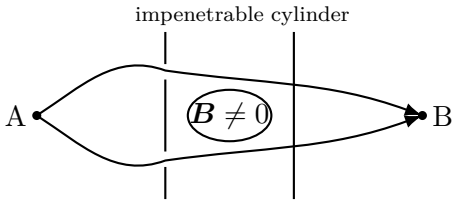
$$\frac{1}{2m}\left(\hat{\boldsymbol{p}}-\frac{e}{c}\boldsymbol{A}\right)^2\psi=-\frac{\hbar^2}{2m}\left[\boldsymbol{e}_\rho\frac{\partial}{\partial\rho}+\frac{\boldsymbol{e}_\phi}{\rho}\left(\frac{\partial}{\partial\phi}-\frac{\mathrm{i}eB\rho_a^2}{2\hbar c}\right)+\boldsymbol{e}_z\frac{\partial}{\partial z}\right]^2\psi=E\psi,\quad \rho_a<\rho<\rho_b.$$

Clearly, the magnetic-field term changes the particle's bound-state energies E. Crucially, the magnetic field $\boldsymbol{B}$ is absent from the particle's accessible region, while $\boldsymbol{A}$ is present there. In classical mechanics the motion of a charged particle is determined by $\boldsymbol{E}$ and $\boldsymbol{B}$, whereas in

quantum mechanics a potential without a field strength can still influence the properties of a system. This is a purely quantum effect.

The above is the influence of the AB effect on a bound-state problem.

Now consider the AB effect using the path integral. A charged particle starts at A and passes through two slits in a screen to reach B, where interference occurs. A long solenoid lies between the two slits behind the screen. The magnetic field is inside the solenoid and vanishes outside. For the classical system, the field inside the solenoid cannot affect the interference at B.



For a quantum system, although the field vanishes outside the solenoid, $\boldsymbol{A} \neq 0$. The action in the path integral is written in the source as

$$\begin{aligned} S[\boldsymbol{x}(t)] &= \int_{t_A}^{t_B} \mathrm{d}t\, \frac{1}{2m} \left(\boldsymbol{p} - \frac{e}{c}\boldsymbol{A} \right)^2 \\ &\simeq \int_{t_A}^{t_B} \mathrm{d}t \left[\frac{m}{2} \left(\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} \right)^2 - \frac{e}{c} \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} \cdot \boldsymbol{A} \right] \\ &= S_0[\boldsymbol{x}(t)] - \frac{e}{c} \int_{t_A}^{t_B} \mathrm{d}t\, \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} \cdot \boldsymbol{A} = S_0[\boldsymbol{x}(t)] - \frac{e}{c} \int_{\boldsymbol{x}_A}^{\boldsymbol{x}_B} \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{x}. \end{aligned}$$

The source neglects the term quadratic in $\boldsymbol{A}$. For any path above the solenoid and any path below it, the two contributions to the amplitude $K(\boldsymbol{x}_B, t_B; \boldsymbol{x}_A, t_A)$ have the form

$$\begin{aligned} &\exp\left[-\frac{\mathrm{i}e}{\hbar c} \int_{\boldsymbol{x}_A,\,\text{above}}^{\boldsymbol{x}_B} \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{x} \right] e^{\mathrm{i}S_0[\boldsymbol{x}_{\text{above}}(t)]/\hbar} \\ &\quad + \exp\left[-\frac{\mathrm{i}e}{\hbar c} \int_{\boldsymbol{x}_A,\,\text{below}}^{\boldsymbol{x}_B} \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{x} \right] e^{\mathrm{i}S_0[\boldsymbol{x}_{\text{below}}(t)]/\hbar}. \end{aligned}$$

There is a relative phase that affects the interference fringes:

$$\begin{aligned} \frac{e}{\hbar c} \left[\int_{\text{above}} \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{x} - \int_{\text{below}} \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{x} \right] &= \frac{e}{\hbar c} \oint \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{x} \\ &= \frac{e}{\hbar c} \int (\boldsymbol{\nabla} \times \boldsymbol{A}) \cdot \mathrm{d}\boldsymbol{S} \\ &= \frac{e}{\hbar c} \int \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{S} = \frac{e}{\hbar c} B \pi R^2. \end{aligned}$$

This is independent of the precise paths; Stokes' theorem has been used.

The AB effect shows that in quantum mechanics, even when the field strength vanishes, a nonzero potential can influence the interference of particles.

2.9 Berry Phase

The Aharonov–Bohm effect showed that the relative phase of a quantum amplitude can depend on the geometry of a path even where the magnetic field vanishes. A related geometric phase appears when a Hamiltonian depends on external parameters that vary slowly along a path. In this case the path lies in *parameter space*, rather than necessarily in the particle's physical position space.

1) Instantaneous eigenstates and adiabatic evolution

Let a Hamiltonian depend smoothly on real control parameters $\boldsymbol{R} = (R_1, \ldots, R_d)$:

$$\hat{H}(\boldsymbol{R})\,|n(\boldsymbol{R})\rangle = E_n(\boldsymbol{R})\,|n(\boldsymbol{R})\rangle\,, \qquad \langle n(\boldsymbol{R})|n(\boldsymbol{R})\rangle = 1.$$

Assume that $\boldsymbol{R}(t)$ changes sufficiently slowly and that the chosen eigenvalue remains nondegenerate and separated by a gap from the other energy levels along the path. Under the adiabatic approximation a state prepared in $|n(\boldsymbol{R}(0))\rangle$ follows the corresponding instantaneous eigenspace. Its evolution contains a dynamical phase and a second, geometrical phase:

$$\boxed{|\psi(t)\rangle \simeq e^{-\frac{\mathrm{i}}{\hbar}\int_0^t E_n(\boldsymbol{R}(t'))\,\mathrm{d}t'} e^{\mathrm{i}\gamma_n(t)}\,|n(\boldsymbol{R}(t))\rangle\,.}$$

We determine γ_n from the Schrödinger equation, rather than assuming its form.

Write $|\psi\rangle = e^{-\mathrm{i}\int E_n \mathrm{d}t/\hbar} e^{\mathrm{i}\gamma_n}\,|n\rangle$ and substitute into

$$\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t}\,|\psi\rangle = \hat{H}(\boldsymbol{R}(t))\,|\psi\rangle\,.$$

Differentiation gives the identity within the adiabatic approximation,

$$(E_n - \hbar\dot{\gamma}_n)\,|n\rangle + \mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t}\,|n\rangle = E_n\,|n\rangle\,.$$

Project onto the instantaneous normalized eigenstate to find

$$\boxed{\dot{\gamma}_n(t) = \mathrm{i}\,\langle n(\boldsymbol{R}(t))|\,\frac{\mathrm{d}}{\mathrm{d}t}\,|n(\boldsymbol{R}(t))\rangle\,.}$$

The normalization condition implies that $\langle n|\dot{n}\rangle$ is purely imaginary, so $\dot{\gamma}_n$ is real. Apply the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}\,|n(\boldsymbol{R}(t))\rangle = \sum_\mu \frac{\partial}{\partial R_\mu}\,|n(\boldsymbol{R})\rangle\,\dot{R}_\mu,$$

to rewrite the accumulated phase as an integral along the *parameter path* C:

$$\boxed{\gamma_n(C) = \int_C \boldsymbol{\mathcal{A}}_n(\boldsymbol{R})\cdot\mathrm{d}\boldsymbol{R}, \qquad \boldsymbol{\mathcal{A}}_n(\boldsymbol{R}) = \mathrm{i}\,\langle n(\boldsymbol{R})|\boldsymbol{\nabla}_R n(\boldsymbol{R})\rangle\,.}$$

The vector field $\boldsymbol{\mathcal{A}}_n$ is the *Berry connection*. It depends on how the phase of the instantaneous eigenstate has been chosen.

2) Gauge transformations of the eigenstate

The normalized eigenket is physically unchanged by a parameter-dependent phase. Choose a new convention

$$|n'(\boldsymbol{R})\rangle = e^{-\mathrm{i}\beta(\boldsymbol{R})}\,|n(\boldsymbol{R})\rangle\,, \qquad \beta(\boldsymbol{R}) \in \mathbb{R}.$$

Substituting it into the Berry connection shows explicitly how the additional derivative acts:

$$\begin{aligned}\boldsymbol{\mathcal{A}}'_n &= \mathrm{i}\,\langle n|\,e^{\mathrm{i}\beta}\boldsymbol{\nabla}_R\big(e^{-\mathrm{i}\beta}\,|n\rangle\big)\\ &= \mathrm{i}\,\langle n|\boldsymbol{\nabla}_R n\rangle + \boldsymbol{\nabla}_R\beta\\ &= \boxed{\boldsymbol{\mathcal{A}}_n + \boldsymbol{\nabla}_R\beta}.\end{aligned}$$

Hence an *open-path* integral transforms as

$$\gamma'_n(C) = \gamma_n(C) + \beta(\boldsymbol{R}_f) - \beta(\boldsymbol{R}_i).$$

For a closed path C, with a single-valued choice of phase along the loop, the two endpoint contributions cancel modulo 2π. Thus the Berry phase around a closed cycle is a gauge-invariant quantity:

$$\boxed{e^{\mathrm{i}\gamma_n(C)} \quad \text{is gauge invariant for a closed } C.}$$

The distinction matters: neither the connection at one point nor the Berry phase assigned to an open segment is independently gauge invariant. A physically measured phase comes from interference, from a cyclic state evolution, or from an appropriate endpoint comparison.

3) Berry curvature and geometry of parameter space

For three control parameters define the Berry curvature by

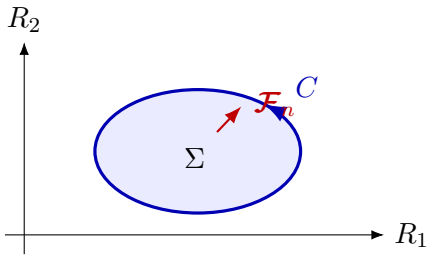
$$\boxed{\boldsymbol{\mathcal{F}}_n(\boldsymbol{R}) = \boldsymbol{\nabla}_R \times \boldsymbol{\mathcal{A}}_n(\boldsymbol{R}).}$$

It is unaffected by the gauge change because $\boldsymbol{\nabla}_R \times \boldsymbol{\nabla}_R \beta = 0$. For a surface Σ whose boundary is C, Stokes' theorem gives

$$\boxed{\gamma_n(C) = \oint_C \boldsymbol{\mathcal{A}}_n \cdot \mathrm{d}\boldsymbol{R} = \int_\Sigma \boldsymbol{\mathcal{F}}_n \cdot \mathrm{d}\boldsymbol{S} \pmod{2\pi},}$$

provided the eigenstate can be chosen smoothly over the surface; singular gauges or degeneracies require appropriate patches. In more than three parameters use the antisymmetric tensor

$$\mathcal{F}_{n,\mu\nu} = \partial_\mu \mathcal{A}_{n,\nu} - \partial_\nu \mathcal{A}_{n,\mu} = \mathrm{i}\left(\langle \partial_\mu n | \partial_\nu n \rangle - \langle \partial_\nu n | \partial_\mu n \rangle\right).$$



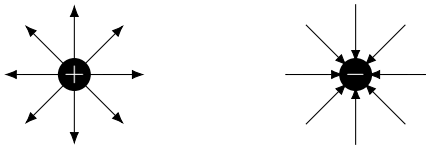
The analogy with the AB effect is now precise: the AB phase involves the electromagnetic vector potential integrated over a path in physical space; the Berry phase involves a connection generated by the *parameter dependence of eigenstates*. Their common mathematical structure is a line integral of a connection, its gauge transformation by a gradient, and a gauge-invariant curvature or closed-loop phase. The Berry curvature is a property of the quantum eigenstate bundle, not an electromagnetic field in ordinary space. This construction will reappear when Bloch-band eigenstates depend on crystal momentum.

2.10 Magnetic Monopoles and Charge Quantization

In classical electromagnetic theory, static electric and magnetic fields are asymmetric. The static electric field has sources but no curl:

$$\boldsymbol{\nabla} \cdot \boldsymbol{E} = 4\pi\rho_e, \qquad \boldsymbol{\nabla} \times \boldsymbol{E} = 0.$$

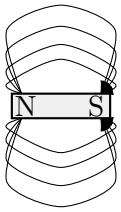
Electric field-line patterns (single-charge configuration):



The static magnetic field has no sources and does have curl:

$$\boldsymbol{\nabla}\cdot\boldsymbol{B}=0,\qquad \boldsymbol{\nabla}\times\boldsymbol{B}=\frac{4\pi}{c}\boldsymbol{J}.$$

The magnetic field-line pattern has a dipole structure:



Question: Can there be an isolated magnetic charge, that is, a magnetic monopole?

Dirac string. If there is a magnetic charge e_m at the origin, the static field it produces satisfies

$$\boldsymbol{\nabla}\cdot\boldsymbol{B}=4\pi e_m\delta^{(3)}(\boldsymbol{r}).$$

Its solution is

$$\boxed{\boldsymbol{B}(\boldsymbol{r})=\frac{e_m}{r^2}\boldsymbol{e}_r.}$$

Can a vector potential $\boldsymbol{A}(\boldsymbol{r})$ be found such that the original relation $\boldsymbol{B}=\boldsymbol{\nabla}\times\boldsymbol{A}$ still holds everywhere? The answer is no.

Proof: If $\boldsymbol{B}=\boldsymbol{\nabla}\times\boldsymbol{A}$, then necessarily

$$\boldsymbol{\nabla}\cdot\boldsymbol{B}=\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A})=0,$$

contradicting the monopole source equation.

Likewise, by Gauss's theorem, for a closed surface surrounding the monopole,

$$\oint\boldsymbol{B}\cdot\mathrm{d}\boldsymbol{S}=4\pi e_m.$$

But if $\boldsymbol{B}=\boldsymbol{\nabla}\times\boldsymbol{A}$, then

$$\oint\boldsymbol{B}\cdot\mathrm{d}\boldsymbol{S}=\oint(\boldsymbol{\nabla}\times\boldsymbol{A})\cdot\mathrm{d}\boldsymbol{S}=\int\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A})\mathrm{d}V=0,$$

contradicting Gauss's theorem. Thus no single globally regular vector potential $\boldsymbol{A}(\boldsymbol{r})$ can produce $\boldsymbol{B}(\boldsymbol{r})=\boldsymbol{\nabla}\times\boldsymbol{A}$.

For example, in spherical coordinates take

$$\boldsymbol{A}=e_m\frac{1-\cos\theta}{r\sin\theta}\boldsymbol{e}_\varphi.$$

For $\theta<\pi-\varepsilon$ (ε infinitesimal), using

$$\begin{aligned}\boldsymbol{\nabla}\times\boldsymbol{A}=&\frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(A_\varphi\sin\theta)-\frac{\partial A_\theta}{\partial\varphi}\right]\boldsymbol{e}_r\\&+\frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\varphi}-\frac{\partial}{\partial r}(rA_\varphi)\right]\boldsymbol{e}_\theta\\&+\frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\theta)-\frac{\partial A_r}{\partial\theta}\right]\boldsymbol{e}_\varphi=\frac{e_m}{r^2}\boldsymbol{e}_r=\boldsymbol{B}.\end{aligned}$$

However, $\boldsymbol{A}$ diverges at $\theta=\pi$. The negative z axis therefore contains a singular string, called the Dirac string. Its singular contribution means that the distributional curl of this globally extended singular potential does not equal the monopole field alone.

Quantization of electric charge. Although a single vector potential satisfying $\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}$ cannot cover all space, such potentials can be found in separate regions. For example, take

$$\boxed{\begin{aligned} \boldsymbol{A}_{\mathrm{I}} &= e_m \frac{1-\cos\theta}{r\sin\theta}\boldsymbol{e}_\varphi, \qquad \theta < \pi - \varepsilon, \\ \boldsymbol{A}_{\mathrm{II}} &= -e_m \frac{1+\cos\theta}{r\sin\theta}\boldsymbol{e}_\varphi, \qquad \theta > \varepsilon. \end{aligned}}$$

$\boldsymbol{A}_{\mathrm{I}}$ diverges at $\theta = \pi$, while $\boldsymbol{A}_{\mathrm{II}}$ diverges at $\theta = 0$. With this choice of potentials,

$$\boldsymbol{\nabla} \times \boldsymbol{A} = \frac{e_m}{r^2}\boldsymbol{e}_r = \boldsymbol{B}$$

in each nonsingular region.

In their overlap, since both potentials produce the same magnetic field, they are related by a gauge transformation Λ:

$$\boldsymbol{A}_{\mathrm{II}} - \boldsymbol{A}_{\mathrm{I}} = -\frac{2e_m}{r\sin\theta}\boldsymbol{e}_\varphi = \boldsymbol{\nabla}\Lambda.$$

In spherical coordinates,

$$\boldsymbol{\nabla}\Lambda = \frac{\partial\Lambda}{\partial r}\boldsymbol{e}_r + \frac{1}{r}\frac{\partial\Lambda}{\partial\theta}\boldsymbol{e}_\theta + \frac{1}{r\sin\theta}\frac{\partial\Lambda}{\partial\varphi}\boldsymbol{e}_\varphi.$$

Hence

$$\boxed{\Lambda(\boldsymbol{r}) = -2e_m\varphi.}$$

Consider a particle of electric charge e moving in the monopole field. The wavefunctions ψ_{I} and ψ_{II} corresponding to $\boldsymbol{A}_{\mathrm{I}}$ and $\boldsymbol{A}_{\mathrm{II}}$ satisfy in the overlapping region

$$\psi_{\mathrm{II}} = e^{\mathrm{i}e\Lambda/(\hbar c)}\psi_{\mathrm{I}} = e^{-2\mathrm{i}ee_m\varphi/(\hbar c)}\psi_{\mathrm{I}}.$$

By single-valuedness of the wavefunction, ψ_{II} must be single valued in the overlap. For example, at $\theta = \pi/2$,

$$\psi_{\mathrm{II}}(r, \theta = \pi/2, \varphi = 0) = \psi_{\mathrm{II}}(r, \theta = \pi/2, \varphi = 2\pi).$$

Consequently,

$$\frac{2ee_m}{\hbar c} = \pm n, \qquad n = 0, \pm 1, \pm 2, \ldots.$$

This is electric-charge quantization:

$$\boxed{e = \pm\frac{\hbar c}{2e_m}n.}$$

There is a smallest unit of electric charge.

Quantum mechanics does not require the existence of magnetic monopoles. If a magnetic monopole exists, however, electric charge must be quantized and there is a smallest unit of electric charge.

Chapter 3

Angular Momentum Theory

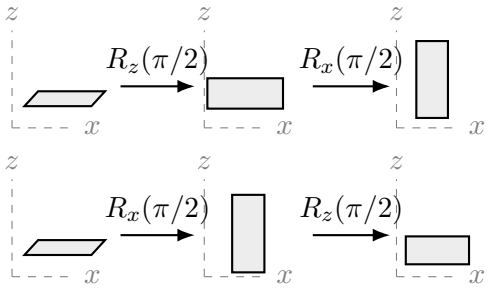
3.1 Rotations and Angular Momentum

Rotations and angular momentum have a unique position in quantum mechanics.

1) Classical rotations

Two rotations about the same axis commute. For instance, rotating first through 30° about the Z axis and then through 60° is exactly equivalent to performing the 60° rotation first and the 30° rotation second.

However, two rotations about different axes do not commute. As shown in the figure, a counterclockwise rotation through 90° about the Z axis followed by a 90° rotation about the X axis gives a result different from performing the X rotation first and the Z rotation second.



The upper and lower sequences show the different final orientations produced by reversing the order of the two rotations.

A three-dimensional vector $\boldsymbol{V} = V_x\boldsymbol{e}_x + V_y\boldsymbol{e}_y + V_z\boldsymbol{e}_z$ becomes $\boldsymbol{V}' = V_x'\boldsymbol{e}_x + V_y'\boldsymbol{e}_y + V_z'\boldsymbol{e}_z$ after a rotation $R(\varphi)$. In matrix form,

$$\boldsymbol{V}' = R(\varphi)\boldsymbol{V}.$$

Here $R(\varphi)$ is a 3×3 matrix. Obviously, rotation does not change the length of a vector:

$$|\boldsymbol{V}'| = |\boldsymbol{V}|.$$

The rotation matrix through an angle φ about the Z axis is easily obtained:

$$R_z(\varphi) = \begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

For an infinitesimal rotation ϵ,

$$R_z(\epsilon) = \begin{pmatrix} 1-\epsilon^2/2 & -\epsilon & 0 \\ \epsilon & 1-\epsilon^2/2 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

where terms higher than the second order in ϵ have been omitted. Similarly,

$$R_x(\epsilon) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1-\epsilon^2/2 & -\epsilon \\ 0 & \epsilon & 1-\epsilon^2/2 \end{pmatrix}, \qquad R_y(\epsilon) = \begin{pmatrix} 1-\epsilon^2/2 & 0 & \epsilon \\ 0 & 1 & 0 \\ -\epsilon & 0 & 1-\epsilon^2/2 \end{pmatrix}.$$

Clearly,

$$R_x(\epsilon)R_y(\epsilon) - R_y(\epsilon)R_x(\epsilon) = \begin{pmatrix} 0 & -\epsilon^2 & 0 \\ \epsilon^2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = R_z(\epsilon^2) - \mathbf{1}.$$

This expression shows that rotations about X and Y do not commute. In general,

$$R_i(\epsilon)R_j(\epsilon) - R_j(\epsilon)R_i(\epsilon) = \epsilon_{ijk}\big[R_k(\epsilon^2) - \mathbf{1}\big].$$

2) Quantum rotations

Let the quantum rotation operator corresponding to the classical rotation matrix $R(\varphi)$ be $\hat{D}(\varphi)$:

$$|\alpha\rangle_\varphi = \hat{D}(\varphi)\,|\alpha\rangle\,.$$

Its representation is a matrix $D(\varphi)$.

First discuss the general properties of the rotation operator. The generator of an infinitesimal spatial translation is the momentum operator:

$$\hat{T}(\mathrm{d}\boldsymbol{x}) = \mathbf{1} - \frac{\mathrm{i}}{\hbar}\hat{\boldsymbol{p}}\cdot\mathrm{d}\boldsymbol{x} \simeq \exp\!\left(-\frac{\mathrm{i}}{\hbar}\hat{\boldsymbol{p}}\cdot\mathrm{d}\boldsymbol{x}\right).$$

Spatial-translation invariance implies momentum conservation. For an infinitesimal time evolution, the generator is the Hamiltonian:

$$\hat{U}(t+\mathrm{d}t, t) = \mathbf{1} - \frac{\mathrm{i}}{\hbar}\hat{H}\mathrm{d}t \simeq \exp\!\left(-\frac{\mathrm{i}}{\hbar}\hat{H}\mathrm{d}t\right).$$

Time-translation invariance implies energy conservation. For a rotation, classical mechanics tells us that its generator is the corresponding conserved mechanical quantity: angular momentum. Let $\hat{\boldsymbol{J}}$ be the generator of the quantum rotation operator $\hat{D}(\varphi)$. For an infinitesimal rotation $\mathrm{d}\varphi$ about the direction $\boldsymbol{e}_n$,

$$\hat{D}(\mathrm{d}\varphi) = \mathbf{1} - \frac{\mathrm{i}}{\hbar}\hat{\boldsymbol{J}}\cdot\boldsymbol{e}_n\mathrm{d}\varphi \simeq \exp\!\left(-\frac{\mathrm{i}}{\hbar}\hat{\boldsymbol{J}}\cdot\boldsymbol{e}_n\mathrm{d}\varphi\right).$$

Because rotations about one axis commute, a finite rotation can be regarded as a sequence of infinitesimal rotations. Thus

$$\boxed{\hat{D}(\varphi) = \exp\!\left(-\frac{\mathrm{i}}{\hbar}\hat{\boldsymbol{J}}\cdot\boldsymbol{e}_n\varphi\right).}$$

Assume that $\hat{D}$ in quantum mechanics and R in classical mechanics satisfy the same commutation relations:

$$[\hat{D}_i(\epsilon), \hat{D}_j(\epsilon)] = \epsilon_{ijk}[\hat{D}_k(\epsilon^2) - \mathbf{1}].$$

Retaining terms through ϵ^2,

$$\left[\mathbf{1} - \frac{\mathrm{i}}{\hbar}\hat{J}_i\epsilon - \frac{\hat{J}_i^2\epsilon^2}{2\hbar^2},\ \mathbf{1} - \frac{\mathrm{i}}{\hbar}\hat{J}_j\epsilon - \frac{\hat{J}_j^2\epsilon^2}{2\hbar^2}\right] = \epsilon_{ijk}\left(\mathbf{1} - \frac{\mathrm{i}}{\hbar}\hat{J}_k\epsilon^2 - \mathbf{1}\right).$$

Hence

$$\boxed{[\hat{J}_i, \hat{J}_j] = \mathrm{i}\hbar\epsilon_{ijk}\hat{J}_k,} \qquad \hat{\boldsymbol{J}} \times \hat{\boldsymbol{J}} = \mathrm{i}\hbar\hat{\boldsymbol{J}}.$$

This is the fundamental angular-momentum commutation relation in quantum mechanics.

3) Expectation values of angular momentum

Consider a rotation by φ about Z. The state changes as

$$|\alpha\rangle \longrightarrow |\alpha\rangle_\varphi = \hat{D}_z(\varphi)\,|\alpha\rangle = e^{-\mathrm{i}\hat{J}_z\varphi/\hbar}\,|\alpha\rangle\,.$$

The expectation value of $\hat{J}_x$ changes from

$$\langle\hat{J}_x\rangle_0 = \left\langle\alpha\middle|\hat{J}_x\middle|\alpha\right\rangle$$

to

$$\langle\hat{J}_x\rangle_\varphi = \left\langle\alpha_\varphi\middle|\hat{J}_x\middle|\alpha_\varphi\right\rangle = \langle\alpha|\,e^{\mathrm{i}\hat{J}_z\varphi/\hbar}\hat{J}_x e^{-\mathrm{i}\hat{J}_z\varphi/\hbar}\,|\alpha\rangle\,.$$

Expand $e^{\pm\mathrm{i}\hat{J}_z\varphi/\hbar}$ in a power series and use

$$[\hat{J}_z, \hat{J}_x] = \mathrm{i}\hbar\hat{J}_y.$$

We obtain

$$e^{\mathrm{i}\hat{J}_z\varphi/\hbar}\hat{J}_x e^{-\mathrm{i}\hat{J}_z\varphi/\hbar} = \hat{J}_x\cos\varphi - \hat{J}_y\sin\varphi.$$

Therefore

$$\langle\hat{J}_x\rangle_\varphi = \langle\hat{J}_x\rangle_0\cos\varphi - \langle\hat{J}_y\rangle_0\sin\varphi.$$

Similarly,

$$\langle\hat{J}_y\rangle_\varphi = \langle\hat{J}_y\rangle_0\cos\varphi + \langle\hat{J}_x\rangle_0\sin\varphi, \qquad \langle\hat{J}_z\rangle_\varphi = \langle\hat{J}_z\rangle_0.$$

If $\langle\hat{J}_x\rangle$, $\langle\hat{J}_y\rangle$ and $\langle\hat{J}_z\rangle$ are regarded as three components of $\langle\hat{\boldsymbol{J}}\rangle$, then

$$\boxed{\langle\hat{\boldsymbol{J}}\rangle_\varphi = R_z(\varphi)\langle\hat{\boldsymbol{J}}\rangle_0.}$$

This obeys the same transformation law as a vector mechanical quantity in classical mechanics.

3.2 Spin Angular Momentum

1) Spin precession (relation between the rotation operator and the time-evolution operator)

We have previously discussed the time evolution of spin precession about the Z axis. In the particle's own coordinate frame there is no kinetic energy, only the interaction of its spin with the external magnetic field:

$$\hat{H} = \omega\hat{s}_z, \qquad \omega = -\frac{eB}{mc}, \qquad \hat{U}(t,0) = e^{-\mathrm{i}\hat{H}t/\hbar} = e^{-\mathrm{i}\omega\hat{s}_z t/\hbar}.$$

Suppose the initial state ($t = 0$) is the eigenstate $\left|s_x^+\right\rangle$ of $\hat{s}_x$. Then at any time t the expectation values are

$$\langle\hat{s}_x\rangle_t = \frac{\hbar}{2}\cos\omega t, \qquad \langle\hat{s}_y\rangle_t = \frac{\hbar}{2}\sin\omega t, \qquad \langle\hat{s}_z\rangle_t = 0.$$

Compare $\hat{U}(t,0) = e^{-\mathrm{i}\omega\hat{s}_z t/\hbar}$ with the rotation operator about Z, $\hat{D}_z(\varphi) = e^{-\mathrm{i}\hat{s}_z\varphi/\hbar}$. If $\varphi = \omega t$, the angle φ corresponds one-to-one with t. From the relation between angular-momentum expectation values before and after a rotation,

$$\begin{cases} \langle\hat{s}_x\rangle_\varphi = \langle\hat{s}_x\rangle_0 \cos\varphi - \langle\hat{s}_y\rangle_0 \sin\varphi, \\ \langle\hat{s}_y\rangle_\varphi = \langle\hat{s}_y\rangle_0 \cos\varphi + \langle\hat{s}_x\rangle_0 \sin\varphi, \\ \langle\hat{s}_z\rangle_\varphi = \langle\hat{s}_z\rangle_0, \end{cases}$$

we obtain for any initial state

$$\begin{cases} \langle\hat{s}_x\rangle_t = \langle\hat{s}_x\rangle_0 \cos\omega t - \langle\hat{s}_y\rangle_0 \sin\omega t, \\ \langle\hat{s}_y\rangle_t = \langle\hat{s}_y\rangle_0 \cos\omega t + \langle\hat{s}_x\rangle_0 \sin\omega t, \\ \langle\hat{s}_z\rangle_t = \langle\hat{s}_z\rangle_0. \end{cases}$$

These expressions are the expectation values of spin angular momentum at time t for an arbitrary initial state.

For $t = 2\pi/\omega$,

$$\langle\hat{s}_x\rangle_t = \langle\hat{s}_x\rangle_0, \quad \langle\hat{s}_y\rangle_t = \langle\hat{s}_y\rangle_0, \quad \langle\hat{s}_z\rangle_t = \langle\hat{s}_z\rangle_0.$$

Thus $2\pi/\omega$ is called the spin-precession period.

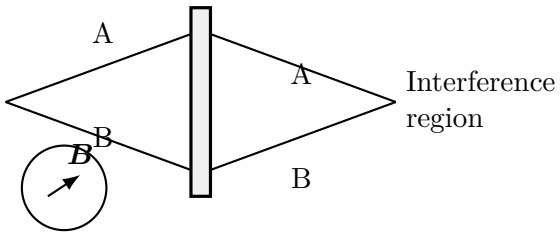
Expand an arbitrary initial state in the eigenstates $|s_z^\pm\rangle$ of $\hat{s}_z$. Under a rotation about Z,

$$\begin{aligned} |\alpha\rangle &= \left|s_z^+\right\rangle \left\langle s_z^+\middle|\alpha\right\rangle + |s_z^-\rangle \langle s_z^-|\alpha\rangle \\ &\longrightarrow |\alpha\rangle_\varphi = e^{-\mathrm{i}\hat{s}_z\varphi/\hbar}|\alpha\rangle = e^{-\mathrm{i}\varphi/2}\left|s_z^+\right\rangle \left\langle s_z^+\middle|\alpha\right\rangle + e^{\mathrm{i}\varphi/2}|s_z^-\rangle \langle s_z^-|\alpha\rangle . \end{aligned}$$

When $\varphi = 2\pi$, or $t = 2\pi/\omega$, $|\alpha\rangle_\varphi = -|\alpha\rangle$; the state does not return to itself! Only after $\varphi = 4\pi$ does it return. Unlike the period of the expectation values, $2\pi/\omega$, the period of the state is $4\pi/\omega$.

2) Neutron interference experiment (experimental verification of spin precession)

The neutron has spin 1/2. A neutron beam with almost fixed energy is split into two beams, which travel along paths A and B to an interference region. Along path B a region containing a nonzero magnetic field is inserted, as illustrated.



The magnetic moment of the neutron is

$$\mu_n = \frac{g_n e\hbar}{2m_n c}, \qquad g_n = -1.91,$$

and the spin-precession frequency is

$$\omega = \frac{g_n eB}{m_n c}.$$

(Note: for an electron, $g_e = -1$, and the precession frequency is $\omega = g_e eB/(m_e c)$ in the source's convention.) From the time evolution

$$\hat{U}(t,0) = e^{-\mathrm{i}\hat{H}t/\hbar} = e^{-\mathrm{i}\omega\hat{s}_z t/\hbar},$$

the state of the neutron following path B acquires an additional phase factor $e^{\pm \mathrm{i}\omega T/2}$ after passing through the field region. Here T is the time the neutron takes to traverse that region:

$$T = l/v_n = lm_n/p_n = lm_n\lambda/\hbar.$$

The interference consequently has a term

$$\cos\left(\frac{\omega T}{2} + \delta\right),$$

where δ is the phase difference between paths A and B in the absence of the magnetic field.

Changing the external field B changes the Larmor frequency and produces a periodic change in the interference intensity. The frequency change corresponding to one full period obeys

$$\frac{\Delta\omega\, T}{2} = 2\pi,$$

corresponding to

$$\boxed{\Delta B = \frac{4\pi m_n c}{g_n e T}.}$$

Such changes in interference intensity caused by varying the magnetic field have been confirmed experimentally.

3) Pauli matrices

Introduce the Pauli operator $\hat{\boldsymbol{\sigma}}$:

$$\hat{\boldsymbol{s}} = \frac{\hbar}{2}\hat{\boldsymbol{\sigma}}.$$

The eigenvalues of $\hat{\sigma}_i$ are ± 1; $\hat{\sigma}_i^2 = \mathbf{1}$, and $\hat{\boldsymbol{\sigma}}^{\,2} = 3\mathbf{1}$. Using the commutation relations

$$[\hat{\sigma}_i, \hat{\sigma}_j] = 2\mathrm{i}\epsilon_{ijk}\hat{\sigma}_k,$$

we have, for example,

$$\begin{aligned}\hat{\sigma}_x\hat{\sigma}_y + \hat{\sigma}_y\hat{\sigma}_x &= \frac{1}{2\mathrm{i}}(\hat{\sigma}_y\hat{\sigma}_z - \hat{\sigma}_z\hat{\sigma}_y)\hat{\sigma}_y + \frac{1}{2\mathrm{i}}\hat{\sigma}_y(\hat{\sigma}_y\hat{\sigma}_z - \hat{\sigma}_z\hat{\sigma}_y) \\ &= \frac{1}{2\mathrm{i}}(-\hat{\sigma}_z\hat{\sigma}_y^2 + \hat{\sigma}_y^2\hat{\sigma}_z) = \frac{1}{2\mathrm{i}}(-\hat{\sigma}_z + \hat{\sigma}_z) = 0.\end{aligned}$$

Hence the anticommutation relations are

$$\boxed{\{\hat{\sigma}_i, \hat{\sigma}_j\} = 2\delta_{ij}\mathbf{1}.}$$

In the $\hat{\sigma}_z$ representation,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The eigenstates of $\hat{\sigma}_z$ are

$$\chi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \chi_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

By completeness, any spin state has the form $\chi = c_+\chi_+ + c_-\chi_-$.

Consider rotations of a spin state. In the s_z representation,

$$D_n(\varphi) = e^{-\mathrm{i}\hat{\boldsymbol{s}}\cdot\boldsymbol{e}_n\varphi/\hbar} = e^{-\mathrm{i}\boldsymbol{\sigma}\cdot\boldsymbol{e}_n\varphi/2}.$$

It is straightforward to prove for arbitrary vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ that

$$(\boldsymbol{\sigma}\cdot\boldsymbol{a})(\boldsymbol{\sigma}\cdot\boldsymbol{b}) = \boldsymbol{a}\cdot\boldsymbol{b} + \mathrm{i}\boldsymbol{\sigma}\cdot(\boldsymbol{a}\times\boldsymbol{b}).$$

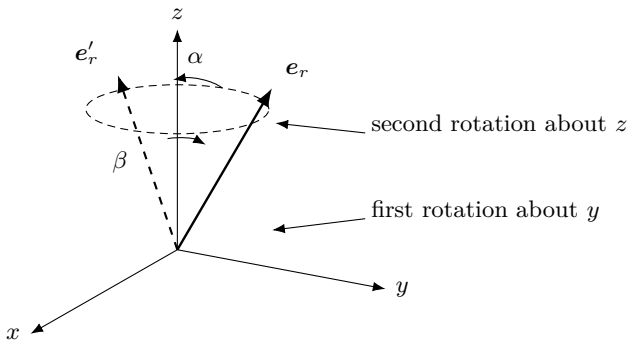
In particular, $(\boldsymbol{\sigma}\cdot\boldsymbol{a})^2 = |\boldsymbol{a}|^2$, and

$$(\boldsymbol{\sigma}\cdot\boldsymbol{e}_n)^r = \begin{cases} \mathbf{1}, & r \text{ even}, \\ \boldsymbol{\sigma}\cdot\boldsymbol{e}_n, & r \text{ odd}. \end{cases}$$

Therefore

$$\boxed{D_n(\varphi) = \cos\frac{\varphi}{2}\,\mathbf{1} - \mathrm{i}\boldsymbol{\sigma}\cdot\boldsymbol{e}_n \sin\frac{\varphi}{2}.}$$

Example: any spin state can be generated by rotating an eigenstate of s_z (or σ_z). Suppose that the orientation of an arbitrary spin state χ is $\boldsymbol{e}_n$, whose polar and azimuthal angles are β and α. Assume it has eigenvalue $+1$:



$$(\boldsymbol{\sigma}\cdot\boldsymbol{e}_n)\chi = \chi.$$

It may be obtained by rotating the eigenstate $\sigma_z\chi_+ = \chi_+$, $\chi_+ = (1,0)^{\mathsf{T}}$, through an angle β about Y and then through an angle α about Z:

$$\begin{aligned}
\chi &= D_z(\alpha)D_y(\beta)\chi_+ = e^{-\mathrm{i}\sigma_z\alpha/2}e^{-\mathrm{i}\sigma_y\beta/2}\chi_+ \\
&= \left(\cos\frac{\alpha}{2} - \mathrm{i}\sigma_z\sin\frac{\alpha}{2}\right)\left(\cos\frac{\beta}{2} - \mathrm{i}\sigma_y\sin\frac{\beta}{2}\right)\chi_+ \\
&= \begin{pmatrix} e^{-\mathrm{i}\alpha/2} & 0 \\ 0 & e^{\mathrm{i}\alpha/2} \end{pmatrix}\begin{pmatrix} \cos(\beta/2) & -\sin(\beta/2) \\ \sin(\beta/2) & \cos(\beta/2) \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
&= \begin{pmatrix} \cos(\beta/2)e^{-\mathrm{i}\alpha/2} \\ \sin(\beta/2)e^{\mathrm{i}\alpha/2} \end{pmatrix} = \cos\frac{\beta}{2}e^{-\mathrm{i}\alpha/2}\chi_+ + \sin\frac{\beta}{2}e^{\mathrm{i}\alpha/2}\chi_-.
\end{aligned}$$

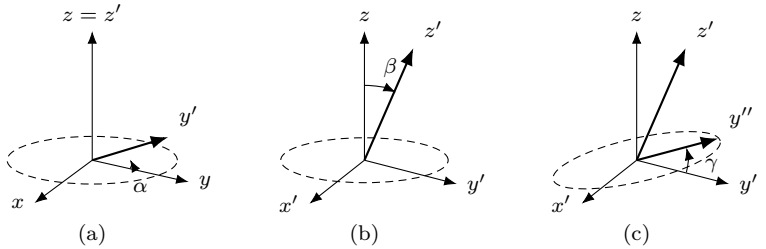
4) Euler rotations

We now discuss how an arbitrary rotation can be expressed as a product of rotations about the X, Y and Z axes.

Let (X, Y, Z) be fixed coordinates and (X', Y', Z') a frame that moves with the vector. In analytical mechanics, an arbitrary rigid-body rotation can be described by three Euler angles. Initially the two frames coincide. First rotate counterclockwise about Z by α, as shown in (a); Y' then no longer coincides with Y. Next rotate about Y' by β, as in (b), after which Z' no longer coincides with Z. Finally rotate about Z' by γ, as in (c); the Y' axis changes again, becoming Y''. Thus an arbitrary rigid-body rotation can be written as

$$R(\alpha,\beta,\gamma) = R_{z'}(\gamma)R_{y'}(\beta)R_z(\alpha).$$

The angles α, β, γ are the Euler angles.



Two of these three Euler rotations are about moving axes, y' and z', which is inconvenient for calculations: rotations generated by angular momentum are defined with respect to the fixed frame. We must therefore express rotations about moving axes as rotations about fixed axes. The preceding three rotations are equivalent to three rotations about the fixed axes:

$$R(\alpha, \beta, \gamma) = R_z(\alpha) R_y(\beta) R_z(\gamma).$$

The corresponding quantum rotation operator factorizes as

$$\hat{D}(\alpha, \beta, \gamma) = \hat{D}_z(\alpha) \hat{D}_y(\beta) \hat{D}_z(\gamma) = e^{-\mathrm{i}\hat{J}_z \alpha/\hbar} e^{-\mathrm{i}\hat{J}_y \beta/\hbar} e^{-\mathrm{i}\hat{J}_z \gamma/\hbar}.$$

For spin 1/2,

$$D(\alpha, \beta, \gamma) = e^{-\mathrm{i}\sigma_z \alpha/2} e^{-\mathrm{i}\sigma_y \beta/2} e^{-\mathrm{i}\sigma_z \gamma/2}.$$

In the s_z representation,

$$\begin{aligned} D(\alpha, \beta, \gamma) &= \left(\cos\frac{\alpha}{2} - \mathrm{i}\sigma_z \sin\frac{\alpha}{2}\right)\left(\cos\frac{\beta}{2} - \mathrm{i}\sigma_y \sin\frac{\beta}{2}\right)\left(\cos\frac{\gamma}{2} - \mathrm{i}\sigma_z \sin\frac{\gamma}{2}\right) \\ &= \begin{pmatrix} e^{-\mathrm{i}\alpha/2} & 0 \\ 0 & e^{\mathrm{i}\alpha/2} \end{pmatrix} \begin{pmatrix} \cos(\beta/2) & -\sin(\beta/2) \\ \sin(\beta/2) & \cos(\beta/2) \end{pmatrix} \begin{pmatrix} e^{-\mathrm{i}\gamma/2} & 0 \\ 0 & e^{\mathrm{i}\gamma/2} \end{pmatrix} \\ &= \begin{pmatrix} e^{-\mathrm{i}(\alpha+\gamma)/2}\cos(\beta/2) & -e^{-\mathrm{i}(\alpha-\gamma)/2}\sin(\beta/2) \\ e^{\mathrm{i}(\alpha-\gamma)/2}\sin(\beta/2) & e^{\mathrm{i}(\alpha+\gamma)/2}\cos(\beta/2) \end{pmatrix}. \end{aligned}$$

3.3 Ensembles and the Density Operator

We consider the spin properties of a many-particle system.

1) Pure ensembles and mixed ensembles

Identical physical systems form an ensemble; for example, an ensemble may consist of particles with identical spins. The spin state of a spin-1/2 particle, with azimuthal and polar angles α, β, is

$$\chi(\alpha, \beta) = c_+(\alpha, \beta)\chi_+ + c_-(\alpha, \beta)\chi_- = \cos\frac{\beta}{2} e^{-\mathrm{i}\alpha/2}\chi_+ + \sin\frac{\beta}{2} e^{\mathrm{i}\alpha/2}\chi_-,$$

where χ_+, χ_- are eigenstates of $\hat{s}_z$ and

$$\frac{c_+}{c_-} = \frac{\cos(\beta/2)}{e^{\mathrm{i}\alpha}\sin(\beta/2)}.$$

If all particles are in the same state, e.g., their spins point in the same direction, the ensemble is called a *pure ensemble* or a polarized system. If particles are in different states, e.g., their spins do not all point in one direction, the ensemble is called a *mixed ensemble*. For example, if 70% of particles have spin up and 30% spin down, the system is partially polarized and forms a partially mixed ensemble. In a system with completely random spin directions, the probabilities of spin up and spin down are each 50%. Their overall contributions cancel and there is no net spin polarization.

2) Ensemble averages and the density operator

For a fixed state $|\alpha\rangle$, the average of a mechanical quantity $\hat{A}$ is

$$\langle A\rangle_\alpha = \left\langle \alpha \middle| \hat{A} \middle| \alpha \right\rangle .$$

This is a *quantum average*. Insert any complete basis B:

$$\langle A\rangle_\alpha = \sum_{b,b'} \langle \alpha|b'\rangle \left\langle b' \middle| \hat{A} \middle| b \right\rangle \langle b|\alpha\rangle .$$

The ensemble average is

$$\langle A\rangle_\rho = \sum_\alpha w_\alpha \langle A\rangle_\alpha,$$

where w_α is the probability that the system occupies state $|\alpha\rangle$. It satisfies the normalization condition

$$\sum_\alpha w_\alpha = 1.$$

This is a *statistical average*. Since

$$\begin{aligned}\langle A\rangle_\rho &= \sum_{\alpha,b,b'} w_\alpha \langle \alpha|b'\rangle \left\langle b' \middle| \hat{A} \middle| b \right\rangle \langle b|\alpha\rangle \\ &= \sum_{b,b'} \left(\sum_\alpha w_\alpha \langle b|\alpha\rangle \langle \alpha|b'\rangle \right) \left\langle b' \middle| \hat{A} \middle| b \right\rangle ,\end{aligned}$$

we define the density operator by

$$\boxed{\hat{\rho} = \sum_\alpha w_\alpha |\alpha\rangle \langle \alpha| .}$$

Its matrix elements in the B representation are

$$\rho_{bb'} = \langle b|\hat{\rho}|b'\rangle = \sum_\alpha w_\alpha \langle b|\alpha\rangle \langle \alpha|b'\rangle .$$

Therefore

$$\boxed{\langle A\rangle_\rho = \sum_{b,b'} \langle b|\hat{\rho}|b'\rangle \left\langle b' \middle| \hat{A} \middle| b \right\rangle = \sum_b \left\langle b \middle| \hat{\rho}\hat{A} \middle| b \right\rangle \equiv \mathrm{tr}(\hat{\rho}\hat{A}).}$$

This fundamental equation of quantum statistical mechanics includes both quantum and statistical averaging. The trace is unchanged by a change of representation, so the formula is independent of the chosen representation.

In a continuous representation, e.g., the position basis, the matrix elements are

$$\rho_{\boldsymbol{x}\boldsymbol{x}'} = \langle \boldsymbol{x}|\hat{\rho}|\boldsymbol{x}'\rangle = \sum_\alpha w_\alpha \langle \boldsymbol{x}|\alpha\rangle \langle \alpha|\boldsymbol{x}'\rangle = \sum_\alpha w_\alpha \psi_\alpha(\boldsymbol{x})\psi_\alpha^*(\boldsymbol{x}').$$

The ensemble average becomes

$$\langle A\rangle_\rho = \mathrm{tr}(\hat{\rho}\hat{A}) = \int \mathrm{d}^3x \left\langle \boldsymbol{x} \middle| \hat{\rho}\hat{A} \middle| \boldsymbol{x} \right\rangle .$$

The density matrix satisfies the normalization condition

$$\begin{aligned}\mathrm{tr}\,\hat{\rho} &= \sum_{\alpha,b} w_\alpha \langle b|\alpha\rangle \langle \alpha|b\rangle \\ &= \sum_\alpha w_\alpha \langle \alpha| \left(\sum_b |b\rangle \langle b| \right) |\alpha\rangle && \text{(completeness)} \\ &= \sum_\alpha w_\alpha \langle \alpha|\alpha\rangle && \text{(quantum-state normalization)} \\ &= \sum_\alpha w_\alpha = 1 && \text{(statistical normalization).}\end{aligned}$$

Suppose the density operator has an eigenstate $|\theta\rangle$:

$$\hat{\rho}\,|\theta\rangle = \theta\,|\theta\rangle\,, \qquad \hat{\rho}^2\,|\theta\rangle = \theta^2\,|\theta\rangle\,.$$

For a pure ensemble all systems are in the same state $|n\rangle$:

$$w_\alpha = \begin{cases} 1, & \alpha = n, \\ 0, & \alpha \neq n. \end{cases}$$

Then

$$\hat{\rho} = |n\rangle\langle n|\,, \qquad \hat{\rho}^2 = |n\rangle\langle n|n\rangle\langle n| = |n\rangle\langle n| = \hat{\rho},$$

so $\theta^2 = \theta$ and its eigenvalues are $\theta = 0, 1$. If $|n\rangle$ is a basis state of the chosen representation, the density matrix is diagonal:

$$\rho_{bb'} = \langle b|n\rangle\langle n|b'\rangle = \delta_{bn}\delta_{b'n}, \qquad \rho = \begin{pmatrix} 0 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 0 \end{pmatrix}.$$

Only one matrix entry is 1; all others are 0.

For a completely mixed ensemble, all possible states occur with equal probabilities $w_\alpha = w_0 = 1/N$, where N is the number of states. Then

$$\rho_{bb'} = \sum_\alpha w_\alpha \langle b|\alpha\rangle\langle\alpha|b'\rangle = \frac{1}{N}\langle b|b'\rangle = \frac{1}{N}\delta_{bb'},$$

and the density matrix is proportional to the identity:

$$\rho = \frac{1}{N}\begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}.$$

Example 1: pure ensemble. Suppose that the completely polarized state is $|n\rangle = |+\rangle$, with $\hat{s}_z\,|\pm\rangle = \pm(\hbar/2)\,|\pm\rangle$. Then

$$\hat{\rho} = |+\rangle\langle+|\,.$$

In the s_z basis ρ is a 2×2 matrix:

$$\rho_{++} = \langle+|+\rangle\langle+|+\rangle = 1, \quad \rho_{+-} = \langle+|+\rangle\langle+|-\rangle = 0,$$

$$\rho_{-+} = \langle-|+\rangle\langle+|+\rangle = 0, \quad \rho_{--} = \langle-|+\rangle\langle+|-\rangle = 0, \qquad \rho = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

If the completely polarized state is

$$|n\rangle = \left|s_x^+\right\rangle = \frac{1}{\sqrt{2}}\,|+\rangle + \frac{1}{\sqrt{2}}\,|-\rangle\,, \qquad \hat{s}_x\left|s_x^+\right\rangle = \frac{\hbar}{2}\left|s_x^+\right\rangle,$$

then $\hat{\rho} = \left|s_x^+\right\rangle\left\langle s_x^+\right|$. In the s_z representation,

$$\rho_{++} = \left\langle+\middle|s_x^+\right\rangle\left\langle s_x^+\middle|+\right\rangle = \frac{1}{2}, \quad \rho_{+-} = \left\langle+\middle|s_x^+\right\rangle\left\langle s_x^+\middle|-\right\rangle = \frac{1}{2},$$

$$\rho_{-+} = \left\langle-\middle|s_x^+\right\rangle\left\langle s_x^+\middle|+\right\rangle = \frac{1}{2}, \quad \rho_{--} = \left\langle-\middle|s_x^+\right\rangle\left\langle s_x^+\middle|-\right\rangle = \frac{1}{2}, \qquad \rho = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Example 2: density matrix for a completely unpolarized system. With $N = 2$,

$$\hat{\rho} = \frac{1}{2} |+\rangle \langle +| + \frac{1}{2} |-\rangle \langle -| .$$

In the s_z representation,

$$\rho_{++} = \frac{1}{2}, \quad \rho_{+-} = 0, \quad \rho_{-+} = 0, \quad \rho_{--} = \frac{1}{2}, \qquad \rho = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The density matrix of a completely mixed ensemble is proportional to the identity. The expectation values in a completely unpolarized ensemble obviously cancel:

$$\langle s_i \rangle_\rho = \mathrm{tr}(\hat{\rho} \hat{s}_i) = \frac{1}{2} \mathrm{tr}(\hat{s}_i).$$

Inserting the spin matrices s_x, s_y, s_z gives $\langle s_i \rangle_\rho = 0$.

Example 3: density matrix for a partially polarized system. Let

$$\hat{\rho} = \frac{3}{4} |+\rangle \langle +| + \frac{1}{4} \left| s_x^+ \right\rangle \left\langle s_x^+ \right| .$$

In the s_z representation,

$$\rho_{++} = \frac{3}{4} + \frac{1}{4}\frac{1}{2} = \frac{7}{8}, \qquad \rho_{+-} = \rho_{-+} = \rho_{--} = \frac{1}{8},$$

so that

$$\rho = \frac{1}{8} \begin{pmatrix} 7 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \langle s_x \rangle_\rho = \frac{\hbar}{8}, \quad \langle s_y \rangle_\rho = 0, \quad \langle s_z \rangle_\rho = \frac{3\hbar}{8}.$$

3) Evolution of the ensemble

How does the density operator evolve in time? If the statistical properties of the ensemble do not vary in time, its probability distribution w_α is constant and

$$\hat{\rho}(t) = \sum_\alpha w_\alpha |\alpha, t\rangle \langle \alpha, t| .$$

By the state evolution equations

$$\mathrm{i}\hbar \frac{\partial}{\partial t} |\alpha, t\rangle = \hat{H} |\alpha, t\rangle , \qquad -\mathrm{i}\hbar \frac{\partial}{\partial t} \langle \alpha, t| = \langle \alpha, t| \hat{H},$$

we obtain

$$\begin{aligned} \mathrm{i}\hbar \frac{\partial \hat{\rho}}{\partial t} &= \sum_\alpha w_\alpha \left(\hat{H} |\alpha, t\rangle \langle \alpha, t| - |\alpha, t\rangle \langle \alpha, t| \hat{H} \right) \\ &= -[\hat{\rho}, \hat{H}]. \end{aligned}$$

This is the density operator's time-evolution equation. It resembles the equation of motion of mechanical quantities in the Heisenberg picture, differing by a minus sign; however $\hat{\rho}$ is not an observable operator but an operator constructed from states in the Schrödinger picture.

3.4 Composite Systems and Generalized Quantum Measurements

The preceding discussion described pure and mixed ensembles by a density operator $\hat{\rho}$. We now ask how a measurement can be modeled as a physical interaction. The essential observation is that the system of interest and the measuring device together form a composite quantum system. Unitary evolution of that larger system, followed by a readout of the probe, generates measurement probabilities and conditional state changes of the measured subsystem.

1) Composite systems and reduced states

Let two subsystems A, B have Hilbert spaces $\mathcal{H}_A, \mathcal{H}_B$. The composite Hilbert space is

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B.$$

For bases $\{|i\rangle_A\}$ and $\{|j\rangle_B\}$, an arbitrary pure composite state has the form

$$|\Psi_{AB}\rangle = \sum_{ij} c_{ij} \, |i\rangle_A \otimes |j\rangle_B \,, \qquad \sum_{ij} |c_{ij}|^2 = 1.$$

If $|\Psi_{AB}\rangle$ cannot be factored as a product $|\psi_A\rangle \otimes |\psi_B\rangle$, the two subsystems are entangled. The density operator of the full system is $\hat{\rho}_{AB} = |\Psi_{AB}\rangle \langle\Psi_{AB}|$; the state of subsystem A is obtained by *partial trace*,

$$\boxed{\hat{\rho}_A = \mathrm{Tr}_B \, \hat{\rho}_{AB} = \sum_j {}_B\langle j| \, \hat{\rho}_{AB} \, |j\rangle_B \,.}$$

Expectation values of every observable acting only on A can be computed equivalently as

$$\mathrm{Tr}_{AB}[\hat{\rho}_{AB}(\hat{A} \otimes \hat{I}_B)] = \mathrm{Tr}_A(\hat{\rho}_A \hat{A}).$$

For example, for the entangled Bell state

$$\left|\Phi^+\right\rangle_{AB} = \frac{|00\rangle + |11\rangle}{\sqrt{2}},$$

the reduced state is

$$\hat{\rho}_A = \frac{1}{2}(|0\rangle \langle 0| + |1\rangle \langle 1|) = \frac{1}{2}\hat{I}_A.$$

The composite state is pure even though its subsystem is maximally mixed. The purity $\mathrm{Tr}\,\hat{\rho}^2$ equals one for a pure state and is smaller for a mixed state; it is preserved under unitary evolution of the closed composite system but need not be preserved under a partial trace.

2) Projective measurement and conditional states

A projective measurement of an observable $\hat{A} = \sum_j a_j \hat{Q}_j$ uses mutually orthogonal projectors $\hat{Q}_j \hat{Q}_k = \delta_{jk} \hat{Q}_j$ and $\sum_j \hat{Q}_j = \hat{I}$. As established earlier, outcome j has probability

$$p_j = \mathrm{Tr}(\hat{Q}_j \hat{\rho}),$$

and the selectively updated state for a Lüders measurement is

$$\hat{\rho}_j = \frac{\hat{Q}_j \hat{\rho} \hat{Q}_j}{p_j} \quad (p_j > 0).$$

If the outcome is ignored, the state is $\sum_j \hat{Q}_j \hat{\rho} \hat{Q}_j$. General laboratory measurements need not be described by orthogonal projectors on the system alone; this is the motivation for the system–probe construction.

3) A measurement as a system–probe interaction

Denote the measured system by S and the probe or meter by M. Prepare an initially uncorrelated state

$$|\Psi_{SM}(0)\rangle = |\psi\rangle_S \otimes |\theta\rangle_M \,.$$

For an interaction of duration τ, the combined evolution is unitary:

$$|\Psi_{SM}(\tau)\rangle = \hat{U}_{SM}\,|\psi\rangle_S\,|\theta\rangle_M\,, \qquad \hat{U}_{SM} = \mathcal{T}\exp\!\Big(-\frac{\mathrm{i}}{\hbar}\int_0^\tau \hat{H}_{SM}(t)\,\mathrm{d}t\Big)\,.$$

When the interaction is sufficiently short and strong, the free subsystem Hamiltonians can be neglected relative to the interaction Hamiltonian during this interval. A general interaction correlates the system and probe. We then make a projective measurement on the *probe* in an orthonormal basis $\{|r_j\rangle_M\}$.

Define the induced operator on the system Hilbert space by

$$\boxed{\hat{M}_j = {}_M\langle r_j|\,\hat{U}_{SM}\,|\theta\rangle_M\,.}$$

Projecting the composite state onto $|r_j\rangle_M$ gives the unnormalized system state $\hat{M}_j\,|\psi\rangle_S$. Its squared norm is the probability of outcome j:

$$\boxed{p_j = \Big\langle\psi\Big|\hat{M}_j^\dagger\hat{M}_j\Big|\psi\Big\rangle\,, \qquad |\psi_j\rangle = \frac{\hat{M}_j\,|\psi\rangle}{\sqrt{p_j}}.}$$

Completeness of the probe readout basis and unitarity of $\hat{U}_{SM}$ imply

$$\sum_j \hat{M}_j^\dagger\hat{M}_j = {}_M\langle\theta|\,\hat{U}_{SM}^\dagger(\hat{I}_S\otimes\sum_j|r_j\rangle_M\langle r_j|)\hat{U}_{SM}\,|\theta\rangle_M = \hat{I}_S.$$

The system's state generally changes even if no direct projective measurement was performed on it. This is *measurement backaction*. For an ideal rank-one probe readout conditioned on j, the post-readout pure composite state is the product $|\psi_j\rangle_S\otimes|r_j\rangle_M$, whereas the pre-readout state may have been entangled.

4) POVMs and quantum instruments

The positive operators

$$\boxed{\hat{E}_j = \hat{M}_j^\dagger\hat{M}_j, \qquad \hat{E}_j\geq 0, \qquad \sum_j\hat{E}_j = \hat{I}_S}$$

are called *effects*, and their complete collection forms a positive operator-valued measure (POVM). They determine outcome probabilities, $p_j = \mathrm{Tr}(\hat{\rho}_S\hat{E}_j)$. A projective measurement is the special case $\hat{M}_j = \hat{E}_j = \hat{Q}_j$.

For a mixed probe state $\hat{\rho}_M = \sum_k q_k\,|\theta_k\rangle\langle\theta_k|$, the same recorded result j may arise from several unobserved probe components. Define the Kraus operators

$$\hat{M}_{jk} = \sqrt{q_k}\,{}_M\langle r_j|\,\hat{U}_{SM}\,|\theta_k\rangle_M\,.$$

Now

$$\boxed{\begin{aligned}\hat{E}_j &= \sum_k \hat{M}_{jk}^\dagger\hat{M}_{jk},\\ p_j &= \mathrm{Tr}(\hat{\rho}_S\hat{E}_j),\\ \hat{\rho}_{S|j} &= \frac{\sum_k \hat{M}_{jk}\hat{\rho}_S\hat{M}_{jk}^\dagger}{p_j}.\end{aligned}}$$

The *nonselective* state after the measurement is

$$\boxed{\hat{\rho}'_S = \sum_{jk}\hat{M}_{jk}\hat{\rho}_S\hat{M}_{jk}^\dagger, \qquad \mathrm{Tr}\,\hat{\rho}'_S = 1.}$$

The individual outcome maps $\mathcal{I}_j(\hat{\rho}) = \sum_k \hat{M}_{jk}\hat{\rho}\hat{M}_{jk}^\dagger$ comprise a quantum *instrument*; their sum is completely positive and trace-preserving. An effect $\hat{E}_j$ specifies a probability but *not, by itself,* the conditional state: different instruments can have identical POVM effects.

5) A qubit interacting with a qubit meter

As an explicit example, prepare a system qubit and a two-level meter as

$$|\psi\rangle_S = a_0\,|0\rangle + a_1\,|1\rangle\,, \qquad |\theta\rangle_M = \frac{|\downarrow\rangle + |\uparrow\rangle}{\sqrt{2}},$$

with $|a_0|^2 + |a_1|^2 = 1$. To allow the meter preparation to vary, first write $|\theta\rangle_M = b_0\,|\downarrow\rangle + b_1\,|\uparrow\rangle$ with $|b_0|^2 + |b_1|^2 = 1$. We will subsequently choose $b_0 = b_1 = 1/\sqrt{2}$ to obtain a particularly transparent measurement. The handout uses the interaction Hamiltonian

$$\boxed{\hat{H}_{\rm int} = \mathrm{i}g(|1\rangle\langle 1| - |0\rangle\langle 0|) \otimes (|\uparrow\rangle\langle\downarrow| - |\downarrow\rangle\langle\uparrow|),}$$

which is Hermitian because the operator in the second factor is anti-Hermitian. Introduce $\sigma_y^{(M)} = \mathrm{i}(|\uparrow\rangle\langle\downarrow| - |\downarrow\rangle\langle\uparrow|)$ and $\theta = g\tau/\hbar$. Then

$$\hat{U}_{SM} = e^{-\mathrm{i}\theta(|1\rangle\langle 1| - |0\rangle\langle 0|)\otimes\sigma_y^{(M)}}.$$

For a general meter preparation b_0, b_1, direct evaluation of this unitary and projection of the meter onto $|\downarrow\rangle$ or $|\uparrow\rangle$ yield

$$\begin{aligned}
\hat{M}_\downarrow &= (b_0\cos\theta + b_1\sin\theta)\,|0\rangle\langle 0| + (b_0\cos\theta - b_1\sin\theta)\,|1\rangle\langle 1|\,,\\
\hat{M}_\uparrow &= (b_1\cos\theta - b_0\sin\theta)\,|0\rangle\langle 0| + (b_1\cos\theta + b_0\sin\theta)\,|1\rangle\langle 1|\,.
\end{aligned}$$

These two operators satisfy the completeness relation for any normalized b_0, b_1. Specialize now to $b_0 = b_1 = 1/\sqrt{2}$. For $|m_\pm\rangle = (|\downarrow\rangle \pm |\uparrow\rangle)/\sqrt{2}$, the state after the interaction is

$$\begin{aligned}
|\Psi_{SM}(\tau)\rangle = a_0\,|0\rangle\,(\cos\theta\,|m_+\rangle + \sin\theta\,|m_-\rangle)\\
+ a_1\,|1\rangle\,(\cos\theta\,|m_+\rangle - \sin\theta\,|m_-\rangle).
\end{aligned}$$

A measurement of the meter in the $\{|\downarrow\rangle\,, |\uparrow\rangle\}$ basis induces the system operators

$$\boxed{\begin{aligned}
\hat{M}_\downarrow &= \tfrac{1}{\sqrt{2}}\big[(\cos\theta + \sin\theta)\,|0\rangle\langle 0| + (\cos\theta - \sin\theta)\,|1\rangle\langle 1|\big],\\
\hat{M}_\uparrow &= \tfrac{1}{\sqrt{2}}\big[(\cos\theta - \sin\theta)\,|0\rangle\langle 0| + (\cos\theta + \sin\theta)\,|1\rangle\langle 1|\big].
\end{aligned}}$$

Set $\eta = \sin(2\theta)$. The two POVM effects are

$$\hat{E}_\downarrow = \tfrac{1}{2}[(1+\eta)\,|0\rangle\langle 0| + (1-\eta)\,|1\rangle\langle 1|], \qquad \hat{E}_\uparrow = \hat{I} - \hat{E}_\downarrow.$$

They are positive and sum to the identity for $|\eta| \le 1$. When $\theta = \pi/4$, the measurement is projective:

$$\hat{M}_\downarrow = |0\rangle\langle 0|\,, \quad \hat{M}_\uparrow = |1\rangle\langle 1|\,, \quad |\Psi_{SM}\rangle = a_0\,|0\rangle\,|\downarrow\rangle + a_1\,|1\rangle\,|\uparrow\rangle\,.$$

The outcomes identify the computational-basis state, with probabilities $|a_0|^2$ and $|a_1|^2$. At $\theta = 3\pi/4$ the two projector outcomes are exchanged (up to an irrelevant phase): the meter result $\downarrow$ selects system state $|1\rangle$, and $\uparrow$ selects $|0\rangle$. When $\theta = 0$, both effects are $\hat{I}/2$: the result contains no information about the system and the conditional state is unchanged. Intermediate $0 < \theta < \pi/4$ implements a tunable *unsharp* measurement, whose backaction and information gain vary continuously with the interaction strength and time. The result follows entirely from a unitary system–probe interaction and an ordinary projective readout of the meter.

3.5 Eigenvalues and Eigenstates of Angular Momentum

We study the eigenvalues and eigenstates of a general angular momentum. The general definition of angular momentum $\hat{\boldsymbol{J}}$ is its commutation relation,

$$[\hat{J}_i, \hat{J}_j] = \mathrm{i}\hbar\epsilon_{ijk}\hat{J}_k.$$

A mechanical quantity satisfying this relation is called an angular momentum. Orbital angular momentum $\hat{\boldsymbol{L}}$ and spin angular momentum are special cases. We employ an algebraic method, analogous to that used for the harmonic oscillator, to find the eigenvalues of $\hat{\boldsymbol{J}}^2$ and $\hat{J}_z$.

1) Definition of angular momentum squared

Define the operator describing the magnitude of angular momentum:

$$\hat{\boldsymbol{J}}^2 = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2 = \hat{J}_i\hat{J}_i.$$

It commutes with angular momentum in any direction:

$$\begin{aligned}[\hat{\boldsymbol{J}}^2, \hat{J}_j] &= \hat{J}_i[\hat{J}_i, \hat{J}_j] + [\hat{J}_i, \hat{J}_j]\hat{J}_i \\ &= \mathrm{i}\hbar\epsilon_{ijk}\hat{J}_i\hat{J}_k + \mathrm{i}\hbar\epsilon_{ijk}\hat{J}_k\hat{J}_i \\ &= \mathrm{i}\hbar\epsilon_{ijk}\hat{J}_i\hat{J}_k + \mathrm{i}\hbar\epsilon_{kji}\hat{J}_i\hat{J}_k = 0.\end{aligned}$$

Here we used the antisymmetry of the Levi–Civita tensor, $\epsilon_{ijk} = -\epsilon_{kji}$.

Because $\hat{\boldsymbol{J}}^2$ and $\hat{J}_z$ commute, they possess common eigenstates. Since $\hat{\boldsymbol{J}}^2$ is not determined by $\hat{J}_z$ alone, at least two quantum numbers are required. Denote a common eigenstate by $|\lambda, m\rangle$; quantum numbers are conventionally dimensionless. Taking the dimensions of the operators into account,

$$\hat{\boldsymbol{J}}^2 |\lambda, m\rangle = \lambda\hbar^2 |\lambda, m\rangle, \qquad \hat{J}_z |\lambda, m\rangle = m\hbar |\lambda, m\rangle.$$

The question is: $\lambda = ?$, $m = ?$

2) Construct a new set of operators

Replace $\hat{J}_x, \hat{J}_y, \hat{J}_z$ by

$$\hat{J}_+ = \hat{J}_x + \mathrm{i}\hat{J}_y, \qquad \hat{J}_- = \hat{J}_x - \mathrm{i}\hat{J}_y = \hat{J}_+^\dagger, \qquad \hat{J}_z.$$

It follows that

$$\begin{aligned}\hat{J}_+\hat{J}_- &= \hat{\boldsymbol{J}}^2 - \hat{J}_z^2 + \hbar\hat{J}_z, \\ \hat{J}_-\hat{J}_+ &= \hat{\boldsymbol{J}}^2 - \hat{J}_z^2 - \hbar\hat{J}_z, \\ \hat{\boldsymbol{J}}^2 &= \frac{1}{2}(\hat{J}_+\hat{J}_- + \hat{J}_-\hat{J}_+) + \hat{J}_z^2.\end{aligned}$$

The new commutation relations are

$$[\hat{J}_+, \hat{J}_-] = 2\hbar\hat{J}_z, \qquad [\hat{J}_-, \hat{J}_z] = \hbar\hat{J}_-, \qquad [\hat{J}_+, \hat{J}_z] = -\hbar\hat{J}_+,$$
$$[\hat{\boldsymbol{J}}^2, \hat{J}_+] = [\hat{\boldsymbol{J}}^2, \hat{J}_-] = [\hat{\boldsymbol{J}}^2, \hat{J}_z] = 0.$$

3) Positivity bound

Take an expectation value in $|\lambda, m\rangle$:

$$\begin{aligned}\left\langle \lambda, m \middle| \hat{\boldsymbol{J}}^2 - \hat{J}_z^2 \middle| \lambda, m \right\rangle &= \frac{1}{2}\left\langle \lambda, m \middle| \hat{J}_+\hat{J}_- + \hat{J}_-\hat{J}_+ \middle| \lambda, m \right\rangle, \\ (\lambda - m^2)\hbar^2 \langle \lambda, m | \lambda, m \rangle &= \frac{1}{2}\left\langle \lambda, m \middle| \hat{J}_+\hat{J}_- \middle| \lambda, m \right\rangle + \frac{1}{2}\left\langle \lambda, m \middle| \hat{J}_-\hat{J}_+ \middle| \lambda, m \right\rangle \geq 0.\end{aligned}$$

Thus $\lambda \geq m^2$.

4) Raising and lowering operators

Using their commutators with $\hat{\boldsymbol{J}}^2$ and $\hat{J}_z$,

$$\begin{cases} \hat{\boldsymbol{J}}^2(\hat{J}_+ |\lambda, m\rangle) = \lambda\hbar^2(\hat{J}_+ |\lambda, m\rangle), \\ \hat{J}_z(\hat{J}_+ |\lambda, m\rangle) = (\hbar\hat{J}_+ + \hat{J}_+\hat{J}_z) |\lambda, m\rangle = (m+1)\hbar(\hat{J}_+ |\lambda, m\rangle), \end{cases}$$

and

$$\begin{cases} \hat{\boldsymbol{J}}^2(\hat{J}_- |\lambda, m\rangle) = \lambda\hbar^2(\hat{J}_- |\lambda, m\rangle), \\ \hat{J}_z(\hat{J}_- |\lambda, m\rangle) = (-\hbar\hat{J}_- + \hat{J}_-\hat{J}_z) |\lambda, m\rangle = (m-1)\hbar(\hat{J}_- |\lambda, m\rangle). \end{cases}$$

Thus if $|\lambda, m\rangle$ is a common eigenstate of $\hat{\boldsymbol{J}}^2$ and $\hat{J}_z$, so are $\hat{J}_+ |\lambda, m\rangle$ and $\hat{J}_- |\lambda, m\rangle$, when nonzero. Their eigenvalues form the sequence

Common eigenstate	$\hat{J}_z$ eigenvalue	$\hat{\boldsymbol{J}}^2$ eigenvalue
$\vdots$	$\vdots$	$\vdots$
$(\hat{J}_+)^2 \|\lambda, m\rangle$	$(m+2)\hbar$	$\lambda\hbar^2$
$\hat{J}_+ \|\lambda, m\rangle$	$(m+1)\hbar$	$\lambda\hbar^2$
$\|\lambda, m\rangle$	$m\hbar$	$\lambda\hbar^2$
$\hat{J}_- \|\lambda, m\rangle$	$(m-1)\hbar$	$\lambda\hbar^2$
$(\hat{J}_-)^2 \|\lambda, m\rangle$	$(m-2)\hbar$	$\lambda\hbar^2$
$\vdots$	$\vdots$	$\vdots$

Therefore $\hat{J}_-$ is called a lowering operator and $\hat{J}_+$ a raising operator. Since the bound $\lambda \geq m^2$ limits m, there must be largest and smallest values of m, denoted j and j'. The values of m form the sequence $j', j'+1, \ldots, j-1, j$, and $\lambda \geq j^2, \lambda \geq j'^2$.

5) Determine the upper and lower bounds

For the largest value j,

$$\hat{J}_+ |\lambda, j\rangle = 0,$$

otherwise $j\hbar$ would not be the greatest eigenvalue of $\hat{J}_z$. Using

$$0 = \hat{J}_-\hat{J}_+ |\lambda, j\rangle = (\hat{\boldsymbol{J}}^2 - \hat{J}_z^2 - \hbar\hat{J}_z) |\lambda, j\rangle = (\lambda - j^2 - j)\hbar^2 |\lambda, j\rangle ,$$

we obtain $\lambda = j(j+1)$.

For the smallest value j',

$$\hat{J}_- |\lambda, j'\rangle = 0,$$

otherwise $j'\hbar$ would not be the smallest eigenvalue. Similarly,

$$0 = \hat{J}_+\hat{J}_- |\lambda, j'\rangle = (\lambda - j'^2 + j')\hbar^2 |\lambda, j'\rangle ,$$

so $\lambda = j'(j'-1)$. Since $j(j+1) = j'(j'-1)$,

$$j' = \begin{cases} j+1, \\ -j. \end{cases}$$

The choice $j' = j+1 > j$ contradicts the fact that j is the maximum and j' the minimum. Thus $j' = -j$, and

$$\hat{\boldsymbol{J}}^2 |j, m\rangle = j(j+1)\hbar^2 |j, m\rangle , \qquad \hat{J}_z |j, m\rangle = m\hbar |j, m\rangle ,$$

where $m = -j, -j+1, \ldots, j-1, j$ comprises $2j+1$ values. We relabel $|\lambda, m\rangle$ as $|j, m\rangle$. The remaining question is: which values of j are possible?

6) Allowed angular-momentum quantum numbers

Applying $\hat{J}_-$ to $|j,j\rangle$ exactly $2j$ times produces $|j,-j\rangle$; applying $\hat{J}_+$ to $|j,-j\rangle$ exactly $2j$ times produces $|j,j\rangle$. Therefore $2j$ is zero or a positive integer, so j may be zero, a positive integer, or a positive half-integer.

In summary,

$$\boxed{\begin{aligned}\hat{\boldsymbol{J}}^{\,2}\,|j,m\rangle &= j(j+1)\hbar^2\,|j,m\rangle\,, \quad j = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots,\\ \hat{J}_z\,|j,m\rangle &= m\hbar\,|j,m\rangle\,, \quad m = -j, -j+1, \ldots, j \quad (2j+1 \text{ values}).\end{aligned}}$$

For j zero or a positive integer, $\hat{\boldsymbol{J}}$ can be explained by orbital angular momentum $\hat{\boldsymbol{L}}$. What is the physical meaning of $\hat{\boldsymbol{J}}$ when j is a positive half-integer? The angular-momentum commutation relation itself establishes the possibility of a new type of angular momentum: spin angular momentum.

7) Matrix elements of raising and lowering operators

Because the eigenvalue pair (j,m) labels a unique eigenstate in an irreducible fixed-j multiplet,

$$\hat{J}_+\,|j,m\rangle = a_{jm}\,|j,m+1\rangle\,, \qquad \hat{J}_-\,|j,m\rangle = b_{jm}\,|j,m-1\rangle\,.$$

Taking Hermitian conjugates,

$$\langle j,m|\,\hat{J}_- = a^*_{jm}\,\langle j,m+1|\,, \qquad \langle j,m|\,\hat{J}_+ = b^*_{jm}\,\langle j,m-1|\,.$$

Consequently,

$$\begin{aligned}|a_{jm}|^2 &= \Big\langle j,m\Big|\hat{J}_-\hat{J}_+\Big|j,m\Big\rangle = \Big\langle j,m\Big|\hat{\boldsymbol{J}}^{\,2} - \hat{J}_z^2 - \hbar\hat{J}_z\Big|j,m\Big\rangle\\ &= [j(j+1) - m^2 - m]\hbar^2.\end{aligned}$$

Choose the phase convention

$$a_{jm} = \hbar\sqrt{j(j+1) - m(m+1)} = \hbar\sqrt{(j-m)(j+m+1)}.$$

Similarly,

$$b_{jm} = \hbar\sqrt{(j+m)(j-m+1)}.$$

Thus

$$\boxed{\begin{aligned}\hat{J}_+\,|j,m\rangle &= \hbar\sqrt{(j-m)(j+m+1)}\,|j,m+1\rangle\,,\\ \hat{J}_-\,|j,m\rangle &= \hbar\sqrt{(j+m)(j-m+1)}\,|j,m-1\rangle\,.\end{aligned}}$$

Because

$$\hat{J}_x = \tfrac{1}{2}(\hat{J}_+ + \hat{J}_-), \qquad \hat{J}_y = \tfrac{1}{2\mathrm{i}}(\hat{J}_+ - \hat{J}_-),$$

we obtain

$$\begin{aligned}\hat{J}_x\,|j,m\rangle &= \frac{\hbar}{2}\sqrt{(j-m)(j+m+1)}\,|j,m+1\rangle + \frac{\hbar}{2}\sqrt{(j+m)(j-m+1)}\,|j,m-1\rangle\,,\\ \hat{J}_y\,|j,m\rangle &= \frac{\hbar}{2\mathrm{i}}\sqrt{(j-m)(j+m+1)}\,|j,m+1\rangle - \frac{\hbar}{2\mathrm{i}}\sqrt{(j+m)(j-m+1)}\,|j,m-1\rangle\,.\end{aligned}$$

8) Common representation of $\hat{\boldsymbol{J}}^2$ and $\hat{J}_z$

The basis consists of $|j,m\rangle$. For fixed j, the eigenvalue of $\hat{\boldsymbol{J}}^2$ is fixed, and the eigenstates of $\hat{J}_z$ span a subspace of dimension $D = 2j+1$. The matrix elements are

$$\Big\langle j,n\Big|\hat{\boldsymbol{J}}^2\Big|j,m\Big\rangle = j(j+1)\hbar^2\delta_{nm}, \qquad \Big\langle j,n\Big|\hat{J}_z\Big|j,m\Big\rangle = m\hbar\delta_{nm},$$

$$\Big\langle j,n\Big|\hat{J}_x\Big|j,m\Big\rangle = \frac{\hbar}{2}\sqrt{(j-m)(j+m+1)}\delta_{n,m+1} + \frac{\hbar}{2}\sqrt{(j+m)(j-m+1)}\delta_{n,m-1},$$
$$\Big\langle j,n\Big|\hat{J}_y\Big|j,m\Big\rangle = \frac{\hbar}{2\mathrm{i}}\sqrt{(j-m)(j+m+1)}\delta_{n,m+1} - \frac{\hbar}{2\mathrm{i}}\sqrt{(j+m)(j-m+1)}\delta_{n,m-1}.$$

These are the matrix representations of $\hat{J}_x, \hat{J}_y, \hat{J}_z$ in the common eigenbasis.

Example 1: $j = 1/2$ **(electron spin).** In this case

$$\hat{\boldsymbol{J}}^2 = \frac{3}{4}\hbar^2, \qquad m = \pm\frac{1}{2}, \qquad D = 2.$$

Choose the first basis state $|1/2,1/2\rangle$ and the second $|1/2,-1/2\rangle$. Then

$$J_z = \begin{pmatrix}\hbar/2 & 0\\ 0 & -\hbar/2\end{pmatrix} = \frac{\hbar}{2}\sigma_z, \quad J_x = \begin{pmatrix}0 & \hbar/2\\ \hbar/2 & 0\end{pmatrix} = \frac{\hbar}{2}\sigma_x,$$

$$J_y = \begin{pmatrix}0 & -\mathrm{i}\hbar/2\\ \mathrm{i}\hbar/2 & 0\end{pmatrix} = \frac{\hbar}{2}\sigma_y.$$

If an eigenstate of $\hat{J}_z$ has components $(a,b)^{\mathsf{T}}$, its eigenvalue equation is

$$\begin{pmatrix}\hbar/2 & 0\\ 0 & -\hbar/2\end{pmatrix}\begin{pmatrix}a\\ b\end{pmatrix} = m\hbar\begin{pmatrix}a\\ b\end{pmatrix}.$$

After normalization, the eigenstate is $(1,0)^{\mathsf{T}}$ for $m = 1/2$ and $(0,1)^{\mathsf{T}}$ for $m = -1/2$.

Example 2: $j = 1$ **(photon spin).** Here $\hat{\boldsymbol{J}}^2 = 2\hbar^2$, $m = -1, 0, 1$, and $D = 3$. Choose the ordered basis $|1,1\rangle$, $|1,0\rangle$, $|1,-1\rangle$:

$$J_x = \frac{\hbar}{\sqrt{2}}\begin{pmatrix}0 & 1 & 0\\ 1 & 0 & 1\\ 0 & 1 & 0\end{pmatrix}, \qquad J_y = \frac{\hbar}{\sqrt{2}}\begin{pmatrix}0 & -\mathrm{i} & 0\\ \mathrm{i} & 0 & -\mathrm{i}\\ 0 & \mathrm{i} & 0\end{pmatrix}, \qquad J_z = \hbar\begin{pmatrix}1 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & -1\end{pmatrix}.$$

The eigenstates of J_z have components

$$m = 1: \begin{pmatrix}1\\0\\0\end{pmatrix}, \quad m = 0: \begin{pmatrix}0\\1\\0\end{pmatrix}, \quad m = -1: \begin{pmatrix}0\\0\\1\end{pmatrix}.$$

9) Matrix representation of rotation operators

The rotation operator is

$$\hat{D}_n(\varphi) = \exp\!\left(-\frac{\mathrm{i}}{\hbar}\hat{\boldsymbol{J}}\cdot\boldsymbol{e}_n\varphi\right).$$

Since

$$[\hat{\boldsymbol{J}}^2, \hat{\boldsymbol{J}}] = 0, \qquad [\hat{\boldsymbol{J}}^2, \hat{D}_n(\varphi)] = 0,$$

$\hat{D}_n(\varphi)$ and $\hat{\boldsymbol{J}}^2$ have common eigenstates. In the $\hat{\boldsymbol{J}}^2$ representation, the rotation matrix is block diagonal; in the common basis of $\hat{\boldsymbol{J}}^2$ and $\hat{J}_z$,

$$D^{jj'}_{mm'} = \left\langle j, m \middle| \hat{D}(\varphi) \middle| j', m' \right\rangle = 0 \qquad (j \neq j').$$

But $[\hat{J}_z, \hat{D}_n(\varphi)] \neq 0$ in general, so within a fixed-j subspace D has off-diagonal entries. Since any rotation can be written in terms of Euler angles α, β, γ, the matrix elements in a fixed-j subspace obey

$$\begin{aligned} D^{j}_{mm'} &= \left\langle j, m \middle| \hat{D}_n(\varphi) \middle| j, m' \right\rangle = \left\langle j, m \middle| e^{-\mathrm{i}\hat{\boldsymbol{J}}\cdot\boldsymbol{e}_n\varphi/\hbar} \middle| j, m' \right\rangle \\ &= \left\langle j, m \middle| e^{-\mathrm{i}\hat{J}_z\alpha/\hbar} e^{-\mathrm{i}\hat{J}_y\beta/\hbar} e^{-\mathrm{i}\hat{J}_z\gamma/\hbar} \middle| j, m' \right\rangle \\ &= e^{-\mathrm{i}(m\alpha+m'\gamma)} \left\langle j, m \middle| e^{-\mathrm{i}\hat{J}_y\beta/\hbar} \middle| j, m' \right\rangle \\ &= e^{-\mathrm{i}(m\alpha+m'\gamma)} d^{j}_{mm'}(\beta). \end{aligned}$$

Thus D is block diagonal in j, with nonzero matrix elements generally occurring only within the same j block.

For $j = 1/2$, we previously found

$$d^{1/2}(\beta) = \begin{pmatrix} \cos(\beta/2) & -\sin(\beta/2) \\ \sin(\beta/2) & \cos(\beta/2) \end{pmatrix}.$$

For $j = 1$, in the common representation,

$$J_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -\mathrm{i} & 0 \\ \mathrm{i} & 0 & -\mathrm{i} \\ 0 & \mathrm{i} & 0 \end{pmatrix}.$$

One can verify

$$e^{-\mathrm{i}J_y\beta/\hbar} = \mathbf{1} - \frac{\mathrm{i}}{\hbar}\sin\beta\, J_y + \left(\frac{\mathrm{i}}{\hbar}\right)^2 (1-\cos\beta) J_y^2,$$

which yields

$$\boxed{d^1(\beta) = \frac{1}{2} \begin{pmatrix} 1+\cos\beta & -\sqrt{2}\sin\beta & 1-\cos\beta \\ \sqrt{2}\sin\beta & 2\cos\beta & -\sqrt{2}\sin\beta \\ 1-\cos\beta & \sqrt{2}\sin\beta & 1+\cos\beta \end{pmatrix}.}$$

3.6 Orbital Angular Momentum

1) Eigenvalues and eigenstates

The classical orbital angular momentum and its quantum operator are

$$\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p} \quad \longrightarrow \quad \hat{\boldsymbol{L}} = \hat{\boldsymbol{r}} \times \hat{\boldsymbol{p}}, \qquad \hat{L}_i = \epsilon_{ijk}\hat{x}_j\hat{p}_k \quad (i,j,k = 1,2,3).$$

From the fundamental commutation relations

$$[\hat{x}_i, \hat{x}_j] = 0, \qquad [\hat{p}_i, \hat{p}_j] = 0, \qquad [\hat{x}_i, \hat{p}_j] = \mathrm{i}\hbar\delta_{ij},$$

one can show that

$$[\hat{L}_i, \hat{L}_j] = \mathrm{i}\hbar\epsilon_{ijk}\hat{L}_k.$$

Thus the orbital angular-momentum operators satisfy the general definition of angular momentum.

Because this is orbital angular momentum, we calculate eigenvalues and eigenstates in the coordinate representation:

$$\hat{\boldsymbol{r}} = \boldsymbol{r}, \qquad \hat{\boldsymbol{p}} = -\mathrm{i}\hbar\boldsymbol{\nabla},$$
$$\hat{\boldsymbol{L}} = -\mathrm{i}\hbar\boldsymbol{r}\times\boldsymbol{\nabla}, \qquad \hat{L}_i = -\mathrm{i}\hbar\epsilon_{ijk}x_j\frac{\partial}{\partial x_k}.$$

It is convenient to consider rotations in spherical coordinates, $(x, y, z) \to (r, \theta, \varphi)$. The operators are

$$\hat{L}_x = \mathrm{i}\hbar\left(\sin\varphi\frac{\partial}{\partial\theta} + \cot\theta\cos\varphi\frac{\partial}{\partial\varphi}\right),$$
$$\hat{L}_y = -\mathrm{i}\hbar\left(\cos\varphi\frac{\partial}{\partial\theta} - \cot\theta\sin\varphi\frac{\partial}{\partial\varphi}\right),$$
$$\hat{L}_z = -\mathrm{i}\hbar\frac{\partial}{\partial\varphi},$$
$$\hat{\boldsymbol{L}}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right].$$

This shows that $\hat{\boldsymbol{L}}$ depends only on the angles θ and φ.

First consider the eigenvalue equation of $\hat{L}_z$. As $\hat{L}_z$ does not depend on θ,

$$\hat{L}_z\Phi(\varphi) = L_z\Phi(\varphi), \qquad -\mathrm{i}\hbar\frac{\mathrm{d}\Phi}{\mathrm{d}\varphi} = L_z\Phi(\varphi).$$

The solution is

$$\Phi(\varphi) = A\exp(\mathrm{i}L_z\varphi/\hbar).$$

The single-valuedness condition

$$\Phi(\varphi) = \Phi(\varphi + 2\pi)$$

implies $L_z = m\hbar$, where $m = 0, \pm1, \ldots, \pm\infty$, and $\Phi(\varphi) = Ae^{\mathrm{i}m\varphi}$. Normalize:

$$\int_0^{2\pi}|\Phi(\varphi)|^2\mathrm{d}\varphi = 1 \quad\Longrightarrow\quad A = \frac{1}{\sqrt{2\pi}}.$$

Thus

$$\boxed{\Phi(\varphi) = \frac{e^{\mathrm{i}m\varphi}}{\sqrt{2\pi}}.}$$

Next consider the eigenvalue equation of $\hat{\boldsymbol{L}}^2$:

$$\hat{\boldsymbol{L}}^2Y(\theta,\varphi) = L^2Y(\theta,\varphi).$$

This is

$$\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2} + \frac{L^2}{\hbar^2}\right]Y_{lm}(\theta,\varphi) = 0.$$

Using separation of variables and the single-valuedness and finiteness conditions, one finds

$$L^2 = l(l+1)\hbar^2, \qquad l = 0, 1, 2, \ldots.$$

The eigenstates are spherical harmonics

$$Y_{lm}(\theta,\varphi) = N_{lm}P_l^m(\cos\theta)e^{\mathrm{i}m\varphi},$$

where $P_l^m(x)$ is the associated Legendre polynomial. The magnetic quantum number is

$$m = 0, \pm1, \pm2, \ldots, \pm l,$$

with $2l+1$ values. The eigenvalue L^2 depends only on l whereas the eigenstates also depend on m, so the degeneracy is $g = 2l+1$.

The normalization condition is

$$\int_0^{2\pi} \mathrm{d}\varphi \int_0^{\pi} \sin\theta \mathrm{d}\theta \, |Y_{lm}(\theta,\varphi)|^2 = 1,$$

which gives the normalization factor

$$N_{lm} = \sqrt{\frac{(l-m)!(2l+1)}{(l+m)!4\pi}}.$$

Because the φ-dependent factor in Y_{lm} is an eigenstate of $\hat{L}_z$, the spherical harmonics are common eigenstates of $\hat{\boldsymbol{L}}^2$ and $\hat{L}_z$:

$$\boxed{\begin{aligned} \hat{\boldsymbol{L}}^2 Y_{lm}(\theta,\varphi) &= l(l+1)\hbar^2 Y_{lm}(\theta,\varphi), \\ \hat{L}_z Y_{lm}(\theta,\varphi) &= m\hbar Y_{lm}(\theta,\varphi), \end{aligned}}$$

where $l = 0, 1, 2, \ldots$ and $m = 0, \pm 1, \ldots, \pm l$. The component of angular momentum is constrained by its total magnitude, so m cannot extend to $\pm\infty$ for fixed l.

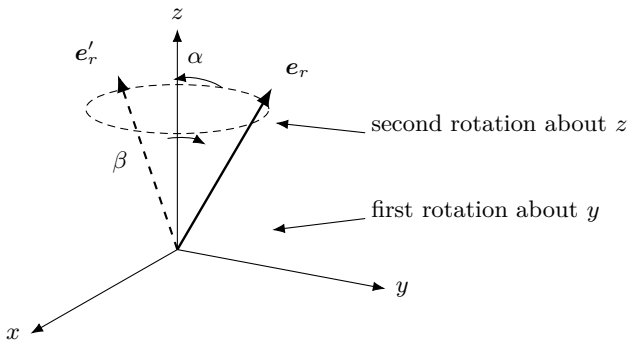
This shows that orbital angular momentum realizes the general angular-momentum algebra in coordinate space. It is only one form of angular momentum: l is restricted to zero and positive integers and cannot be a positive half-integer.

2) Spatial rotation operator

The orbital-angular-momentum eigenstate is the projection of $|l,m\rangle$ onto a radial position basis state $|\boldsymbol{e}_r(\theta,\varphi)\rangle$:

$$Y_{lm}(\theta,\varphi) = \langle \boldsymbol{e}_r(\theta,\varphi)|l,m\rangle \, .$$

The unit vector $\boldsymbol{e}_r(\theta,\varphi)$ can be obtained from $\boldsymbol{e}_z(0,\varphi)$ by two rotations, first by θ about the y axis and then by φ about the z axis:



$$|\boldsymbol{e}_r(\theta,\varphi)\rangle = \hat{D}_z(\varphi)\hat{D}_y(\theta)\,|\boldsymbol{e}_z(0,\varphi)\rangle \, .$$

Insert the complete basis:

$$|\boldsymbol{e}_r(\theta,\varphi)\rangle = \sum_{m'} \hat{D}_z(\varphi)\hat{D}_y(\theta)\,|l,m'\rangle\,\langle l,m'|\boldsymbol{e}_z(0,\varphi)\rangle \, .$$

Hence

$$\begin{aligned} Y_{lm}^*(\theta,\varphi) &= \langle l,m|\boldsymbol{e}_r(\theta,\varphi)\rangle \\ &= \sum_{m'} \left\langle l,m\middle|\hat{D}_z(\varphi)\hat{D}_y(\theta)\middle|l,m'\right\rangle \langle l,m'|\boldsymbol{e}_z(0,\varphi)\rangle \, . \end{aligned}$$

Since

$$\langle l, m' | \boldsymbol{e}_z(0,\varphi)\rangle = Y^*_{lm'}(0,\varphi) = \sqrt{\frac{2l+1}{4\pi}}\delta_{m'0},$$

and

$$\langle l, m|\, \hat{D}_z(\varphi) = \langle l, m|\, e^{-\mathrm{i}\hat{L}_z\varphi/\hbar} = e^{-\mathrm{i}m\varphi}\, \langle l, m|\,,$$

we obtain

$$\boxed{Y^*_{lm}(\theta,\varphi) = e^{-\mathrm{i}m\varphi}\sqrt{\frac{2l+1}{4\pi}}\, d^l_{m0}(\theta), \qquad d^l_{mm'}(\theta) = \left\langle l, m\middle|\hat{D}_y(\theta)\middle| l, m'\right\rangle.}$$

A spherical harmonic in any direction can thus be obtained by rotating a spherical harmonic along the Z direction.

3.7 Coupling of Two Angular Momenta

Consider the addition of two arbitrary independent angular momenta $\hat{\boldsymbol{J}}_1$ and $\hat{\boldsymbol{J}}_2$:

$$\hat{\boldsymbol{J}} = \hat{\boldsymbol{J}}_1 + \hat{\boldsymbol{J}}_2.$$

They satisfy

$$[\hat{J}_{1i}, \hat{J}_{1j}] = \mathrm{i}\hbar\epsilon_{ijk}\hat{J}_{1k}, \qquad [\hat{J}_{2i}, \hat{J}_{2j}] = \mathrm{i}\hbar\epsilon_{ijk}\hat{J}_{2k}, \qquad [\hat{J}_{1i}, \hat{J}_{2j}] = 0.$$

The respective eigenvalues are

$$J_1^2 = j_1(j_1+1)\hbar^2, \quad J_{1z} = m_1\hbar, \quad m_1 = -j_1, \ldots, j_1,$$
$$J_2^2 = j_2(j_2+1)\hbar^2, \quad J_{2z} = m_2\hbar, \quad m_2 = -j_2, \ldots, j_2.$$

It is easy to prove

$$[\hat{J}_i, \hat{J}_j] = \mathrm{i}\hbar\epsilon_{ijk}\hat{J}_k.$$

Thus $\hat{\boldsymbol{J}}$ is again an angular-momentum operator, called the total angular momentum, with eigenvalues

$$J^2 = j(j+1)\hbar^2, \qquad J_z = m\hbar, \qquad m = -j, \ldots, j.$$

The problem is to find the relation between $|j, m\rangle$ and $|j_1, m_1\rangle\,|j_2, m_2\rangle$.

1) Two representations

Uncoupled representation. Since the two angular momenta commute with one another, the four operators $\hat{\boldsymbol{J}}_1^2$, $\hat{J}_{1z}$, $\hat{\boldsymbol{J}}_2^2$, $\hat{J}_{2z}$ have common eigenvectors $|j_1 m_1 j_2 m_2\rangle$, which form the uncoupled representation:

$$\begin{aligned}
\hat{\boldsymbol{J}}_1^2\,|j_1 m_1 j_2 m_2\rangle &= j_1(j_1+1)\hbar^2\,|j_1 m_1 j_2 m_2\rangle\,,\\
\hat{J}_{1z}\,|j_1 m_1 j_2 m_2\rangle &= m_1\hbar\,|j_1 m_1 j_2 m_2\rangle\,,\\
\hat{\boldsymbol{J}}_2^2\,|j_1 m_1 j_2 m_2\rangle &= j_2(j_2+1)\hbar^2\,|j_1 m_1 j_2 m_2\rangle\,,\\
\hat{J}_{2z}\,|j_1 m_1 j_2 m_2\rangle &= m_2\hbar\,|j_1 m_1 j_2 m_2\rangle\,.
\end{aligned}$$

But $[\hat{\boldsymbol{J}}^2, \hat{J}_{1z}] \neq 0$ and $[\hat{\boldsymbol{J}}^2, \hat{J}_{2z}] \neq 0$. Consequently, $\hat{\boldsymbol{J}}^2$ is not diagonal in this basis, making its eigenvalues inconvenient to obtain directly.

Coupled representation. Since

$$[\hat{\boldsymbol{J}}^2, \hat{\boldsymbol{J}}_1^2] = [\hat{\boldsymbol{J}}^2, \hat{\boldsymbol{J}}_2^2] = [\hat{\boldsymbol{J}}^2, \hat{J}_z] = 0,$$

and the remaining relevant pairs also commute, the four operators $\hat{\boldsymbol{J}}_1^2$, $\hat{\boldsymbol{J}}_2^2$, $\hat{\boldsymbol{J}}^2$, and $\hat{J}_z$ have common eigenvectors $|j_1 j_2 j m\rangle$, which form the coupled representation:

$$\begin{aligned}
\hat{\boldsymbol{J}}_1^2 \,|j_1 j_2 j m\rangle &= j_1(j_1+1)\hbar^2 \,|j_1 j_2 j m\rangle\,, \\
\hat{\boldsymbol{J}}_2^2 \,|j_1 j_2 j m\rangle &= j_2(j_2+1)\hbar^2 \,|j_1 j_2 j m\rangle\,, \\
\hat{\boldsymbol{J}}^2 \,|j_1 j_2 j m\rangle &= j(j+1)\hbar^2 \,|j_1 j_2 j m\rangle\,, \\
\hat{J}_z \,|j_1 j_2 j m\rangle &= m\hbar \,|j_1 j_2 j m\rangle\,.
\end{aligned}$$

Because the eigenvalues and eigenvectors of the four uncoupled operators are known, obtaining those of total angular momentum requires connecting the two representations by a change of basis.

From completeness of the uncoupled basis,

$$\sum_{m_1,m_2} |j_1 m_1 j_2 m_2\rangle \langle j_1 m_1 j_2 m_2| = \mathbf{1},$$

the transformation is

$$\boxed{|j_1 j_2 j m\rangle = \sum_{m_1,m_2} |j_1 m_1 j_2 m_2\rangle \langle j_1 m_1 j_2 m_2 | j_1 j_2 j m\rangle\,.}$$

The overlaps $\langle j_1 m_1 j_2 m_2 | j_1 j_2 j m\rangle$ are the Clebsch–Gordan (CG) coefficients.

2) Eigenvalues (j, m) of the coupled representation

Apply $\hat{J}_z = \hat{J}_{1z} + \hat{J}_{2z}$ to the expansion:

$$\begin{aligned}
m\hbar \,|j_1 j_2 j m\rangle &= \sum_{m_1,m_2} \langle j_1 m_1 j_2 m_2 | j_1 j_2 j m\rangle \,(\hat{J}_{1z} + \hat{J}_{2z})\,|j_1 m_1 j_2 m_2\rangle \\
&= \sum_{m_1,m_2} (m_1+m_2)\hbar \,\langle j_1 m_1 j_2 m_2 | j_1 j_2 j m\rangle \,|j_1 m_1 j_2 m_2\rangle\,.
\end{aligned}$$

Thus

$$\sum_{m_1,m_2} (m - m_1 - m_2)\hbar \,\langle j_1 m_1 j_2 m_2 | j_1 j_2 j m\rangle \,|j_1 m_1 j_2 m_2\rangle = 0.$$

The uncoupled basis vectors are linearly independent. Therefore for each term either the CG coefficient is zero or $m = m_1 + m_2$. We seek nonzero CG coefficients, and hence

$$\boxed{m = m_1 + m_2.}$$

Next consider the possible values of j. Let $j_{\min} \le j \le j_{\max}$. The maximum is

$$j_{\max} = m_{\max} = (m_1)_{\max} + (m_2)_{\max} = j_1 + j_2.$$

The dimension of the uncoupled representation is

$$D = (2j_1+1)(2j_2+1),$$

while the dimension of the coupled representation is

$$D = \sum_{j=j_{\min}}^{j_{\max}} (2j+1) = (j_{\max}+1)^2 - j_{\min}^2.$$

Equating the dimensions,

$$(2j_1+1)(2j_2+1) = (j_1+j_2+1)^2 - j_{\min}^2,$$

gives $j_{\min}^2 = (j_1 - j_2)^2$ and $j_{\min} = |j_1 - j_2|$. Thus for fixed (j_1, j_2),

$$\boxed{\begin{gathered} J^2 = j(j+1)\hbar^2, \qquad j = |j_1 - j_2|, |j_1 - j_2| + 1, \ldots, j_1 + j_2, \\ J_z = m\hbar, \qquad m = m_1 + m_2. \end{gathered}}$$

3) Eigenstates $|j_1 j_2 j m\rangle$ of the coupled representation

The central task is to calculate the CG coefficients. We do not undertake a general treatment; standard tables are available.

Example 1: spin and orbital angular momentum. Let $\hat{\boldsymbol{J}}_1 = \hat{\boldsymbol{L}}$, $\hat{\boldsymbol{J}}_2 = \hat{\boldsymbol{S}}$, and $\hat{\boldsymbol{J}} = \hat{\boldsymbol{L}} + \hat{\boldsymbol{S}}$. The coupled basis is $\left|l, \frac{1}{2}; j, m\right\rangle$ and the uncoupled basis is $\left|l, m_l; \frac{1}{2}, m_s\right\rangle$. The transformation reads

$$\left|l, \tfrac{1}{2}; j, m\right\rangle = \sum_{m_l, m_s} C_{m_l m_s} \left|l, m_l; \tfrac{1}{2}, m_s\right\rangle = \sum_{m_l} \left[A_{m_l} \left|l, m_l; \tfrac{1}{2}, \tfrac{1}{2}\right\rangle + B_{m_l} \left|l, m_l; \tfrac{1}{2}, -\tfrac{1}{2}\right\rangle\right].$$

Because $m = m_l + m_s$,

$$m_l = m - m_s = \begin{cases} m - \frac{1}{2}, & m_s = \frac{1}{2}, \\ m + \frac{1}{2}, & m_s = -\frac{1}{2}, \end{cases}$$

there are only two terms:

$$\boxed{\left|l, \tfrac{1}{2}; j, m\right\rangle = A \left|l, m - \tfrac{1}{2}; \tfrac{1}{2}, \tfrac{1}{2}\right\rangle + B \left|l, m + \tfrac{1}{2}; \tfrac{1}{2}, -\tfrac{1}{2}\right\rangle.}$$

We determine the coefficients A, B.

In the spin subspace of the uncoupled representation,

$$\begin{aligned} \hat{\boldsymbol{J}}^2 &= \hat{\boldsymbol{L}}^2 + \hat{\boldsymbol{S}}^2 + 2\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{S}} \\ &= \begin{pmatrix} \hat{\boldsymbol{L}}^2 + \frac{3}{4}\hbar^2 + \hbar \hat{L}_z & \hbar \hat{L}_- \\ \hbar \hat{L}_+ & \hat{\boldsymbol{L}}^2 + \frac{3}{4}\hbar^2 - \hbar \hat{L}_z \end{pmatrix}. \end{aligned}$$

Here $\hat{L}_\pm = \hat{L}_x \pm \mathrm{i}\hat{L}_y$ are the orbital raising and lowering operators. The uncoupled states are represented as

$$\left|l, m - \tfrac{1}{2}; \tfrac{1}{2}, \tfrac{1}{2}\right\rangle = \begin{pmatrix} \left|l, m - \frac{1}{2}\right\rangle \\ 0 \end{pmatrix}, \qquad \left|l, m + \tfrac{1}{2}; \tfrac{1}{2}, -\tfrac{1}{2}\right\rangle = \begin{pmatrix} 0 \\ \left|l, m + \frac{1}{2}\right\rangle \end{pmatrix}.$$

Insert the expansion into the eigenvalue equation

$$\hat{\boldsymbol{J}}^2 \left|l, \tfrac{1}{2}; j, m\right\rangle = j(j+1)\hbar^2 \left|l, \tfrac{1}{2}; j, m\right\rangle.$$

It becomes

$$\begin{aligned} j(j+1)\hbar^2 \big(A \left|l, m - \tfrac{1}{2}; \tfrac{1}{2}, \tfrac{1}{2}\right\rangle + B \left|l, m + \tfrac{1}{2}; \tfrac{1}{2}, -\tfrac{1}{2}\right\rangle\big) = (\hat{\boldsymbol{L}}^2 + \hat{\boldsymbol{S}}^2 + 2\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{S}}) \\ \times \big(A \left|l, m - \tfrac{1}{2}; \tfrac{1}{2}, \tfrac{1}{2}\right\rangle + B \left|l, m + \tfrac{1}{2}; \tfrac{1}{2}, -\tfrac{1}{2}\right\rangle\big). \end{aligned}$$

In the spin subspace this gives

$$\begin{cases} A\left(\hat{\boldsymbol{L}}^2 + \frac{3}{4}\hbar^2 + \hbar \hat{L}_z - j(j+1)\hbar^2\right) \left|l, m - \frac{1}{2}\right\rangle + B\hbar \hat{L}_- \left|l, m + \frac{1}{2}\right\rangle = 0, \\ A\hbar \hat{L}_+ \left|l, m - \frac{1}{2}\right\rangle + B\left(\hat{\boldsymbol{L}}^2 + \frac{3}{4}\hbar^2 - \hbar \hat{L}_z - j(j+1)\hbar^2\right) \left|l, m + \frac{1}{2}\right\rangle = 0. \end{cases}$$

Using the orbital-angular-momentum eigenvalues and ladder-operator coefficients yields

$$\begin{cases} \left[l(l+1) + \frac{3}{4} + \left(m - \frac{1}{2}\right) - j(j+1)\right] A + \sqrt{\left(l - m + \frac{1}{2}\right)\left(l + m + \frac{1}{2}\right)}\, B = 0, \\ \sqrt{\left(l + m + \frac{1}{2}\right)\left(l - m + \frac{1}{2}\right)}\, A + \left[l(l+1) + \frac{3}{4} - \left(m + \frac{1}{2}\right) - j(j+1)\right] B = 0. \end{cases}$$

These equations determine A, B. The allowed total angular momenta are $j = l - \frac{1}{2}$ and $j = l + \frac{1}{2}$ (where permitted).

For $j = l + \frac{1}{2}$, normalization yields

$$\boxed{\left|l, \tfrac{1}{2}; j, m\right\rangle = \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}\left|l, m-\tfrac{1}{2}; \tfrac{1}{2}, \tfrac{1}{2}\right\rangle + \sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}\left|l, m+\tfrac{1}{2}; \tfrac{1}{2}, -\tfrac{1}{2}\right\rangle.}$$

For $j = l - \frac{1}{2}$,

$$\boxed{\left|l, \tfrac{1}{2}; j, m\right\rangle = -\sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}\left|l, m-\tfrac{1}{2}; \tfrac{1}{2}, \tfrac{1}{2}\right\rangle + \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}\left|l, m+\tfrac{1}{2}; \tfrac{1}{2}, -\tfrac{1}{2}\right\rangle.}$$

Example 2: coupling two electron spins. Let

$$\hat{\boldsymbol{s}} = \hat{\boldsymbol{s}}_1 + \hat{\boldsymbol{s}}_2, \qquad s_1 = s_2 = \frac{1}{2}, \qquad m_1, m_2 = \pm\frac{1}{2}, \qquad s = 0, 1.$$

The uncoupled basis states are

$$\begin{aligned} &\left|\tfrac{1}{2}, \tfrac{1}{2}\right\rangle\left|\tfrac{1}{2}, \tfrac{1}{2}\right\rangle, \quad \left|\tfrac{1}{2}, \tfrac{1}{2}\right\rangle\left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle, \\ &\left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle\left|\tfrac{1}{2}, \tfrac{1}{2}\right\rangle, \quad \left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle\left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle. \end{aligned}$$

The coupled basis is $\left|\frac{1}{2}, \frac{1}{2}; s, m\right\rangle$ with $(s, m) = (1,1), (1,0), (1,-1), (0,0)$. As in Example 1, coupled states can be expressed in the uncoupled basis.

For $s = 1$, the *spin triplet* is

$$\begin{aligned} \left|\tfrac{1}{2}, \tfrac{1}{2}; 1, 1\right\rangle &= \left|\tfrac{1}{2}, \tfrac{1}{2}\right\rangle\left|\tfrac{1}{2}, \tfrac{1}{2}\right\rangle, \\ \left|\tfrac{1}{2}, \tfrac{1}{2}; 1, 0\right\rangle &= \frac{1}{\sqrt{2}}\left(\left|\tfrac{1}{2}, \tfrac{1}{2}\right\rangle\left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle + \left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle\left|\tfrac{1}{2}, \tfrac{1}{2}\right\rangle\right), \\ \left|\tfrac{1}{2}, \tfrac{1}{2}; 1, -1\right\rangle &= \left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle\left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle. \end{aligned}$$

For $s = 0$, the *spin singlet* is

$$\boxed{\left|\tfrac{1}{2}, \tfrac{1}{2}; 0, 0\right\rangle = \frac{1}{\sqrt{2}}\left(\left|\tfrac{1}{2}, \tfrac{1}{2}\right\rangle\left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle - \left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle\left|\tfrac{1}{2}, \tfrac{1}{2}\right\rangle\right).}$$

3.8 Spin Correlations and Bell's Inequality

1) Correlations in spin measurements

The spin singlet of two electrons may be expanded in the eigenstates of $\hat{s}_{1z}$ and $\hat{s}_{2z}$:

$$\left|\tfrac{1}{2}, \tfrac{1}{2}; 0, 0\right\rangle = \frac{1}{\sqrt{2}}\Big(\left|s_{1z}^{+}\right\rangle\left|s_{2z}^{-}\right\rangle - \left|s_{1z}^{-}\right\rangle\left|s_{2z}^{+}\right\rangle\Big).$$

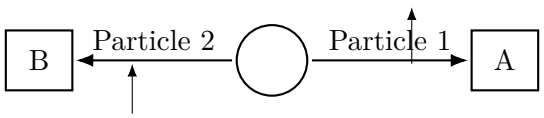
Because

$$|s_z^{\pm}\rangle = \frac{1}{\sqrt{2}}\Big(\left|s_x^{+}\right\rangle \pm |s_x^{-}\rangle\Big),$$

the spin singlet can also be expanded in the eigenstates of $\hat{s}_{1x}, \hat{s}_{2x}$:

$$\left|\tfrac{1}{2}, \tfrac{1}{2}; 0, 0\right\rangle = \frac{1}{\sqrt{2}}\Big(\left|s_{1x}^{-}\right\rangle\left|s_{2x}^{+}\right\rangle - \left|s_{1x}^{+}\right\rangle\left|s_{2x}^{-}\right\rangle\Big).$$

As illustrated below, let two electrons in the singlet state move from a central source to detectors A and B on opposite sides. Detector A first measures the spin of electron 1; immediately afterwards detector B measures the spin of electron 2.



If A measures s_z for electron 1 and B measures s_x for electron 2, then the probabilities for electron 2 to have s_x^+ or s_x^- are each 50%: there is no correlation between these measurement outcomes.

If A measures s_x for electron 1 and B also measures s_x for electron 2, the spin of electron 2 is determined by the result at A. When electron 1 is found to have s_x^+, electron 2 has s_x^-; when electron 1 has s_x^-, electron 2 has s_x^+. The two measurements are correlated.

If A does not measure electron 1, while B measures s_x for electron 2, then electron 2 has s_x^+ and s_x^- with equal probability, 50% each. No correlation with an A measurement is observed.

2) Einstein locality and Bell's inequality

The spin-measurement correlations discussed above suggest that what is measured for particle 2 at B is related to the physical quantity previously measured for particle 1 at A, despite a large separation between A and B.

Many physicists found this long-range correlation puzzling. Einstein and collaborators proposed the principle of locality: the behavior of system 2 should not depend on the state of distant system 1 (the EPR paradox). To resolve the puzzle, many proposed hidden-variable theories: quantum mechanics results from a theory containing additional parameters that are not observed (hidden variables). We will not discuss a particular hidden-variable theory, but consider how experiments can distinguish the implications of Einstein locality from quantum-mechanical predictions.

Consider two particles. Assume a particle is of a definite type such that measuring s_z gives s_z^+ and measuring s_x gives s_x^+, irrespective of the other distant particle's state, as required by the locality assumption in this model. Denote this type by (s_z^+, s_x^+). Of course s_z and s_x are still not measured simultaneously. To maintain the total spin singlet, its partner must have the opposite type (s_z^-, s_x^-). Continuing in this way, the possible pairs are

Particle 1	Particle 2
(s_z^+, s_x^+)	(s_z^-, s_x^-)
(s_z^+, s_x^-)	(s_z^-, s_x^+)
(s_z^-, s_x^+)	(s_z^+, s_x^-)
(s_z^-, s_x^-)	(s_z^+, s_x^+)

This reproduces the spin-measurement correlations described above. For example, measuring s_z for particle 1 at A and s_x for particle 2 at B allows both s_x^+ and s_x^-; measuring s_x for both particles gives opposite results. Thus two measurement directions do not reveal a difference between these predictions.

Continue to consider two particles, but now measure spin along three directions $\boldsymbol{e}_a, \boldsymbol{e}_b, \boldsymbol{e}_c$, i.e.

$$s_a = \boldsymbol{s} \cdot \boldsymbol{e}_a, \qquad s_b = \boldsymbol{s} \cdot \boldsymbol{e}_b, \qquad s_c = \boldsymbol{s} \cdot \boldsymbol{e}_c.$$

The directions need not be mutually perpendicular. A particle of type (s_a^+, s_b^+, s_c^+) means it yields s_a^+ if s_a is measured, s_b^+ if s_b is measured, and s_c^+ if s_c is measured (not all simultaneously). Requiring the two-particle system to be a spin singlet, the locality model gives the following

types for particles 1 and 2 measured at A and B:

Particle 1	Particle 2	Count
(s_a^+, s_b^+, s_c^+)	(s_a^-, s_b^-, s_c^-)	N_1
(s_a^+, s_b^+, s_c^-)	(s_a^-, s_b^-, s_c^+)	N_2
(s_a^+, s_b^-, s_c^+)	(s_a^-, s_b^+, s_c^-)	N_3
(s_a^+, s_b^-, s_c^-)	(s_a^-, s_b^+, s_c^+)	N_4
(s_a^-, s_b^+, s_c^+)	(s_a^+, s_b^-, s_c^-)	N_5
(s_a^-, s_b^+, s_c^-)	(s_a^+, s_b^-, s_c^+)	N_6
(s_a^-, s_b^-, s_c^+)	(s_a^+, s_b^+, s_c^-)	N_7
(s_a^-, s_b^-, s_c^-)	(s_a^+, s_b^+, s_c^+)	N_8

The probability that particle 1 at A is found in s_a^+ and particle 2 at B in s_b^+ is

$$P(s_a^+, s_b^+) = \frac{N_3 + N_4}{\sum_{i=1}^8 N_i}.$$

Similarly,

$$P(s_a^+, s_c^+) = \frac{N_2 + N_4}{\sum_{i=1}^8 N_i}, \qquad P(s_c^+, s_b^+) = \frac{N_3 + N_7}{\sum_{i=1}^8 N_i}.$$

Since

$$N_3 + N_4 \le (N_3 + N_7) + (N_2 + N_4),$$

we have

$$\boxed{P(s_a^+, s_b^+) \le P(s_a^+, s_c^+) + P(s_c^+, s_b^+).}$$

This is Bell's inequality in the form used here; it expresses a consequence of Einstein locality under the stated assignments.

3) Quantum mechanics and Bell's inequality

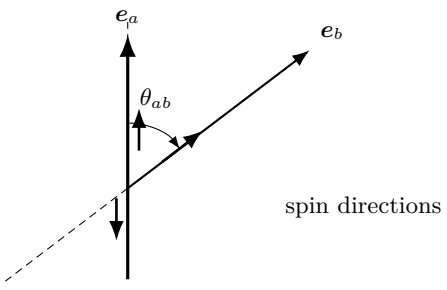
The preceding inequality is based on Einstein locality. Is it satisfied by quantum mechanics? We calculate the three probabilities in the inequality using quantum mechanics.

When particle 1 is measured along direction a at A and found in s_a^+, we wish to determine the probability that particle 2 is measured along b at B and found in s_b^+. For the spin singlet, particle 2 is in s_a^-. Using the rotation of spin states discussed previously, $|s_a^-\rangle$ can be expressed in the eigenbasis of $\hat{s}_b$. Let θ_{ab} be the angle between $\boldsymbol{e}_b$ and $\boldsymbol{e}_a$, with relative azimuth α_{ab}:

$$\left|s_a^-\right\rangle = \cos\frac{\pi - \theta_{ab}}{2} e^{-\mathrm{i}\alpha_{ab}/2} \left|s_b^+\right\rangle + \sin\frac{\pi - \theta_{ab}}{2} e^{\mathrm{i}\alpha_{ab}/2} \left|s_b^-\right\rangle.$$

Thus, conditional on s_a^- for particle 2, the probability of s_b^+ is

$$\cos^2\frac{\pi - \theta_{ab}}{2} = \sin^2\frac{\theta_{ab}}{2}.$$



Accounting for the probability 1/2 that particle 1 is in s_a^+,

$$\boxed{P(s_a^+, s_b^+) = \frac{1}{2}\sin^2\frac{\theta_{ab}}{2}.}$$

Likewise,

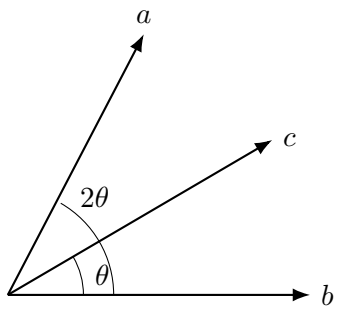
$$P(s_a^+, s_c^+) = \frac{1}{2}\sin^2\frac{\theta_{ac}}{2}, \qquad P(s_c^+, s_b^+) = \frac{1}{2}\sin^2\frac{\theta_{cb}}{2}.$$

Bell's inequality would therefore require, for any three directions $\boldsymbol{e}_a, \boldsymbol{e}_b, \boldsymbol{e}_c$,

$$\boxed{\sin^2\frac{\theta_{ab}}{2} \le \sin^2\frac{\theta_{ac}}{2} + \sin^2\frac{\theta_{cb}}{2}.}$$

However, this inequality is not always satisfied. Let the three directions lie in one plane and let

$$\theta_{ab} = 2\theta, \qquad \theta_{ac} = \theta_{cb} = \theta.$$



For $0 < \theta < \pi/2$, Bell's inequality reads

$$\sin^2\theta \le 2\sin^2\frac{\theta}{2}.$$

This fails, for example, at $\theta = \pi/4$: it would demand the impossible numerical relation $0.5 \le 0.292$. Thus the quantum-mechanical calculation and Bell's inequality are inconsistent.

4) Experimental results

Many precise experiments have shown violations of Bell's inequality, with results agreeing with quantum-mechanical predictions within experimental errors.

Nevertheless, the interpretation of quantum mechanics, especially measurement, remains a question that has not been completely resolved.

3.9 Quantum Teleportation: An Application of Entanglement and Measurement

The Bell correlations developed above and the general measurement formalism in this chapter can be combined into a communication protocol. *Quantum teleportation* transfers an unknown qubit state using one previously shared entangled pair and two classical bits. The protocol moves the quantum *state*, not the physical particle carrying it.

1) Bell states and an unknown input qubit

Use the computational basis $|0\rangle, |1\rangle$ and the Pauli operators

$$\sigma_x = |0\rangle\langle 1| + |1\rangle\langle 0|, \qquad \sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|.$$

The four Bell states are

$$|\Phi^{\pm}\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}},$$
$$|\Psi^{\pm}\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}.$$

They form an orthonormal basis of two qubits. One may obtain them from computational-basis states with a Hadamard gate on the first qubit and a controlled-NOT (CNOT) gate from the first qubit to the second: a Hadamard creates a superposition; the CNOT correlates the target with its control. More precisely,

$$\boxed{|\beta_{xy}\rangle = \mathrm{CNOT}(\hat{U}_{\mathrm{H}} \otimes \hat{I})\,|x, y\rangle = \frac{|0, y\rangle + (-1)^x\,|1, 1-y\rangle}{\sqrt{2}}, \quad x, y \in \{0, 1\}.}$$

This convention gives $|\beta_{00}\rangle = |\Phi^+\rangle$, $|\beta_{01}\rangle = |\Psi^+\rangle$, $|\beta_{10}\rangle = |\Phi^-\rangle$, and $|\beta_{11}\rangle = |\Psi^-\rangle$.

Let Alice hold an unknown normalized state

$$|\psi\rangle_1 = \alpha\,|0\rangle_1 + \beta\,|1\rangle_1, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Alice and Bob have previously shared the entangled state $|\Phi^+\rangle_{23}$, with qubit 2 at Alice's location and qubit 3 at Bob's location. Their initial three-qubit state is

$$|\Psi_0\rangle = |\psi\rangle_1 \otimes \left|\Phi^+\right\rangle_{23} = \frac{1}{\sqrt{2}}\big[\alpha(|000\rangle + |011\rangle) + \beta(|100\rangle + |111\rangle)\big].$$

The ordering of the three tensor factors is always $1, 2, 3$.

2) The two-gate derivation

Alice first applies a CNOT with qubit 1 as the control and qubit 2 as the target. The target flips only when the control is $|1\rangle$:

$$|\Psi_1\rangle = \frac{1}{\sqrt{2}}\big[\alpha(|000\rangle + |011\rangle) + \beta(|110\rangle + |101\rangle)\big].$$

Next Alice applies the Hadamard gate $\hat{U}_{\mathrm{H}}$ to qubit 1, with

$$\hat{U}_{\mathrm{H}}\,|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad \hat{U}_{\mathrm{H}}\,|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Grouping the resulting three-qubit amplitudes according to Alice's two-qubit state gives

$$\boxed{\begin{aligned}|\Psi_2\rangle = \frac{1}{2}\{\ & |00\rangle_{12}\,(\alpha\,|0\rangle + \beta\,|1\rangle)_3 \\ & + |01\rangle_{12}\,(\alpha\,|1\rangle + \beta\,|0\rangle)_3 \\ & + |10\rangle_{12}\,(\alpha\,|0\rangle - \beta\,|1\rangle)_3 \\ & + |11\rangle_{12}\,(\alpha\,|1\rangle - \beta\,|0\rangle)_3\}.\end{aligned}}$$

The expression contains the entire protocol. Alice measures both qubits in the computational basis. Each two-bit result occurs with probability $1/4$, independent of the unknown amplitudes α, β. Conditioned on this outcome, Bob's qubit is one of four states related to $|\psi\rangle$ by Pauli operations.

3) Classical communication and the correction

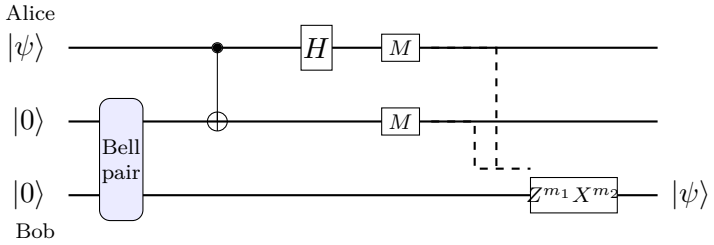
Write Alice's two measured bits as $m_1 m_2$. Bob applies the appropriate correction:

Alice's result	Bob's conditional state	Correction
00	$\|\psi\rangle$	$\hat{I}$
01	$\sigma_x \|\psi\rangle$	σ_x
10	$\sigma_z \|\psi\rangle$	σ_z
11	$\sigma_x\sigma_z \|\psi\rangle$	$\sigma_z\sigma_x$

The corrections are summarized, up to irrelevant global phases, by

$$\boxed{\hat{U}_{\text{corr}} = \sigma_z^{m_1}\sigma_x^{m_2}.}$$

Because the Pauli operators square to the identity, after Bob receives Alice's two classical bits his final state is $|\psi\rangle$. The prior entangled resource is consumed by the measurement.



In the circuit the pair shared before the protocol is represented by the initialization of qubits 2 and 3; the dashed lines indicate the two classical outcomes sent to Bob. The CNOT control is on qubit 1 and its target is qubit 2.

4) Causality and the no-cloning principle

Without Alice's classical message, Bob must average over her possible outcomes. His density operator is

$$\hat{\rho}_3 = \frac{1}{4}(\hat{\rho} + \sigma_x\hat{\rho}\sigma_x + \sigma_z\hat{\rho}\sigma_z + \sigma_x\sigma_z\hat{\rho}\sigma_z\sigma_x) = \frac{1}{2}\hat{I},$$

where $\hat{\rho} = |\psi\rangle\langle\psi|$. Bob's local state is independent of α, β before he receives the classical bits, so the protocol cannot transmit information faster than light.

Teleportation also respects the no-cloning theorem. Alice measures and thereby changes her original input system; after the full protocol the unknown state is available at Bob's site, with no additional intact copy left at Alice's site. One shared Bell pair plus two classical bits is therefore sufficient to transfer a single arbitrary qubit state, but the entanglement is a consumed resource and classical communication remains necessary.

Chapter 4

Approximation Methods

4.1 The Idea of Stationary Perturbation Theory

The Schrödinger equation is

$$\mathrm{i}\hbar\frac{\partial}{\partial t}\left|\psi,t\right\rangle = \hat{H}\left|\psi,t\right\rangle .$$

Only a very small number of physical problems can be solved exactly. For most actual problems, the construction of $\hat{H}$ is so complicated that an exact solution is impossible; approximate methods must be used. Different problems call for different methods of approximation.

When $\hat{H}$ does not depend on time, separation of variables gives a stationary state

$$\left|\psi_n,t\right\rangle = \left|n\right\rangle e^{-\mathrm{i}E_n t/\hbar},$$

where the energy E_n and the energy eigenstate $|n\rangle$ satisfy the stationary Schrödinger equation

$$\hat{H}\left|n\right\rangle = E_n\left|n\right\rangle .$$

Write

$$\hat{H} = \hat{H}^{(0)} + \hat{H}^{(1)}.$$

The following are required:

1. $\hat{H}^{(0)}$ contains the main part of $\hat{H}$, so $\hat{H}^{(1)}$ is small. Since both $\hat{H}^{(0)}$ and $\hat{H}^{(1)}$ are operators, their relative magnitudes can be understood from the classical correspondence principle. Strictly speaking, one compares the matrix elements of the two operators, as discussed below.

2. The stationary equation for $\hat{H}^{(0)}$,

$$\hat{H}^{(0)}\left|n\right\rangle^{(0)} = E_n^{(0)}\left|n\right\rangle^{(0)},$$

   can be solved exactly.

Expand E_n and $|n\rangle$:

$$E_n = E_n^{(0)} + E_n^{(1)} + E_n^{(2)} + \cdots , \qquad \left|n\right\rangle = \left|n\right\rangle^{(0)} + \left|n\right\rangle^{(1)} + \left|n\right\rangle^{(2)} + \cdots .$$

Insert these into the stationary Schrödinger equation:

$$\begin{aligned}(\hat{H}^{(0)} + \hat{H}^{(1)})(\left|n\right\rangle^{(0)} + \left|n\right\rangle^{(1)} + \left|n\right\rangle^{(2)} + \cdots) \\ = (E_n^{(0)} + E_n^{(1)} + E_n^{(2)} + \cdots)(\left|n\right\rangle^{(0)} + \left|n\right\rangle^{(1)} + \left|n\right\rangle^{(2)} + \cdots).\end{aligned}$$

The equation is then solved successively, order by order.

Zeroth-order approximation:

$$\hat{H}^{(0)}\,|n\rangle^{(0)} = E_n^{(0)}\,|n\rangle^{(0)} \quad \longrightarrow \quad E_n^{(0)},\ |n\rangle^{(0)}\,.$$

First-order approximation:

$$\hat{H}^{(0)}\,|n\rangle^{(1)} + \hat{H}^{(1)}\,|n\rangle^{(0)} = E_n^{(0)}\,|n\rangle^{(1)} + E_n^{(1)}\,|n\rangle^{(0)}\,,$$

or

$$(\hat{H}^{(0)} - E_n^{(0)})\,|n\rangle^{(1)} = -(\hat{H}^{(1)} - E_n^{(1)})\,|n\rangle^{(0)} \quad \longrightarrow \quad E_n^{(1)},\ |n\rangle^{(1)}\,.$$

Second-order approximation:

$$(\hat{H}^{(0)} - E_n^{(0)})\,|n\rangle^{(2)} = -(\hat{H}^{(1)} - E_n^{(1)})\,|n\rangle^{(1)} + E_n^{(2)}\,|n\rangle^{(0)} \quad \longrightarrow \quad E_n^{(2)},\ |n\rangle^{(2)}\,.$$

And so forth. In general, only the first nonzero correction is calculated.

Perturbation theory is based on the eigenvalues $E_n^{(0)}$ and eigenstates $|n\rangle^{(0)}$ of $\hat{H}^{(0)}$. In obtaining higher-order corrections, it is necessary to consider whether $E_n^{(0)}$ is degenerate. We first consider the nondegenerate case.

4.2 Nondegenerate Perturbation Theory

1) First-order correction

Expand the first-order equation in the $\hat{H}^{(0)}$ representation:

$$|n\rangle^{(1)} = \sum_i |i\rangle^{(0)}\,{}^{(0)}\Big\langle i\Big|n^{(1)}\Big\rangle\,,$$

$$\sum_i (\hat{H}^{(0)} - E_n^{(0)})\,|i\rangle^{(0)}\,{}^{(0)}\Big\langle i\Big|n^{(1)}\Big\rangle = -(\hat{H}^{(1)} - E_n^{(1)})\,|n\rangle^{(0)}\,.$$

Multiply on the left by ${}^{(0)}\langle m|$ to obtain

$$\begin{aligned}\sum_i (E_i^{(0)} - E_n^{(0)})\delta_{mi}\,{}^{(0)}\Big\langle i\Big|n^{(1)}\Big\rangle &= -{}^{(0)}\Big\langle m\Big|\hat{H}^{(1)}\Big|n^{(0)}\Big\rangle + E_n^{(1)}\delta_{mn} \\ &= -H_{mn}^{(1)} + E_n^{(1)}\delta_{mn}.\end{aligned}$$

Here $H_{mn}^{(1)} = {}^{(0)}\Big\langle m\Big|\hat{H}^{(1)}\Big|n^{(0)}\Big\rangle$ is the matrix element of $\hat{H}^{(1)}$ in the $\hat{H}^{(0)}$ representation. The orthonormality of the eigenstates of $\hat{H}^{(0)}$ has already been used. Thus

$$(E_m^{(0)} - E_n^{(0)})\,{}^{(0)}\Big\langle m\Big|n^{(1)}\Big\rangle = -H_{mn}^{(1)} + E_n^{(1)}\delta_{mn}.$$

For $m = n$,

$$E_n^{(1)} = H_{nn}^{(1)}.$$

For $m \neq n$,

$${}^{(0)}\Big\langle m\Big|n^{(1)}\Big\rangle = \frac{H_{mn}^{(1)}}{E_n^{(0)} - E_m^{(0)}}.$$

To obtain the full expansion

$$|n\rangle^{(1)} = \sum_m |m\rangle^{(0)}\,{}^{(0)}\Big\langle m\Big|n^{(1)}\Big\rangle\,,$$

we must also determine ${}^{(0)}\Big\langle n\Big|n^{(1)}\Big\rangle$.

Normalize to first order, neglecting terms of second and higher order:

$$\begin{aligned} 1 = \langle n|n\rangle &= ({}^{(0)}\langle n| + {}^{(1)}\langle n|)(|n\rangle^{(0)} + |n\rangle^{(1)}) \\ &\simeq \left\langle n^{(0)}\middle|n^{(0)}\right\rangle + \left\langle n^{(1)}\middle|n^{(0)}\right\rangle + \left\langle n^{(0)}\middle|n^{(1)}\right\rangle . \end{aligned}$$

Since $\left\langle n^{(0)}\middle|n^{(0)}\right\rangle = 1$,

$$\left\langle n^{(0)}\middle|n^{(1)}\right\rangle + \left\langle n^{(1)}\middle|n^{(0)}\right\rangle = 0,$$

$$\left\langle n^{(0)}\middle|n^{(1)}\right\rangle + (\left\langle n^{(0)}\middle|n^{(1)}\right\rangle)^* = 0.$$

The real part of $\left\langle n^{(0)}\middle|n^{(1)}\right\rangle$ must vanish; it has only an imaginary part. Write

$$\left\langle n^{(0)}\middle|n^{(1)}\right\rangle = \mathrm{i}\alpha.$$

Therefore

$$\begin{aligned} |n\rangle &= |n\rangle^{(0)} + \sum_m |m\rangle^{(0)} \left\langle m^{(0)}\middle|n^{(1)}\right\rangle \\ &= |n\rangle^{(0)} + \mathrm{i}\alpha\,|n\rangle^{(0)} + \sum_{m\neq n} \frac{H^{(1)}_{mn}}{E^{(0)}_n - E^{(0)}_m}\,|m\rangle^{(0)} \\ &\simeq e^{\mathrm{i}\alpha}\,|n\rangle^{(0)} + \sum_{m\neq n} \frac{H^{(1)}_{mn}}{E^{(0)}_n - E^{(0)}_m}\,|m\rangle^{(0)} \\ &\simeq e^{\mathrm{i}\alpha}\left(|n\rangle^{(0)} + \sum_{m\neq n} \frac{H^{(1)}_{mn}}{E^{(0)}_n - E^{(0)}_m}\,|m\rangle^{(0)}\right) \quad \text{(to first order)}. \end{aligned}$$

The contribution $\left\langle n^{(0)}\middle|n^{(1)}\right\rangle = \mathrm{i}\alpha$ to the first-order correction merely supplies a phase factor and does not affect the probability distribution. One may choose $\alpha = 0$.

The solution of the stationary Schrödinger equation to first order is thus

$$\boxed{\begin{aligned} &E_n = E^{(0)}_n + H^{(1)}_{nn}, \\ &|n\rangle = |n\rangle^{(0)} + \sum_{m\neq n} \frac{H^{(1)}_{mn}}{E^{(0)}_n - E^{(0)}_m}\,|m\rangle^{(0)}. \end{aligned}}$$

2) Second-order correction

Expand the second-order equation in the $\hat{H}^{(0)}$ representation:

$$|n\rangle^{(2)} = \sum_i |i\rangle^{(0)} \left\langle i^{(0)}\middle|n^{(2)}\right\rangle ,$$

$$\begin{aligned} &\sum_i (\hat{H}^{(0)} - E^{(0)}_n)\,|i\rangle^{(0)} \left\langle i^{(0)}\middle|n^{(2)}\right\rangle \\ &\quad = -\sum_{i\neq n} (\hat{H}^{(1)} - E^{(1)}_n)\,|i\rangle^{(0)} \left\langle i^{(0)}\middle|n^{(1)}\right\rangle + E^{(2)}_n\,|n\rangle^{(0)} . \end{aligned}$$

Multiplying by ${}^{(0)}\langle m|$ gives

$$\sum_i (E_i^{(0)} - E_n^{(0)})\delta_{mi} \left\langle i^{(0)} \middle| n^{(2)} \right\rangle = -\sum_{i\neq n} (H_{mi}^{(1)} - E_n^{(1)}\delta_{mi}) \left\langle i^{(0)} \middle| n^{(1)} \right\rangle + E_n^{(2)}\delta_{mn},$$

$$(E_m^{(0)} - E_n^{(0)}) \left\langle m^{(0)} \middle| n^{(2)} \right\rangle = -\sum_{i\neq n} H_{mi}^{(1)} \left\langle i^{(0)} \middle| n^{(1)} \right\rangle + E_n^{(1)} \left\langle m^{(0)} \middle| n^{(1)} \right\rangle + E_n^{(2)}\delta_{mn}.$$

For $m = n$, taking $\left\langle n^{(0)} \middle| n^{(1)} \right\rangle = \mathrm{i}\alpha = 0$,

$$E_n^{(2)} = \sum_{i\neq n} H_{ni}^{(1)} \left\langle i^{(0)} \middle| n^{(1)} \right\rangle = \sum_{i\neq n} \frac{H_{ni}^{(1)} H_{in}^{(1)}}{E_n^{(0)} - E_i^{(0)}}$$
$$= \sum_{i\neq n} \frac{|H_{in}^{(1)}|^2}{E_n^{(0)} - E_i^{(0)}}.$$

For $m \neq n$, the coefficient $\left\langle m^{(0)} \middle| n^{(2)} \right\rangle$ can also be found (see the textbook and calculate it yourself).

The approximate solution of the stationary Schrödinger equation through second order is

$$\boxed{\begin{aligned} E_n &= E_n^{(0)} + H_{nn}^{(1)} + \sum_{i\neq n} \frac{|H_{in}^{(1)}|^2}{E_n^{(0)} - E_i^{(0)}}, \\ |n\rangle &= |n\rangle^{(0)} + \sum_{i\neq n} \frac{H_{in}^{(1)}}{E_n^{(0)} - E_i^{(0)}} |i\rangle^{(0)} + |n\rangle^{(2)}. \end{aligned}}$$

3) Discussion of convergence

The condition for perturbation theory to be applicable is convergence, namely,

$$\left| \frac{H_{in}^{(1)}}{E_n^{(0)} - E_i^{(0)}} \right| \ll 1, \quad \text{i.e.,} \quad |H_{in}^{(1)}| \ll |E_n^{(0)} - E_i^{(0)}|.$$

This is the meaning of $\hat{H}^{(1)}$ being much smaller than $\hat{H}^{(0)}$: in the $H^{(0)}$ representation, its matrix elements are much smaller than the differences between the corresponding energy levels.

It follows that:

1. The above method applies only when $\hat{H}^{(0)}$ has a discrete spectrum. For a continuous spectrum, $|E_n^{(0)} - E_i^{(0)}| \to 0$ and the above inequality cannot hold.
2. It cannot be used when $\hat{H}^{(0)}$ has degeneracies. In that case, $|E_n^{(0)} - E_i^{(0)}| = 0$ and the inequality cannot hold either.

Example: A charged harmonic oscillator in an external electric field. Let the electric charge be q and the electric-field strength be $\mathcal{E}$. The Hamiltonian is

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{1}{2}m\omega^2 x^2 - q\mathcal{E}x.$$

For a weak electric field $\mathcal{E}$, take

$$\hat{H} = \hat{H}^{(0)} + \hat{H}^{(1)}, \quad \hat{H}^{(0)} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{1}{2}m\omega^2 x^2, \quad \hat{H}^{(1)} = -q\mathcal{E}x.$$

Zeroth-order approximation:

$$\hat{H}^{(0)}\,|n\rangle^{(0)} = E_n^{(0)}\,|n\rangle^{(0)}\,, \quad E_n^{(0)} = \left(n+\frac{1}{2}\right)\hbar\omega, \quad \left\langle x^{(0)}\middle|n^{(0)}\right\rangle = \psi_n^{(0)}(x).$$

The harmonic-oscillator eigenfunctions are Hermite-polynomial functions, with discrete, non-degenerate energies and separated variables.

First-order correction:

$$E_n^{(1)} = H_{nn}^{(1)} = -q\mathcal{E}x_{nn},$$

$$x_{nm} = {}^{(0)}\left\langle n\middle|x\middle|m^{(0)}\right\rangle = \sqrt{\frac{\hbar}{m\omega}}\left(\sqrt{\frac{n+1}{2}}\delta_{n+1,m} + \sqrt{\frac{n}{2}}\delta_{n-1,m}\right), \quad x_{nn}=0,\ E_n^{(1)}=0.$$

For the first-order change of the state,

$$\begin{aligned}
|n\rangle^{(1)} &= \sum_{m\neq n}\frac{H_{mn}^{(1)}}{E_n^{(0)}-E_m^{(0)}}\,|m\rangle^{(0)} = -q\mathcal{E}\sum_{m\neq n}\frac{x_{mn}}{E_n^{(0)}-E_m^{(0)}}\,|m\rangle^{(0)} \\
&= -q\mathcal{E}\sqrt{\frac{\hbar}{2m\omega}}\left(\frac{\sqrt{n+1}}{E_n^{(0)}-E_{n+1}^{(0)}}\,|n+1\rangle^{(0)} + \frac{\sqrt{n}}{E_n^{(0)}-E_{n-1}^{(0)}}\,|n-1\rangle^{(0)}\right) \\
&= \frac{q\mathcal{E}}{\sqrt{2m\hbar\omega^3}}\left(\sqrt{n+1}\,|n+1\rangle^{(0)} - \sqrt{n}\,|n-1\rangle^{(0)}\right).
\end{aligned}$$

In the coordinate representation,

$$\psi_n^{(1)}(x) = \left\langle x\middle|n^{(1)}\right\rangle = \frac{q\mathcal{E}}{\sqrt{2m\hbar\omega^3}}\left[\sqrt{n+1}\,\psi_{n+1}^{(0)}(x) - \sqrt{n}\,\psi_{n-1}^{(0)}(x)\right].$$

Second-order correction:

$$\begin{aligned}
E_n^{(2)} &= \sum_{m\neq n}\frac{|H_{mn}^{(1)}|^2}{E_n^{(0)}-E_m^{(0)}} \\
&= \frac{q^2\mathcal{E}^2\hbar}{m\omega}\sum_{m\neq n}\frac{\left|\sqrt{(n+1)/2}\,\delta_{m,n+1} + \sqrt{n/2}\,\delta_{m,n-1}\right|^2}{E_n^{(0)}-E_m^{(0)}} \\
&= -\frac{q^2\mathcal{E}^2}{2m\omega^2}.
\end{aligned}$$

Thus, keeping the first two nonzero corrections,

$$\boxed{\begin{aligned}
E_n &= \left(n+\frac{1}{2}\right)\hbar\omega - \frac{q^2\mathcal{E}^2}{2m\omega^2}, \\
\psi_n(x) &\simeq \psi_n^{(0)}(x) + \frac{q\mathcal{E}}{\sqrt{2m\hbar\omega^3}}\left(\sqrt{n+1}\,\psi_{n+1}^{(0)}(x) - \sqrt{n}\,\psi_{n-1}^{(0)}(x)\right).
\end{aligned}}$$

Can an exact solution be found? Start from

$$\left(-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{1}{2}m\omega^2x^2 - q\mathcal{E}x\right)\psi(x) = E\psi(x).$$

Complete the square:

$$\left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{1}{2}m\omega^2\left(x - \frac{q\mathcal{E}}{m\omega^2}\right)^2\right]\psi(x) = \left(E + \frac{q^2\mathcal{E}^2}{2m\omega^2}\right)\psi(x).$$

Put $\xi = x - q\mathcal{E}/(m\omega^2)$; then

$$\left(-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}\xi^2} + \frac{1}{2}m\omega^2\xi^2\right)\psi(\xi) = \left(E + \frac{q^2\mathcal{E}^2}{2m\omega^2}\right)\psi(\xi).$$

The solution is

$$\boxed{\begin{aligned} E_n &= \left(n + \frac{1}{2}\right)\hbar\omega - \frac{q^2\mathcal{E}^2}{2m\omega^2}, \\ \psi_n(\xi) &= \psi_n^{(0)}\left(x - \frac{q\mathcal{E}}{m\omega^2}\right), \end{aligned}}$$

where $\psi_n^{(0)}$ is a Hermite-polynomial function.

The energy computed through second order is already exact.

4.3 Degenerate Perturbation Theory

Suppose

$$\hat{H}^{(0)}\,|n,i\rangle^{(0)} = E_n^{(0)}\,|n,i\rangle^{(0)}\,, \qquad i = 1,2,\ldots,a.$$

The zeroth-order energy has a-fold degeneracy. Schmidt orthogonalization or the introduction of a complete set of mechanical quantities can guarantee mutual orthonormality:

$${}^{(0)}\langle n,i|n,j\rangle^{(0)} = \delta_{ij}.$$

Question: Which one of the states $|n,i\rangle^{(0)}$ should be chosen as the wavefunction for $E_n^{(0)}$?

Take the general form

$$|n\rangle^{(0)} = \sum_{i=1}^{a} c_i\,|n,i\rangle^{(0)}\,.$$

It is still an eigenstate of $\hat{H}^{(0)}$ belonging to $E_n^{(0)}$.

Question: How should the c_i be chosen?

Substitute $|n\rangle^{(0)}$ into the first-order approximation equation:

$$(\hat{H}^{(0)} - E_n^{(0)})\,|n\rangle^{(1)} = -(\hat{H}^{(1)} - E_n^{(1)})\,|n\rangle^{(0)}\,,$$

so that

$$(\hat{H}^{(0)} - E_n^{(0)})\,|n\rangle^{(1)} = -\sum_{i=1}^{a} c_i(\hat{H}^{(1)} - E_n^{(1)})\,|n,i\rangle^{(0)}\,.$$

Multiply on the left by ${}^{(0)}\langle n,j|$:

$$0 = \sum_{i=1}^{a} c_i(H_{ji}^{(1)} - E_n^{(1)}\delta_{ji}), \qquad H_{ji}^{(1)} = {}^{(0)}\left\langle n,j\middle|\hat{H}^{(1)}\middle|n,i\right\rangle^{(0)}\,.$$

This is a homogeneous system of equations for the coefficients c_i (a equations, $j = 1,\ldots,a$, determining a coefficients, $i = 1,\ldots,a$). For a nonzero solution, its determinant must vanish:

$$\begin{vmatrix} H_{11}^{(1)} - E_n^{(1)} & H_{12}^{(1)} & \cdots & H_{1a}^{(1)} \\ H_{21}^{(1)} & H_{22}^{(1)} - E_n^{(1)} & \cdots & H_{2a}^{(1)} \\ \vdots & \vdots & \ddots & \vdots \\ H_{a1}^{(1)} & H_{a2}^{(1)} & \cdots & H_{aa}^{(1)} - E_n^{(1)} \end{vmatrix} = 0.$$

The secular equation therefore determines a roots $E_{ni}^{(1)}$ of the first-order energy correction:

$$E_n^{(0)} \longrightarrow E_{ni} = E_n^{(0)} + E_{ni}^{(1)}, \qquad i = 1, \ldots, a.$$

If the a roots of the secular equation are distinct, the degeneracy is completely removed; if roots repeat, it is only partially removed.

Note. If $\hat{H}^{(1)}$ is diagonal in the degenerate subspace, $E_{ni}^{(1)} = H_{ii}^{(1)}$.

Substituting each $E_{ni}^{(1)}$ back into the homogeneous equations gives a set of coefficients $\{c_{i1}, c_{i2}, \ldots, c_{ia}\}$, and hence the zeroth-order wavefunction of $\hat{H}$ corresponding to E_{ni}:

$$E_{ni} = E_n^{(0)} + E_{ni}^{(1)}, \qquad |n_i\rangle^{(0)} = \sum_{j=1}^{a} c_{ij} \, |n, j\rangle^{(0)} .$$

Higher-order corrections can be found in the references.

Example 1: A doubly degenerate system. Let $E_n^{(0)}$ have basis states $|n, 1\rangle^{(0)}$ and $|n, 2\rangle^{(0)}$. The secular equation in this two-dimensional subspace is

$$\begin{vmatrix} H_{11}^{(1)} - E_n^{(1)} & H_{12}^{(1)} \\ H_{21}^{(1)} & H_{22}^{(1)} - E_n^{(1)} \end{vmatrix} = 0, \quad H_{ij}^{(1)} = {}^{(0)}\Big\langle n, i \Big| \hat{H}^{(1)} \Big| n, j \Big\rangle^{(0)}, \quad i, j = 1, 2.$$

Thus

$$(E_n^{(1)})^2 - (H_{11}^{(1)} + H_{22}^{(1)})E_n^{(1)} + H_{11}^{(1)} H_{22}^{(1)} - |H_{12}^{(1)}|^2 = 0,$$

$$E_n^{(1)} = \frac{1}{2}\left[H_{11}^{(1)} + H_{22}^{(1)} \pm \sqrt{(H_{11}^{(1)} - H_{22}^{(1)})^2 + 4|H_{12}^{(1)}|^2} \right].$$

If $H_{12}^{(1)} = 0$, then

$$E_{n1}^{(1)} = H_{11}^{(1)}, \quad E_{n2}^{(1)} = H_{22}^{(1)}, \quad E_{n1} = E_n^{(0)} + H_{11}^{(1)}, \quad E_{n2} = E_n^{(0)} + H_{22}^{(1)}.$$

Example 2: The hydrogen atom in an external electric field (Stark effect). Choose the z axis along the external electric field $\boldsymbol{\mathcal{E}}$:

$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{e^2}{r} + e\boldsymbol{\mathcal{E}} \cdot \boldsymbol{r} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{e^2}{r} + e\mathcal{E} r \cos\theta.$$

For a weak field $\mathcal{E}$,

$$\hat{H}^{(0)} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{e^2}{r}, \qquad \hat{H}^{(1)} = e\mathcal{E} r \cos\theta.$$

Zeroth-order approximation:

$$E_n^{(0)} = -\frac{\mu e^4}{2n^2\hbar^2}, \qquad \psi_{nlm}(\boldsymbol{r}), \quad l = 0, \ldots, n-1, \quad m = -l, \ldots, l,$$

with degeneracy

$$g = \sum_{l=0}^{n-1} (2l + 1) = n^2.$$

For $n = 2$, the zeroth-order energy $E_2^{(0)}$ is fourfold degenerate, with states

$$\phi_1 = \psi_{200}, \qquad \phi_2 = \psi_{210}, \qquad \phi_3 = \psi_{211}, \qquad \phi_4 = \psi_{21,-1}.$$

To find the first-order splitting, calculate the matrix elements

$$H_{ij}^{(1)} = \int \mathrm{d}^3 r \, \phi_i^*(\boldsymbol{r}) \, \hat{H}^{(1)} \phi_j(\boldsymbol{r}), \qquad i, j = 1, 2, 3, 4.$$

The matrix element is

$$H_{12}^{(1)} = H_{21}^{(1)} = -3e\mathcal{E}a_0,$$

with all other entries zero. The secular equation in the four-dimensional subspace is

$$\begin{vmatrix} -E_2^{(1)} & -3e\mathcal{E}a_0 & 0 & 0 \\ -3e\mathcal{E}a_0 & -E_2^{(1)} & 0 & 0 \\ 0 & 0 & -E_2^{(1)} & 0 \\ 0 & 0 & 0 & -E_2^{(1)} \end{vmatrix} = 0.$$

Hence

$$E_{21}^{(1)} = 3e\mathcal{E}a_0, \qquad E_{22}^{(1)} = -3e\mathcal{E}a_0, \qquad E_{23}^{(1)} = E_{24}^{(1)} = 0.$$

The original fourfold-degenerate level splits into one level shifted upward by $3e\mathcal{E}a_0$, a twofold-degenerate level with zero shift, and one level shifted downward by $3e\mathcal{E}a_0$. The degenerate perturbation is partially removed.

Example 3: Relativistic correction for the hydrogen atom. The nonrelativistic kinetic energy is

$$T = \frac{p^2}{2m}.$$

The relativistic kinetic energy is

$$\begin{aligned} T &= \sqrt{p^2c^2 + m^2c^4} - mc^2 = mc^2\sqrt{1 + \frac{p^2}{m^2c^2}} - mc^2 \\ &= \frac{p^2}{2m} - \frac{p^4}{8m^3c^2} + \cdots . \end{aligned}$$

Keeping only the p^4 correction gives the relativistic Hamiltonian

$$\hat{H} = \sqrt{\hat{p}^2c^2 + m^2c^4} - mc^2 - \frac{e^2}{r} \simeq \frac{\hat{p}^2}{2m} - \frac{e^2}{r} - \frac{\hat{p}^4}{8m^3c^2}.$$

Take

$$\hat{H}^{(0)} = \frac{\hat{p}^2}{2m} - \frac{e^2}{r}, \qquad \hat{H}^{(1)} = -\frac{\hat{p}^4}{8m^3c^2}.$$

The zeroth-order energy is $E_n^{(0)}$ and the n^2 eigenstates are $|nlm\rangle$, with wavefunctions

$$\langle \boldsymbol{r}|nlm\rangle = \psi_{nlm}(\boldsymbol{r}) = R_{nl}(r)Y_{lm}(\theta,\varphi).$$

In the n^2-dimensional subspace,

$$H_{nlm,n'l'm'}^{(1)} = -\frac{1}{8m^3c^2}\left\langle nlm\middle|\hat{p}^4\middle|n'l'm'\right\rangle.$$

The zeroth-order Schrödinger equation is

$$\left(\frac{\hat{p}^2}{2m} + V(r)\right)|nlm\rangle = E_n^{(0)}\,|nlm\rangle\,, \qquad \hat{p}^2\,|nlm\rangle = 2m(E_n^{(0)} - V(r))\,|nlm\rangle\,.$$

Using this relation, the matrix elements become

$$\begin{aligned} H_{nlm,n'l'm'}^{(1)} &= -\frac{1}{2mc^2}\left\langle nlm\middle|(E_n^{(0)} - V(r))^2\middle|n'l'm'\right\rangle \\ &= -\frac{1}{2mc^2}\left[(E_n^{(0)})^2\delta_{nn'}\delta_{ll'}\delta_{mm'} - 2E_n^{(0)}\,\langle nlm|V(r)|n'l'm'\rangle + \left\langle nlm\middle|V^2(r)\middle|n'l'm'\right\rangle\right]. \end{aligned}$$

Since $V(r)$ depends only on r and not on the angles,

$$\langle nlm|V(r)|n'l'm'\rangle = \int \mathrm{d}^3r\, R_{nl}(r)Y^*_{lm}(\theta,\varphi)\left(-\frac{e^2}{r}\right)R_{n'l'}(r)Y_{l'm'}(\theta,\varphi) = -\frac{e^2}{n^2a_0}\delta_{ll'}\delta_{mm'}$$

within a fixed principal-energy subspace. Similarly,

$$\left\langle nlm\middle|V^2(r)\middle|nl'm'\right\rangle = \frac{e^4}{(l+1/2)n^3a_0^2}\delta_{ll'}\delta_{mm'}.$$

Therefore the first-order matrix is diagonal:

$$H^{(1)}_{nlm,nl'm'} = H^{(1)}_{nl}\delta_{ll'}\delta_{mm'},$$

$$H^{(1)}_{nl} = -\frac{1}{2mc^2}\left[(E^{(0)}_n)^2 + \frac{2e^2}{n^2a_0}E^{(0)}_n + \frac{e^4}{(l+1/2)n^3a_0^2}\right].$$

This shows that the eigenstates of $\hat{H}^{(0)}$ are also eigenstates of $\hat{H}^{(1)}$ in this subspace, and the secular equation reduces to $E^{(1)}_{nl} = H^{(1)}_{nl}$:

$$E^{(0)}_n \quad (n^2\text{-fold degenerate}) \longrightarrow E_{nl} = E^{(0)}_n + H^{(1)}_{nl} \quad ((2l+1)\text{-fold degenerate}).$$

Example 4: The fine structure of the spectrum. Consider the spin–orbit coupling of the electron:

$$\hat{H} = \hat{H}^{(0)} + \hat{H}^{(1)}, \qquad \hat{H}^{(0)} = -\frac{\hbar^2}{2\mu}\nabla^2 + V(r),$$

$$\hat{H}^{(1)} = \frac{1}{2\mu^2 r}\frac{\mathrm{d}V(r)}{\mathrm{d}r}\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{S}} \equiv \xi(r)\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{S}}.$$

Zeroth-order states: the system is in an energy eigenstate $E^{(0)}_{nl}$. For a given l, the energy level has $2(2l+1)$ degeneracy. For this level the basis may be chosen as

$$\{|n,l,m,s_z\rangle\}, \qquad |n,l,j,m_j\rangle\,,$$

using either the $\{\hat{H}^{(0)}, \hat{\boldsymbol{L}}^2, \hat{L}_z, \hat{\boldsymbol{S}}^2, \hat{S}_z\}$ representation or the $\{\hat{H}^{(0)}, \hat{\boldsymbol{J}}^2, \hat{\boldsymbol{L}}^2, \hat{J}_z\}$ representation. The latter describes an eigenstate $|nljm_j\rangle$.

The zeroth-order energy is degenerate, so find the matrix elements of $\hat{H}^{(1)}$. Which representation should be used?

In the uncoupled representation, $[\hat{H}^{(1)}, \hat{L}_z] \neq 0$ and $[\hat{H}^{(1)}, \hat{S}_z] \neq 0$, so $|nlms_z\rangle$ is not an eigenstate of $\hat{H}^{(1)}$. Its matrix elements need not be diagonal and the secular equation is more complicated.

In the coupled representation, $[\hat{H}^{(1)}, \hat{\boldsymbol{J}}^2] = 0$ and $[\hat{H}^{(1)}, \hat{J}_z] = 0$. Thus $|nljm_j\rangle$ is an eigenstate of $\hat{H}^{(1)}$, whose matrix is diagonal. The first-order energy correction is

$$\begin{aligned}
E^{(1)}_{nlj} &= \left\langle nljm_j\middle|\hat{H}^{(1)}\middle|nljm_j\right\rangle \\
&= \left\langle nljm_j\middle|\xi(r)\hat{\boldsymbol{L}}\cdot\hat{\boldsymbol{S}}\middle|nljm_j\right\rangle \\
&= \left\langle nljm_j\middle|\xi(r)\frac{1}{2}(\hat{\boldsymbol{J}}^2 - \hat{\boldsymbol{L}}^2 - \hat{\boldsymbol{S}}^2)\middle|nljm_j\right\rangle \\
&= \frac{\hbar^2}{2}[j(j+1) - l(l+1) - s(s+1)]\int_0^\infty R^2_{nl}(r)\xi(r)r^2\mathrm{d}r.
\end{aligned}$$

The level changes as

$$E^{(0)}_{nl} \quad (2(2l+1)\text{-fold degenerate}) \longrightarrow E_{nlj} = E^{(0)}_{nl} + E^{(1)}_{nlj} \quad ((2j+1)\text{-fold degenerate}).$$

For example, when $n = 2$,

$$E_2^{(0)} \quad \text{(8-fold degenerate)} \longrightarrow \begin{cases} E_2^{(0)} + E_{2,1,3/2}^{(1)} & \text{(4-fold)}, \\ E_2^{(0)} + E_{2,1,1/2}^{(1)} & \text{(2-fold)}, \\ E_2^{(0)} + E_{2,0,1/2}^{(1)} & \text{(2-fold)}. \end{cases}$$

Note: when $l = 0$, $j = s$ and $E_{2,0,1/2}^{(1)} = 0$.

4.4 Variational Method

The variational method is used to solve the stationary equation

$$\hat{H} \left| n \right\rangle = E_n \left| n \right\rangle .$$

For an arbitrary state,

$$\left| \psi \right\rangle = \sum_n c_n \left| n \right\rangle ,$$

the energy expectation value is

$$\begin{aligned} \langle E \rangle = \left\langle \psi \middle| \hat{H} \middle| \psi \right\rangle &= \sum_{n,m} c_n^* c_m \left\langle n \middle| \hat{H} \middle| m \right\rangle \\ &= \sum_{n,m} c_n^* c_m E_n \delta_{nm} = \sum_n |c_n|^2 E_n \geq E_0. \end{aligned}$$

This is the basis of the variational method: take different trial functions $|\psi\rangle$, calculate $\langle E \rangle = \left\langle \psi \middle| \hat{H} \middle| \psi \right\rangle$, and choose the lowest $\langle E \rangle$ to approximate the ground-state energy E_0.

Method. From the physical properties of the system, guess a trial wavefunction $|\psi(\lambda)\rangle$ containing a parameter λ. Calculate

$$\langle E \rangle(\lambda) = \left\langle \psi(\lambda) \middle| \hat{H} \middle| \psi(\lambda) \right\rangle .$$

Solve

$$\frac{\mathrm{d}\langle E \rangle(\lambda)}{\mathrm{d}\lambda} = 0, \qquad \frac{\mathrm{d}^2\langle E \rangle(\lambda)}{\mathrm{d}\lambda^2} > 0 \quad \longrightarrow \quad \lambda_0,$$

which gives the minimum $\langle E \rangle(\lambda_0)$. The ground-state energy is approximately

$$E_0 \simeq \langle E \rangle(\lambda_0).$$

Example: The helium atom.

$$\begin{aligned} \hat{H} &= -\frac{\hbar^2}{2\mu}\nabla_1^2 - \frac{\hbar^2}{2\mu}\nabla_2^2 - \frac{2e^2}{r_1} - \frac{2e^2}{r_2} + \frac{e^2}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|} \\ &= \hat{H}^{(0)} + \frac{e^2}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|}. \end{aligned}$$

Here the atomic nucleus is approximately considered fixed. In the ground state of $\hat{H}^{(0)}$,

$$E_0^{(0)} = -\frac{4e^2}{a_0},$$

$$\psi_0(\boldsymbol{r}_1, \boldsymbol{r}_2) = \psi_{100}(\boldsymbol{r}_1)\psi_{100}(\boldsymbol{r}_2), \qquad \psi_{100}(\boldsymbol{r}) = \frac{1}{\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} e^{-Zr/a_0}, \qquad Z = 2.$$

Taking the interaction between the two electrons into account, the screening by one electron reduces the effective nuclear charge felt by the other to $Z < 2$. Treat Z as a parameter, and use $\psi_0(\boldsymbol{r}_1, \boldsymbol{r}_2, Z)$ as the trial wavefunction for $\hat{H}$. Since $\hat{H}$ itself is independent of Z,

$$\begin{aligned}\langle E\rangle(Z) &= \int \mathrm{d}^3 r_1 \mathrm{d}^3 r_2 \, \psi_0^*(\boldsymbol{r}_1, \boldsymbol{r}_2, Z) \hat{H} \psi_0(\boldsymbol{r}_1, \boldsymbol{r}_2, Z) \\ &= \left(2Z^2 - \frac{27}{4}Z\right) \frac{e^2}{2a_0}.\end{aligned}$$

From

$$\frac{\mathrm{d}\langle E\rangle(Z)}{\mathrm{d}Z} = 0, \qquad \frac{\mathrm{d}^2\langle E\rangle(Z)}{\mathrm{d}Z^2} > 0,$$

one obtains

$$Z_0 = \frac{27}{16} < 2.$$

The ground-state energy is

$$E_0 \simeq \langle E\rangle(Z_0) = -2.85\frac{e^2}{a_0},$$

with corresponding trial wavefunction $\psi_0(\boldsymbol{r}_1, \boldsymbol{r}_2, Z_0)$.

Perturbation theory can also be used to obtain the correction to the ground-state energy. Take

$$\hat{H}^{(1)} = \frac{e^2}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|}.$$

The ground state is nondegenerate; nondegenerate perturbation theory gives

$$E_0^{(1)} = \int \mathrm{d}^3 r_1 \mathrm{d}^3 r_2 \, \psi_0^*(\boldsymbol{r}_1, \boldsymbol{r}_2) \frac{e^2}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|} \psi_0(\boldsymbol{r}_1, \boldsymbol{r}_2) = \frac{5e^2}{4a_0}.$$

Therefore

$$E_0^{(0)} + E_0^{(1)} = -\frac{4e^2}{a_0} + \frac{5e^2}{4a_0} = -2.75\frac{e^2}{a_0}.$$

The experimental ground-state energy of helium is $-2.904e^2/a_0$. Thus the variational result is closer to the experimental value and is better.

4.5 Strong-Coupling Schrödinger Equation

In 1999, T. D. Lee and collaborators proposed a method for the strongly coupled Schrödinger equation. We give a brief introduction using the Yukawa potential as an example.

The Yukawa potential is

$$V(r) = -g^2 \frac{e^{-\alpha r}}{r}, \qquad g > 1 \quad \text{(strong coupling)}.$$

The energy eigenvalue equation is

$$\left(-\frac{\hbar^2}{2\mu}\nabla^2 + V(r)\right)\psi(\boldsymbol{r}) = E\psi(\boldsymbol{r}).$$

An exact analytic solution is unavailable. Set $\hbar = 1$ to simplify the calculation. The ground-state wavefunction does not depend on the angular coordinates. Let

$$\psi(r) = e^{-S(r)}.$$

Expand the eigenvalue E and the exponent function $S(r)$ in powers of $1/g^2$:

$$E = g^4 E_0 + g^2 E_1 + E_2 + \cdots, \qquad S = g^2 S_0 + S_1 + g^{-2} S_2 + \cdots.$$

The purpose of this expansion is to eliminate V from the equations for E_0 and S_0. Substitute it into the stationary Schrödinger equation, and compare the coefficients of equal powers of g:

$$\begin{aligned}
\text{Zeroth-order equation:} \quad & (\nabla S_0)^2 = -2mE_0, \\
\text{First-order equation:} \quad & \nabla S_0 \cdot \nabla S_1 = \frac{1}{2}\nabla^2 S_0 - m\left(\frac{e^{-\alpha r}}{r} + E_1\right), \\
\text{Second-order equation:} \quad & \nabla S_0 \cdot \nabla S_2 = -\frac{1}{2}(\nabla S_1)^2 + \frac{1}{2}\nabla^2 S_1 - mE_2, \\
& \vdots
\end{aligned}$$

The solution of the zeroth-order equation is

$$S_0(r) = \sqrt{-2mE_0}\, r.$$

The positive square root is chosen in view of the boundary condition that the wavefunction $\psi(r)$ remain finite at infinity. Substituting $S_0(r)$ into the first-order equation gives

$$\sqrt{-2mE_0}\,\frac{\mathrm{d}S_1}{\mathrm{d}r} = \frac{1}{r}(\sqrt{-2mE_0} - me^{-\alpha r}) - mE_1.$$

The wavefunction ψ and its first derivative must be continuous. Thus $S_0, S_1, S_2, \ldots$ and their first derivatives must be continuous, requiring $\mathrm{d}S_1/\mathrm{d}r$ to be nonsingular at $r = 0$ (since S_0 and its first derivative are continuous). Therefore

$$\lim_{r\to 0} \frac{\sqrt{-2mE_0} - me^{-\alpha r}}{r} \quad \text{must be finite.}$$

It follows that

$$E_0 = -\frac{m}{2}, \qquad S_0(r) = mr,$$

$$S_1(r) = \int_0^r \mathrm{d}r' \left[\frac{1 - e^{-\alpha r'}}{r'} - E_1\right].$$

Substitute into the second-order equation and require $\mathrm{d}S_2/\mathrm{d}r$ to be nonsingular at $r = 0$. This gives

$$E_1 = \alpha.$$

The successive solution gives the ground-state energy

$$E = -\frac{m}{2}g^4 + \alpha g^2 - \frac{3\alpha^2}{4m} + \frac{\alpha^2}{2m^2}g^{-2} + \cdots.$$

If $\alpha = 0$, the Yukawa potential reduces to the Coulomb potential. In this case,

$$\boxed{\begin{aligned} E &= -\frac{m}{2}g^4, \\ \psi(r) &= e^{-g^2 S_0(r)} = e^{-mg^2 r}. \end{aligned}}$$

This is exactly the same as the exact ground-state solution of the hydrogen-atom problem.

4.6 Time-Dependent Perturbation Theory

The Schrödinger equation is

$$\mathrm{i}\hbar\frac{\partial}{\partial t}\left|\psi(t)\right\rangle = \hat{H}(t)\left|\psi(t)\right\rangle.$$

The Hamiltonian can be written as

$$\hat{H}(t) = \hat{H}^{(0)} + \hat{H}^{(1)}(t), \qquad \hat{H}^{(0)}\left|m\right\rangle = E_m\left|m\right\rangle.$$

Question. When $t \le 0$, the Hamiltonian is $\hat{H} = \hat{H}^{(0)}$, independent of time, and the system may be in a stationary state. Assume that at $t = 0$ it is in the stationary state $|n\rangle$. For $t > 0$, $\hat{H}(t)$ depends on time; the system is no longer stably in the stationary state $|n\rangle$, but has certain probabilities of being in other stationary states. At time t the state is

$$\left|\psi(t)\right\rangle = \sum_m a_m(t)\left|m\right\rangle e^{-\mathrm{i}E_m t/\hbar}.$$

The probability of the stationary state $|m\rangle$ is $|a_m(t)|^2$. For a time-dependent perturbation, the quantity to be found is the probability that, after a macroscopic time ($t \to \infty$), the system has made a transition from the initial stationary state $|n\rangle$ to another stationary state $|m\rangle$:

$$\lim_{t\to\infty} W_{n\to m}(t) = \lim_{t\to\infty}|a_m(t)|^2.$$

1) Transition equations

Substitute the expansion of $|\psi(t)\rangle$ into the Schrödinger equation:

$$\begin{aligned}&\mathrm{i}\hbar\sum_m \dot{a}_m\left|m\right\rangle e^{-\mathrm{i}E_m t/\hbar} + \sum_m a_m E_m\left|m\right\rangle e^{-\mathrm{i}E_m t/\hbar}\\ &= \sum_m a_m\hat{H}^{(0)}\left|m\right\rangle e^{-\mathrm{i}E_m t/\hbar} + \sum_m a_m\hat{H}^{(1)}(t)\left|m\right\rangle e^{-\mathrm{i}E_m t/\hbar}.\end{aligned}$$

The second term on the left cancels the first term on the right, giving

$$\mathrm{i}\hbar\sum_m \dot{a}_m\left|m\right\rangle e^{-\mathrm{i}E_m t/\hbar} = \sum_m a_m\hat{H}^{(1)}(t)\left|m\right\rangle e^{-\mathrm{i}E_m t/\hbar}.$$

Multiply on the left by $\langle k|$:

$$\mathrm{i}\hbar\dot{a}_k e^{-\mathrm{i}E_k t/\hbar} = \sum_m a_m H^{(1)}_{km}(t)e^{-\mathrm{i}E_m t/\hbar}, \quad H^{(1)}_{km}(t) = \left\langle k\middle|\hat{H}^{(1)}(t)\middle|m\right\rangle.$$

Here $H^{(1)}_{km}(t)$ is the matrix element of $\hat{H}^{(1)}(t)$ in the $\hat{H}^{(0)}$ representation. Thus

$$\boxed{\begin{aligned}&\mathrm{i}\hbar\dot{a}_k = \sum_m a_m H^{(1)}_{km}(t)e^{\mathrm{i}\omega_{km}t},\\ &a_k(0) = \delta_{kn}, \qquad \omega_{km} = \frac{E_k - E_m}{\hbar}.\end{aligned}}$$

2) Perturbative solution

Write

$$a_k(t) = a_k^{(0)}(t) + a_k^{(1)}(t) + \cdots.$$

The transition equation becomes

$$\mathrm{i}\hbar(\dot{a}_k^{(0)} + \dot{a}_k^{(1)} + \cdots) = \sum_m (a_m^{(0)} + a_m^{(1)} + \cdots)H^{(1)}_{km}(t)e^{\mathrm{i}\omega_{km}t}.$$

The zeroth-order equation is

$$\mathrm{i}\hbar\dot{a}_k^{(0)}(t) = 0, \qquad a_k^{(0)}(0) = \delta_{kn},$$

with solution

$$a_k^{(0)}(t) = \delta_{kn}.$$

The first-order equation is

$$\mathrm{i}\hbar\dot{a}_k^{(1)}(t) = \sum_m a_m^{(0)}(t) H_{km}^{(1)}(t) e^{\mathrm{i}\omega_{km}t} = H_{kn}^{(1)}(t) e^{\mathrm{i}\omega_{kn}t}.$$

Its solution is

$$\boxed{a_k^{(1)}(t) = \frac{1}{\mathrm{i}\hbar}\int_0^t H_{kn}^{(1)}(t') e^{\mathrm{i}\omega_{kn}t'}\,\mathrm{d}t'.}$$

For a transition from $|n\rangle$ to $|k\rangle$ $(k \neq n)$, $a_k(t) \simeq a_k^{(1)}(t)$, so the transition probability is

$$W_{n\to k}(t) \simeq |a_k^{(1)}(t)|^2.$$

3) Periodic perturbation

Take

$$\hat{H}^{(1)}(t) = \hat{A}\cos\omega t = \hat{F}(e^{\mathrm{i}\omega t} + e^{-\mathrm{i}\omega t}),$$

where $\hat{A}$, $\hat{F}$ are independent of t. The first-order transition amplitude is

$$\begin{aligned} a_m^{(1)}(t) &= \frac{1}{\mathrm{i}\hbar}\int_0^t H_{mn}^{(1)}(t') e^{\mathrm{i}\omega_{mn}t'}\,\mathrm{d}t' \\ &= \frac{F_{mn}}{\mathrm{i}\hbar}\int_0^t \left[e^{\mathrm{i}(\omega_{mn}+\omega)t'} + e^{\mathrm{i}(\omega_{mn}-\omega)t'}\right]\mathrm{d}t', \end{aligned}$$

where ω_{mn} is the transition frequency and ω is the driving frequency. Integrating gives

$$a_m^{(1)}(t) = -\frac{F_{mn}}{\hbar}\left[\frac{e^{\mathrm{i}(\omega_{mn}+\omega)t} - 1}{\omega_{mn}+\omega} + \frac{e^{\mathrm{i}(\omega_{mn}-\omega)t} - 1}{\omega_{mn}-\omega}\right].$$

When $\omega \neq \pm\omega_{mn}$, $a_m^{(1)}(t)$ oscillates with t and does not increase monotonically with time. When $\omega \to \pm\omega_{mn}$, the nearly resonant term dominates:

$$a_m^{(1)}(t) \simeq -\frac{F_{mn}}{\hbar}\frac{e^{\mathrm{i}(\omega_{mn}\mp\omega)t} - 1}{\omega_{mn}\mp\omega}.$$

The transition probability is

$$W_{n\to m}(t) = |a_m^{(1)}(t)|^2 \simeq \frac{4|F_{mn}|^2}{\hbar^2}\frac{\sin^2[(\omega_{mn}\mp\omega)t/2]}{(\omega_{mn}\mp\omega)^2}.$$

Since

$$\lim_{t\to\infty}\frac{\sin^2(xt)}{\pi t x^2} = \delta(x),$$

it follows that

$$\lim_{t\to\infty} W_{n\to m}(t) \simeq \frac{2\pi t|F_{mn}|^2}{\hbar^2}\delta(\omega_{mn}\mp\omega) = \frac{2\pi t|F_{mn}|^2}{\hbar}\delta(E_m - E_n \mp \hbar\omega).$$

It grows linearly with t. The δ function expresses energy conservation, illustrated schematically below:



The transition rate is

$$\frac{\mathrm{d}W_{n\to m}(t)}{\mathrm{d}t} = \frac{2\pi}{\hbar^2}|F_{mn}|^2\delta(\omega_{mn} \mp \omega).$$

A real perturbation is not a strictly monochromatic field but has a frequency range $\Delta\omega$. The total transition rate is

$$\begin{aligned}\frac{\mathrm{d}W_{n\to m}(t)}{\mathrm{d}t} &= \frac{2\pi}{\hbar^2}|F_{mn}|^2\int_{\Delta\omega}\delta(\omega_{mn}\mp\omega)\mathrm{d}\omega \\ &= \begin{cases}\dfrac{2\pi}{\hbar^2}|F_{mn}|^2, & \Delta\omega \text{ contains } |\omega_{mn}|, \\ 0, & \Delta\omega \text{ does not contain } |\omega_{mn}|.\end{cases}\end{aligned}$$

Example. At $t = 0$ a hydrogen atom is in its ground state $|100\rangle$. A perturbation

$$\hat{H}^{(1)}(t) = -e\mathcal{E}_0 z\,\delta(t)$$

is applied. Find the probability for transitions to all excited states.

Solution. The first-order amplitude is

$$\begin{aligned}a_m^{(1)}(t) &= \frac{1}{\mathrm{i}\hbar}\int_0^t H_{mn}^{(1)}(t')e^{\mathrm{i}\omega_{mn}t'}\mathrm{d}t' \\ &= -\frac{e\mathcal{E}_0 z_{mn}}{\mathrm{i}\hbar}\int_0^t e^{\mathrm{i}\omega_{mn}t'}\delta(t')\mathrm{d}t' \\ &= -\frac{e\mathcal{E}_0 z_{mn}}{\mathrm{i}\hbar}.\end{aligned}$$

Here $|n\rangle = |100\rangle$ is the ground state and $|m\rangle = |nlm\rangle$ $(n \neq 1)$ are excited states. The probability of a transition from the ground state to any excited state is

$$W = \sum_{nlm}{}' \left(\frac{e\mathcal{E}_0}{\hbar}\right)^2 |\,\langle nlm|z|100\rangle\,|^2,$$

where the prime means that the ground state is excluded from the sum. Since

$$z = r\cos\theta,$$

the corresponding matrix elements are

$$\langle nlm|z|n'l'm'\rangle = \langle nl|r|n'l'\rangle\,\langle lm|\cos\theta|l'm'\rangle\,.$$

The action of $\cos\theta$ is

$$\cos\theta\,|lm\rangle = \sqrt{\frac{(l+1)^2-m^2}{(2l+1)(2l+3)}}\,|l+1,m\rangle + \sqrt{\frac{l^2-m^2}{(2l-1)(2l+1)}}\,|l-1,m\rangle\,.$$

Therefore, the conditions for $\langle lm|\cos\theta|l'm'\rangle$ to be nonzero (the selection rules for transitions) are

$$\Delta l = l - l' = \pm 1.$$

Thus $\langle 100|z|100\rangle = 0$, and

$$\begin{aligned} W &= \sum_{nlm} \left(\frac{e\mathcal{E}_0}{\hbar}\right)^2 |\langle nlm|z|100\rangle|^2 \\ &= \left(\frac{e\mathcal{E}_0}{\hbar}\right)^2 \sum_{nlm} \langle 100|z|nlm\rangle \langle nlm|z|100\rangle \\ &= \left(\frac{e\mathcal{E}_0}{\hbar}\right)^2 \left\langle 100\middle|z^2\middle|100\right\rangle . \end{aligned}$$

Since

$$\left\langle 100\middle|z^2\middle|100\right\rangle = a_0^2,$$

where a_0 is the Bohr radius, the result is

$$\boxed{W = \left(\frac{e\mathcal{E}_0 a_0}{\hbar}\right)^2}.$$

The probability for the hydrogen atom to remain in the ground state is therefore

$$\boxed{1 - W = 1 - \left(\frac{e\mathcal{E}_0 a_0}{\hbar}\right)^2}.$$

4.7 Interaction Picture and Perturbation Theory

1) Interaction picture

First review the Schrödinger picture and the Heisenberg picture.

Schrödinger picture: The state depends on time t:

$$|\alpha, t\rangle_{\rm S} = \hat{U}(t)\,|\alpha, 0\rangle_{\rm S}\,, \qquad \hat{U}(t) = e^{-{\rm i}\hat{H}t/\hbar}.$$

A mechanical quantity does not depend explicitly on time: $\hat{A}_{\rm S}$.

The dynamical equation is

$${\rm i}\hbar\frac{\partial}{\partial t}\,|\alpha, t\rangle_{\rm S} = \hat{H}\,|\alpha, t\rangle_{\rm S}\,.$$

Heisenberg picture: The state does not depend on time: $|\alpha\rangle_{\rm H}$.

The mechanical quantities depend on time:

$$\hat{A}_{\rm H}(t) = \hat{U}^\dagger(t)\hat{A}_{\rm H}(0)\hat{U}(t).$$

The dynamical equation is

$$\frac{{\rm d}\hat{A}_{\rm H}}{{\rm d}t} = \frac{1}{{\rm i}\hbar}[\hat{A}_{\rm H}, \hat{H}].$$

The Hamiltonian $\hat{H}$ is the same in both pictures.

Now introduce the interaction picture. In the Schrödinger picture, separate the Hamiltonian into two parts:

$$\hat{H} = \hat{H}^{(0)} + \hat{H}^{(1)}.$$

Define the relation between states in the interaction and Schrödinger pictures by

$$|\alpha, t\rangle_{\rm I} = e^{{\rm i}\hat{H}^{(0)}t/\hbar}\,|\alpha, t\rangle_{\rm S}\,.$$

The expectation value of a mechanical quantity $\hat{O}$ is independent of the picture:

$$\begin{aligned}{}_{\mathrm{I}}\Big\langle \alpha,t\Big|\hat{O}_{\mathrm{I}}\Big|\alpha,t\Big\rangle_{\mathrm{I}} &= {}_{\mathrm{S}}\Big\langle \alpha,t\Big|\hat{O}_{\mathrm{S}}\Big|\alpha,t\Big\rangle_{\mathrm{S}} \\ &= {}_{\mathrm{I}}\langle \alpha,t|\, e^{\mathrm{i}\hat{H}^{(0)}t/\hbar}\hat{O}_{\mathrm{S}}e^{-\mathrm{i}\hat{H}^{(0)}t/\hbar}\,|\alpha,t\rangle_{\mathrm{I}}\,.\end{aligned}$$

Hence mechanical quantities are related by

$$\boxed{\hat{O}_{\mathrm{I}} = e^{\mathrm{i}\hat{H}^{(0)}t/\hbar}\hat{O}_{\mathrm{S}}e^{-\mathrm{i}\hat{H}^{(0)}t/\hbar}.}$$

Note: $\hat{H}^{(0)}$ is the same in the Schrödinger and interaction pictures, but $\hat{H}^{(1)}$ is not.

In the interaction picture, both states and mechanical quantities depend on time. The equation of motion for states is

$$\begin{aligned}\mathrm{i}\hbar\frac{\partial}{\partial t}\,|\alpha,t\rangle_{\mathrm{I}} &= -\hat{H}^{(0)}e^{\mathrm{i}\hat{H}^{(0)}t/\hbar}\,|\alpha,t\rangle_{\mathrm{S}} + e^{\mathrm{i}\hat{H}^{(0)}t/\hbar}\mathrm{i}\hbar\frac{\partial}{\partial t}\,|\alpha,t\rangle_{\mathrm{S}} \\ &= -e^{\mathrm{i}\hat{H}^{(0)}t/\hbar}\hat{H}^{(0)}\,|\alpha,t\rangle_{\mathrm{S}} + e^{\mathrm{i}\hat{H}^{(0)}t/\hbar}\hat{H}\,|\alpha,t\rangle_{\mathrm{S}} \\ &= e^{\mathrm{i}\hat{H}^{(0)}t/\hbar}\hat{H}^{(1)}_{\mathrm{S}}\,|\alpha,t\rangle_{\mathrm{S}} \\ &= e^{\mathrm{i}\hat{H}^{(0)}t/\hbar}\hat{H}^{(1)}_{\mathrm{S}}e^{-\mathrm{i}\hat{H}^{(0)}t/\hbar}e^{\mathrm{i}\hat{H}^{(0)}t/\hbar}\,|\alpha,t\rangle_{\mathrm{S}} \\ &= \hat{H}^{(1)}_{\mathrm{I}}\,|\alpha,t\rangle_{\mathrm{I}}\,.\end{aligned}$$

The equation of motion for an operator in the interaction picture is

$$\begin{aligned}\mathrm{i}\hbar\frac{\partial}{\partial t}\hat{O}_{\mathrm{I}} &= -\hat{H}^{(0)}\hat{O}_{\mathrm{I}} + \hat{O}_{\mathrm{I}}\hat{H}^{(0)} \\ &= [\hat{O}_{\mathrm{I}},\hat{H}^{(0)}].\end{aligned}$$

Although both states and mechanical quantities depend on time in the interaction picture (only states depend on time in the Schrödinger picture, and only mechanical quantities depend on time in the Heisenberg picture), the time evolution of the states depends only on $\hat{H}^{(1)}$, while the time evolution of the mechanical quantities depends only on $\hat{H}^{(0)}$. This is very useful in perturbative expansions.

2) Perturbative expansion

The time evolution of a state in the interaction picture is

$$|\alpha,t\rangle_{\mathrm{I}} = \hat{U}(t,t_0)\,|\alpha,t_0\rangle_{\mathrm{I}}\,.$$

The equation of motion for the interaction-picture state gives

$$\mathrm{i}\hbar\frac{\partial}{\partial t}\hat{U}(t,t_0) = \hat{H}^{(1)}_{\mathrm{I}}(t)\hat{U}(t,t_0),$$

subject to

$$\hat{U}(t_0,t_0) = \mathbf{1}.$$

Its integral solution is

$$\hat{U}(t,t_0) = \mathbf{1} - \frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\mathrm{d}t'\,\hat{H}^{(1)}_{\mathrm{I}}(t')\hat{U}(t',t_0).$$

By iteration,

$$\begin{aligned}\hat{U}(t,t_0) = \mathbf{1} &- \frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\mathrm{d}t'\,\hat{H}^{(1)}_{\mathrm{I}}(t') \\ &+ \left(-\frac{\mathrm{i}}{\hbar}\right)^2\int_{t_0}^{t}\mathrm{d}t'\int_{t_0}^{t'}\mathrm{d}t''\,\hat{H}^{(1)}_{\mathrm{I}}(t')\hat{H}^{(1)}_{\mathrm{I}}(t'') + \cdots\,.\end{aligned}$$

For the third term on the right,

$$\begin{aligned}
&\int_{t_0}^{t} \mathrm{d}t' \int_{t_0}^{t'} \mathrm{d}t''\, \hat{H}_{\mathrm{I}}^{(1)}(t')\hat{H}_{\mathrm{I}}^{(1)}(t'') \\
&= \frac{1}{2!}\left[\int_{t_0}^{t} \mathrm{d}t' \int_{t_0}^{t'} \mathrm{d}t''\, \hat{H}_{\mathrm{I}}^{(1)}(t')\hat{H}_{\mathrm{I}}^{(1)}(t'') + \int_{t_0}^{t} \mathrm{d}t'' \int_{t_0}^{t''} \mathrm{d}t'\, \hat{H}_{\mathrm{I}}^{(1)}(t'')\hat{H}_{\mathrm{I}}^{(1)}(t')\right] \\
&= \frac{1}{2!}\int_{t_0}^{t} \mathrm{d}t' \int_{t_0}^{t} \mathrm{d}t''\, \mathcal{T}\left(\hat{H}_{\mathrm{I}}^{(1)}(t')\hat{H}_{\mathrm{I}}^{(1)}(t'')\right).
\end{aligned}$$

The second term on the right of the first equality is obtained by interchanging the integration variables. The time-ordered product $\mathcal{T}$ is defined by

$$\begin{aligned}
&\mathcal{T}\left(\hat{H}_{\mathrm{I}}^{(1)}(t')\hat{H}_{\mathrm{I}}^{(1)}(t'')\right) \\
&\quad = \hat{H}_{\mathrm{I}}^{(1)}(t')\hat{H}_{\mathrm{I}}^{(1)}(t'')\theta(t'-t'') + \hat{H}_{\mathrm{I}}^{(1)}(t'')\hat{H}_{\mathrm{I}}^{(1)}(t')\theta(t''-t').
\end{aligned}$$

The operator corresponding to the later time always stands to the left.

Thus the solution for the time-evolution operator can be written as

$$\boxed{\begin{aligned}
\hat{U}(t,t_0) = \sum_{n=0}^{\infty} \frac{1}{n!}\left(-\frac{\mathrm{i}}{\hbar}\right)^n \int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t} \mathrm{d}t_2 \cdots \int_{t_0}^{t} \mathrm{d}t_n \\
\times \mathcal{T}\left(\hat{H}_{\mathrm{I}}^{(1)}(t_1)\hat{H}_{\mathrm{I}}^{(1)}(t_2)\cdots\hat{H}_{\mathrm{I}}^{(1)}(t_n)\right).
\end{aligned}}$$

This is the perturbative expansion of the evolution operator for states in the interaction picture, and it is the foundation of perturbation theory in quantum mechanics and quantum field theory.

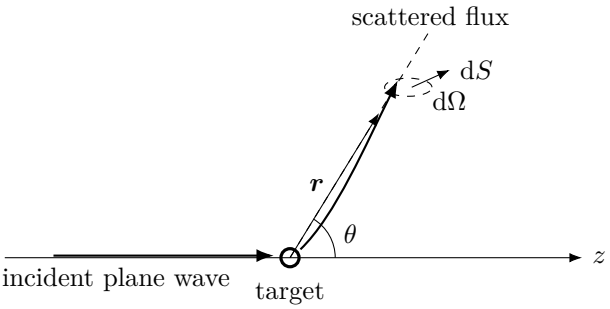
Chapter 5

Scattering Theory

5.1 General Description

The perturbation method discussed above is mainly suitable for discrete spectra, that is, for the case of separated energy levels. How should one solve continuous-spectrum problems? A typical physical problem involving a continuous spectrum is scattering.

1) Differential scattering cross section



An incident plane wave is changed into a spherical wave after scattering. The problem is to find the probability per unit time that an incident particle is scattered into a unit solid angle in the direction Ω, denoted by $\sigma(\theta, \varphi)$. The quantity $\sigma(\theta, \varphi)$ depends on the nature of the target and of the interaction. By measuring $\sigma(\theta, \varphi)$, we may learn about the internal structure of the target or discover new particles.

Scattering is divided into elastic and inelastic scattering:

- **Elastic scattering:** The energy of the particle does not change before and after scattering; only its direction of motion changes.
- **Inelastic scattering:** After scattering, the properties of the particle change; new particles may even be created.

We consider only elastic scattering, in the rest frame of the target particle. Let the incident plane wave be

$$\psi_1(\boldsymbol{r}) = Ae^{\mathrm{i}kz}.$$

The probability-current density of the incident particle (probability crossing unit area per unit time) is

$$J_z = \frac{\mathrm{i}\hbar}{2\mu}\left(\psi_1\frac{\partial\psi_1^*}{\partial z} - \psi_1^*\frac{\partial\psi_1}{\partial z}\right) = \frac{\hbar k}{\mu}|A|^2.$$

The probability per unit time that these incident particles are scattered into the direction $\mathrm{d}\Omega$ is

$$\mathrm{d}N = J_z\sigma(\theta, \varphi)\mathrm{d}\Omega.$$

Analyze the dimension of σ: since $[\mathrm{d}N] = T^{-1}$, $[J_z] = L^{-2}T^{-1}$ and $[\mathrm{d}\Omega] = 1$, it follows that

$$[\sigma] = L^2.$$

It has the dimension of area and therefore $\sigma(\theta, \varphi)$ is called the *differential scattering cross section*.

2) General method for calculating $\sigma(\theta, \varphi)$

As $r \to \infty$, $V(\boldsymbol{r}) \to 0$, and $\psi(\boldsymbol{r})$ consists of two parts: the incident plane wave propagating in the z direction without being affected by the interaction, and an outgoing scattered spherical wave propagating in the $\boldsymbol{r}$ direction,

$$\psi(\boldsymbol{r}) \xrightarrow{r\to\infty} \psi_1(\boldsymbol{r}) + \psi_2(\boldsymbol{r}) = Ae^{\mathrm{i}kz} + Af(\theta, \varphi)\frac{e^{\mathrm{i}kr}}{r}.$$

Note: (1) The spherical wave depends on θ and φ; $f(\theta, \varphi)$ can be chosen so that ψ_2 and ψ_1 have the same constant A. The specific form of $f(\theta, \varphi)$ is determined by the interaction. (2) For elastic scattering, the energy and the magnitude of k remain unchanged, while its direction changes.

The probability-current density of the scattered particles is

$$J_r = \frac{\mathrm{i}\hbar}{2\mu}\left(\psi_2\frac{\partial\psi_2^*}{\partial r} - \psi_2^*\frac{\partial\psi_2}{\partial r}\right) = \frac{\hbar k}{\mu}|A|^2\frac{|f(\theta,\varphi)|^2}{r^2} = J_z\frac{|f(\theta,\varphi)|^2}{r^2}.$$

The probability per unit time scattered into $\mathrm{d}\Omega$ is

$$\mathrm{d}N = J_r\mathrm{d}S = J_r r^2\mathrm{d}\Omega = J_z|f(\theta,\varphi)|^2\mathrm{d}\Omega.$$

Comparing with $\mathrm{d}N = J_z\sigma(\theta,\varphi)\mathrm{d}\Omega$, we obtain

$$\boxed{\sigma(\theta,\varphi) = |f(\theta,\varphi)|^2}.$$

Here $f(\theta, \varphi)$ is called the *scattering amplitude*.

Conclusion: The general method for calculating the differential cross section $\sigma(\theta, \varphi)$ is:

1. Solve the specific Schrödinger equation to obtain $\psi(\boldsymbol{r})$.
2. Compare its asymptotic form as $r \to \infty$ with the standard asymptotic solution

$$\psi(\boldsymbol{r}) \xrightarrow{r\to\infty} Ae^{\mathrm{i}kz} + Af(\theta,\varphi)\frac{e^{\mathrm{i}kr}}{r}$$

   to find $f(\theta, \varphi)$.
3. Obtain $\sigma(\theta, \varphi) = |f(\theta, \varphi)|^2$.

5.2 Green-Function Method

1) Integral equation

The stationary Schrödinger equation is

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + V(\boldsymbol{r})\right)\psi(\boldsymbol{r}) = E\psi(\boldsymbol{r}).$$

In elastic scattering the energy is conserved. E is the known incident energy, and only $\psi(\boldsymbol{r})$ needs to be determined. (In a bound-state problem, the bound-state condition is used to determine the allowed values of E.) Set

$$k^2 = \frac{2mE}{\hbar^2}, \qquad U(\boldsymbol{r}) = \frac{2m}{\hbar^2}V(\boldsymbol{r}).$$

Then

$$(\nabla^2 + k^2)\psi(\boldsymbol{r}) = U(\boldsymbol{r})\psi(\boldsymbol{r}),$$

which has a continuous source (potential) $U(\boldsymbol{r})$.

The idea of the Green-function method is to solve first the stationary equation corresponding to a point source $\delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}')$,

$$(\nabla_{\boldsymbol{r}}^2 + k^2)G(\boldsymbol{r}, \boldsymbol{r}') = \delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}'),$$

thus obtaining the Green function $G(\boldsymbol{r}, \boldsymbol{r}')$, and then solving the stationary equation

$$\begin{aligned}(\nabla^2 + k^2)\psi(\boldsymbol{r}) &= U(\boldsymbol{r})\psi(\boldsymbol{r}) \\ &= \int \mathrm{d}^3 r'\, U(\boldsymbol{r}')\psi(\boldsymbol{r}')\delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}') \\ &= \int \mathrm{d}^3 r'\, U(\boldsymbol{r}')\psi(\boldsymbol{r}')(\nabla_{\boldsymbol{r}}^2 + k^2)G(\boldsymbol{r}, \boldsymbol{r}').\end{aligned}$$

Therefore one solution of the stationary equation is

$$\psi(\boldsymbol{r}) = \int \mathrm{d}^3 r'\, U(\boldsymbol{r}')\psi(\boldsymbol{r}')G(\boldsymbol{r}, \boldsymbol{r}').$$

Taking into account the solution to the sourceless equation

$$(\nabla^2 + k^2)\psi^{(0)}(\boldsymbol{r}) = 0, \qquad \psi^{(0)}(\boldsymbol{r}) = Ae^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}},$$

the general solution of the stationary equation is

$$\boxed{\psi(\boldsymbol{r}) = Ae^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} + \int \mathrm{d}^3 r'\, U(\boldsymbol{r}')\psi(\boldsymbol{r}')G(\boldsymbol{r}, \boldsymbol{r}')}.$$

This satisfies the requirement

$$\lim_{V\to 0} \psi(\boldsymbol{r}) = \psi^{(0)}(\boldsymbol{r}).$$

It is an integral equation for $\psi(\boldsymbol{r})$, completely equivalent to the differential form of the stationary equation.

The integral equation has advantages: (1) The boundary condition is already included in the equation (the incident plane wave $Ae^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}$); (2) an iterative method may be used to solve it.

2) Green functions

Perform a Fourier transformation,

$$G(\boldsymbol{r}, \boldsymbol{r}') = \frac{1}{(2\pi)^3}\int \mathrm{d}^3 k'\, g(\boldsymbol{k}', \boldsymbol{r}')e^{\mathrm{i}\boldsymbol{k}'\cdot\boldsymbol{r}},$$
$$\delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}') = \frac{1}{(2\pi)^3}\int \mathrm{d}^3 k'\, e^{\mathrm{i}\boldsymbol{k}'\cdot(\boldsymbol{r}-\boldsymbol{r}')}.$$

Substitution into the equation satisfied by $G(\boldsymbol{r}, \boldsymbol{r}')$ gives

$$g(\boldsymbol{k}', \boldsymbol{r}') = \frac{e^{-\mathrm{i}\boldsymbol{k}'\cdot\boldsymbol{r}'}}{k^2 - k'^2}.$$

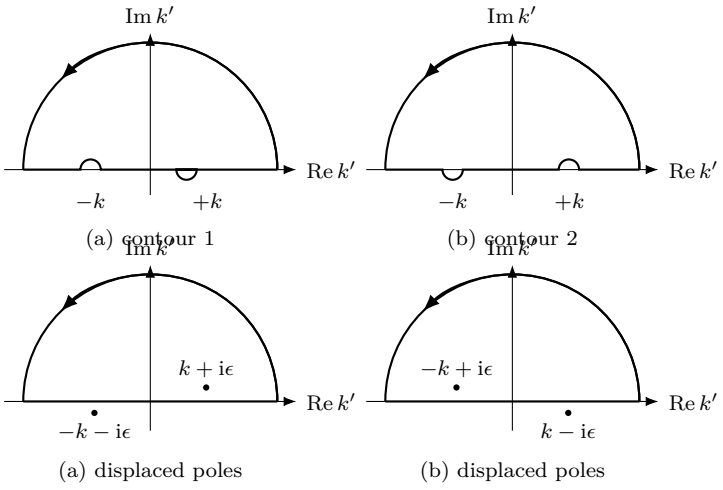
Hence

$$G(\boldsymbol{r},\boldsymbol{r}') = G(\boldsymbol{r}-\boldsymbol{r}') = \frac{1}{(2\pi)^3}\int_{-\infty}^{\infty} \mathrm{d}^3k'\,\frac{e^{\mathrm{i}\boldsymbol{k}'\cdot(\boldsymbol{r}-\boldsymbol{r}')}}{k^2-k'^2}.$$

Take $\boldsymbol{r}-\boldsymbol{r}'$ as the polar axis. After angular integration, with $R=|\boldsymbol{r}-\boldsymbol{r}'|$,

$$\begin{aligned} G(\boldsymbol{r}-\boldsymbol{r}') &= \frac{1}{4\pi^2 R}\int_{-\infty}^{\infty}\mathrm{d}k'\,\frac{k'\sin(k'R)}{k^2-k'^2} \\ &= \frac{1}{4\pi^2\mathrm{i}R}\int_{-\infty}^{\infty}\mathrm{d}k'\,\frac{k'e^{\mathrm{i}k'R}}{k^2-k'^2} \\ &= -\frac{1}{8\pi^2\mathrm{i}R}\int_{-\infty}^{\infty}\mathrm{d}k'\left(\frac{e^{\mathrm{i}k'R}}{k'-k}+\frac{e^{\mathrm{i}k'R}}{k'+k}\right). \end{aligned}$$

The part of the integrand involving $\cos(k'R)$ is an odd function of k' and integrates to zero. The integral has two poles, $k'=\pm k$, and is treated by complex-variable methods.



(a) contour 1 (b) contour 2

(a) displaced poles (b) displaced poles

In the complex k' plane, consider contour 1 closed in the upper half-plane (left). The first pole contributes and the second does not. On the semicircle,

$$e^{\mathrm{i}k'R} = e^{\mathrm{i}\,\mathrm{Re}\,k'R}e^{-\,\mathrm{Im}\,k'R};$$

since $\mathrm{Im}\,k'>0$ in the upper half-plane, this tends to zero as $R\to\infty$. The residue theorem gives

$$\boxed{G^{+}(\boldsymbol{r}-\boldsymbol{r}') = -\frac{1}{4\pi}\frac{e^{\mathrm{i}k|\boldsymbol{r}-\boldsymbol{r}'|}}{|\boldsymbol{r}-\boldsymbol{r}'|}},$$

called the *retarded Green function.* For contour 2 in the upper half-plane (right), the first pole does not contribute and the second does:

$$\boxed{G^{-}(\boldsymbol{r}-\boldsymbol{r}') = -\frac{1}{4\pi}\frac{e^{-\mathrm{i}k|\boldsymbol{r}-\boldsymbol{r}'|}}{|\boldsymbol{r}-\boldsymbol{r}'|}},$$

called the *advanced Green function.*

A contour containing both poles gives what is called the causal Green function. The contours above each make small semicircular indentations around the poles. Equivalently, one may displace the two poles slightly upward or downward and use a straight integration path (bottom diagrams in the source figure).

Retarded, advanced, and causal Green functions are all solutions of

$$(\nabla_{\boldsymbol{r}}^2 + k^2)G(\boldsymbol{r}, \boldsymbol{r}') = \delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}').$$

Substituting the retarded and advanced Green functions into the integral equation for $\psi(\boldsymbol{r})$, for example, gives

$$\boxed{\psi^{\pm}(\boldsymbol{r}) = Ae^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} - \frac{1}{4\pi}\int \mathrm{d}^3r'\, U(\boldsymbol{r}')\psi^{\pm}(\boldsymbol{r}')\frac{e^{\pm \mathrm{i}k|\boldsymbol{r}-\boldsymbol{r}'|}}{|\boldsymbol{r}-\boldsymbol{r}'|}}.$$

5.3 Lippmann–Schwinger Equation

1) Lippmann–Schwinger equation in Hilbert space

Let

$$\hat{H} = \hat{H}_0 + \hat{V}, \qquad \hat{H}_0 = \frac{\hat{\boldsymbol{p}}^2}{2m}.$$

The incident plane wave is $|\varphi\rangle$, with

$$\hat{H}_0 |\varphi\rangle = E|\varphi\rangle, \qquad E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}.$$

The scattered wave is $|\psi\rangle$. In elastic scattering the energy is unchanged:

$$\hat{H}|\psi\rangle = (\hat{H}_0 + \hat{V})|\psi\rangle = E|\psi\rangle, \qquad (E - \hat{H}_0)|\psi\rangle = \hat{V}|\psi\rangle.$$

The formal solution satisfying the initial condition $\lim_{V\to 0}|\psi\rangle = |\varphi\rangle$ is

$$|\psi\rangle = |\varphi\rangle + \frac{1}{E - \hat{H}_0}\hat{V}|\psi\rangle.$$

However, the operator $(E - \hat{H}_0)^{-1}$ diverges when acting on eigenstates of $\hat{H}_0$. To avoid the divergence, extend the energy into the complex plane and write the formal solution as

$$\boxed{|\psi^{\pm}\rangle = |\varphi\rangle + \frac{1}{E - \hat{H}_0 \pm \mathrm{i}\epsilon}\hat{V}|\psi^{\pm}\rangle, \quad \epsilon \to 0^+.}$$

This is the Lippmann–Schwinger equation in Hilbert space.

Enter the coordinate representation:

$$\langle \boldsymbol{r}|\psi^{\pm}\rangle = \langle \boldsymbol{r}|\varphi\rangle + \int \mathrm{d}^3r' \left\langle \boldsymbol{r}\middle|\frac{1}{E - \hat{H}_0 \pm \mathrm{i}\epsilon}\middle|\boldsymbol{r}'\right\rangle \left\langle \boldsymbol{r}'\middle|\hat{V}\middle|\psi^{\pm}\right\rangle.$$

Because

$$\left\langle \boldsymbol{r}\middle|\frac{1}{E - \hat{H}_0 \pm \mathrm{i}\epsilon}\middle|\boldsymbol{r}'\right\rangle = \int \mathrm{d}^3p'\mathrm{d}^3p''\, \langle \boldsymbol{r}|\boldsymbol{p}'\rangle \left\langle \boldsymbol{p}'\middle|\frac{1}{E - \hat{H}_0 \pm \mathrm{i}\epsilon}\middle|\boldsymbol{p}''\right\rangle \langle \boldsymbol{p}''|\boldsymbol{r}'\rangle,$$

$$\left\langle \boldsymbol{p}'\middle|\frac{1}{E - \hat{H}_0 \pm \mathrm{i}\epsilon}\middle|\boldsymbol{p}''\right\rangle = \frac{\delta^{(3)}(\boldsymbol{p}' - \boldsymbol{p}'')}{E - E' \pm \mathrm{i}\epsilon},$$

$$\langle \boldsymbol{r}|\boldsymbol{p}'\rangle = \frac{e^{\mathrm{i}\boldsymbol{p}'\cdot\boldsymbol{r}/\hbar}}{(2\pi\hbar)^{3/2}}, \qquad \langle \boldsymbol{p}''|\boldsymbol{r}'\rangle = \frac{e^{-\mathrm{i}\boldsymbol{p}''\cdot\boldsymbol{r}'/\hbar}}{(2\pi\hbar)^{3/2}},$$

we have

$$\left\langle \boldsymbol{r}\left|\frac{1}{E-\hat{H}_0\pm \mathrm{i}\epsilon}\right|\boldsymbol{r}'\right\rangle = \int \frac{\mathrm{d}^3p'}{(2\pi\hbar)^3}\frac{e^{\mathrm{i}\boldsymbol{p}'\cdot(\boldsymbol{r}-\boldsymbol{r}')/\hbar}}{E-E'\pm \mathrm{i}\epsilon} = \int \frac{\mathrm{d}^3k'}{(2\pi)^3}\frac{e^{\mathrm{i}\boldsymbol{k}'\cdot(\boldsymbol{r}-\boldsymbol{r}')}}{E-E'\pm \mathrm{i}\epsilon}.$$

By the same contour deformation used to define the Green function, moving a pole slightly upward or downward makes the integration path straight. With $E=\hbar^2k^2/(2m)$,

$$G^{\pm}(\boldsymbol{r}-\boldsymbol{r}') = \int \frac{\mathrm{d}^3k'}{(2\pi)^3}\frac{e^{\mathrm{i}\boldsymbol{k}'\cdot(\boldsymbol{r}-\boldsymbol{r}')}}{k^2-k'^2\pm \mathrm{i}0},$$
$$\left\langle \boldsymbol{r}\left|\frac{1}{E-\hat{H}_0\pm \mathrm{i}\epsilon}\right|\boldsymbol{r}'\right\rangle = \frac{2m}{\hbar^2}G^{\pm}(\boldsymbol{r}-\boldsymbol{r}').$$

Hence the Lippmann–Schwinger equation in the coordinate representation becomes

$$\langle \boldsymbol{r}|\psi^{\pm}\rangle = \langle \boldsymbol{r}|\varphi\rangle + \frac{2m}{\hbar^2}\int \mathrm{d}^3r'\, G^{\pm}(\boldsymbol{r}-\boldsymbol{r}')\left\langle \boldsymbol{r}'\left|\hat{V}\right|\psi^{\pm}\right\rangle.$$

Since

$$\left\langle \boldsymbol{r}'\left|\hat{V}\right|\psi^{\pm}\right\rangle = \int \mathrm{d}^3r''\,\left\langle \boldsymbol{r}'\left|\hat{V}\right|\boldsymbol{r}''\right\rangle\langle \boldsymbol{r}''|\psi^{\pm}\rangle = \int \mathrm{d}^3r''\, V(\boldsymbol{r}')\delta^{(3)}(\boldsymbol{r}'-\boldsymbol{r}'')\langle \boldsymbol{r}''|\psi^{\pm}\rangle = V(\boldsymbol{r}')\langle \boldsymbol{r}'|\psi^{\pm}\rangle,$$

we obtain

$$\boxed{\langle \boldsymbol{r}|\psi^{\pm}\rangle = \langle \boldsymbol{r}|\varphi\rangle + \int \mathrm{d}^3r'\, U(\boldsymbol{r}')G^{\pm}(\boldsymbol{r}-\boldsymbol{r}')\langle \boldsymbol{r}'|\psi^{\pm}\rangle.}$$

This is precisely the integral equation corresponding to the coordinate-space Schrödinger equation obtained in the preceding section by the Green-function method. It is the coordinate representation of the Lippmann–Schwinger equation.

2) Dyson equation

In the general Lippmann–Schwinger equation,

$$|\psi^{\pm}\rangle = |\varphi\rangle + \frac{1}{E-\hat{H}_0\pm \mathrm{i}\epsilon}\hat{V}|\psi^{\pm}\rangle,$$

let

$$\hat{G}_0^{\pm} = \frac{1}{E-\hat{H}_0\pm \mathrm{i}\epsilon}.$$

The coordinate representation $\left\langle \boldsymbol{r}\left|\hat{G}_0^{\pm}\right|\boldsymbol{r}'\right\rangle$ is a Green function, and thus $\hat{G}_0^{\pm}$ is called the *free Green operator*, related to $\hat{H}_0$.

The Lippmann–Schwinger equation can also be rewritten as

$$(E-\hat{H}_0\pm \mathrm{i}\epsilon)|\psi^{\pm}\rangle = (E-\hat{H}_0\pm \mathrm{i}\epsilon)|\varphi\rangle + \hat{V}|\psi^{\pm}\rangle,$$
$$(E-\hat{H}\pm \mathrm{i}\epsilon)|\psi^{\pm}\rangle = (E-\hat{H}\pm \mathrm{i}\epsilon)|\varphi\rangle + \hat{V}|\varphi\rangle,$$
$$|\psi^{\pm}\rangle = |\varphi\rangle + \frac{1}{E-\hat{H}\pm \mathrm{i}\epsilon}\hat{V}|\varphi\rangle.$$

Note that $|\varphi\rangle$ is an eigenstate of $\hat{H}_0$, not of $\hat{H}$. Define

$$\hat{G}^{\pm} = \frac{1}{E - \hat{H} \pm \mathrm{i}\epsilon}$$

as the *full Green operator*, related to $\hat{H}$.

From

$$(E - \hat{H} \pm \mathrm{i}\epsilon)\hat{G}^{\pm} = \mathbf{1},$$
$$(E - \hat{H}_0 \pm \mathrm{i}\epsilon)\hat{G}^{\pm} - \hat{V}\hat{G}^{\pm} = \mathbf{1},$$

we have

$$\boxed{\hat{G}^{\pm} = \hat{G}_0^{\pm} + \hat{G}_0^{\pm}\hat{V}\hat{G}^{\pm}}.$$

This is the *Dyson equation*, which describes the relation between the free and full Green operators. It has important applications in quantum statistical mechanics and quantum field theory.

3) Scattering amplitude

Consider the state $|\psi^+\rangle$ associated with the retarded Green function G^+. The coordinate-space Lippmann–Schwinger equation gives

$$\psi^+(\boldsymbol{r}) = Ae^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} - \frac{1}{4\pi}\int \mathrm{d}^3r'\, U(\boldsymbol{r}')\psi^+(\boldsymbol{r}')\frac{e^{\mathrm{i}k|\boldsymbol{r}-\boldsymbol{r}'|}}{|\boldsymbol{r}-\boldsymbol{r}'|}.$$

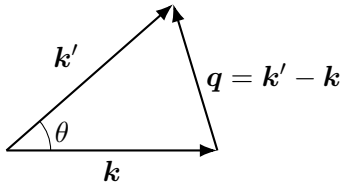
Because the effective region of the potential $V(\boldsymbol{r}')$ is finite, when $r \to \infty$,

$$|\boldsymbol{r}'| \ll r,$$
$$|\boldsymbol{r}-\boldsymbol{r}'| = \sqrt{(\boldsymbol{r}-\boldsymbol{r}')^2} = \sqrt{r^2 + r'^2 - 2\boldsymbol{r}\cdot\boldsymbol{r}'}$$
$$\simeq r\left(1 - 2\frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^2}\right)^{1/2} \simeq r - \frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r}.$$

Consequently,

$$\psi^+(\boldsymbol{r}) \xrightarrow{r\to\infty} Ae^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} - \frac{1}{4\pi}\frac{2m}{\hbar^2}\frac{e^{\mathrm{i}kr}}{r}\int \mathrm{d}^3r'\, e^{-\mathrm{i}\boldsymbol{k}'\cdot\boldsymbol{r}'}V(\boldsymbol{r}')\psi^+(\boldsymbol{r}').$$

In the phase, $|\boldsymbol{r}-\boldsymbol{r}'|$ is retained to first order; in the denominator only the zeroth-order term is kept. The scattered wavevector is $\boldsymbol{k}' = k\boldsymbol{r}/r$, with $|\boldsymbol{k}'| = k$.



Comparing with the standard asymptotic solution

$$\psi(\boldsymbol{r}) = Ae^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} + Af(\theta,\varphi)\frac{e^{\mathrm{i}kr}}{r},$$

we obtain the scattering amplitude

$$\begin{aligned} f(\theta,\varphi) &= -\frac{(2\pi)^3}{4\pi}\frac{2m}{\hbar^2}\int \mathrm{d}^3r'\,\frac{e^{-\mathrm{i}\boldsymbol{k}'\cdot\boldsymbol{r}'}}{(2\pi)^{3/2}}V(\boldsymbol{r}')\psi^+(\boldsymbol{r}') \\ &= -\frac{(2\pi)^3}{4\pi}\frac{2m}{\hbar^2}\int \mathrm{d}^3r'\,\langle\boldsymbol{k}'|\boldsymbol{r}'\rangle\, V(\boldsymbol{r}')\left\langle\boldsymbol{r}'\middle|\psi^+\right\rangle \\ &= -\frac{(2\pi)^3}{4\pi}\frac{2m}{\hbar^2}\left\langle\boldsymbol{k}'\middle|\hat{V}\middle|\psi^+\right\rangle \equiv f(\boldsymbol{k},\boldsymbol{k}'). \end{aligned}$$

Here we have taken

$$\langle\boldsymbol{r}|\varphi\rangle = \frac{e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}}{(2\pi)^{3/2}}.$$

Define the transition operator $\hat{T}$ by

$$\hat{V}\left|\psi^+\right\rangle \equiv \hat{T}\left|\varphi\right\rangle.$$

Then the scattering amplitude is

$$\boxed{f(\boldsymbol{k},\boldsymbol{k}') = -\frac{(2\pi)^3}{4\pi}\frac{2m}{\hbar^2}\left\langle\boldsymbol{k}'\middle|\hat{T}\middle|\boldsymbol{k}\right\rangle.}$$

The physical meaning of the transition matrix element $\left\langle\boldsymbol{k}'\middle|\hat{T}\middle|\boldsymbol{k}\right\rangle$ is clear: it is the transition amplitude from the plane wave $|\boldsymbol{k}\rangle$ to the free state $|\boldsymbol{k}'\rangle$.

If one considers the advanced Green function G^-, the resulting $\psi^-(\boldsymbol{r})$ is a superposition of a plane wave and an inward-propagating spherical wave, rather than a solution of the outgoing scattering problem. Advanced and causal Green functions are relevant to time evolution in dynamical systems.

4) Optical theorem

The forward scattering amplitude is

$$f(0,\varphi) = f(\boldsymbol{k},\boldsymbol{k}) = -\frac{(2\pi)^3}{4\pi}\frac{2m}{\hbar^2}\left\langle\boldsymbol{k}\middle|\hat{V}\middle|\psi^+\right\rangle.$$

From the Lippmann–Schwinger equation (with $|\varphi\rangle = |\boldsymbol{k}\rangle$),

$$\begin{aligned} \left|\psi^+\right\rangle &= |\varphi\rangle + \frac{1}{E-\hat{H}_0+\mathrm{i}\epsilon}\hat{V}\left|\psi^+\right\rangle, \\ \langle\boldsymbol{k}| &= \left\langle\psi^+\right| - \left\langle\psi^+\right|\hat{V}\frac{1}{E-\hat{H}_0-\mathrm{i}\epsilon}, \end{aligned}$$

so that

$$\begin{aligned} \left\langle\boldsymbol{k}\middle|\hat{V}\middle|\psi^+\right\rangle &= \left(\left\langle\psi^+\right| - \left\langle\psi^+\right|\hat{V}\frac{1}{E-\hat{H}_0-\mathrm{i}\epsilon}\right)\hat{V}\left|\psi^+\right\rangle \\ &= \left\langle\psi^+\middle|\hat{V}\middle|\psi^+\right\rangle - \left\langle\psi^+\right|\hat{V}\frac{1}{E-\hat{H}_0-\mathrm{i}\epsilon}\hat{V}\left|\psi^+\right\rangle. \end{aligned}$$

Using

$$\frac{1}{E-\hat{H}_0-\mathrm{i}\epsilon} = \mathrm{Pr}\,\frac{1}{E-\hat{H}_0} + \mathrm{i}\pi\delta(E-\hat{H}_0),$$

and noting that $\hat{V}$ and $\hat{V}(\mathrm{Pr}(E-\hat{H}_0)^{-1})\hat{V}$ are Hermitian operators, whose expectation values in $\left|\psi^+\right\rangle$ are real, we find

$$\begin{aligned}
\mathrm{Im}\left\langle \boldsymbol{k}\middle|\hat{V}\middle|\psi^+\right\rangle &= -\pi\left\langle\psi^+\middle|\,\hat{V}\delta(E-\hat{H}_0)\hat{V}\,\middle|\psi^+\right\rangle \\
&= -\pi\int \mathrm{d}^3k'\,\left\langle\psi^+\middle|\hat{V}\middle|\boldsymbol{k}'\right\rangle\left\langle\boldsymbol{k}'\middle|\hat{V}\middle|\psi^+\right\rangle\delta(E-\hbar^2k'^2/2m) \\
&= -\pi\int \mathrm{d}^3k'\,\left|\left\langle\boldsymbol{k}'\middle|\hat{V}\middle|\psi^+\right\rangle\right|^2\delta\left(E-\frac{\hbar^2k'^2}{2m}\right) \\
&= -\frac{\pi mk}{\hbar^2}\int \mathrm{d}\Omega'\,\left|\left\langle\boldsymbol{k}'\middle|\hat{V}\middle|\psi^+\right\rangle\right|^2.
\end{aligned}$$

Because

$$\sigma(\boldsymbol{k}',\boldsymbol{k}) = |f(\boldsymbol{k}',\boldsymbol{k})|^2 = \frac{(2\pi)^6}{16\pi^2}\frac{4m^2}{\hbar^4}\left|\left\langle\boldsymbol{k}'\middle|\hat{V}\middle|\psi^+\right\rangle\right|^2, \qquad \sigma_{\mathrm{tot}} = \int \mathrm{d}\Omega\,\sigma(\theta,\varphi),$$

we obtain

$$\mathrm{Im}\left\langle \boldsymbol{k}\middle|\hat{V}\middle|\psi^+\right\rangle = -\frac{\hbar^2k}{16\pi^3m}\int \mathrm{d}\Omega'\,\sigma(\theta,\varphi) = -\frac{\hbar^2k}{16\pi^3m}\sigma_{\mathrm{tot}}.$$

Therefore

$$\begin{aligned}
\mathrm{Im}\,f(\theta=0) &= -\frac{(2\pi)^3}{4\pi}\frac{2m}{\hbar^2}\,\mathrm{Im}\left\langle \boldsymbol{k}\middle|\hat{V}\middle|\psi^+\right\rangle \\
&= \frac{k}{4\pi}\sigma_{\mathrm{tot}}.
\end{aligned}$$

This relation between the forward scattering amplitude and the total scattering cross section is called the *optical theorem*:

$$\boxed{\mathrm{Im}\,f(0) = \frac{k}{4\pi}\sigma_{\mathrm{tot}}}.$$

5.4 Born Series

1) First Born approximation

For the integral equation and scattering amplitude,

$$\psi^+(\boldsymbol{r}) \xrightarrow{r\to\infty} Ae^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} - \frac{1}{4\pi}\frac{2m}{\hbar^2}\frac{e^{\mathrm{i}kr}}{r}\int \mathrm{d}^3r'\,e^{-\mathrm{i}\boldsymbol{k}'\cdot\boldsymbol{r}'}V(\boldsymbol{r}')\psi^+(\boldsymbol{r}'),$$

$$f(\theta,\varphi) = -\frac{(2\pi)^3}{4\pi}\frac{2m}{\hbar^2}\int \mathrm{d}^3r'\,\frac{e^{-\mathrm{i}\boldsymbol{k}'\cdot\boldsymbol{r}'}}{(2\pi)^{3/2}}V(\boldsymbol{r}')\psi^+(\boldsymbol{r}'),$$

if the potential $V(\boldsymbol{r})$ is weak (incident energy $E \gg V$), one may solve iteratively. The zeroth-order approximation is

$$\psi^{(0)}(\boldsymbol{r}) = \frac{e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}}{(2\pi)^{3/2}}.$$

The first Born approximation then gives

$$\begin{aligned}
f(\theta,\varphi) &= -\frac{(2\pi)^3}{4\pi}\frac{2m}{\hbar^2}\int \mathrm{d}^3r'\,\frac{e^{-\mathrm{i}\boldsymbol{k}'\cdot\boldsymbol{r}'}}{(2\pi)^{3/2}}V(\boldsymbol{r}')\frac{e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}'}}{(2\pi)^{3/2}} \\
&= -\frac{1}{4\pi}\int \mathrm{d}^3r'\,e^{\mathrm{i}(\boldsymbol{k}-\boldsymbol{k}')\cdot\boldsymbol{r}'}U(\boldsymbol{r}').
\end{aligned}$$

For low-energy scattering (still satisfying $E \gg V$),

$$q = |\boldsymbol{k} - \boldsymbol{k}'| = \sqrt{k^2 + k'^2 - 2kk'\cos\theta}$$
$$= \sqrt{2k^2(1-\cos\theta)} = \sqrt{4k^2\sin^2\frac{\theta}{2}} = 2k\sin\frac{\theta}{2}.$$

If q is small, the phase may be regarded as constant over the effective region of the interaction, and contributes no change to the cross section. Hence

$$\boxed{f(\theta,\varphi) = -\frac{1}{4\pi}\int \mathrm{d}^3r'\, U(\boldsymbol{r}').}$$

Example. Consider low-energy scattering by

$$V(r) = \begin{cases} V_0, & r \le a, \\ 0, & r > a. \end{cases}$$

Then

$$f(\theta,\varphi) = -\frac{1}{4\pi}\int \mathrm{d}^3r\, U(r) = -\frac{m}{2\pi\hbar^2}\int \mathrm{d}^3r\, V(r)$$
$$= -\frac{mV_0}{2\pi\hbar^2}\int_{r\le a}\mathrm{d}^3r = -\frac{mV_0}{2\pi\hbar^2}\frac{4\pi}{3}a^3.$$

The differential scattering cross section is

$$\sigma(\theta,\varphi) = |f(\theta,\varphi)|^2 = \left(\frac{2mV_0a^3}{3\hbar^2}\right)^2.$$

The total cross section is

$$\sigma_{\mathrm{tot}} = \int \sigma(\theta,\varphi)\mathrm{d}\Omega = 4\pi\left(\frac{2mV_0a^3}{3\hbar^2}\right)^2.$$

For general scattering by a central potential, $V(\boldsymbol{r}) = V(r)$,

$$f(\theta,\varphi) = -\frac{1}{4\pi}\int_0^\infty \mathrm{d}r'\, r'^2U(r')\int_0^\pi \mathrm{d}\theta'\,\sin\theta' e^{-\mathrm{i}qr'\cos\theta'}\int_0^{2\pi}\mathrm{d}\varphi'$$
$$= -\frac{1}{q}\int_0^\infty \mathrm{d}r'\, U(r')r'\sin(qr') = -\frac{2m}{\hbar^2 q}\int_0^\infty \mathrm{d}r'\, V(r')r'\sin(qr').$$

Note: The Born approximation applies to scattering by a weak potential.

2) Born series

The coordinate-space Lippmann–Schwinger equation is

$$\psi(\boldsymbol{r}) = \psi^{(0)}(\boldsymbol{r}) + \int \mathrm{d}^3r'\,\psi(\boldsymbol{r}')U(\boldsymbol{r}')G(\boldsymbol{r},\boldsymbol{r}').$$

Zeroth-order approximation:

$$\psi^{(0)}(\boldsymbol{r}) = Ae^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}.$$

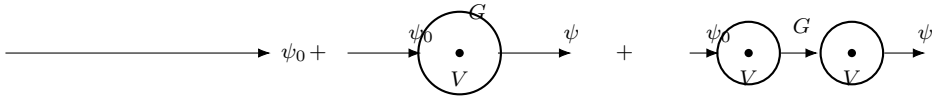
First-order approximation:

$$\psi^{(1)}(\boldsymbol{r}) = \psi^{(0)}(\boldsymbol{r}) + \int \mathrm{d}^3r'\,\psi^{(0)}(\boldsymbol{r}')U(\boldsymbol{r}')G(\boldsymbol{r},\boldsymbol{r}').$$

Second-order approximation:

$$\begin{aligned}\psi^{(2)}(\boldsymbol{r}) &= \psi^{(0)}(\boldsymbol{r}) + \int \mathrm{d}^3r'\, \psi^{(1)}(\boldsymbol{r}')U(\boldsymbol{r}')G(\boldsymbol{r},\boldsymbol{r}') \\ &= \psi^{(0)}(\boldsymbol{r}) + \int \mathrm{d}^3r'\, \psi^{(0)}(\boldsymbol{r}')U(\boldsymbol{r}')G(\boldsymbol{r},\boldsymbol{r}') \\ &\quad + \int \mathrm{d}^3r'\mathrm{d}^3r''\, \psi^{(0)}(\boldsymbol{r}')U(\boldsymbol{r}')G(\boldsymbol{r}'',\boldsymbol{r}')U(\boldsymbol{r}'')G(\boldsymbol{r},\boldsymbol{r}'').\end{aligned}$$

The above series may be represented by diagrams:



The Green function is the function that propagates interactions between two points.

5.5 Partial-Wave Method

Find the scattering amplitude in a central potential field.

1) Idea

The incident plane wave $Ae^{\mathrm{i}kz}$ is a common eigenstate of the free-particle observables $\{\hat{H}, \hat{\boldsymbol{p}}, \hat{L}_z\}$, with eigenvalues

$$E = \frac{\hbar^2k^2}{2\mu}, \qquad p_x = p_y = 0, \qquad p_z = \hbar k, \qquad L_z = 0.$$

For a central potential, $[\hat{\boldsymbol{p}}, \hat{H}] \neq 0$, so momentum $\boldsymbol{p}$ is not conserved, while $[\hat{\boldsymbol{L}}^2, \hat{H}] = 0$: orbital angular momentum is conserved. A commuting set of conserved observables is $\{\hat{H}, \hat{\boldsymbol{L}}^2, \hat{L}_z\}$. Thus a plane wave is not an eigenstate of the conserved-observable set for a central field, and it is inconvenient for describing directly the relation between incoming and outgoing states.

The partial-wave method: Under conservation of energy (unchanged k) and conservation of $\hat{L}_z$ (unchanged $m = 0$), expand the incident plane wave in eigenstates of $\hat{\boldsymbol{L}}^2$, $P_l(\cos\theta)$ (a partial-wave expansion). The scattering of a plane wave is thereby converted into the sum of scatterings of eigenstates (partial waves) of the conserved-observable set.

2) Partial-wave expansion of the plane wave

$$\boxed{e^{\mathrm{i}kz} = e^{\mathrm{i}kr\cos\theta} = \sum_{l=0}^{\infty}(2l+1)\mathrm{i}^l j_l(kr)P_l(\cos\theta).}$$

For the spherical Bessel function of order l,

$$j_l(kr) \xrightarrow{r\to\infty} \frac{1}{kr}\sin\left(kr - \frac{l\pi}{2}\right).$$

Hence

$$e^{\mathrm{i}kz} \xrightarrow{r\to\infty} \sum_{l=0}^{\infty}(2l+1)\mathrm{i}^l\frac{1}{kr}\sin\left(kr - \frac{l\pi}{2}\right)P_l(\cos\theta).$$

The standard asymptotic solution is

$$\begin{aligned}&\psi(\boldsymbol{r}) \xrightarrow{r\to\infty} Ae^{\mathrm{i}kz} + Af(\theta)\frac{e^{\mathrm{i}kr}}{r} \\ &\qquad = A\sum_{l=0}^{\infty}(2l+1)\mathrm{i}^l\frac{1}{kr}\sin\left(kr - \frac{l\pi}{2}\right)P_l(\cos\theta) + Af(\theta)\frac{e^{\mathrm{i}kr}}{r}.\end{aligned}$$

3) Partial-wave expansion of the state in a central field

A particular solution of the stationary Schrödinger equation for a central field is

$$\psi_{nlm}(r,\theta,\varphi) = R_{nl}(r)Y_{lm}(\theta,\varphi).$$

The general solution is

$$\psi(r,\theta,\varphi) = \sum_{n,l,m} B_{nlm}R_{nl}(r)Y_{lm}(\theta,\varphi).$$

The general solution satisfying conservation of energy (k unchanged) and $\hat{L}_z$ ($m=0$ unchanged) is

$$\psi(r,\theta) = \sum_{l=0}^{\infty} B_l R_l(r) P_l(\cos\theta).$$

Here $R_l(r)$ is determined by the radial equation

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}R_l}{\mathrm{d}r}\right) + \left(k^2 - \frac{2\mu}{\hbar^2}V(r) - \frac{l(l+1)}{r^2}\right)R_l = 0.$$

As $r\to\infty$, $V(r)\to 0$, and

$$\frac{\mathrm{d}^2(rR_l)}{\mathrm{d}r^2} + k^2(rR_l) = 0.$$

Thus

$$R_l(r\to\infty) = \frac{C_l}{kr}\sin\left(kr - \frac{l\pi}{2} + \delta_l\right).$$

The two undetermined constants C_l and δ_l depend on the particular central potential $V(r)$; $-l\pi/2$ is introduced for convenience in calculation. The asymptotic solution is therefore

$$\psi(r,\theta) \xrightarrow{r\to\infty} A\sum_{l=0}^{\infty}\frac{A_l}{kr}\sin\left(kr - \frac{l\pi}{2} + \delta_l\right)P_l(\cos\theta), \qquad AA_l = B_lC_l.$$

Comparing with the standard asymptotic solution gives

$$\sum_{l=0}^{\infty}(2l+1)\mathrm{i}^l\frac{1}{kr}\sin\left(kr - \frac{l\pi}{2}\right)P_l(\cos\theta) + f(\theta)\frac{e^{\mathrm{i}kr}}{r} = \sum_{l=0}^{\infty}\frac{A_l}{kr}\sin\left(kr - \frac{l\pi}{2} + \delta_l\right)P_l(\cos\theta).$$

Use

$$\sin x = \frac{e^{\mathrm{i}x} - e^{-\mathrm{i}x}}{2\mathrm{i}}.$$

Then

$$0 = \frac{e^{\mathrm{i}kr}}{2\mathrm{i}kr}\left[2\mathrm{i}kf(\theta) + \sum_{l=0}^{\infty}(2l+1)\mathrm{i}^l e^{-\mathrm{i}l\pi/2}P_l(\cos\theta) - \sum_{l=0}^{\infty}A_l e^{\mathrm{i}(\delta_l - l\pi/2)}P_l(\cos\theta)\right] - \frac{e^{-\mathrm{i}kr}}{2\mathrm{i}kr}\left[\sum_{l=0}^{\infty}(2l+1)\mathrm{i}^l e^{\mathrm{i}l\pi/2}P_l(\cos\theta) - \sum_{l=0}^{\infty}A_l e^{-\mathrm{i}(\delta_l - l\pi/2)}P_l(\cos\theta)\right].$$

For this to hold for arbitrary r, the coefficients of $e^{\mathrm{i}kr}$ and $e^{-\mathrm{i}kr}$ must vanish separately. We find

$$A_l = (2l+1)\mathrm{i}^l e^{\mathrm{i}\delta_l},$$

and

$$\boxed{f(\theta) = \sum_{l=0}^{\infty} \frac{2l+1}{k} P_l(\cos\theta) e^{\mathrm{i}\delta_l} \sin\delta_l}.$$

Define the partial-wave scattering amplitude

$$\boxed{f_l(\theta) = \frac{1}{k}(2l+1) P_l(\cos\theta) e^{\mathrm{i}\delta_l} \sin\delta_l}.$$

The total scattering amplitude is the sum of the partial-wave amplitudes:

$$f(\theta) = \sum_{l=0}^{\infty} f_l(\theta).$$

The differential scattering cross section is

$$\boxed{\sigma(\theta) = \left| \sum_{l=0}^{\infty} f_l(\theta) \right|^2}.$$

The partial-wave scattering amplitudes depend only on δ_l. The key is to find the phase shifts δ_l:

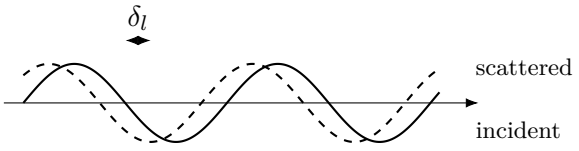
1. Substitute $V(r)$ into the radial equation and obtain $R_l(r)$.
2. Compare $\lim_{r\to\infty} R_l(r)$ with $C_l(kr)^{-1}\sin(kr - l\pi/2 + \delta_l)$ to find δ_l.

The physical meaning of δ_l is as follows. The partial wave before scattering is

$$\frac{1}{kr}\sin\left(kr - \frac{l\pi}{2}\right),$$

whereas the partial wave after scattering is

$$\frac{1}{kr}\sin\left(kr - \frac{l\pi}{2} + \delta_l\right).$$



Thus scattering involves the same partial wave before and after, with only an overall phase shift δ_l.

4) Convergence

A semiclassical estimate of the convergence of the sum gives $L \sim pr$. Suppose the potential $V(r)$ has effective radius a. Scattering occurs only for $L < pa$, so that partial waves satisfying approximately $\sqrt{l(l+1)} < ka$ contribute. When the incident energy $E(k)$ is small, one need retain only low-order partial waves, such as the s wave.

Conclusion: The partial-wave method is suitable for low-energy scattering by a central field.

Example. Scattering of a low-energy particle by a spherically symmetric attractive potential well:

$$V(r) = \begin{cases} -V_0, & r \le a, \\ 0, & r > a. \end{cases}$$

The radial equation is

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}R_l}{\mathrm{d}r}\right)+\left(k^2-\frac{2\mu}{\hbar^2}V(r)-\frac{l(l+1)}{r^2}\right)R_l=0.$$

Consider only the s partial wave ($l=0$). Then

$$\begin{cases}\dfrac{\mathrm{d}^2(rR_0)}{\mathrm{d}r^2}+k'^2(rR_0)=0, & r\le a,\\ \dfrac{\mathrm{d}^2(rR_0)}{\mathrm{d}r^2}+k^2(rR_0)=0, & r>a,\end{cases}\qquad k'^2=k^2+\frac{2\mu}{\hbar^2}V_0.$$

The solution is

$$R_0(r)=\begin{cases}\dfrac{C_0'}{k'r}\sin(k'r+B_0'), & r\le a,\\ \dfrac{C_0}{kr}\sin(kr+B_0), & r>a.\end{cases}$$

Requiring $R_0(r)$ to remain finite at $r=0$ gives $B_0'=0$. Furthermore, continuity of $R_0(r)$ and $\mathrm{d}R_0/\mathrm{d}r$ at $r=a$ gives

$$B_0=\arctan\left(\frac{k}{k'}\tan(k'a)\right)-ka,$$
$$C_0'=C_0\frac{k'}{k}\frac{\sin(ka+B_0)}{\sin(k'a)}.$$

Compare

$$\lim_{r\to\infty}R_0(r)=\frac{C_0}{kr}\sin(kr+B_0)$$

with the standard asymptotic form for a central potential,

$$\lim_{r\to\infty}R_0(r)=\frac{C_0}{kr}\sin(kr+\delta_0).$$

The s-wave phase shift is

$$\boxed{\delta_0=B_0}.$$

Since $P_0(\cos\theta)=1$,

$$f_0(\theta)=\frac{1}{k}e^{\mathrm{i}\delta_0}\sin\delta_0,$$

$$\sigma(\theta)=|f(\theta)|^2\simeq|f_0(\theta)|^2=\frac{\sin^2\delta_0}{k^2}.$$

At low energy, $k\to0$,

$$\arctan\left(\frac{k}{k'}\tan k'a\right)\longrightarrow\frac{k}{k'}\tan k'a,\qquad k'^2\longrightarrow\frac{2\mu V_0}{\hbar^2}.$$

Therefore

$$\delta_0\simeq\frac{k}{k'}\tan k'a-ka=ka\left(\frac{\tan k'a}{k'a}-1\right),$$

and

$$\boxed{\sigma(\theta)\simeq\frac{\sin^2\delta_0}{k^2}\simeq\frac{\delta_0^2}{k^2}\simeq a^2\left(\frac{\tan k'a}{k'a}-1\right)^2}.$$

Since only the s partial wave has been considered, σ is independent of θ. The total scattering cross section is

$$\boxed{\sigma_{\mathrm{tot}}=\int\sigma(\theta)\mathrm{d}\Omega=4\pi a^2\left(\frac{\tan k'a}{k'a}-1\right)^2}.$$

Chapter 6

Identical Particles and Permutation Symmetry

6.1 Identical Particles and Permutation Symmetry

In a system of identical particles, all particles have the same intrinsic properties: mass, charge, spin, and any other labels that identify the species. The spin–statistics connection is established in relativistic quantum field theory: integer-spin particles obey Bose–Einstein statistics, while half-integer-spin particles obey Fermi–Dirac statistics. Our immediate purpose is to understand how these two classes arise in the state space of a many-particle system. The analysis begins with permutations of particle labels.

1) States, observables, and permutations

Consider N identical particles with particle labels $1, \dots, N$. Their Hamiltonian is symmetric under exchanging the particle labels, including the associated spatial and spin coordinates. A many-body wavefunction is written

$$\Psi(1, 2, \dots, N),$$

where each argument collectively denotes the coordinates of one particle. Define the exchange operator $\hat{P}_{ij}$ by

$$\hat{P}_{ij}\Psi(1, \dots, i, \dots, j, \dots, N) = \Psi(1, \dots, j, \dots, i, \dots, N).$$

Exchanging two particles twice restores the original state, hence

$$\hat{P}_{ij}^2 = \hat{I}, \qquad \hat{P}_{ij}^{-1} = \hat{P}_{ij}.$$

The Hamiltonian commutes with every permutation operator:

$$\boxed{[\hat{H}, \hat{P}] = 0, \qquad P \in S_N.}$$

Here S_N is the symmetric group consisting of the $N!$ permutations of N objects. Any permutation is a product of elementary transpositions; its parity is even or odd according to the parity of the number of transpositions in such a decomposition. Different decompositions have the same parity.

On the many-particle Hilbert space each permutation is represented by a unitary operator. To see this, let the inner product be the integral over all particle coordinates. Relabeling dummy integration variables gives

$$\begin{aligned}\left\langle \Psi \middle| \hat{P}\Phi \right\rangle &= \int \mathrm{d}1 \cdots \mathrm{d}N \, \Psi^*(1, \dots, N)(\hat{P}\Phi)(1, \dots, N) \\ &= \left\langle \hat{P}^{-1}\Psi \middle| \Phi \right\rangle,\end{aligned}$$

so $\hat{P}^\dagger = \hat{P}^{-1}$. In particular, $\hat{P}_{ij}$ is Hermitian as well as unitary, and its eigenvalues are $+1$ or -1.

An observable that treats identical particles identically is invariant under relabeling, and therefore

$$\boxed{\hat{P}^{-1}\hat{A}\hat{P} = \hat{A}, \qquad [\hat{A}, \hat{P}] = 0 \quad (P \in S_N).}$$

The converse follows immediately by multiplying the commutator by $\hat{P}^{-1}$. Invariance of the Hamiltonian means that permutation symmetry is preserved under time evolution: for $\hat{U}(t) = e^{-\mathrm{i}\hat{H}t/\hbar}$,

$$\hat{P}\hat{U}(t)\left|\Psi\right\rangle = \hat{U}(t)\hat{P}\left|\Psi\right\rangle.$$

Therefore, if a state is symmetric or antisymmetric at the initial time, its symmetry character remains unchanged.

2) Examples: two and three particles

For two particles, $S_2 = \{I, P_{12}\}$. Given an arbitrary wavefunction $\Psi(1,2)$, its symmetric and antisymmetric combinations are

$$\Psi_\pm(1,2) \propto \Psi(1,2) \pm \Psi(2,1),$$

which satisfy $\hat{P}_{12}\Psi_\pm = \pm\Psi_\pm$ whenever the combinations are nonzero. The normalization depends on the overlap of the two terms; it is derived explicitly in the next section.

The three-particle example reveals an important additional possibility. The six elements of S_3 are the identity, three exchanges, and two cyclic permutations. For instance,

$$\hat{P}_{12}\hat{P}_{23}\hat{P}_{12} = \hat{P}_{13}, \qquad \hat{P}_{12}\hat{P}_{23} \neq \hat{P}_{23}\hat{P}_{12}.$$

These operators need not commute. Hence a basis of simultaneous eigenstates of *all* exchange operators generally does not exist in the full distinguishable-particle tensor-product space.

Take three orthonormal one-particle orbitals $\left|a\right\rangle, \left|b\right\rangle, \left|c\right\rangle$. Acting with all elements of S_3 on $\left|abc\right\rangle := \left|a\right\rangle \otimes \left|b\right\rangle \otimes \left|c\right\rangle$ produces six orthonormal ordered products. This six-dimensional permutation space contains an invariant symmetric combination

$$\left|\Psi_\mathrm{s}\right\rangle = \frac{1}{\sqrt{6}}(\left|abc\right\rangle + \left|acb\right\rangle + \left|bac\right\rangle + \left|bca\right\rangle + \left|cab\right\rangle + \left|cba\right\rangle),$$

and an invariant antisymmetric combination

$$\left|\Psi_\mathrm{a}\right\rangle = \frac{1}{\sqrt{6}}(\left|abc\right\rangle - \left|acb\right\rangle - \left|bac\right\rangle + \left|bca\right\rangle + \left|cab\right\rangle - \left|cba\right\rangle).$$

They transform as $\hat{P}\left|\Psi_\mathrm{s}\right\rangle = \left|\Psi_\mathrm{s}\right\rangle$ and $\hat{P}\left|\Psi_\mathrm{a}\right\rangle = \mathrm{sgn}(P)\left|\Psi_\mathrm{a}\right\rangle$.

There are also states in this six-dimensional space that transform into *linear combinations of one another* under permutations. These form two copies of the two-dimensional standard irreducible representation of S_3. Two orthonormal vectors in the remaining four-dimensional complement are

$$\left|u_1\right\rangle = \frac{1}{\sqrt{6}}(2\left|abc\right\rangle - \left|bca\right\rangle - \left|cab\right\rangle),$$
$$\left|u_2\right\rangle = \frac{1}{\sqrt{2}}(\left|bca\right\rangle - \left|cab\right\rangle).$$

Permutation operators mix vectors in this four-dimensional complement. By choosing appropriate two-dimensional invariant subspaces, each permutation is represented by a 2×2 matrix

rather than a single overall sign. More abstractly, the six-dimensional regular representation decomposes as

$$\mathbf{6} \simeq \mathbf{1}_{\mathrm{s}} \oplus \mathbf{1}_{\mathrm{a}} \oplus \mathbf{2} \oplus \mathbf{2}, \qquad 1^2 + 1^2 + 2^2 = 6.$$

This example explains why permutation-group symmetry alone permits mathematical representations beyond the completely symmetric and completely antisymmetric ones. In the usual nonrelativistic treatment of ordinary identical bosons and fermions, we select the corresponding one-dimensional sectors as the physical exchange-symmetry postulate. Its relation to particle spin is supplied by the spin–statistics theorem under its relativistic assumptions.

3) Permutation-invariant subspaces

A subspace $\mathcal{V}$ is invariant under a representation of S_N if $\hat{P}\mathcal{V} \subseteq \mathcal{V}$ for every $P \in S_N$. Symmetric and antisymmetric states constitute two especially simple invariant subspaces. The physical action of a permutation never changes measurable expectation values:

$$\left\langle \hat{P}\Psi \middle| \hat{A} \middle| \hat{P}\Psi \right\rangle = \left\langle \Psi \middle| \hat{P}^\dagger \hat{A} \hat{P} \middle| \Psi \right\rangle = \left\langle \Psi \middle| \hat{A} \middle| \Psi \right\rangle .$$

This is the operational meaning of particle indistinguishability. The next section constructs normalized many-particle states in the two physical exchange-symmetry sectors.

6.2 Completely Symmetric and Antisymmetric States

Let $\{|i\rangle\}$ be an orthonormal one-particle basis. An ordered basis vector in the N-particle tensor-product space is

$$|i_1, \ldots, i_N\rangle := |i_1\rangle_1 \otimes \cdots \otimes |i_N\rangle_N .$$

Write the sum of the permutation operators as

$$\hat{\mathcal{S}}_\pm = \frac{1}{\sqrt{N!}} \sum_{P \in S_N} [\mathrm{sgn}(P)]_\pm \hat{P},$$

where $[\mathrm{sgn}(P)]_+ = 1$ and $[\mathrm{sgn}(P)]_- = \mathrm{sgn}(P)$. We deliberately use $\mathcal{S}$ for the unnormalized sum with the square-root prefactor; normalized projectors are introduced below. Changing variables $Q = PP'$ in the group sum yields

$$\hat{P}'\hat{\mathcal{S}}_+ = \hat{\mathcal{S}}_+\hat{P}' = \hat{\mathcal{S}}_+, \qquad \hat{P}'\hat{\mathcal{S}}_- = \hat{\mathcal{S}}_-\hat{P}' = \mathrm{sgn}(P')\hat{\mathcal{S}}_-.$$

Consequently $\hat{\mathcal{S}}_+ |i_1, \ldots, i_N\rangle$ is symmetric and $\hat{\mathcal{S}}_- |i_1, \ldots, i_N\rangle$ is antisymmetric. They are also orthogonal for $N > 1$: an exchange P_{12} preserves the first state and reverses the sign of the second, so their overlap equals its own negative.

1) The symmetrization and antisymmetrization projectors

Introduce the actual orthogonal projectors

$$\boxed{\hat{\mathcal{P}}_\pm = \frac{1}{N!} \sum_{P \in S_N} [\mathrm{sgn}(P)]_\pm \hat{P}.}$$

Their defining properties follow directly from the group multiplication law:

$$\hat{\mathcal{P}}_\pm^\dagger = \hat{\mathcal{P}}_\pm, \qquad \hat{\mathcal{P}}_\pm^2 = \hat{\mathcal{P}}_\pm, \qquad \hat{\mathcal{P}}_+\hat{\mathcal{P}}_- = 0 \quad (N > 1).$$

For example, in the square of the plus projector, for each fixed $R \in S_N$ there are $N!$ pairs (P, Q) with $PQ = R$; therefore

$$\hat{\mathcal{P}}_+^2 = \frac{1}{(N!)^2} \sum_{P,Q} \hat{PQ} = \frac{1}{N!} \sum_R \hat{R} = \hat{\mathcal{P}}_+.$$

For the minus projector the factor $\mathrm{sgn}(P)\,\mathrm{sgn}(Q) = \mathrm{sgn}(PQ)$ yields the identical argument. Every completely symmetric state can be written as a linear combination of projected ordered basis states, since the identity resolution can be inserted before projection.

2) Normalization and repeated orbital occupation

Suppose the ordered list $(i_1, \ldots, i_N)$ contains the orbital i precisely n_i times, with $\sum_i n_i = N$. For bosons, there are $N!/\prod_i n_i!$ distinct permutations of that list, and every distinct ordered product occurs $\prod_i n_i!$ times in the full permutation sum. The norm is therefore

$$\left\| \sum_{P \in S_N} \hat{P} \, |i_1, \ldots, i_N\rangle \right\|^2 = N! \prod_i n_i!.$$

Thus the normalized Bose state is

$$\boxed{|\{n_i\}\rangle_{\mathrm{B}} = \frac{1}{\sqrt{N! \prod_i n_i!}} \sum_{P \in S_N} \hat{P} \, |i_1, \ldots, i_N\rangle \,.}$$

In terms of the handout's $\hat{\mathcal{S}}_+$, this is $|\{n_i\}\rangle_{\mathrm{B}} = \hat{\mathcal{S}}_+ \, |i_1, \ldots, i_N\rangle \, / \sqrt{\prod_i n_i!}$. This factorial is essential whenever an orbital is occupied more than once.

For fermions, if $i_a = i_b$ for $a \neq b$, the exchanged ordered product is unchanged but the antisymmetrization sign is reversed, giving zero. Therefore all occupied orbitals must be distinct. For distinct $i_1, \ldots, i_N$, every ordered product in the permutation sum is orthogonal to every other, and

$$\boxed{|i_1, \ldots, i_N\rangle_{\mathrm{F}} = \frac{1}{\sqrt{N!}} \sum_{P \in S_N} \mathrm{sgn}(P) \hat{P} \, |i_1, \ldots, i_N\rangle \,.}$$

In coordinate representation the same expression is the normalized Slater determinant,

$$\Psi_{\mathrm{F}}(x_1, \ldots, x_N) = \frac{1}{\sqrt{N!}} \det \begin{pmatrix} \varphi_{i_1}(x_1) & \cdots & \varphi_{i_N}(x_1) \\ \vdots & \ddots & \vdots \\ \varphi_{i_1}(x_N) & \cdots & \varphi_{i_N}(x_N) \end{pmatrix}.$$

The variable x may include spin as well as position. Both constructions provide complete orthonormal bases of their respective symmetric and antisymmetric N-particle sectors. The next chapter assembles sectors with different N into Fock space and derives operator rules directly from these states.

Chapter 7

Bosons, Fermions, and Second Quantization

7.1 Bosons: Fock Space and Creation and Annihilation Operators

A completely symmetric N-particle state can be labeled by the *occupation numbers* n_i of one-particle orbitals $|i\rangle$. The actual particle labels have disappeared from this description. Define

$$|n_1, n_2, \ldots\rangle_{\mathrm{B}}\,, \qquad n_i = 0, 1, 2, \ldots, \qquad \sum_i n_i = N,$$

by the normalized symmetrization in the preceding chapter. Because different occupation lists are orthogonal,

$${}_{\mathrm{B}}\langle n_1, n_2, \ldots | n_1', n_2', \ldots\rangle_{\mathrm{B}} = \prod_i \delta_{n_i n_i'}.$$

For a fixed N these states span the symmetric N-particle Hilbert space $\mathcal{H}_N^{(\mathrm{B})}$. Combining the spaces for every particle number leads to the Bose Fock space,

$$\boxed{\mathcal{F}_{\mathrm{B}} = \bigoplus_{N=0}^{\infty} \mathcal{H}_N^{(\mathrm{B})}.}$$

The $N = 0$ subspace contains a distinguished normalized state, the *vacuum*, written $|0\rangle = |0, 0, \ldots\rangle$. A state with several occupied orbitals belongs to one fixed-N sector; a general Fock-space state may be a superposition of sectors.

1) Creation and annihilation operators

The occupation-number basis makes the effect of adding or removing one particle in orbital i explicit. Define

$$\boxed{\begin{aligned} \hat{a}_i^\dagger\,|\ldots, n_i, \ldots\rangle_{\mathrm{B}} &= \sqrt{n_i + 1}\,|\ldots, n_i + 1, \ldots\rangle_{\mathrm{B}}\,, \\ \hat{a}_i\,|\ldots, n_i, \ldots\rangle_{\mathrm{B}} &= \sqrt{n_i}\,|\ldots, n_i - 1, \ldots\rangle_{\mathrm{B}}\,. \end{aligned}}$$

The second formula gives zero when $n_i = 0$. The square-root factors follow from the normalized symmetric states, including the factorial $n_i!$. In particular,

$$\hat{a}_i\,|0\rangle = 0, \qquad |i\rangle = \hat{a}_i^\dagger\,|0\rangle\,, \qquad |n_1, n_2, \ldots\rangle_{\mathrm{B}} = \prod_i \frac{(\hat{a}_i^\dagger)^{n_i}}{\sqrt{n_i!}}\,|0\rangle\,.$$

The order of factors is immaterial for bosons. To check the normalization of the last expression, use the norm relation

$$\left\|\hat{a}_i^\dagger \left|n_i\right\rangle\right\|^2 = n_i + 1 \quad \Longrightarrow \quad \left\|(\hat{a}_i^\dagger)^{n_i} \left|0\right\rangle\right\|^2 = n_i!.$$

Thus the operator algebra reproduces the explicit permutation sum without referring to labels of individual particles.

2) Derivation of Bose commutation relations

For different modes $i \neq j$, creation of particles in modes i, j can be carried out in either order. Applying the preceding formulas twice yields

$$\hat{a}_i \hat{a}_j \left|\{n_k\}\right\rangle = \sqrt{n_i n_j} \left|\ldots, n_i - 1, \ldots, n_j - 1, \ldots\right\rangle = \hat{a}_j \hat{a}_i \left|\{n_k\}\right\rangle .$$

The same argument applies to two creation operators and to $\hat{a}_i \hat{a}_j^\dagger$ for $i \neq j$. For one fixed mode,

$$\hat{a}_i \hat{a}_i^\dagger \left|\{n_k\}\right\rangle = (n_i + 1) \left|\{n_k\}\right\rangle ,$$
$$\hat{a}_i^\dagger \hat{a}_i \left|\{n_k\}\right\rangle = n_i \left|\{n_k\}\right\rangle .$$

Their difference is the identity on every occupation-number basis state. We have therefore established

$$\boxed{[\hat{a}_i, \hat{a}_j] = 0, \quad [\hat{a}_i^\dagger, \hat{a}_j^\dagger] = 0, \quad [\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij} \hat{I}.}$$

Here $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$. The positive sign in the last relation is characteristic of bosonic statistics.

3) Particle-number operators and independent particles

Define the occupation-number operator for mode i by

$$\hat{n}_i = \hat{a}_i^\dagger \hat{a}_i, \qquad \hat{n}_i \left|\{n_k\}\right\rangle = n_i \left|\{n_k\}\right\rangle .$$

Summing over all modes gives the total particle-number operator,

$$\boxed{\hat{N} = \sum_i \hat{n}_i = \sum_i \hat{a}_i^\dagger \hat{a}_i.}$$

For a noninteracting system with one-particle energies ϵ_i, the many-particle Hamiltonian acts on an occupation state as

$$\hat{H}_0 \left|\{n_i\}\right\rangle = \left(\sum_i n_i \epsilon_i\right) \left|\{n_i\}\right\rangle ,$$

which is reproduced exactly by

$$\boxed{\hat{H}_0 = \sum_i \epsilon_i \hat{a}_i^\dagger \hat{a}_i.}$$

This formula is a first example of a sum over individually identical one-particle operators represented entirely in terms of occupation modes.

4) General one-particle operator

Consider a one-particle observable $\hat{t}$ and the additive many-particle operator

$$\hat{\mathcal{O}} = \sum_{\alpha=1}^{N} \hat{t}_\alpha, \qquad t_{ij} = \left\langle i \middle| \hat{t} \middle| j \right\rangle .$$

The completeness of the one-particle basis gives

$$\hat{t} = \sum_{ij} t_{ij} \left|i\right\rangle \left\langle j\right| , \qquad \hat{\mathcal{O}} = \sum_{ij} t_{ij} \left(\sum_{\alpha=1}^{N} \left|i\right\rangle_\alpha \left\langle j\right|_\alpha \right) .$$

The operator in parentheses removes one particle from $|j\rangle$ and returns one in $|i\rangle$. On a bosonic occupation state with $i \neq j$, there are n_j possible removals and a corresponding symmetrization factor, which together produce

$$\begin{aligned} \left(\sum_\alpha \left|i\right\rangle_\alpha \left\langle j\right|_\alpha \right) \left|\ldots, n_i, \ldots, n_j, \ldots\right\rangle_{\mathrm{B}} &= \sqrt{n_j(n_i+1)}\, \left|\ldots, n_i+1, \ldots, n_j-1, \ldots\right\rangle_{\mathrm{B}} \\ &= \hat{a}_i^\dagger \hat{a}_j \left|\{n_k\}\right\rangle_{\mathrm{B}} . \end{aligned}$$

For $i = j$ it simply counts n_i particles. Since the actions agree on a complete Fock basis,

$$\boxed{\hat{\mathcal{O}} = \sum_{ij} t_{ij} \hat{a}_i^\dagger \hat{a}_j .}$$

The noninteracting Hamiltonian above follows by setting $t_{ij} = \epsilon_i \delta_{ij}$. Notice the operator order: first the particle is removed from j, and then one is created in i.

5) Two-particle interactions

For a symmetric pair potential $V(x_1, x_2) = V(x_2, x_1)$, the first-quantized operator counts each unordered pair once:

$$\hat{V} = \sum_{\alpha<\beta} V(\hat{x}_\alpha, \hat{x}_\beta) = \frac{1}{2} \sum_{\alpha\neq\beta} V(\hat{x}_\alpha, \hat{x}_\beta).$$

Introduce the *unsymmetrized* orbital matrix elements

$$V_{ij;kl} = \int \mathrm{d}x\, \mathrm{d}y\, \varphi_i^*(x) \varphi_j^*(y) V(x,y) \varphi_k(x) \varphi_l(y).$$

They obey $V_{ij;kl} = V_{ji;lk}$ for a symmetric interaction. Acting on an occupation state with the ordered product $\hat{a}_l \hat{a}_k$ removes the two incoming particles, and $\hat{a}_i^\dagger \hat{a}_j^\dagger$ inserts the two outgoing particles. Summing over the orbital indices exactly reconstructs the matrix elements of $\sum_{\alpha\neq\beta} V_{\alpha\beta}$; the factor $1/2$ removes double counting. Thus

$$\boxed{\hat{V} = \frac{1}{2} \sum_{ijkl} V_{ij;kl} \hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_l \hat{a}_k .}$$

The ordering of k, l on the right is deliberate and will be retained for fermions. This formula uses unsymmetrized matrix elements; antisymmetrizing those matrix elements would change the appropriate prefactor and indexing conventions.

7.2 Fermions: Antisymmetric States and Operator Algebra

Fermionic many-particle states are completely antisymmetric under exchanging any two particles. From the preceding chapter, the normalized wavefunction made of N distinct single-particle orbitals is a Slater determinant:

$$\Psi_{\rm F}(x_1,\ldots,x_N)=\frac{1}{\sqrt{N!}}\det[\varphi_{i_a}(x_b)]_{a,b=1}^{N}.$$

If any two orbitals coincide, two rows of the determinant are identical and the wavefunction vanishes. This is the Pauli exclusion principle: the occupation number of each one-particle state is either zero or one.

1) Fermionic Fock space and ordered occupation states

Choose a fixed ordering $1<2<\cdots$ of the one-particle orbitals. The antisymmetric states with different occupation lists form an orthonormal basis,

$$|n_1,n_2,\ldots\rangle_{\rm F}\,,\qquad n_i\in\{0,1\},\qquad N=\sum_i n_i.$$

The full fermionic Fock space is the direct sum

$$\mathcal{F}_{\rm F}=\bigoplus_{N=0}^{\infty}\mathcal{H}_N^{(\rm F)},$$

with vacuum $|0\rangle$. The fixed ordering is essential: exchanging the order of fermion creation operators changes the sign of the state.

Define creation operators by the ordered convention

$$\boxed{|n_1,n_2,\ldots\rangle_{\rm F}=\prod_{i=1}^{\infty}(\hat a_i^\dagger)^{n_i}\,|0\rangle\,,}$$

where the product is taken in increasing order of i, from left to right. Since adding a particle to an occupied orbital would produce a repeated column in the Slater determinant,

$$(\hat a_i^\dagger)^2=0.$$

Exchanging two distinct orbitals reverses the Slater determinant sign, hence

$$\hat a_i^\dagger\hat a_j^\dagger=-\hat a_j^\dagger\hat a_i^\dagger\qquad(i\neq j).$$

2) Fermionic creation and annihilation operators

To put a newly created operator into the fixed orbital order, it must pass every occupied lower-index mode. Define the parity count

$$\nu_i=\sum_{j<i}n_j.$$

The creation and annihilation actions are then

$$\boxed{\begin{aligned}\hat a_i^\dagger\,|\ldots,n_i,\ldots\rangle_{\rm F}&=(1-n_i)(-1)^{\nu_i}\,|\ldots,n_i+1,\ldots\rangle_{\rm F}\,,\\ \hat a_i\,|\ldots,n_i,\ldots\rangle_{\rm F}&=n_i(-1)^{\nu_i}\,|\ldots,n_i-1,\ldots\rangle_{\rm F}\,.\end{aligned}}$$

Thus creation at an already occupied mode and annihilation at an empty mode both give zero. For the vacuum, $\hat a_i\,|0\rangle=0$ as in the bosonic case. The sign is the number of already occupied orbitals passed when inserting or removing a fermion.

3) Derivation of anticommutation relations

Apply the operators in the two possible orders to an arbitrary occupation state. For the same mode,

$$\hat{a}_i\hat{a}_i^\dagger \left|\{n_k\}\right\rangle_{\mathrm{F}} = (1-n_i)\left|\{n_k\}\right\rangle_{\mathrm{F}},$$
$$\hat{a}_i^\dagger\hat{a}_i \left|\{n_k\}\right\rangle_{\mathrm{F}} = n_i \left|\{n_k\}\right\rangle_{\mathrm{F}}.$$

Their sum is the identity. For distinct $i<j$, exchanging the order of the two operations changes the number of occupied orbitals through which one operator passes by one. Therefore

$$\hat{a}_i\hat{a}_j^\dagger \left|\{n_k\}\right\rangle_{\mathrm{F}} = -\hat{a}_j^\dagger\hat{a}_i \left|\{n_k\}\right\rangle_{\mathrm{F}}, \quad i\neq j,$$

and analogously for two creation or two annihilation operators. Define the anticommutator $\{\hat{A},\hat{B}\} = \hat{A}\hat{B}+\hat{B}\hat{A}$. We have

$$\boxed{\{\hat{a}_i,\hat{a}_j\}=0, \quad \{\hat{a}_i^\dagger,\hat{a}_j^\dagger\}=0, \quad \{\hat{a}_i,\hat{a}_j^\dagger\}=\delta_{ij}\hat{I}.}$$

In particular, the nilpotence of the creation operator follows again from $\{\hat{a}_i^\dagger,\hat{a}_i^\dagger\}=0$. The occupation-number operator has the same formal expression as for bosons,

$$\hat{n}_i = \hat{a}_i^\dagger\hat{a}_i, \qquad \hat{N} = \sum_i \hat{a}_i^\dagger\hat{a}_i,$$

but its eigenvalues are restricted to $0,1$.

4) One-particle operators and the importance of order

Take the additive operator $\hat{\mathcal{O}} = \sum_{\alpha=1}^{N}\hat{t}_\alpha$, and let $t_{ij} = \left\langle i\middle|\hat{t}\middle|j\right\rangle$. As for bosons, each term $|i\rangle_\alpha \langle j|_\alpha$ removes one occupied orbital j and replaces it by i. In the ordered Slater determinant, the signs from removing and inserting exactly match those in $\hat{a}_i^\dagger\hat{a}_j$. Thus on every antisymmetric basis state,

$$\sum_{\alpha=1}^{N} |i\rangle_\alpha \langle j|_\alpha = \hat{a}_i^\dagger\hat{a}_j,$$

and hence

$$\boxed{\hat{\mathcal{O}} = \sum_{ij} t_{ij}\hat{a}_i^\dagger\hat{a}_j.}$$

No special parity factor is inserted outside the operators: the fermionic sign is already contained in the creation and annihilation algebra.

For a symmetric two-particle interaction, apply the same reasoning with two incoming and two outgoing orbitals. Use the unsymmetrized matrix elements $V_{ij;kl}$ defined in the boson section. With the explicit operator order

$$\hat{a}_i^\dagger\hat{a}_j^\dagger\hat{a}_l\hat{a}_k,$$

particles are first annihilated from k and l, and then created in j and i, with all signs supplied by anticommutation. One obtains

$$\boxed{\hat{V} = \frac{1}{2}\sum_{ijkl} V_{ij;kl}\hat{a}_i^\dagger\hat{a}_j^\dagger\hat{a}_l\hat{a}_k.}$$

The factor $1/2$ counts each *unordered particle pair* once. It is independent of Bose or Fermi statistics when $V_{ij;kl}$ is defined by unsymmetrized orbital integrals.

5) Unified form for bosons and fermions

Both particle species have the same *normal-ordered* form of a general one- plus two-body Hamiltonian:

$$\boxed{\begin{aligned}\hat{H} = &\sum_{ij} h_{ij}\hat{a}_i^\dagger \hat{a}_j \\ &+ \frac{1}{2}\sum_{ijkl} V_{ij;kl}\hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_l \hat{a}_k,\end{aligned}}$$

where $h_{ij} = \left\langle i\middle|\hat{p}^2/(2m) + \hat{U}\middle|j\right\rangle$. The entire difference between the statistics resides in the canonical commutator or anticommutator relations and the allowed occupation numbers:

	Bosons	Fermions
n_i	$0, 1, 2, \ldots$	$0, 1$
$\hat{a}_i\hat{a}_j^\dagger \mp \hat{a}_j^\dagger\hat{a}_i$	δ_{ij}	δ_{ij}

Here the minus sign applies to bosons and the plus sign to fermions. For a one-body potential U, its matrix elements are included in h_{ij}; interactions between two particles are included in $V_{ij;kl}$. This is the general result toward which the lecture derivations have been building.

Chapter 8

Relativistic Quantum Mechanics

8.1 Relativistic Equations of Motion

Difficulties of nonrelativistic quantum mechanics:

1) It has no relativistic covariance (momentum and energy are not treated symmetrically, nor are space and time).

2) Spin cannot be introduced self-consistently within the theory.

3) Probability is conserved, and there is no particle creation or annihilation.

Passing from nonrelativistic to relativistic quantum mechanics is not simply the replacement

$$E = \frac{\boldsymbol{p}^2}{2m} \quad \longrightarrow \quad E^2 = \boldsymbol{p}^2 c^2 + m^2 c^4.$$

In the course of its development, a theory for relativistic many-particle systems—quantum field theory—was eventually established, with relativistic quantum mechanics representing an intermediate stage.

Relativistic quantum mechanics is an important tool for studying some quantum systems (for example, graphene). More importantly, it provides a foundation for establishing quantum field theory.

1) Natural units

There are three independent dimensions commonly used in physics: length $[L]$, time $[T]$, and energy $[E]$. In relativistic quantum mechanics, especially in the quantum field theory developed from it, a new system of units is introduced. Take the speed of light c as the unit of velocity and $\hbar$ as the unit of action. Then only one independent unit remains, which may be chosen to be energy $[E]$.

Thus in the new unit system one may set $\hbar = c = 1$, so that $\hbar$ and c no longer appear explicitly in calculations.

2) Klein–Gordon equation

Consider the wave equation for a free particle.

Nonrelativistic energy:

$$E = \frac{\boldsymbol{p}^2}{2m}.$$

Relativistic energy:

$$E^2 = \boldsymbol{p}^2 + m^2.$$

Apply the canonical quantization prescription

$$E \longrightarrow \mathrm{i}\frac{\partial}{\partial t}, \qquad \boldsymbol{p} \longrightarrow -\mathrm{i}\boldsymbol{\nabla}.$$

One obtains the nonrelativistic Schrödinger equation

$$\mathrm{i}\frac{\partial \psi}{\partial t} = -\frac{\boldsymbol{\nabla}^2}{2m}\psi,$$

and the relativistic Klein–Gordon equation

$$(\partial^\mu \partial_\mu + m^2)\psi = 0, \qquad \partial^\mu \partial_\mu = \partial_t^2 - \boldsymbol{\nabla}^2.$$

Multiply the Klein–Gordon equation on the left by ψ^*, and multiply its complex conjugate on the right by ψ:

$$\psi^*(\partial^\mu \partial_\mu + m^2)\psi = 0, \qquad [(\partial^\mu \partial_\mu + m^2)\psi^*]\psi = 0.$$

Subtracting these two equations gives

$$\begin{aligned} \psi^*(\partial^\mu \partial_\mu \psi) - (\partial^\mu \partial_\mu \psi^*)\psi = 0, \\ \partial^\mu \left[\psi^*(\partial_\mu \psi) - (\partial_\mu \psi^*)\psi\right] = 0. \end{aligned}$$

This is a conserved-current (continuity) equation:

$$\partial^\mu j_\mu = 0, \qquad j_\mu = \psi^*(\partial_\mu \psi) - (\partial_\mu \psi^*)\psi.$$

Alternatively,

$$\partial_t \rho + \boldsymbol{\nabla} \cdot \boldsymbol{j} = 0,$$

where

$$\rho = \frac{\mathrm{i}}{2m}\left(\psi^* \frac{\partial \psi}{\partial t} - \frac{\partial \psi^*}{\partial t}\psi\right), \qquad \boldsymbol{j} = \frac{1}{2\mathrm{i}m}\left(\psi^* \boldsymbol{\nabla}\psi - \psi \boldsymbol{\nabla}\psi^*\right).$$

However, ρ is not positive definite, so it cannot be guaranteed that $\rho \geq 0$. This is because the Klein–Gordon equation is second order in time. Both ψ and $\partial\psi/\partial t$ are independent initial conditions; consequently, the above ρ may be negative. This is the problem of a non-positive-definite probability in the Klein–Gordon equation. A way to solve the negative-probability difficulty is to treat the Klein–Gordon equation as a field equation (quantum field theory). Then the current is an electric current rather than a probability current, and the charge density can be negative. By contrast, the Schrödinger equation is first order in time, and $\rho = \psi^*\psi$ is positive definite.

Furthermore, a plane wave of arbitrary energy E,

$$\psi = e^{-\mathrm{i}p^\mu x_\mu} = e^{-\mathrm{i}(Et - \boldsymbol{p}\cdot\boldsymbol{x})},$$

is a solution of the Klein–Gordon equation provided that $E^2 = \boldsymbol{p}^2 + m^2$. Hence plane waves with both

$$E = +\sqrt{\boldsymbol{p}^2 + m^2} \qquad \text{and} \qquad E = -\sqrt{\boldsymbol{p}^2 + m^2}$$

solve the equation. This is the negative-energy difficulty of the Klein–Gordon equation. In classical mechanics a particle initially in a positive-energy state has no transitions and therefore no negative-energy difficulty. In quantum mechanics, transitions can lead to the negative-energy difficulty. Note that the Schrödinger equation has no negative-energy difficulty. A way to solve this problem is to introduce antiparticles (quantum field theory).

3) Dirac equation

To address the negative-probability difficulty arising from a second-order time derivative, consider an equation that is first order in time. Relativistic covariance then requires treating the spatial derivatives as first order as well. Dirac therefore proposed the equation of motion for the wavefunction

$$(\mathrm{i}\gamma^\mu\partial_\mu - m)\psi = 0.$$

Here ψ is a four-component wavefunction in Dirac space (as shown below, the four components correspond to positive and negative energy and the two spin directions); it is called a *Dirac spinor wavefunction.* The γ^μ form a Lorentz four-vector of 4×4 matrices acting on Dirac space:

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix} \quad (i = 1, 2, 3).$$

They satisfy

$$\begin{aligned}
\{\gamma_\mu, \gamma_\nu\} &= \gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = 2g_{\mu\nu}I, \\
(\gamma^\mu)^\dagger &= \gamma^0\gamma^\mu\gamma^0, \\
\gamma^\mu\gamma_\mu &= g^\mu{}_\mu I = 4I, \\
\gamma^\mu\gamma^\nu\gamma_\mu &= -2\gamma^\nu, \\
\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu &= 4g^{\nu\rho}I, \\
\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu &= -2\gamma^\sigma\gamma^\rho\gamma^\nu.
\end{aligned}$$

$g_{\mu\nu}$ is the metric tensor of four-dimensional Minkowski space:

$$g_{\mu\nu} = g^{\mu\nu} = \mathrm{diag}(1, -1, -1, -1), \qquad g^\mu{}_\nu = g^{\mu\sigma}g_{\sigma\nu} = \mathrm{diag}(1, 1, 1, 1).$$

Since

$$\begin{aligned}
(\mathrm{i}\gamma^\mu\partial_\mu + m)(\mathrm{i}\gamma^\nu\partial_\nu - m)\psi &= 0, \\
-(\gamma^\mu\gamma^\nu\partial_\mu\partial_\nu + m^2)\psi &= 0, \\
\left[-\frac{1}{2}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu)\partial_\mu\partial_\nu - m^2\right]\psi &= 0, \\
\left[-\frac{1}{2}(2g^{\mu\nu})\partial_\mu\partial_\nu - m^2\right]\psi &= 0, \\
(\partial^\mu\partial_\mu + m^2)\psi &= 0,
\end{aligned}$$

each component of the four-vector ψ satisfies the Klein–Gordon equation. However, since the spinor also satisfies the Dirac equation, additional conditions apply. Thus positive- and negative-energy plane-wave solutions satisfying $E^2 = \boldsymbol{p}^2 + m^2$ are both solutions of the Dirac equation, showing that the Dirac equation also has the negative-energy difficulty.

Take the Hermitian conjugate of the Dirac equation:

$$\psi^\dagger(\mathrm{i}\gamma^{\mu\dagger}\overleftarrow{\partial}_\mu + m) = 0.$$

Using $(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0$, $(\gamma^0)^2 = I$, and defining the Dirac adjoint

$$\overline{\psi} = \psi^\dagger\gamma^0,$$

one obtains

$$\overline{\psi}(\mathrm{i}\gamma^\mu\overleftarrow{\partial}_\mu + m) = 0.$$

Multiplying the last equation on the right by ψ and the Dirac equation on the left by $\overline{\psi}$, and adding them gives

$$\overline{\psi}\gamma^\mu(\overleftarrow{\partial}_\mu + \partial_\mu)\psi = 0,$$
$$\partial_\mu(\overline{\psi}\gamma^\mu\psi) = 0.$$

Define the current

$$j^\mu = \overline{\psi}\gamma^\mu\psi.$$

The continuity equation becomes

$$\partial_\mu j^\mu = 0, \qquad j^0 = \rho = \overline{\psi}\gamma^0\psi = \psi^\dagger\psi \geq 0 \quad \text{(positive definite)}, \qquad \boldsymbol{j} = \overline{\psi}\boldsymbol{\gamma}\psi.$$

This shows that the Dirac equation does not suffer from a non-positive-definite probability.

8.2 Relativistic Invariance

1) Lorentz invariance

Consider two coordinate systems O and O', in which the Dirac equations are respectively

$$\mathrm{i}\gamma^\mu\frac{\partial}{\partial x^\mu}\psi(x) - m\psi(x) = 0, \qquad \mathrm{i}\gamma^\mu\frac{\partial}{\partial x'^\mu}\psi'(x') - m\psi'(x') = 0.$$

Assume that the coordinates x^μ and wavefunction ψ are related in the two coordinate systems by

$$x' = \Lambda x, \qquad \psi'(x') = S(\Lambda)\psi(x).$$

Here Λ and $S(\Lambda)$ are 4×4 matrices and are not functions of the coordinates. Since Lorentz scalars remain invariant under a transformation,

$$x'_\mu x'^\mu = x_\mu x^\mu,$$

the matrix Λ satisfies

$$\Lambda^\nu{}_\mu\Lambda^\mu{}_\rho = g^\nu{}_\rho$$

(in metric-contracted index notation). Taking determinants of the Lorentz-invariance condition gives

$$(\det\Lambda)^2 = 1, \qquad \det\Lambda = \pm 1.$$

A transformation with $\det\Lambda = +1$ is called a proper Lorentz transformation; one with $\det\Lambda = -1$ is called an improper Lorentz transformation.

Under the above transformation, the Dirac equation in the O' system is

$$\mathrm{i}\gamma^\mu\frac{\partial x^\nu}{\partial x'^\mu}\frac{\partial}{\partial x^\nu}S\psi(x) - mS\psi(x) = 0,$$
$$\mathrm{i}\gamma^\mu(\Lambda^{-1})^\nu{}_\mu\frac{\partial}{\partial x^\nu}S\psi(x) - mS\psi(x) = 0.$$

Multiplying this equation from the left by S^{-1} gives

$$\mathrm{i}S^{-1}\gamma^\mu S(\Lambda^{-1})^\nu{}_\mu\frac{\partial\psi(x)}{\partial x^\nu} - m\psi(x) = 0.$$

For the Dirac equation to be Lorentz invariant, i.e. for the O' and O equations to be equivalent, the transformation S must satisfy

$$S\gamma^\mu S^{-1} = (\Lambda^{-1})^\mu{}_\nu\gamma^\nu, \qquad \text{or} \qquad S^{-1}\gamma^\mu S = \Lambda^\mu{}_\nu\gamma^\nu.$$

This is the restriction on the transformation S imposed by relativistic invariance of the Dirac equation.

For example, consider an infinitesimal transformation

$$\Lambda^{\mu}{}_{\nu} = g^{\mu}{}_{\nu} + \omega^{\mu}{}_{\nu}, \qquad \omega^{\mu\nu} = -\omega^{\nu\mu},$$

where $\omega^{\mu\nu}$ is completely antisymmetric in its two indices, and

$$(\Lambda^{-1})^{\mu}{}_{\nu} = g^{\mu}{}_{\nu} - \omega^{\mu}{}_{\nu}, \qquad \omega^{\mu}{}_{\nu} = \omega^{\mu\sigma} g_{\sigma\nu}.$$

Take

$$S = I - \frac{\mathrm{i}}{4}\sigma_{\mu\nu}\omega^{\mu\nu}, \qquad S^{-1} = I + \frac{\mathrm{i}}{4}\sigma_{\mu\nu}\omega^{\mu\nu},$$

where the $\sigma_{\mu\nu}$ are 4×4 matrices to be determined. Substituting into the condition on S, one obtains

$$\begin{aligned}
\left(I - \frac{\mathrm{i}}{4}\sigma_{\mu\nu}\omega^{\mu\nu}\right)\gamma^{\rho}\left(I + \frac{\mathrm{i}}{4}\sigma_{\mu\nu}\omega^{\mu\nu}\right) &= \gamma^{\nu}(g^{\rho}{}_{\nu} - \omega^{\rho}{}_{\nu}), \\
\frac{\mathrm{i}}{4}(\gamma^{\rho}\sigma_{\mu\nu} - \sigma_{\mu\nu}\gamma^{\rho})\omega^{\mu\nu} &= -\gamma_{\nu}\omega^{\rho\nu}, \\
\frac{\mathrm{i}}{4}[\gamma^{\rho}, \sigma_{\mu\nu}]\omega^{\mu\nu} &= \frac{1}{2}(-\gamma_{\nu}\omega^{\rho\nu} - \gamma_{\mu}\omega^{\mu\rho}) \\
&= -\frac{1}{2}(\gamma_{\nu}g^{\rho}{}_{\mu} - \gamma_{\mu}g^{\rho}{}_{\nu})\omega^{\mu\nu}.
\end{aligned}$$

Hence $\sigma_{\mu\nu}$ can be chosen as

$$\sigma_{\mu\nu} = \frac{\mathrm{i}}{2}[\gamma_{\mu}, \gamma_{\nu}].$$

Clearly $\sigma_{\mu\nu} = -\sigma_{\nu\mu}$ is an antisymmetric tensor matrix.

2) Spin

Particle spin is automatically contained in the Dirac equation.

For an infinitesimal Lorentz transformation,

$$\psi'(x') = \left(I - \frac{\mathrm{i}}{4}\sigma_{\mu\nu}\omega^{\mu\nu}\right)\psi(x) = \left(I - \frac{\mathrm{i}}{4}\sigma_{\mu\nu}\omega^{\mu\nu}\right)\psi(\Lambda^{-1}x').$$

Performing a Taylor expansion,

$$\begin{aligned}
\psi'(x') &= \left(I - \frac{\mathrm{i}}{4}\sigma_{\mu\nu}\omega^{\mu\nu}\right)\left(1 + x'_{\mu}\omega^{\mu\nu}\partial'_{\nu}\right)\psi(x'), \\
\psi'(x) &= \left(I - \frac{\mathrm{i}}{4}\sigma_{\mu\nu}\omega^{\mu\nu}\right)(1 + x_{\mu}\omega^{\mu\nu}\partial_{\nu})\,\psi(x) \\
&= \left(I - \frac{\mathrm{i}}{4}\sigma_{\mu\nu}\omega^{\mu\nu} + x_{\mu}\omega^{\mu\nu}\partial_{\nu}\right)\psi(x) \\
&= \left[I - \frac{\mathrm{i}}{4}\sigma_{\mu\nu}\omega^{\mu\nu} + \frac{1}{2}(x_{\mu}\partial_{\nu} - x_{\nu}\partial_{\mu})\omega^{\mu\nu}\right]\psi(x).
\end{aligned}$$

Terms of second order in the infinitesimal parameters have been neglected.

For an infinitesimal rotation,

$$\psi'(x) = \left(I - \frac{\mathrm{i}}{2}J_{\mu\nu}\omega^{\mu\nu}\right)\psi(x),$$

where the angular-momentum tensor $J_{\mu\nu}$ generates rotations. Comparison with the preceding equation gives

$$J_{\mu\nu} = \frac{1}{2}\sigma_{\mu\nu} + \mathrm{i}(x_{\mu}\partial_{\nu} - x_{\nu}\partial_{\mu}).$$

Writing the angular-momentum components as

$$J_1 = J_{23}, \qquad J_2 = J_{31}, \qquad J_3 = J_{12},$$

we obtain the total angular momentum

$$\boldsymbol{J} = \frac{1}{2}\begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix} + \boldsymbol{r} \times \boldsymbol{p} = \frac{1}{2}\boldsymbol{\Sigma} + \boldsymbol{L}, \qquad \boldsymbol{p} = -\mathrm{i}\boldsymbol{\nabla}.$$

Thus the total angular momentum $\boldsymbol{J}$ consists of orbital angular momentum $\boldsymbol{L}$ and spin angular momentum $\boldsymbol{S} = \frac{1}{2}\boldsymbol{\Sigma}$.

The fact that the Dirac equation automatically contains spin is a relativistic effect. Specifically, this is a consequence of the Dirac equation containing first-order derivatives; the Klein–Gordon equation is second order and does not contain spin.

3) Spatial-inversion invariance (P transformation)

As shown above, invariance of the Dirac equation under $\psi'(x') = S\psi(x)$ requires

$$S^{-1}\gamma^\mu S = \Lambda^\mu{}_\nu \gamma^\nu.$$

For continuous transformations we can take $S = I - \frac{\mathrm{i}}{4}\sigma_{\mu\nu}\omega^{\mu\nu}$, where $\sigma_{\mu\nu} = \frac{\mathrm{i}}{2}[\gamma_\mu, \gamma_\nu]$. For a discontinuous transformation, such as parity,

$$x' = \Lambda x, \qquad \Lambda^{\mu\nu} = \mathrm{diag}(1, -1, -1, -1) = g^{\mu\nu}.$$

Choose

$$S = P = \gamma^0.$$

Then

$$S^{-1}\gamma^\mu S = \gamma^0\gamma^\mu\gamma^0 = g^{\mu\mu}\gamma^\mu = \Lambda^\mu{}_\nu\gamma^\nu \quad \text{(no sum in } \mu\text{)}.$$

This satisfies the condition on S, showing that the Dirac equation is invariant under spatial inversion.

4) Time-reversal invariance (T transformation)

For a time-reversal transformation,

$$x' = \Lambda x, \qquad \Lambda^{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1) = -g^{\mu\nu}.$$

Choose

$$S = T = \gamma_1\gamma_2\gamma_3.$$

Since

$$\begin{aligned} SS &= \gamma_1\gamma_2\gamma_3\gamma_1\gamma_2\gamma_3 \\ &= -\gamma_1\gamma_2\gamma_1\gamma_3\gamma_2\gamma_3 = \gamma_1\gamma_1\gamma_2\gamma_3\gamma_2\gamma_3 \\ &= -\gamma_1\gamma_1\gamma_2\gamma_2\gamma_3\gamma_3 = I, \end{aligned}$$

we have $S^{-1} = S$, and

$$S^{-1}\gamma^\mu S = S\gamma^\mu S = S\gamma^\mu\gamma_1\gamma_2\gamma_3 = \Lambda^\mu{}_\nu S\gamma_1\gamma_2\gamma_3\gamma^\nu = \Lambda^\mu{}_\nu SS\gamma^\nu = \Lambda^\mu{}_\nu\gamma^\nu.$$

This satisfies the condition on S and shows that the Dirac equation is time-reversal invariant.

5) Charge-conjugation invariance (C transformation)

Introduce an electromagnetic interaction by minimal coupling. Replace the derivative in the Dirac equation by the covariant derivative

$$\partial_\mu \longrightarrow D_\mu = \partial_\mu + \mathrm{i}eA_\mu,$$

where the electromagnetic potential $A_\mu = (A_0, \boldsymbol{A})$ is an external field, not a wavefunction. The Dirac equation is

$$(\mathrm{i}\gamma^\mu\partial_\mu - e\gamma^\mu A_\mu - m)\psi = 0.$$

It remains invariant under a local gauge transformation

$$\psi(x) \longrightarrow e^{\mathrm{i}e\alpha(x)}\psi(x), \qquad A_\mu(x) \longrightarrow A_\mu(x) - \partial_\mu\alpha(x).$$

Take the Hermitian conjugate of the Dirac equation and multiply from the right by γ^0:

$$\overline{\psi}\left(\mathrm{i}\gamma^\mu \overleftarrow{\partial}_\mu + e\gamma^\mu A_\mu + m\right) = 0.$$

Transpose:

$$\left(\mathrm{i}\gamma^{\mu\mathsf{T}}\partial_\mu + e\gamma^{\mu\mathsf{T}}A_\mu + m\right)\overline{\psi}^{\mathsf{T}} = 0.$$

Multiply on the left by the transformation matrix C:

$$\left(\mathrm{i}C\gamma^{\mu\mathsf{T}}C^{-1}\partial_\mu + eC\gamma^{\mu\mathsf{T}}C^{-1}A_\mu + m\right)C\overline{\psi}^{\mathsf{T}} = 0.$$

If C satisfies

$$C\gamma^{\mu\mathsf{T}}C^{-1} = -\gamma^\mu,$$

and we define

$$\psi^c = C\overline{\psi}^{\mathsf{T}},$$

then

$$(\mathrm{i}\gamma^\mu\partial_\mu + e\gamma^\mu A_\mu - m)\psi^c = 0.$$

Compared with the Dirac equation for a particle with charge e, the wavefunction ψ^c describes a particle of charge $-e$. The transformation C is called charge conjugation, or the particle–antiparticle transformation.

Note that C is an internal transformation, not a spacetime transformation.

6) Bilinear covariants

Physically observable quantities should be constructed from the Dirac spinor

$$\psi(x) = \begin{pmatrix}\psi_1(x)\\ \psi_2(x)\\ \psi_3(x)\\ \psi_4(x)\end{pmatrix}, \qquad \overline{\psi}(x) = \psi^\dagger(x)\gamma^0,$$

in the form of Dirac bilinears $\overline{\psi}\Gamma\psi$, with Γ a 4×4 matrix. What are their Lorentz-transformation properties?

First consider the transformation of $\overline{\psi}(x)$. Since

$$\psi'(x') = S\psi(x),$$

we have

$$\psi'^\dagger(x') = \psi^\dagger(x)S^\dagger,$$

and multiplication on the right by γ^0 gives

$$\overline{\psi}'(x') = \overline{\psi}(x)\gamma^0 S^\dagger \gamma^0.$$

Since

$$\gamma^0 \sigma^\dagger_{\mu\nu}\gamma^0 = \sigma_{\mu\nu}, \qquad \gamma^0 S^\dagger \gamma^0 = \gamma^0\left(I + \frac{\mathrm{i}}{4}\sigma^\dagger_{\mu\nu}\omega^{\mu\nu}\right)\gamma^0 = I + \frac{\mathrm{i}}{4}\sigma_{\mu\nu}\omega^{\mu\nu} = S^{-1},$$

it follows that

$$\overline{\psi}'(x') = \overline{\psi}(x)S^{-1}.$$

There are 16 linearly independent 4×4 matrices, which can be taken as

$$\Gamma \in \begin{cases} I & (1), \\ \gamma_\mu & (4), \\ \sigma_{\mu\nu} & (6), \\ \gamma_5 & (1), \\ \gamma_5\gamma_\mu & (4), \end{cases}$$

where

$$\gamma_5 = \mathrm{i}\gamma_0\gamma_1\gamma_2\gamma_3.$$

Its properties include

$$\{\gamma_5, \gamma_\mu\} = 0, \qquad \gamma_5^2 = I, \qquad S^{-1}\gamma_5 S = \gamma_5.$$

Using the transformation properties of Γ,

$$\begin{aligned} S^{-1}IS &= I, \\ S^{-1}\gamma_\mu S &= \Lambda_\mu{}^\nu \gamma_\nu, \\ S^{-1}\sigma_{\mu\nu}S &= \Lambda_\mu{}^\rho \Lambda_\nu{}^\sigma \sigma_{\rho\sigma}, \\ S^{-1}\gamma_5 S &= -\gamma_5, \\ S^{-1}\gamma_5\gamma_\mu S &= -\Lambda_\mu{}^\nu \gamma_5\gamma_\nu, \end{aligned}$$

we see that the 16 independent Dirac bilinears transform as

$$\begin{aligned} \overline{\psi}'(x')\psi'(x') &= \overline{\psi}(x)S^{-1}S\psi(x) = \overline{\psi}(x)\psi(x), && \text{Lorentz scalar,} \\ \overline{\psi}'(x')\gamma^\mu\psi'(x') &= \overline{\psi}(x)S^{-1}\gamma^\mu S\psi(x) = \Lambda^\mu{}_\nu\overline{\psi}(x)\gamma^\nu\psi(x), && \text{Lorentz vector,} \\ \overline{\psi}'(x')\gamma^5\psi'(x') &= \overline{\psi}(x)S^{-1}\gamma^5 S\psi(x) = \det\Lambda\,\overline{\psi}(x)\gamma^5\psi(x), && \text{Lorentz pseudoscalar,} \\ \overline{\psi}'(x')\gamma^5\gamma^\mu\psi'(x') &= \overline{\psi}(x)S^{-1}\gamma^5 S S^{-1}\gamma^\mu S\psi(x) = \det\Lambda\,\Lambda^\mu{}_\nu\overline{\psi}(x)\gamma^5\gamma^\nu\psi(x), && \text{Lorentz pseudovector,} \\ \overline{\psi}'(x')\sigma^{\mu\nu}\psi'(x') &= \overline{\psi}(x)S^{-1}\sigma^{\mu\nu}S\psi(x) = \Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma\overline{\psi}(x)\sigma^{\rho\sigma}\psi(x), && \text{Lorentz tensor.} \end{aligned}$$

The transformation S may be either continuous or a space-reflection transformation.

8.3 Plane-Wave Solutions of the Free Dirac Equation

The Dirac equation for a free particle is

$$(\mathrm{i}\gamma^\mu\partial_\mu - m)\psi(x) = 0.$$

Assume that positive- and negative-energy plane waves take the forms

$$\psi^+(x) = u(k)e^{-\mathrm{i}k^\mu x_\mu}, \qquad \psi^-(x) = v(k)e^{+\mathrm{i}k^\mu x_\mu}, \qquad k_0 = E = \sqrt{m^2 + \boldsymbol{k}^2}.$$

Here $u(k)$ and $v(k)$ are Dirac spinors. Substitution yields

$$(\gamma^\mu k_\mu - m)u(k) = 0, \qquad (\gamma^\mu k_\mu + m)v(k) = 0.$$

First consider the solutions in the particle rest frame, i.e. for $k_\mu = (m, \mathbf{0})$:

$$(\gamma_0 - I)u(m, \mathbf{0}) = 0, \qquad (\gamma_0 + I)v(m, \mathbf{0}) = 0.$$

Choose

$$\gamma_0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \qquad \boldsymbol{\gamma} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ -\boldsymbol{\sigma} & 0 \end{pmatrix}.$$

I is the identity matrix and $\boldsymbol{\sigma}$ the Pauli matrices. Thus each of u and v has two independent solutions:

$$u_\uparrow(m, \mathbf{0}) = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad u_\downarrow(m, \mathbf{0}) = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix},$$

$$v_\uparrow(m, \mathbf{0}) = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad v_\downarrow(m, \mathbf{0}) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Let the Lorentz transformations from the particle rest frame to a moving frame be

$$u_\alpha(k) = S_u u_\alpha(m, \mathbf{0}), \qquad v_\alpha(k) = S_v v_\alpha(m, \mathbf{0}), \qquad \alpha = \uparrow, \downarrow .$$

Substituting the equations satisfied by $u(k)$ and $v(k)$ gives

$$(\gamma^\mu k_\mu - m)S_u u_\alpha(m, \mathbf{0}) = 0, \qquad (\gamma^\mu k_\mu + m)S_v v_\alpha(m, \mathbf{0}) = 0.$$

Note that

$$(\gamma^\mu k_\mu - m)(\gamma^\mu k_\mu + m) = (\gamma^\mu k_\mu + m)(\gamma^\mu k_\mu - m) = k^2 - m^2 = 0.$$

Therefore one may take

$$S_u = A(\gamma^\mu k_\mu + m), \qquad S_v = B(\gamma^\mu k_\mu - m),$$

where A and B are constants. Hence

$$u_\alpha(k) = A(\gamma^\mu k_\mu + m)u_\alpha(m, \mathbf{0}) = A \begin{pmatrix} (E + m)\varphi_\alpha(m, \mathbf{0}) \\ \boldsymbol{\sigma} \cdot \boldsymbol{k}\, \varphi_\alpha(m, \mathbf{0}) \end{pmatrix},$$

$$v_\alpha(k) = B(\gamma^\mu k_\mu - m)v_\alpha(m, \mathbf{0}) = -B \begin{pmatrix} \boldsymbol{\sigma} \cdot \boldsymbol{k}\, \varphi_\alpha(m, \mathbf{0}) \\ (E + m)\varphi_\alpha(m, \mathbf{0}) \end{pmatrix}.$$

Here φ_α is a two-component spinor, with

$$\varphi_\uparrow(m, \mathbf{0}) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \varphi_\downarrow(m, \mathbf{0}) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

To determine A and B from the normalization conditions, define

$$\overline{u}_\alpha = u_\alpha^\dagger \gamma^0, \qquad \overline{v}_\alpha = v_\alpha^\dagger \gamma^0.$$

Taking Hermitian conjugates,

$$\overline{u}_\alpha(k) = \overline{u}_\alpha(m, \mathbf{0})(\gamma^\mu k_\mu + m)A = A\left(\varphi_\alpha^{\mathsf{T}}(m, \mathbf{0})(E + m), -\varphi_\alpha^{\mathsf{T}}(m, \mathbf{0})\boldsymbol{\sigma} \cdot \boldsymbol{k}\right),$$

$$\overline{v}_\alpha(k) = \overline{v}_\alpha(m, \mathbf{0})(\gamma^\mu k_\mu - m)B = -B\left(-\varphi_\alpha^{\mathsf{T}}(m, \mathbf{0})\boldsymbol{\sigma} \cdot \boldsymbol{k}, \varphi_\alpha^{\mathsf{T}}(m, \mathbf{0})(E + m)\right).$$

Define the orthonormality relations

$$\begin{aligned}
\overline{u}_\alpha(k)u_\beta(k) &= \delta_{\alpha\beta}, \\
\overline{v}_\alpha(k)v_\beta(k) &= -\delta_{\alpha\beta}, \\
\overline{u}_\alpha(k)v_\beta(k) &= 0, \\
\overline{v}_\alpha(k)u_\beta(k) &= 0.
\end{aligned}$$

The constants are then

$$\boxed{A = -B = \frac{1}{\sqrt{2m(m+E)}}.}$$